



CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DUNG

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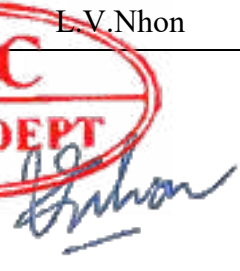
Comply:
ISO 9001:2015

WELDING PROCEDURE REGISTER

(Doc No.: DDC-QAQC-APF26-WPS-001 Rev.00)

DỰ ÁN: TRUNG TÂM TỔ CHỨC HỘI NGHỊ APEC

HẠNG MỤC: TRUNG TÂM HỘI NGHỊ VÀ TRIỂN LÃM (NHÀ S3) - HỆ MÁI NHÔM

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	May 08, 2026	All	IFC	N.Q.Sa	L.V.Nhon
					

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**DAI DUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION**

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WELDING PROCEDURE REGISTER
(Doc. No.: DDDC-QAQC-APF26-WPS-001 Rev.00)**Project:** TRUNG TÂM TỔ CHỨC HỘI NGHỊ APEC
Item: TRUNG TÂM HỘI NGHỊ VÀ TRIỂN LÃM (NHÀ S3) - HỆ MÁI NHÔM

No.	WPS No.	Rev.	Standard	Process	In PQR		Range Qualified										Supporting PQR No.	Remarks	
					Position	Base Metal	Material Group	Thickness (mm)		Diameter (mm) O.D. (*)	Position		Consumable/ Filler Metal						
								Groove	Fillet		Groove	Fillet	AWS Spec.	AWS Class.	F-No.	Size			
1	WPS-DD-GTAW-304	0	AWS D1.6:2017	GTAW	2G	ASTM A240 304	Any M-8 Group I o M-8 Group I	2 to 20 mm	All	All	All	All	All	AWS A5.9	ER308L	6	Ø2.4	PQR-DD-GTAW-304	
2	WPS-DD-FCAW-308	0	AWS D1.6:2017	FCAW	2G	ASTM A240 304	Any M-8 Group I to M-8 Group I	2 to 20 mm	All	All	All	All	All	AWS A5.22	E308LT1-1/-4	6	Ø1.2	PQR-DD-FCAW-308	
3	WPS-DD-FCAW-333	0	AWS D1.6:2017	FCAW	2F	ASTM A240 304 to JIS G3101 SS400	Any M-1 Group I to M-8 Group I	2 to 20 mm	All	All	All	All	All	AWS A5.22	E309LT1-1/-4	6	Ø1.2	PQR-DD-FCAW-333	
4	WPS-DD-FCAW-334	0	AWS D1.6:2017	FCAW	2F	ASTM A240 304 to ASTM A36	Any M-1 Group I to M-8 Group I	2 to 20 mm	All	All	All	All	All	AWS A5.22	E309LT1-1/-4	6	Ø1.2	PQR-DD-FCAW-334	
5	WPS-DD-FCAW-324	0	AWS D1.3:2018	FCAW	1G; 2G 1F; 2F	JIS G3101 SS400	JIS G3101 SS400 ASTM A36 (Any Group I, II)	1-4 mm	1-4 mm	All	F, H	F, H	AWS A5.20	E71T-1C/-9C	6	Ø1.2	PQR-DD-FCAW-324		
6	WPS-DD-GMAW-325	0	AWS D1.3:2018	GMAW	1G 2G 2F; 3F	JIS G3101 SS400	JIS G3101 SS400 ASTM A36 (Any Group I, II)	1-4 mm	0.8-3.2 mm	All	F, H	F, H, V	AWS A.18	ER70S-6	6	Ø0.8	PQR-DD-GMAW-325		
7	WPS-DD-FCAW-283	1	AWS D1.1:2020	FCAW	6G	JIS G3444 STK 400	JIS G3444 STK400 JIS G3101 SS400	3 - 20 mm (Square Groove: 3- 10 mm)	3-Unlimited	O.D≥26.7mm for CJP; any O.D. for PJP and Fillet	All uphill (PJP: T,K,Y)	All uphill	A5.20	E71T-1C	6	Ø1.2	PQR-DD-FCAW-283.1 & 283.2		
8	WPS-DD-FCAW-284	1	AWS D1.1:2020	FCAW	6G	ASTM A53 Gr B	Any Group I	3 - 20 mm (Square Groove: 3- 10 mm)	3-Unlimited	O.D≥26.7mm for CJP; any O.D. for PJP and Fillet	All uphill (PJP: T,K,Y)	All uphill	A5.20	E71T-1C	6	Ø1.2	PQR-DD-FCAW-284.1 & 284.2		

Note:**DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION**Prepared by: Nguyen Quynh Sa
Position: Welding Coordinator
Signature:

Date: 8/May/26

Reviewed by: La Vinh Nhon
Position: D.QA Manager
Signature:

Date: 8/May/26

Comply:
ISO 9001:2015
AWS D1.6:2017

WELDING PROCEDURE

Doc. No.: WPS-DD-GTAW-304
PQR-DD-GTAW-304

Rev	Date	Page	Description	Prepared by	Checked and Approved by
1	Nov 09, 2024	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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WELDING PROCEDURE SPECIFICATION (WPS)

(AWS D1.6/D1.6M:2017 Structural Welding Code-Stainless Steel)



Company Name: DaiDung Metallic Manufacture Construction & Trade Corporation
WPS No.: WPS-DD-GTAW-304.1
Supporting PQR(s): PQR-DD-GTAW-304
CVN Report: -
Rev. No.: A
Date: Nov 9, 2024
Page: 1 of 1

BASE METALS	Specification	Type or Grade	AWS Group No.	BASE METAL THICKNESS	As-Welded (mm)	With PWHT
Base Material	ASTM A240/276/312/403 or equivalent	304/316	Metal Group A,B Table 5.2	CJP Groove Welds	-	-
Welded To		Any M-8		CJP Groove with CVN	-	-
Backing Material	-	-	-	PJP Groove Welds	-	-
Other	-	-	-	Fillet Welds	All	-
				DIAMETER	All	-

JOINT DETAILS		JOINT SKETCH	
Joint Type	T, Corner & Lap Joint		
Weld Type	Fillet Weld		
Root Opening	0 - 2 (mm)		
Root Face	-		
Dihedral	Per drawing		
Weld Size	Per drawing		
POSTWELD HEAT TREATMENT		Welded Joint Details as Figures 5.2 AWS D1.6	
Temperature	-		
Time at Temperature	-		
Other	-		

PROCEDURE			
Weld Layer(s)		All	
Weld Pass(es)		All	
Process		GTAW	
Type (Manual, Mechanized, etc.)		Manual	
Position		All	
Vertical Progression		-	
Filler Metal (AWS Spec.)		AWS A5.9	
AWS Classification		ER308L - Group B Table 5.3	
Diameter (mm)		1.6 - 2.4	
Manufacturer/Trade Name		Not strictly	
Shielding Gas (Composition)		99.9% Argon	
Flow Rate (LPM)		14 - 25	
Nozzle Size		No. 6	
Preheat Temperature (Min.)		15°C (Ambient)	
Interpass Temperature (Max.)		175°C	
Electrical Characteristics			
Current Type & Polarity		DCEN	
Transfer Mode		-	
Power Source Type (cc, cv, etc.) (Amps)		CC	
Amps (A)		90 - 180	
Volts (V)		11 - 14	
Wire Feed Speed		-	
Travel Speed (mm/min.)		85 - 130	
Maximum Heat Input (kJ/mm)		1.4	
Technique			
Stringer or Weave		Stringer or Weave	
Multi or Single Pass (per side)		Multi or Single Pass	
Oscillation (Mech./Auto.)		-	
Number of Electrodes		1	
Contact Tube to Work Dist. (mm)		15 - 25	
Peening		None	
Interpass Cleaning		Grinding and Wire Brush	
Other		This procedure can be used for build up or repair.	

Prepared by Welding Coordinator		Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature			
Date	Nov 9, 2024	Nov 9, 2024	





WELDING PROCEDURE SPECIFICATION (WPS)

(AWS D1.6/D1.6M:2017 Structural Welding Code-Stainless Steel)



Company Name: DaiDung Metallic Manufacture Construction & Trade Corporation
WPS No.: WPS-DD-GTAW-304.2
Supporting PQR(s): PQR-DD-GTAW-304
CVN Report: -
Rev. No.: A
Date: Nov 9, 2024
Page: 1 of 1

BASE METALS	Specification	Type or Grade	AWS Group No.	BASE METAL THICKNESS	As-Welded (mm)	With PWHT
Base Material	ASTM A240/276/312/403 or equivalent	304/316	Metal Group A,B	CJP Groove Welds	-	-
Welded To		Any M-8		CJP Groove with CVN	-	-
Backing Material	-	-	-	PJP Groove Welds	2-20 mm	-
Other	-	-	-	Fillet Welds	-	-
				DIAMETER	All	-

JOINT DETAILS			JOINT SKETCH	
Joint Type	Butt, T & Corner Joint			
Weld Type	Single V	Single Bevel		
	Groove	Groove Weld		
Groove Angle	60° - 70°	45° - 55°		
Root Opening	0 - 2 mm			
Root Face	Per drawing			
Weld Size	Table 5.5			
POSTWELD HEAT TREATMENT				
Temperature	-			
Time at Temperature	-			
Other	-			

Welded Joint Details as Figures 5.3 AWS D1.6

PROCEDURE			
Weld Layer(s)		All	
Weld Pass(es)		All	
Process		GTAW	
Type (Manual, Mechanized, etc.)		Manual	
Position		All	
Vertical Progression		-	
Filler Metal (AWS Spec.)		AWS A5.9	
AWS Classification		ER308L - Group B Table 5.3	
Diameter	(mm)	1.6 - 2.4	
Manufacturer/Trade Name		Not strictly	
Shielding Gas (Composition)		99.9% Argon	
Flow Rate	(LPM)	14 - 25	
Nozzle Size		No. 6	
Preheat Temperature	(Min.)	15°C (Ambient)	
Interpass Temperature	(Max.)	175°C	
Electrical Characteristics			
Current Type & Polarity		DCEN	
Transfer Mode		-	
Power Source Type (cc, cv, etc.)	(Amps)	CC	
Amps	(A)	90 - 180	
Volts	(V)	11 - 14	
Wire Feed Speed		-	
Travel Speed	(mm/min.)	85 - 130	
Maximum Heat Input	(kJ/mm)	1.4	
Technique			
Stringer or Weave		Stringer or Weave	
Multi or Single Pass (per side)		Multi or Single Pass	
Oscillation (Mech./Auto.)		-	
Number of Electrodes		1	
Contact Tube to Work Dist.	(mm)	15 - 25	
Peening		None	
Interpass Cleaning		Grinding and Wire Brush	
Other		This procedure can be used for build up or repair.	

Prepared by Welding Coordinator		Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature			
Date	Nov 9, 2024	Nov 9, 2024	



		WELDING PROCEDURE SPECIFICATION (WPS) (AWS D1.6/D1.6M:2017 Structural Welding Code-Stainless Steel)			
Company Name WPS No. Supporting PQR(s) CVN Report		DaiDung Metallic Manufacture Construction & Trade Corporation WPS-DD-GTAW-304.3 PQR-DD-GTAW-304 -			
		Rev. No.	A	Date:	Nov 9, 2024
					Page: 1 of 2

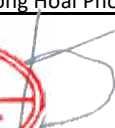
BASE METALS	Specification	Type or Grade	AWS Group No.
Base Material	ASTM A240/276/312/403 or equivalent	304/316	Metal Group A,B Table 5.2
Welded To		Any M-8	
Backing Material	-	-	-
Other	-		

BASE METAL THICKNESS	As-Welded (mm)	With PWHT
CJP Groove Welds	2 - 20	-
CJP Groove with CVN	-	-
PJP Groove Welds	-	-
Fillet Welds	-	-
DIAMETER	All	-

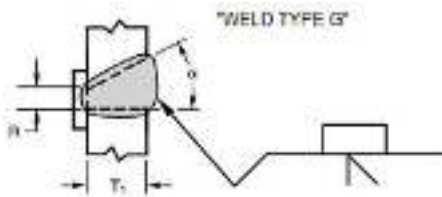
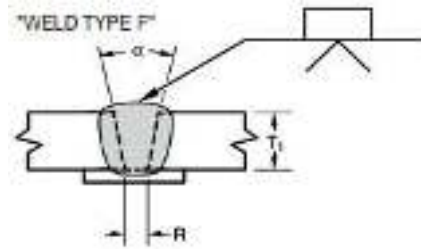
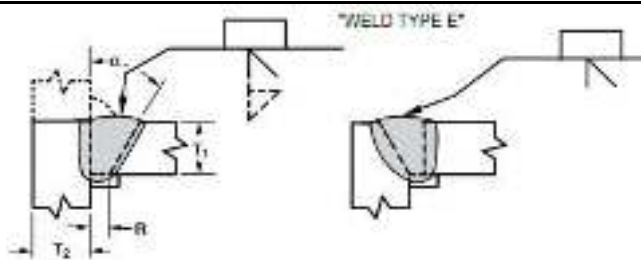
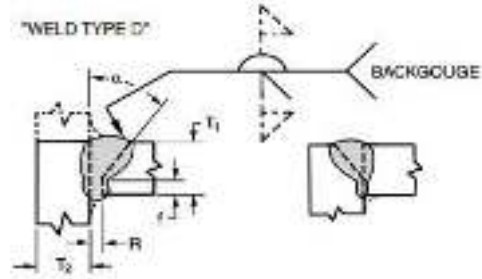
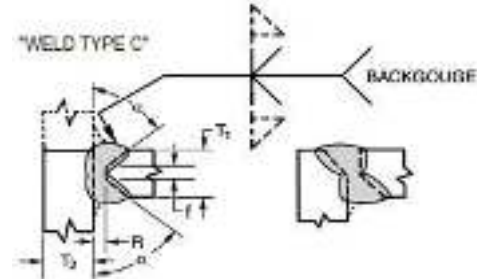
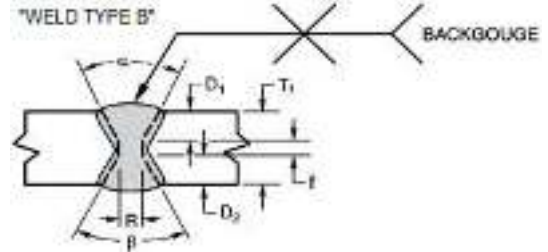
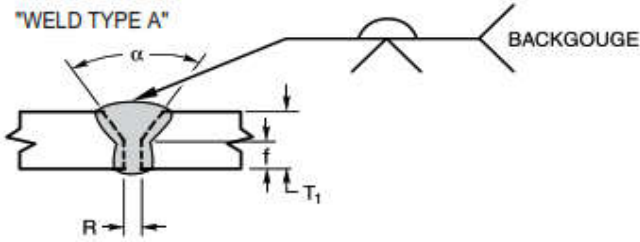
JOINT DETAILS	
Groove Type	See attached sketch for more details
Groove Angle	See attached sketch for more details
Root Opening	See attached sketch for more details
Root Face	See attached sketch for more details
Backgouging	Yes
Method	Grinding

POSTWELD HEAT TREATMENT	
Temperature	-
Time at Temperature	-
Other	-

PROCEDURE			
Weld Layer(s)	All		
Weld Pass(es)	All		
Process	GTAW		
Type (Manual, Mechanized, etc.)	Manual		
Position	All		
Vertical Progression	-		
Filler Metal (AWS Spec.)	AWS A5.9		
AWS Classification	ER308L - Group B Table 5.3		
Diameter (mm)	1.6 - 2.4		
Manufacturer/Trade Name	Not strictly		
Shielding Gas (Composition)	99.9% Argon		
Flow Rate (LPM)	14 - 25		
Nozzle Size	No. 6		
Preheat Temperature (Min.)	15°C (Ambient)		
Interpass Temperature (Max.)	175°C		
Electrical Characteristics			
Current Type & Polarity	DCEN		
Transfer Mode	-		
Power Source Type (cc, cv, etc.) (Amps)	CC		
Amps (A)	90 - 180		
Volts (V)	11 - 14		
Wire Feed Speed	-		
Travel Speed (mm/min.)	85 - 130		
Maximum Heat Input (kJ/mm)	1.4		
Technique			
Stringer or Weave	Stringer or Weave		
Multi or Single Pass (per side)	Multi or Single Pass		
Oscillation (Mech./Auto.)	-		
Number of Electrodes	1		
Contact Tube to Work Dist. (mm)	15 - 25		
Peening	None		
Interpass Cleaning	Grinding and Wire Brush		
Other	This procedure can be used for build up or repair.		

	Prepared by Welding Coordinator	Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature	 		
Date	Nov 9, 2024	Nov 9, 2024	

CJP

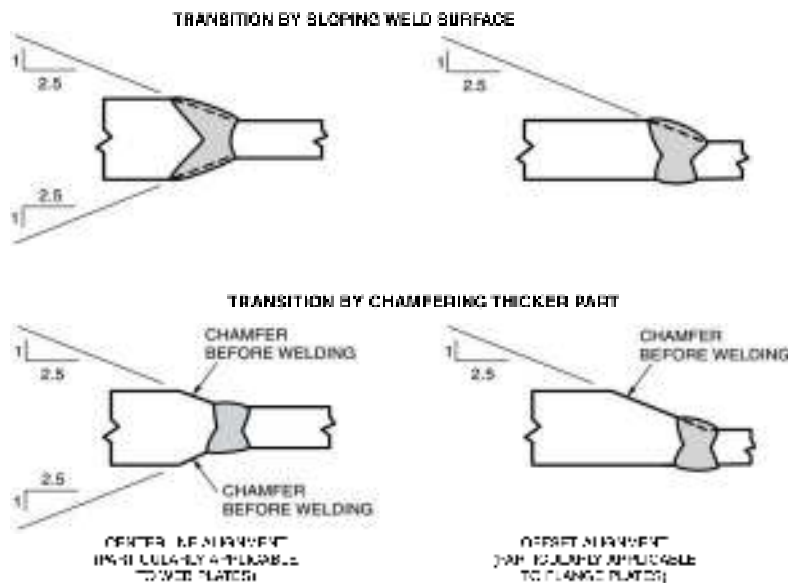


For "WELD TYPE E, F, G":

The permanent backing metal can be used and in conformance with design drawing

Weld Type	Joint Data				Remark
	Root Opening - R (mm)	Root Face - f (mm)	Groove Angel 1 - α	Groove Angle2 - β	
A	0 - 5	0 - 5	$60^{\circ} - 70^{\circ}$	-	Welded Joint Details shall be as described in AWS D1.6, Figure 5.4
B	0 - 5	0 - 5	$60^{\circ} - 70^{\circ}$	$60^{\circ} - 70^{\circ}$	
C	0 - 5	0 - 5	$45^{\circ} - 55^{\circ}$	$45^{\circ} - 55^{\circ}$	
D	0 - 5	0 - 5	$45^{\circ} - 55^{\circ}$	-	
E, F, G	6 - 8	-	$45^{\circ} - 55^{\circ}$	-	

Transition of Splice Joint in Parts of Unequal Section



These Transition shall be as per structure design

Note: Joint data shall be as per the Code AWS D1.6 and in conformance with Project Specification, Drawings.



PROCEDURE QUALIFICATION RECORD (PQR)

(Structural Welding Code-Stainless Steel AWS D1.6/D1.6M:2017)



Company Name: **DaiDung Metallic Manufacture Construction & Trade Corporation**

PQR No.: **PQR-DD-GTAW-304**

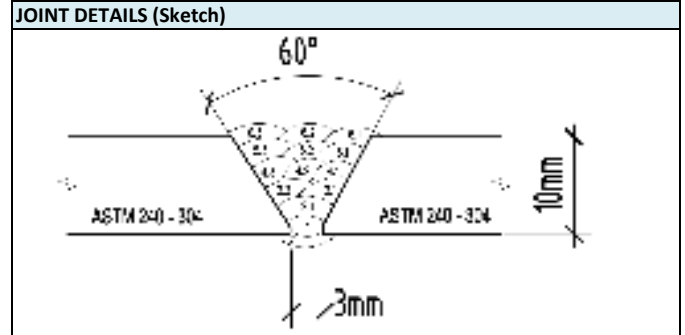
Rev. No. **0**

Date: **Nov 9, 2024**

Page: **1 of 2**

BASE METALS	Specification	Type or Grade	AWS Group No.	Thickness	Size (NPS)	Schedule	Diameter
Base Material	ASTM A240	304	Metal Group B	10 mm	-	-	-
Welded To	ASTM A240	304	Metal Group B	10 mm	-	-	-
Backing Material	Ceramic	-	-	-	-	-	-
Other	-	-	-	-	-	-	-

JOINT DETAILS	
Groove Type	Single V Groove Butt Joint
Groove Angle	60°
Root Opening	3-4 mm
Root Face	1 mm
Backgouging	Yes
Method	Grinding



POSTWELD HEAT TREATMENT	
Temperature	N/a
Time at Temperature	-
Other	-

PROCEDURE					
Weld Layer(s)	1	2 - 5	6	7	
Weld Pass(es)	1	2.1 - 5.3	6.1 - 6.3	7	
Process	GTAW	GTAW	GTAW	GTAW	
Type (Manual, Mechanized, etc.)	Manual				
Position	2G	2G	2G	2G	
Vertical Progression	-	-	-	-	
Filler Metal (AWS Spec.)	AWS A5.9	AWS A5.9	AWS A5.9	AWS A5.9	
AWS Classification	ER308L				
Diameter (mm)	Ø 2.4	Ø 2.4	Ø 2.4	Ø 2.4	
Manufacturer/Trade Name	Hyundai ST-308L				
Shielding Gas (Composition)	Argon (99.9%)	Argon (99.9%)	Argon (99.9%)	Argon (99.9%)	
Flow Rate (LPM)	15-20	15-20	15-20	15-20	
Nozzle Size	No. 6	No. 6	No. 6	No. 6	
Preheat Temperature (Min.)	33 °C (ambient)				
Interpass Temperature (Max.)		135 °C	110°C	45 °C	
Electrical Characteristics					
Current Type & Polarity	DCEN	DCEN	DCEN	DCEN	
Transfer Mode	-	-	-	-	
Power Source Type (cc, cv, etc.)	CC	CC	CC	CC	
Amps (A)	95 - 114	132 - 162	126 - 134	132 - 134	
Volts (V)	11 - 12	11 - 13	11 - 13	11 - 12	
Wire Feed Speed	-	-	-	-	
Travel Speed (mm/min.)	88	109 - 119	108 - 119	115	
Maximum Heat Input (kJ/mm)	0.9	1.2	0.9	0.8	
Technique					
Stringer or Weave	Weave	Weave	Weave	Weave	
Multi or Single Pass (per side)	Multipass	Multipass	Multipass	Multipass	
Oscillation (Mech./Auto.)	-	-	-	-	
Number of Electrodes	1	1	1	1	
Contact Tube to Work Dist.(mm)	20	20	20	20	
Peening	None	None	None	None	
Interpass Cleaning	Wire Brush	Grinding	Wire Brush	Wire Brush	
Other	-	-	-	-	

Prepared by Welding Coordinator		Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature			
Date	Nov 9, 2024	Nov 9, 2024	





PROCEDURE QUALIFICATION RECORD (PQR)

(Structural Welding Code-Stainless Steel AWS D1.6/D1.6M:2017)



PQR No.: PQR-DD-GTAW-304

Rev. No.: 0

Date: Nov 9, 2024

Page: 2 of 2

TEST RESULTS

TESTS

	Type of Tests	Clause/Figure(s) Reference	Acceptance Criteria	Result	Remarks
✓	Visual Inspection	Figure 6.5	6.9.3.1	Acceptable	WPR-DD-GTAW-304
	Radiographic Examination	8.14.1	8.12		
	Ultrasonic Testing	8.14.3	8.13		
	Penetrant Testing	8.14.4	8.10 & Table 8.1		
	Magnetic Particle Testing	8.14.5	8.10 & Table 8.1		
	2 Side Bends	Figure 6.6	6.9.3.2		
	4 Side Bends	Figure 6.6	6.9.3.2		
✓	2 Transverse Root Bends	Figure 6.7 or Figure 6.8	6.9.3.2	Acceptable	INL24/512130-799-1
✓	2 Transverse Face Bends	Figure 6.7 or Figure 6.8	6.9.3.2	Acceptable	INL24/512130-799-1
	2 Longitudinal Face Bends	Figure 6.9	6.9.3.2		
	2 Longitudinal Root Bends	Figure 6.9	6.9.3.2		
✓	2 Tensile Tests	Figure 6.14	6.9.3.3	Acceptable	INL24/512130-799-1
✓	3 Macroetch	Annex I	6.9.3.4	Acceptable	INL24/512130-799-1

TENSILE TEST DETAILS

Specimen Number	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Tensile Load (kN)	Ultimate Unit Stress (MPa)	Type of Failure and Location
TS 1	20.1	9.7	195.1	117.4	602	Ductile/ In weld
TS 2	20.1	9.8	195.8	115.7	591	Ductile/ In weld

TOUGHNESS TEST DETAILS

Specimen Number	Notch Location	Specimen Size (mm)	Test Temperature	Absorbed Energy (J)	Percent Shear	Lateral Expansion	Average (J)

CERTIFICATION

Welder's Name	ID Number	Stamp Number	Tests Conducted by
NGUYEN VAN DEM	-	DDC-WD-0564	Laboratory
-	-	-	SGS
-	-	-	Test Number
			INL24/512130-799-1
			File Number
			-

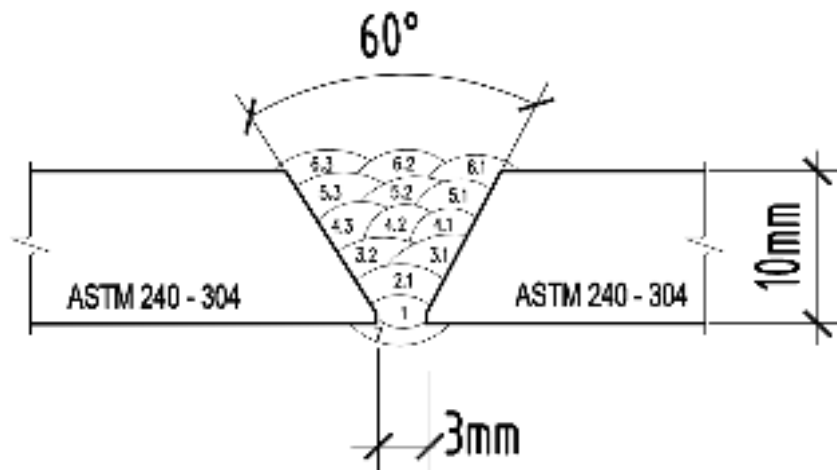
We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Clause 6 of AWS D1.6/D1.6M, 2017 *Structural Welding Code-Steel*.

Prepared by Welding Coordinator	Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)
Signature	 	
Date	Nov 9, 2024	Nov 9, 2024

WELDING PARAMETERS RECORD

Company name DaiDung Metallic Manufacture Construction & Trade Corporation
PQR No. PQR-DD-GTAW-304
Test Coupon DD-GTAW-304
Test position 2G
Base metal:
- Specification ASTM A240, 304
- Dimension PL10x200x600 mm to PL10x200x600 mm
- Heat/Product No. Y210531025-3
- Certificaion No. 20210717107
Filler metal:
- Specification AWS A5.9
- Classification ER308L
- Trade name Hyundai ST-308L
- Batch/Lot No. 1228105402
- Certificates No. 2302013300004
Gas: Argon (99.9%)
Welder Name: NGUYEN VAN DEM
I.D: DDC-WD-0564

Joint Sketch:



Root face: 1 mm
Root gap: 3-4 mm
Groove angle: 60°

Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
1	GTAW	AWS A5.9 ER308L	2.4	DCEN	95 - 114	11 - 12	6	47	600	88	33 (Ambient)	15-20	0.7 - 0.9	8	3	
2	GTAW	AWS A5.9 ER308L	2.4	DCEN	137 - 139	11 - 12	5	23	600	111	46	15-20	0.8 - 0.9	9	2	
3.1	GTAW	AWS A5.9 ER308L	2.4	DCEN	156 - 158	12 - 13	5	21	600	112	55	15-20	1 - 1.1	10	2	
3.2	GTAW	AWS A5.9 ER308L	2.4	DCEN	156 - 158	12 - 13	5	5	600	118	45	15-20	1 - 1	10		
4.1	GTAW	AWS A5.9 ER308L	2.4	DCEN	132 - 134	12 - 13	5	31	600	109	87	15-20	0.9 - 1	10	2	
4.2	GTAW	AWS A5.9 ER308L	2.4	DCEN	132 - 134	12 - 13	5	15	600	114	95	15-20	0.8 - 0.9	10		

Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		Volts (V)	min.						sec.	Width	
4.3	GTAW	AWS A5.9/ ER308L	2.4	DCEN	132 - 134	12 - 13	5	3	600	119	107	15-20	0.8 - 0.9	10		
5.1	GTAW	AWS A5.9 ER308L	2.4	DCEN	159 - 162	12 - 13	5	31	600	109	110	15-20	1.1 - 1.2	11	1	
5.2	GTAW	AWS A5.9 ER308L	2.4	DCEN	159 - 162	12 - 13	5	14	600	115	125	15-20	1 - 1.1	11		
5.3	GTAW	AWS A5.9/ ER308L	2.4	DCEN	159 - 162	12 - 13	5	3	600	119	135	15-20	1 - 1.1	11		
6.1	GTAW	AWS A5.9/ ER308L	2.4	DCEN	126 - 128	12 - 13	5	7	600	117	70	15-20	0.8 - 0.9	11	2	Reinforcement
6.2	GTAW	AWS A5.9 ER308L	2.4	DCEN	126 - 128	12 - 13	5	32	600	108	90	15-20	0.8 - 0.9	11		
6.3	GTAW	AWS A5.9/ ER308L	2.4	DCEN	126 - 128	12 - 13	5	3	600	119	110	15-20	0.8 - 0.8	11		
7	GTAW	AWS A5.9/ ER308L	2.4	DCEN	132 - 134	11 - 12	5	14	600	115	45	15-20	0.8 - 0.8	11	2	Reinforcement
		Prepared by Welding Coordinator				Reviewed and Approved by										
Name		Nguyen Bao Tinh				Truong Hoai Phong Vu										
Signature																
Date		Nov 1, 2024				Nov 1, 2024										

Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
Client's address / Địa chỉ : LÔ SỐ 38, KHU C, ĐƯỜNG D1, KHU CÔNG NGHIỆP AN HẠ, XÃ PHẠM VĂN HAI, HUYỆN BÌNH CHÁNH, THÀNH PHỐ HỒ CHÍ MINH, VIỆT NAM

The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : DD-GTAW-304
Base material: ASTM A240 304
Size: 10x400x600 mm
Welding consumable: AWS A5.9 ER308L
Trade name : Hyundai ST-308L (2.4mm)
Type of Joint: Single V groove, Butt joint
Position 2G
Welder : Nguyen Van Dem - DDC-WD-0564

Project / Dự án : LTIA-PBB (PASSENGER BOARDING BRIDGE)
Quantity / Số lượng mẫu : 01 sample / mẫu
Description / Mô tả mẫu : 600 x 400 x 10 (mm)
Date of receiving / Ngày nhận mẫu : 05-Nov-2024
Date of testing / Ngày thử nghiệm : From / từ 05-Nov-2024 to / đến 07-Nov-2024
Test method / Phương pháp thử : AWS D1.6/D1.6M:2017 Structural Welding Code - Stainless Steel
Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Signed for and on behalf of
SGS Vietnam LTD

Thay mặt công ty

SGS SGS Vietnam Ltd

Chau Tran Khanh Dung
Laboratory Supervisor
Industrial Services

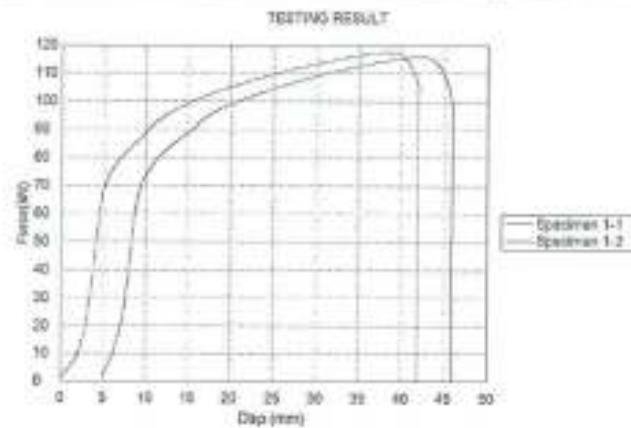
Please extract your time to give us your feedback as this will help our service better. Thank you!



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Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm	
A. Transverse tension test / Thử kéo theo phương ngang	1	2
• Thickness of specimen / Chiều dày mẫu mm	9.72	9.75
• Width of specimen / Chiều rộng mẫu mm	20.07	20.08
• Cross section area / Tiết diện mm ²	195.1	195.8
• Tensile load / Lực kéo đứt kN	117.4	115.7
• Tensile strength / Giới hạn bền kéo MPa	602	591
• Broken position / Vị trí mẫu đứt	In weld	In weld



The relationship between tensile load and deformation / Biểu đồ tương quan giữa lực thử kéo và biến dạng

Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm			
B. Bend test / Thử uốn	1	2	3	4
• Location bend / Vị trí uốn	Face bend Uốn mặt	Face bend Uốn mặt	Root bend Uốn chân	Root bend Uốn chân
• Thickness of specimen / Chiều dày mm	9.76	9.76	9.75	9.76
• Width of specimen / Chiều rộng mm	40.06	40.02	39.99	40.03
• Bending angle / Góc uốn °/ độ	180	180	180	180
• Plunger diameter / Đường kính gối uốn mm	38	38	38	38
• Result / Kết quả	No Crack Không nứt	No Crack Không nứt	No Crack Không nứt	No Crack Không nứt
• Evaluation / Đánh giá	Pass / Đạt	Pass / Đạt	Pass / Đạt	Pass / Đạt

*** TO BE CONTINUED ***

041954

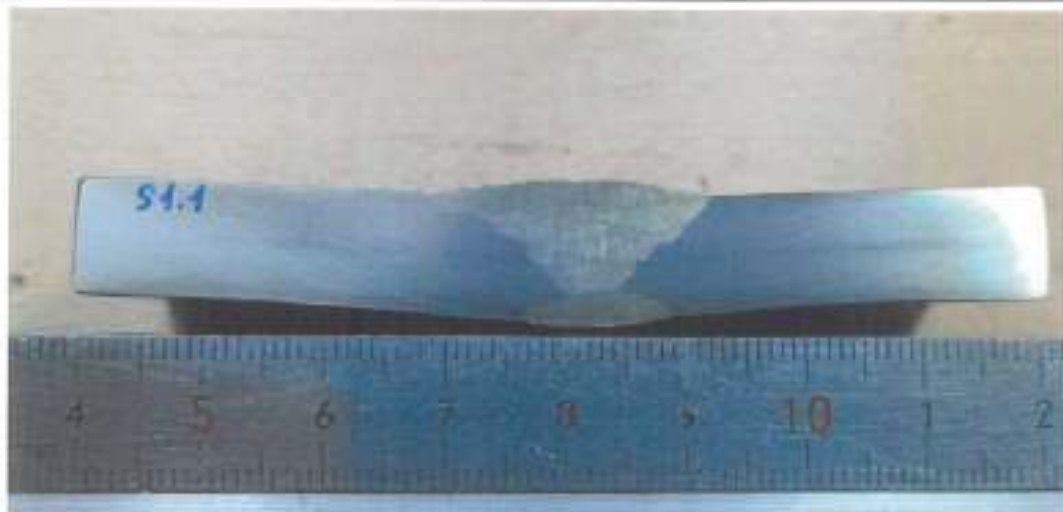
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C. Characteristic / Tên chỉ tiêu : Macro structure / Tổ chức thô đại

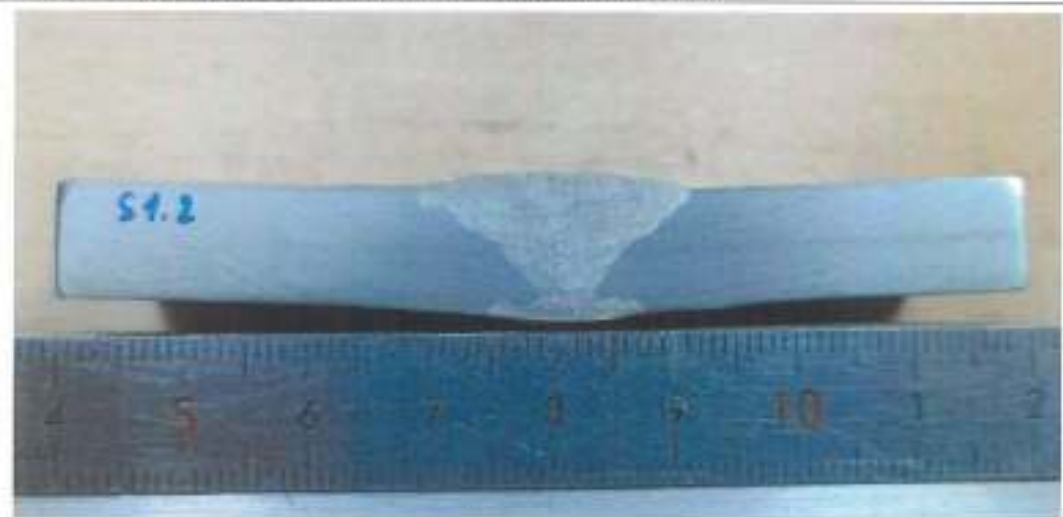
➤ **Specimen 1-1 / Mẫu 1-1 :**

- **Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn.**
- **No defect in weld metal / Trong mối hàn không có khuyết tật.**



➤ **Specimen 1-2 / Mẫu 1-2 :**

- **Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn.**
- **No defect in weld metal / Trong mối hàn không có khuyết tật.**



*** TO BE CONTINUED ***

041855

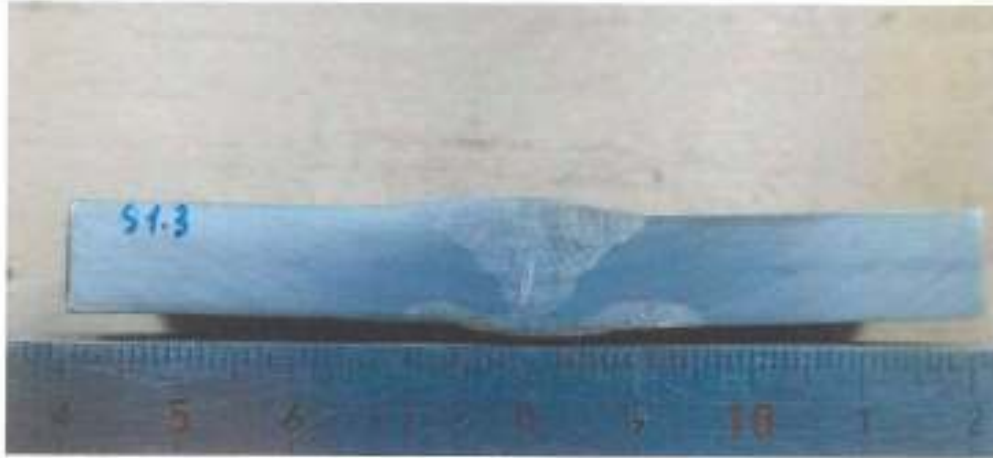
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D. Characteristic / Tên chỉ tiêu : Macro etch [□] / Tổ chức thô đại

➤ **Specimen 1-3 / Mẫu 1-3 :**

- **Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn.**
- **No defect in weld metal / Trong mối hàn không có khuyết tật.**



Original sample photo(s) / Hình ảnh mẫu nhận:



SGS authenticate the photos on original report only / SGS chỉ xác thực các hình ảnh trong báo cáo chính thức

*** END OF THE REPORT ***

041858

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MILL TEST CERTIFICATE

EN 10204/3.1



CNS 112666000

ADD: PT. INDU, Desa Fatmaja, Kecamatan Sukadana, Kabupaten
Korupali, Provinsi Sulawesi Tengah, Indonesia

Tel: 16202256701870

Email: liuy@etalingshan.com

Manufacturer: PT. INDONESIA GUNG CHING STEEL AND STAINLESS
STEEL INDUSTRY

Product		VIB-STAR INTERNATIONAL JOINT STOCK COMPANY		Rolling Factory		SPH+700+LP+CO		Contract No.		0011/000000-000						Certificate No.		20200111007		
Product Name		HOT ROLLED STAINLESS STEEL COIL		Steel Grade		304		Product Standard		ASTM A240						Destination of delivery		For rolling or other application		
No.	Coil No.	QTY	Heat No.	Size (mm)	Weight (kg)	Chemical Composition (%)										Tensile Test				Tolerance
						C	Si	Mn	P	S	Cr	Ni	Al	Fe	As	R _m (MPa)	R _{p0.2} (MPa)	F _{0.01} (MPa)	Elongation A (mm)	
1	0010010011	1	0010000011-0	5.00*1000*0	21.355	0.042	0.41	1.14	0.033	0.002	18.16	8.25	0.030	0.01	0.01	525.8	208.2	91.5	81.5	
2	0010010012	1	0010000012-0	5.00*1000*0	21.160	0.045	1.46	1.16	0.035	0.002	18.11	8.04	0.044	0.04	0.04	545.5	204.4	95.8	96.8	
3	0010010013	1	0010000013-0	5.00*1000*0	21.231	0.040	1.76	1.18	0.032	0.002	18.20	8.01	0.065	0.03	0.03	543.6	206.5	95.8	85.8	
4	0010010014	1	0010000014-0	5.00*1000*0	21.165	0.044	1.41	0.95	0.032	0.002	18.21	8.02	0.049	0.04	0.04	544.3	205.2	95.7	95.7	
5	0010010015	1	0010000015-0	5.00*1000*0	21.217	0.047	1.42	0.94	0.034	0.002	18.18	8.02	0.066	0.04	0.04	543.5	207.8	95.7	98.0	
6	0010010016	1	0010000016-0	5.00*1000*0	21.110	0.041	0.40	0.93	0.037	0.002	18.25	8.04	0.040	0.04	0.04	537.3	205.0	95.4	97.4	
7	0010010017	1	0010000017-0	5.00*1000*0	21.160	0.045	0.45	1.02	0.034	0.002	18.20	8.02	0.040	0.04	0.04	544.5	202.8	95.9	88.2	

Remarks

1. Product marking: steel number, heat No, coil No, size.

2. Surface Quality: ok 3. Shape: ok 4. Dimension: ok 5. Radioactivity Tests: ok 6. Non-metallic inclusions: ok 7. Intergranular corrosion: ok

1. We hereby certify that the material herein described has been manufactured, sampled, tested and inspected in accordance with the requirements of above specifications.

2. We hereby certify Product No-weld repair, Free of mercury, Free from radiation contamination.

3. ISO 9001:2015 Quality management system identified by ZJQC.

4. We declare that our testing center has passed ISO/IEC 17025:2017 accreditation review, Accreditation body: CNAS, accreditation capability range: Spectral analysis and tensile testing, certificate number: CNAS L12639.

5. We certify that the material described below fully conforms to IS 6811:2017 chemical composition and Mechanical properties of product as listed in accordance with the scheme of Testing and Inspection outlined in BIS certification marks license No. CNL-0100045662.

6. Issued in agreement with TÜV SÜD Industrie Service GmbH (April 2018), QMS approved acc. to PECCAD 2000, Annex 1, Para. 4.3 by Notified Body 0026 (Certification no. DQR-0036-03-W 13520-01-MUC).

7. Approved acc. to CPR 305/2011 with certification no. 0036-CPR-M-115-2010.

8. Product heat treatment temperature: 1100°C.

9. The certificate should be sealed by quality control department, add date, inspection stamp or authorized official stamp of sales.

Typed by	徐红叶	
Rectifying	刘	
Date	2021/7/17	

CERTIFIED MATERIAL REPORT

CMR No. : 2302013300004

Date of Issue : 2023-02-01

Customer	GLOBAL TECHNOLOGY COMMERCIAL CORPORA		
Trade Name	ST-308L	Size	2.4mm*1000mm*5kg
Lot No.	1228105402	Purchaser's Order No.	3322_JH_GT
Specification	AWS A5.9 ER308L		
Welding Current	DCEN	Shielding Gas	Ar
		Post Weld Heat Treatment (°C × hr.)	AS WELDED

Mechanical Properties

Weld Metal	Tensile Test			Impact Test (CVN - Joule)				
	YS(MPa)	TS(MPa)	EL(%)	Temp.(°C)	X1	X2	X3	Avg.
	456	585	46.4	-196	54	48	51	51
AWS Spec.	-	-	-	-				

Chemical Composition (%)

* According to EN 10204-2004 : 3.1

WIRE	C	Si	Mn	P	S	Cu	Ni	Cr	Mo		
	0.01	0.39	1.99	0.017	0.011	0.01	10.74	19.76	0.01		
AWS Spec.	≤0.03	0.30 -0.65	1.0 -2.5	≤0.03	≤0.03	≤0.75	9.0 -11.0	19.5 -22.0	≤0.75		

Others

* Q'ty : 300kg

ORIGINAL

We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.

Approved by

K.W. Park

QA Manager Park K. W.



HYUNDAI
WELDING

HYUNDAI WELDING CO., LTD.

Head Office

507, Teheran-ro, Gangnam-gu, Seoul, Republic of Korea

Pohang Plant

100, Daesong-ro, Nam-gu, Pohang, Republic of Korea

Comply:
ISO 9001:2015
AWS D1.6:2017

WELDING PROCEDURE

Doc. No.: WPS-DD-FCAW-308

PQR-DD-FCAW-308

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Nov 09, 2024	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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WELDING PROCEDURE SPECIFICATION (WPS)

(AWS D1.6/D1.6M:2017 Structural Welding Code-Stainless Steel)



Company Name: DaiDung Metallic Manufacture Construction & Trade Corporation
WPS No.: WPS-DD-FCAW-308.1
Supporting PQR(s): PQR-DD-FCAW-308
CVN Report: -
Rev. No.: A
Date: Nov 9, 2024
Page: 1 of 1

BASE METALS	Specification	Type or Grade	AWS Group No.	BASE METAL THICKNESS	As-Welded (mm)	With PWHT
Base Material	ASTM A240/276/312/403 or equivalent	304/316	Metal Group A,B Table 5.2	CJP Groove Welds	-	-
Welded To		Any M-8		CJP Groove with CVN	-	-
Backing Material	-	-	-	PJP Groove Welds	-	-
Other	-	-	-	Fillet Welds	All	-
				DIAMETER	All	-

JOINT DETAILS		JOINT SKETCH	
Joint Type	T, Corner & Lap Joint		
Weld Type	Fillet Weld		
Root Opening	0 - 2 (mm)		
Root Face	-		
Dihedral	Per drawing		
Weld Size	Per drawing		
POSTWELD HEAT TREATMENT		Welded Joint Details as Figures 5.2 AWS D1.6	
Temperature	-		
Time at Temperature	-		
Other	-		

PROCEDURE			
Weld Layer(s)		All	
Weld Pass(es)		All	
Process		FCAW	
Type (Manual, Mechanized, etc.)		Semi-Automatic, Mechanized	
Position		All	
Vertical Progression		-	
Filler Metal (AWS Spec.)		AWS A5.22	
AWS Classification		E308LT1-1/-4; Group B Table 5.3	
Diameter (mm)		1.2	
Manufacturer/Trade Name		Not strictly	
Shielding Gas (Composition)		99.8% CO2	
Flow Rate (LPM)		15 - 25	
Nozzle Size		No. 6	
Preheat Temperature (Min.)		15°C (Ambient)	
Interpass Temperature (Max.)		175°C	
Electrical Characteristics			
Current Type & Polarity		DCEP	
Transfer Mode		-	
Power Source Type (cc, cv, etc.) (Amps)		CV	
Amps (A)		100 - 180	
Volts (V)		26 - 30	
Wire Feed Speed (Amps)		(Amps)	
Travel Speed (mm/min.)		110 - 470	
Maximum Heat Input (kJ/mm)		2.28	
Technique			
Stringer or Weave		Stringer or Weave	
Multi or Single Pass (per side)		Multi or Single Pass	
Oscillation (Mech./Auto.)		-	
Number of Electrodes		1	
Contact Tube to Work Dist. (mm)		15 - 25	
Peening		None	
Interpass Cleaning		Grinding and Wire Brush	
Other	This procedure can be used for build up or repair.		

Prepared by Welding Coordinator		Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature			
Date	Nov 9, 2024	Nov 9, 2024	





WELDING PROCEDURE SPECIFICATION (WPS)

(AWS D1.6/D1.6M:2017 Structural Welding Code-Stainless Steel)



Company Name: DaiDung Metallic Manufacture Construction & Trade Corporation
WPS No.: WPS-DD-FCAW-308.2
Supporting PQR(s): PQR-DD-FCAW-308
CVN Report: -
Rev. No.: A
Date: Nov 9, 2024
Page: 1 of 1

BASE METALS	Specification	Type or Grade	AWS Group No.	BASE METAL THICKNESS	As-Welded (mm)	With PWHT
Base Material	ASTM A240/276/312/403 or equivalent	304/316	Metal Group A,B	CJP Groove Welds	-	-
Welded To		Any M-8		CJP Groove with CVN	-	-
Backing Material	-	-	-	PJP Groove Welds	2-20 mm	-
Other	-	-	-	Fillet Welds	-	-
				DIAMETER	All	-

JOINT DETAILS			JOINT SKETCH	
Joint Type	Butt, T & Corner Joint			
Weld Type	Single V	Single Bevel		
	Groove	Groove Weld		
Groove Angle	60° - 70°	45° - 55°		
Root Opening	0 - 2 mm			
Root Face	Per drawing			
Weld Size	Table 5.5			
POSTWELD HEAT TREATMENT				
Temperature	-			
Time at Temperature	-			
Other	-			

Welded Joint Details as Figures 5.3 AWS D1.6

PROCEDURE			
Weld Layer(s)	All		
Weld Pass(es)	All		
Process	FCAW		
Type (Manual, Mechanized, etc.)	Semi-Automatic, Mechanized		
Position	All		
Vertical Progression	-		
Filler Metal (AWS Spec.)	AWS A5.22		
AWS Classification	E308LT1-1/-4; Group B Table 5.3		
Diameter (mm)	1.2		
Manufacturer/Trade Name	Not strictly		
Shielding Gas (Composition)	99.8% CO2		
Flow Rate (LPM)	15 - 25		
Nozzle Size	No. 6		
Preheat Temperature (Min.)	15°C (Ambient)		
Interpass Temperature (Max.)	175°C		
Electrical Characteristics			
Current Type & Polarity	DCEP		
Transfer Mode	-		
Power Source Type (cc, cv, etc.) (Amps)	CV		
Amps (A)	100 - 180		
Volts (V)	26 - 30		
Wire Feed Speed	(Amps)		
Travel Speed (mm/min.)	110 - 470		
Maximum Heat Input (kJ/mm)	2.28		
Technique			
Stringer or Weave	Stringer or Weave		
Multi or Single Pass (per side)	Multi or Single Pass		
Oscillation (Mech./Auto.)	-		
Number of Electrodes	1		
Contact Tube to Work Dist. (mm)	15 - 25		
Peening	None		
Interpass Cleaning	Grinding and Wire Brush		
Other	This procedure can be used for build up or repair.		

Prepared by Welding Coordinator		Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature			
Date	Nov 9, 2024	Nov 9, 2024	



	WELDING PROCEDURE SPECIFICATION (WPS) (AWS D1.6/D1.6M:2017 Structural Welding Code-Stainless Steel)	
--	---	--

Company Name	DaiDung Metallic Manufacture Construction & Trade Corporation			
WPS No.	WPS-DD-FCAW-308.3	Rev. No.	A	Date: Nov 9, 2024
Supporting PQR(s)	PQR-DD-FCAW-308			Page: 1 of 2
CVN Report	-			

BASE METALS	Specification	Type or Grade	AWS Group No.	BASE METAL THICKNESS	As-Welded (mm)	With PWHT
Base Material	ASTM A240/276/312/403 or equivalent	304/316	Metal Group A,B Table 5.2	CJP Groove Welds	2 - 20	-
Welded To		Any M-8		CJP Groove with CVN	-	-
Backing Material	Ceramic or same base metal	-	-	PJP Groove Welds	-	-
Other	-			Fillet Welds	-	-
				DIAMETER	All	-

JOINT DETAILS	
Groove Type	See attached sketch for more details
Groove Angle	See attached sketch for more details
Root Opening	See attached sketch for more details
Root Face	See attached sketch for more details
Backgouging	Yes
Method	Grinding
POSTWELD HEAT TREATMENT	
Temperature	-
Time at Temperature	-
Other	-

JOINT SKETCH

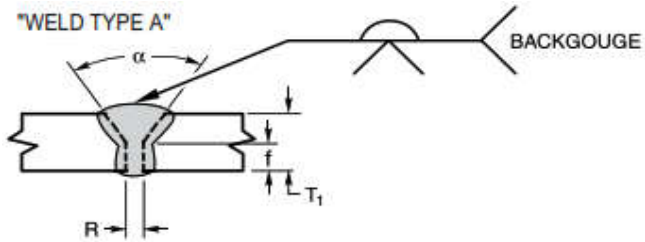
See Joint Sketch in page 2 of 2

PROCEDURE			
Weld Layer(s)		All	
Weld Pass(es)		All	
Process		FCAW	
Type (Manual, Mechanized, etc.)		Semi-Automatic, Mechanized	
Position		All	
Vertical Progression		-	
Filler Metal (AWS Spec.)		AWS A5.22	
AWS Classification		E308LT1-1/-4; Group B Table 5.3	
Diameter (mm)		1.2	
Manufacturer/Trade Name		Not strictly	
Shielding Gas (Composition)		99.8% CO2	
Flow Rate (LPM)		15 - 25	
Nozzle Size		No. 6	
Preheat Temperature (Min.)		15°C (Ambient)	
Interpass Temperature (Max.)		175°C	
Electrical Characteristics			
Current Type & Polarity		DCEP	
Transfer Mode		-	
Power Source Type (cc, cv, etc.) (Amps)		CV	
Amps (A)		100 - 180	
Volts (V)		26 - 30	
Wire Feed Speed		(Amps)	
Travel Speed (mm/min.)		110 - 470	
Maximum Heat Input (kJ/mm)		2.28	
Technique			
Stringer or Weave		Stringer or Weave	
Multi or Single Pass (per side)		Multi or Single Pass	
Oscillation (Mech./Auto.)		-	
Number of Electrodes		1	
Contact Tube to Work Dist. (mm)		15 - 25	
Peening		None	
Interpass Cleaning		Grinding and Wire Brush	
Other	This procedure can be used for build up or repair.		

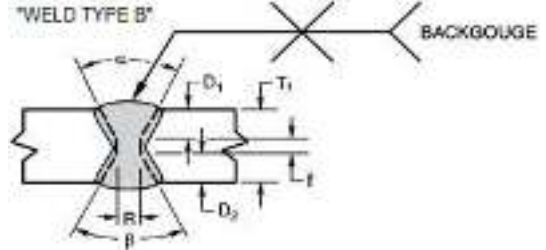
	Prepared by Welding Coordinator	Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature	  		
Date	Nov 9, 2024	Nov 9, 2024	

CJP

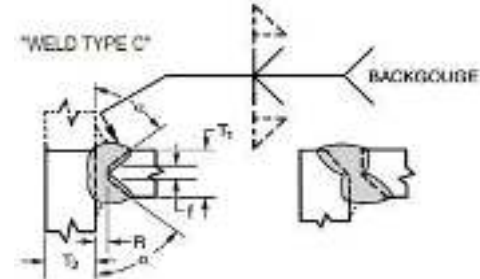
"WELD TYPE A"



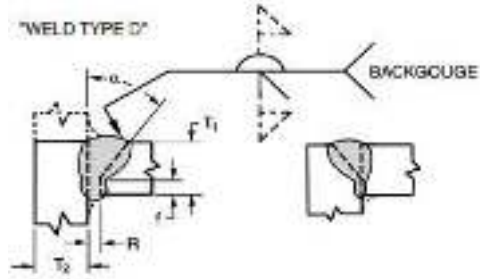
"WELD TYPE B"



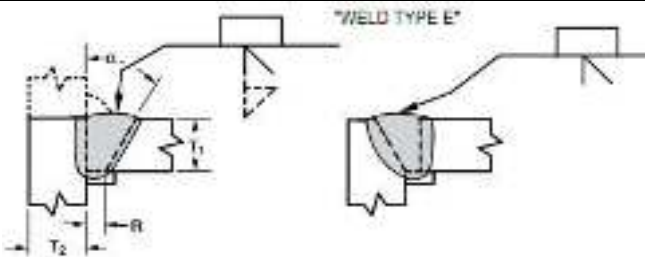
"WELD TYPE C"



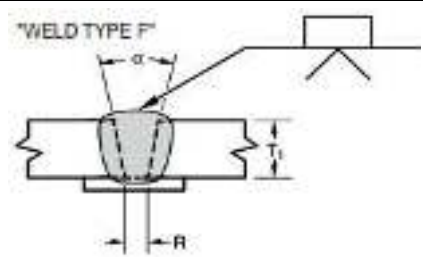
"WELD TYPE D"



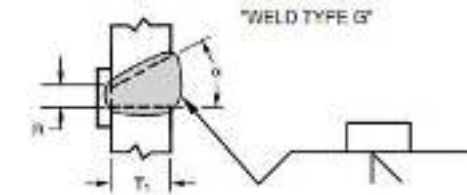
"WELD TYPE E"



"WELD TYPE F"



"WELD TYPE G"

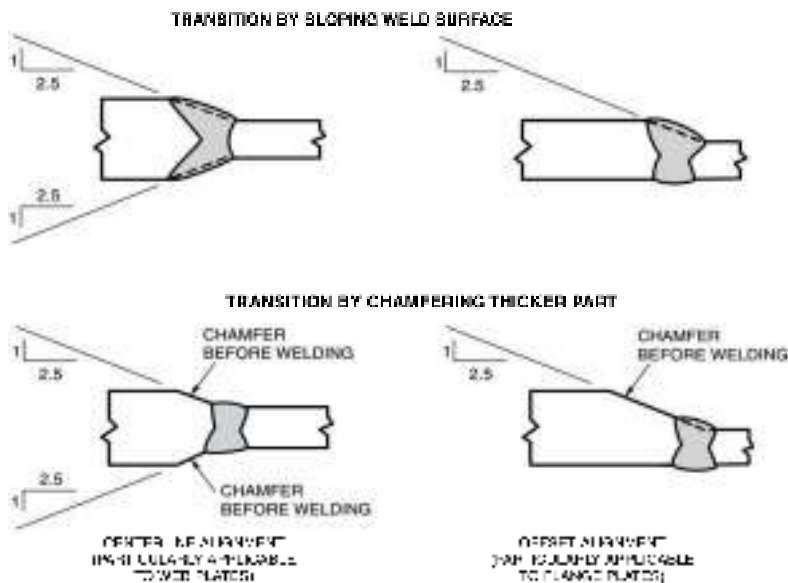


For "WELD TYPE E, F, G":

The permanent backing metal can be used and in conformance with design drawing

Weld Type	Joint Data				Remark
	Root Opening - R (mm)	Root Face - f (mm)	Groove Angel 1 - α	Groove Angle2 - β	
A	0 - 5	0 - 5	$60^{\circ} - 70^{\circ}$	-	Welded Joint Details shall be as described in AWS D1.6, Figure 5.4
B	0 - 5	0 - 5	$60^{\circ} - 70^{\circ}$	$60^{\circ} - 70^{\circ}$	
C	0 - 5	0 - 5	$45^{\circ} - 55^{\circ}$	$45^{\circ} - 55^{\circ}$	
D	0 - 5	0 - 5	$45^{\circ} - 55^{\circ}$	-	
E, F, G	6 - 8	-	$45^{\circ} - 55^{\circ}$	-	

Transition of Splice Joint in Parts of Unequal Section



These Transition shall be as per structure design

Note: Joint data shall be as per the Code AWS D1.6 and in conformance with Project Specification, Drawings.



PROCEDURE QUALIFICATION RECORD (PQR)

(Structural Welding Code-Stainless Steel AWS D1.6/D1.6M:2017)



Company Name: **DaiDung Metallic Manufacture Construction & Trade Corporation**

PQR No.: **PQR-DD-FCAW-308**

Rev. No. **0**

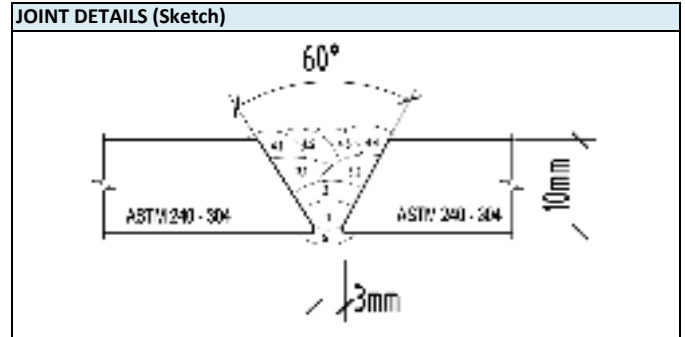
Date: **Nov 9, 2024**

Page: **1 of 2**

BASE METALS	Specification	Type or Grade	AWS Group No.	Thickness	Size (NPS)	Schedule	Diameter
Base Material	ASTM A240	304	Metal Group B	10 mm	-	-	-
Welded To	ASTM A240	304	Metal Group B	10 mm	-	-	-
Backing Material	Ceramic	-	-	-	-	-	-
Other	-	-	-	-	-	-	-

JOINT DETAILS	
Groove Type	Single V Groove Butt Joint
Groove Angle	60°
Root Opening	3-4 mm
Root Face	1 mm
Backgouging	Yes
Method	Grinding

POSTWELD HEAT TREATMENT	
Temperature	N/a
Time at Temperature	-
Other	-



PROCEDURE					
Weld Layer(s)	1	2 - 3	4	5	
Weld Pass(es)	1	2 - 3.2	4.1 - 4.4	5	
Process	FCAW	FCAW	FCAW	FCAW	
Type (Manual, Mechanized, etc.)	Semi-Automatic				
Position	2G	2G	2G	2G	
Vertical Progression	-	-	-	-	
Filler Metal (AWS Spec.)	AWS A5.22	AWS A5.22	AWS A5.22	AWS A5.22	
AWS Classification	E308LT1				
Diameter (mm)	Ø 1.2	Ø 1.2	Ø 1.2	Ø 1.2	
Manufacturer/Trade Name	Hyundai SW-308L Cored				
Shielding Gas (Composition)	CO2 (99.8%)	CO2 (99.8%)	CO2 (99.8%)	CO2 (99.8%)	
Flow Rate (LPM)	15-20	15-20	15-20	15-20	
Nozzle Size	No. 6	No. 6	No. 6	No. 6	
Preheat Temperature (Min.)	34 °C (ambient)				
Interpass Temperature (Max.)		135 °C	114°C	45 °C	
Electrical Characteristics					
Current Type & Polarity	DCEP	DCEP	DCEP	DCEP	
Transfer Mode	-	-	-	-	
Power Source Type (cc, cv, etc.)	CV	CV	CV	CV	
Amps (A)	100 - 130	140 - 160	140 - 160	140 - 160	
Volts (V)	26 - 27	27 - 28	27 - 28	27 - 28	
Wire Feed Speed	-	-	-	-	
Travel Speed (mm/min.)	110	281 - 424	367 - 396	419	
Maximum Heat Input (kJ/mm)	1.9	0.9	0.7	0.6	
Technique					
Stringer or Weave	Weave	Weave	Weave	Weave	
Multi or Single Pass (per side)	Multipass	Multipass	Multipass	Multipass	
Oscillation (Mech./Auto.)	-	-	-	-	
Number of Electrodes	1	1	1	1	
Contact Tube to Work Dist.(mm)	20	20	20	20	
Peening	None	None	None	None	
Interpass Cleaning	Wire Brush	Grinding	Wire Brush	Wire Brush	
Other	-	-	-	-	

Prepared by Welding Coordinator		Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hai Phong Vu (QA Manager)	
Signature			
Date	Nov 9, 2024	Nov 9, 2024	



	PROCEDURE QUALIFICATION RECORD (PQR) (Structural Welding Code-Stainless Steel AWS D1.6/D1.6M:2017)	
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PQR No.: **PQR-DD-FCAW-308**

Rev. No.: **0**

Date: **Nov 9, 2024**

Page: **2 of 2**

TEST RESULTS

TESTS

	Type of Tests	Clause/Figure(s) Reference	Acceptance Criteria	Result	Remarks
✓	Visual Inspection	Figure 6.5	6.9.3.1	Acceptable	WPR-DD-FCAW-308
	Radiographic Examination	8.14.1	8.12		
	Ultrasonic Testing	8.14.3	8.13		
	Penetrant Testing	8.14.4	8.10 & Table 8.1		
	Magnetic Particle Testing	8.14.5	8.10 & Table 8.1		
	2 Side Bends	Figure 6.6	6.9.3.2		
	4 Side Bends	Figure 6.6	6.9.3.2		
✓	2 Transverse Root Bends	Figure 6.7 or Figure 6.8	6.9.3.2	Acceptable	INL24/512130-799-2
✓	2 Transverse Face Bends	Figure 6.7 or Figure 6.8	6.9.3.2	Acceptable	INL24/512130-799-2
	2 Longitudinal Face Bends	Figure 6.9	6.9.3.2		
	2 Longitudinal Root Bends	Figure 6.9	6.9.3.2		
✓	2 Tensile Tests	Figure 6.14	6.9.3.3	Acceptable	INL24/512130-799-2
✓	3 Macroetch	Annex I	6.9.3.4	Acceptable	INL24/512130-799-2

TENSILE TEST DETAILS

Specimen Number	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Tensile Load (kN)	Ultimate Unit Stress (MPa)	Type of Failure and Location
TS 1	20.0	9.8	195.4	119.6	612	Ductile/ In weld
TS 2	20.1	9.8	195.9	119.7	611	Ductile/ In weld

TOUGHNESS TEST DETAILS

Specimen Number	Notch Location	Specimen Size (mm)	Test Temperature	Absorbed Energy (J)	Percent Shear	Lateral Expansion	Average (J)

CERTIFICATION

Welder's Name	ID Number	Stamp Number	Tests Conducted by	
LE QUANG SANG	-	-	Laboratory	SGS
-	-	-	Test Number	INL24/512130-799-2
-	-	-	File Number	-

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Clause 6 of AWS D1.6/D1.6M, 2017 *Structural Welding Code-Steel*.

Prepared by Welding Coordinator	Reviewed and Approved by	
Name La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature 		
Date Nov 9, 2024	Nov 9, 2024	

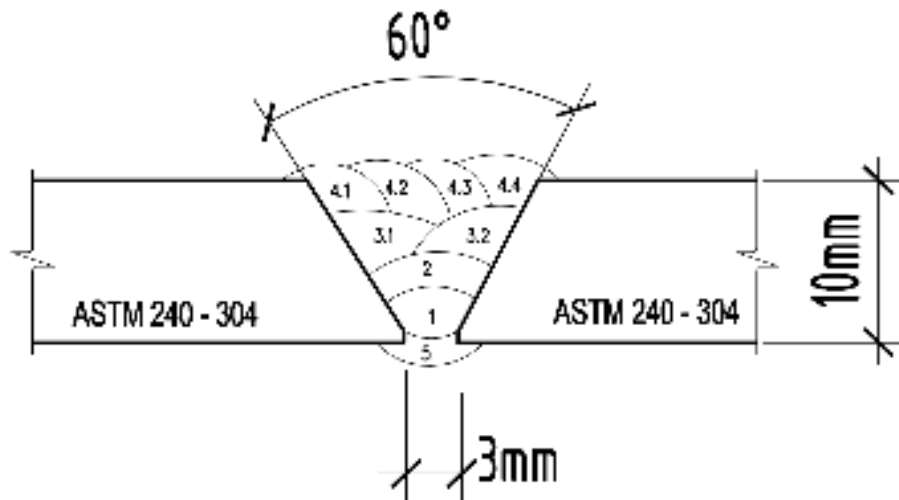


WELDING PARAMETERS RECORD

Company name: DaiDung Metallic Manufacture Construction & Trade Corporation
 PQR No.: PQR-DD-FCAW-308
 Test Coupon: DD-FCAW-308.1
 Test position: 2G
 Base metal:
 - Specification: ASTM A240, TYPE 304
 - Dimension: PL10x200x600 mm to PL10x200x600 mm
 - Heat/Product No.: Y210531025-3
 - Certificate No.: 20210717107
 Filler metal:
 - Specification: AWS A5.22
 - Classification: E308LT1-1
 - Trade name: Hyundai SW-308L Cored
 - Batch/Lot No.: SW23829C1
 - Certificates No.: HDV-2023-0829
 Gas: CO2 (99.8%)
 Welder Name: LE QUANG SANG
 I.D.:

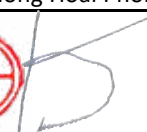
Doc No.: WPR-DD-FCAW-308
 Date: Nov 1, 2024

Joint Sketch:



Root face: 1 mm
 Root gap: 3-4 mm
 Groove angle: 60°

Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
1	FCAW	AWS A5.22 E308LT1-1	1.2	DCEP	100 - 130	26 - 27	5	28	600	110	34 (Ambient)	15-20	1.4 - 1.9	9	4	
2	FCAW	AWS A5.22 E308LT1-1	1.2	DCEP	140 - 160	27 - 28	1	25	600	424	51	15-20	0.5 - 0.6	11	3	
3.1	FCAW	AWS A5.22 E308LT1-1	1.2	DCEP	140 - 160	27 - 28	1	27	600	414	80	15-20	0.5 - 0.6	12	2	
3.2	FCAW	AWS A5.22/ E308LT1-1	1.2	DCEP	140 - 150	27 - 28	2	8	600	281	135	15-20	0.8 - 0.9	12		

Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		Volts (V)	min.						sec.	Width	
4.1	FCAW	AWS A5.22/ E308LT1-1	1.2	DCEP	140 - 160	27 - 28	1	36	600	375	55	15-20	0.6 - 0.7	12	1	Reinforcement 1mm
4.2	FCAW	AWS A5.22 E308LT1-1	1.2	DCEP	140 - 160	27 - 28	1	38	600	367	72	15-20	0.6 - 0.7	12		
4.3	FCAW	AWS A5.22/ E308LT1-1	1.2	DCEP	140 - 160	27 - 28	1	31	600	396	65	15-20	0.6 - 0.7	12		
4.4	FCAW	AWS A5.22/ E308LT1-1	1.2	DCEP	140 - 160	27 - 28	1	35	600	379	114	15-20	0.6 - 0.7	12		
5	FCAW	AWS A5.22/ E308LT1-1	1.2	DCEP	140 - 160	27 - 28	1	26	600	419	45	15-20	0.5 - 0.6	13		
		Prepared by Welding Coordinator				Reviewed and Approved by										
Name		Nguyen Bao Tinh				Truong Hoai Phong Vu										
Signature		 														
Date		Nov 1, 2024				Nov 1, 2024										

Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG

Client's address / Địa chỉ : LÔ SỐ 38, KHU C, ĐƯỜNG D1, KHU CÔNG NGHIỆP AN HẠ, XÃ PHẠM VĂN HAI, HUYỆN BÌNH CHÁNH, THÀNH PHỐ HỒ CHÍ MINH, VIỆT NAM

The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : DD-FCAW-308.1
Base material: ASTM A240 304
Size: 10x400x600 mm
Welding consumable: AWS A5.22 E308LT1-1
Trade name: Hyundai SW-308L Cored (1.2mm)
Type of Joint: Single V groove, Butt joint
Position 2G
Welder: Le Quang Sang

Project / Dự án : LTIA-PBB (PASSENGER BOARDING BRIDGE)

Quantity / Số lượng mẫu : 01 sample / mẫu

Description / Mô tả mẫu : 600 x 400 x 10 (mm)

Date of receiving / Ngày nhận mẫu : 05-Nov-2024

Date of testing / Ngày thử nghiệm : From / từ 05-Nov-2024 to / đến 07-Nov-2024

Test method / Phương pháp thử : AWS D1.6/D1.6M:2017 Structural Welding Code - Stainless Steel

Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Signed for and on behalf of
SGS Vietnam LTD

Thay mặt công ty

SGS SGS Vietnam Ltd

Chau Tran Khanh Dung
Laboratory Supervisor
Industrial Services

Please extract your time to give us your feedback as this will help our service better. Thank you!

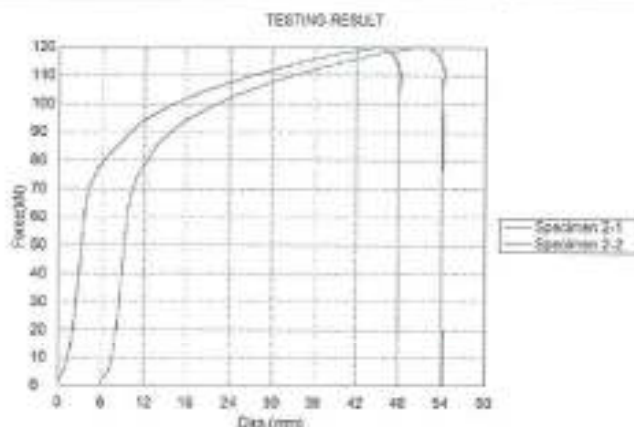


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Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm	
A. Transverse tension test / Thử kéo theo phương ngang	1	2
• Thickness of specimen / Chiều dày mẫu mm	9.76	9.77
• Width of specimen / Chiều rộng mẫu mm	20.02	20.05
• Cross section area / Tiết diện mm ²	195.4	195.9
• Tensile load / Lực kéo đứt kN	119.6	119.7
• Tensile strength / Giới hạn bền kéo MPa	612	611
• Broken position / Vị trí mẫu đứt	In weld	In weld



The relationship between tensile load and deformation / Biểu đồ tương quan giữa lực thử kéo và biến dạng

Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm			
B. Bend test / Thử uốn	1	2	3	4
• Location bend / Vị trí uốn	Face bend Uốn mặt	Face bend Uốn mặt	Root bend Uốn chân	Root bend Uốn chân
• Thickness of specimen / Chiều dày mm	9.79	9.80	9.81	9.80
• Width of specimen / Chiều rộng mm	40.01	40.03	40.01	40.02
• Bending angle / Góc uốn °/ độ	180	180	180	180
• Plunger diameter / Đường kính gối uốn mm	38	38	38	38
• Result / Kết quả	No Crack Không nứt	No Crack Không nứt	No Crack Không nứt	No Crack Không nứt
• Evaluation / Đánh giá	Pass / Đạt	Pass / Đạt	Pass / Đạt	Pass / Đạt

*** TO BE CONTINUED ***

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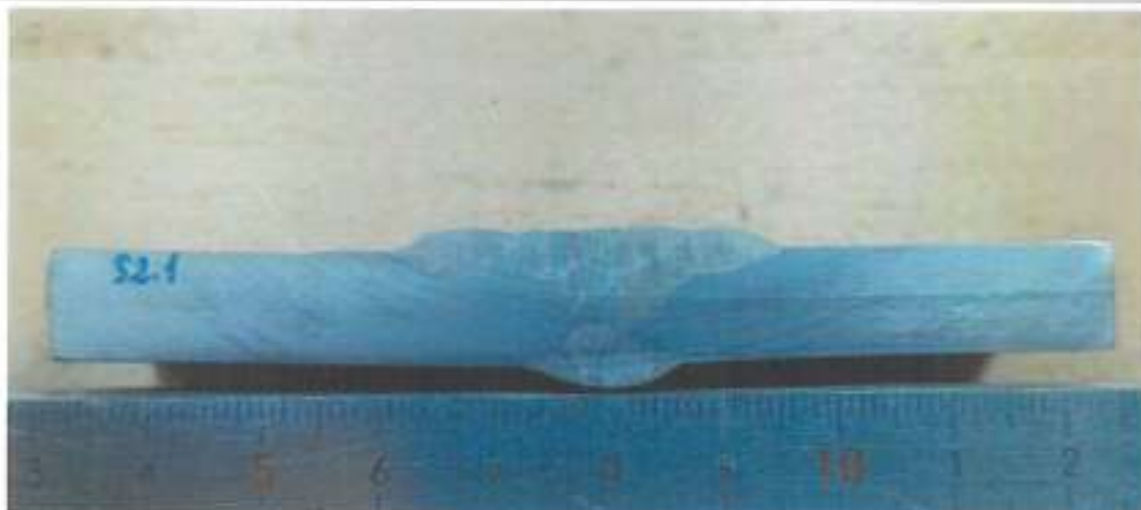
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C. Characteristic / Tên chỉ tiêu : Macro structure / Tổ chức thô đại

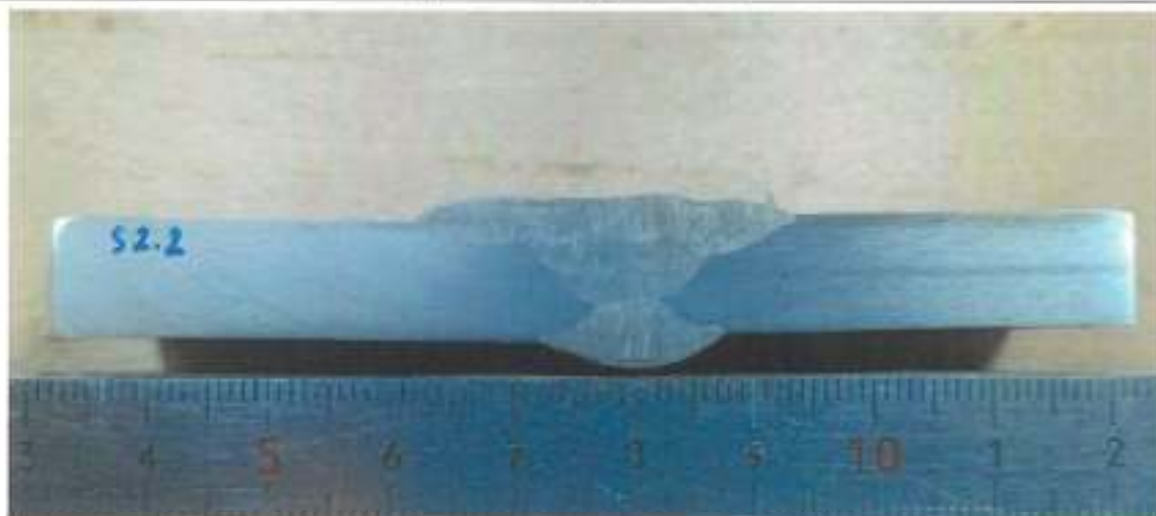
➤ **Specimen 2-1 / Mẫu 2-1 :**

- Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn.
- No defect in weld metal / Trong mối hàn không có khuyết tật.



➤ **Specimen 2-2 / Mẫu 2-2 :**

- Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn.
- No defect in weld metal / Trong mối hàn không có khuyết tật.



*** TO BE CONTINUED ***

041961

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D. Characteristic / Tên chỉ tiêu : Macro etch ⁽¹⁾ / Tổ chức thô đại

➤ **Specimen 2-3 / Mẫu 2-3 :**

- **Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn.**
- **No defect in weld metal / Trong mối hàn không có khuyết tật.**



Original sample photo(s) / Hình ảnh mẫu nhận:



SGS authenticate the photos on original report only / SGS chỉ xác thực các hình ảnh trong báo cáo chính thức.

*** END OF THE REPORT ***

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MILL TEST CERTIFICATE

EN 10204/3.1



CNS 115500000

ADD: PT. INDU, Desa Fatmaja, Kecamatan Sukadadi, Kabupaten
Korowali, Provinsi Sulawesi Tengah, Indonesia

Tel: 16202256701870

Email: liuy@etalingshan.com

Manufacturer: PT. INDONESIA GUNG CHING STEEL AND STAINLESS
STEEL INDUSTRY

Product Code		MILL STAR INTERNATIONAL JOINT STOCK COMPANY		Rolling Process		MILL+HOT+LP+COIL		Contract No.		CCT/ACER/CH-100		Certificate No.		20200111007						
Product Name		HOT ROLLING STAINLESS STEEL COIL		Steel Grade		304		Product Standard		ASTM A240		Description of delivery		For cooling atmosphere						
No.	Coil No.	QTY	Heat No.	Size (mm)	Weight (kg)	Chemical Composition (%)										Tensile Test				Tensile Strength
						C	Si	Mn	P	S	Cr	Ni	Al	Fe	As	R _m (MPa)	R _{p0.2} (MPa)	R _{p0.01} (MPa)	Elongation A (mm)	
1	20200111007	1	7210000021-0	3.05*1550*C	21.355	0.042	0.41	1.14	0.033	0.002	18.16	8.25	0.030	0.01	0.01	525.8	329.2	7	55.5	81.5
2	20200111007	1	7210000021-1	3.05*1550*C	21.355	0.045	1.48	1.16	0.035	0.002	18.11	8.04	0.044	0.04	0.04	545.5	329.4	7	55.6	86.8
3	20200111007	1	7210000021-2	3.05*1550*C	21.355	0.040	1.76	1.18	0.032	0.002	18.20	8.01	0.035	0.03	0.03	545.5	329.4	7	55.1	85.1
4	20200111007	1	7210000021-3	3.05*1550*C	21.355	0.044	1.41	0.95	0.032	0.002	18.21	8.02	0.035	0.04	0.04	545.5	329.4	7	55.7	85.7
5	20200111007	1	7210000021-4	3.05*1550*C	21.355	0.041	1.42	0.94	0.034	0.002	18.16	8.02	0.035	0.04	0.04	545.5	329.4	7	55.2	88.0
6	20200111007	1	7210000021-5	3.05*1550*C	21.355	0.041	1.40	0.94	0.037	0.002	18.25	8.04	0.030	0.04	0.04	545.5	329.4	7	55.4	87.4
7	20200111007	1	7210000021-6	3.05*1550*C	21.355	0.045	1.42	1.02	0.034	0.002	18.20	8.02	0.040	0.04	0.04	545.5	329.4	7	55.9	88.2

Remarks

1. Product marking: steel number, heat No, coil No, size.

2. Surface Quality: ok 3. Shape: ok 4. Dimension: ok 5. Radioactivity Tests: ok 6. Non-metallic inclusions: ok 7. Intergranular corrosion: ok

1. We hereby certify that the material herein described has been manufactured, sampled, tested and inspected in accordance with the requirements of above specifications.

2. We hereby certify Product No-weld repair, Free of mercury, Free from radiation contamination.

3. ISO 9001:2015 Quality management system identified by ZJQC.

4. We declare that our testing center has passed ISO/IEC 17025:2017 accreditation review, Accreditation body: CNAS, accreditation capability range: Spectral analysis and tensile testing, certificate number: CNAS L12639.

5. We certify that the material described below fully conforms to IS 6811:2017 chemical composition and Mechanical properties of product as listed in accordance with the scheme of Testing and Inspection outlined in BIS certification marks license No. CNL-0100045662.

6. Issued in agreement with TÜV SÜD Industrie Service GmbH (April 2018), QMS approved acc. to PECAAD/2020, Annex 1, Para. 4.3 by Notified Body 0026 (Certification no. DQR-0036-03-W 13520-01-MUC).

7. Approved acc. to CPR 305/2011 with certification no. 0036-CPR-M-115-2010.

8. Product heat treatment temperature: 1100°C.

9. The certificate should be sealed by quality control department, add date, inspection stamp or authorized official stamp of sales.

Typed by	徐红叶	
Rectifying	刘	
Date	2021/7/17	

CERTIFIED MATERIAL REPORT

CMR No. : HDV – 2023 – 0829 (Rev.2)

Date of Issue : 2023 – 09 – 04



Customer		G-TECH CORPORATION							
Trade Name		SW-308L Cored			Size		1.2 mm x 15 kg		
Lot No.		SW23829C1			Purchaser's Order No.		40-2023/GT/HW/PO		
Specification		AWS A5.22 E308LT1-1 EN ISO 17633-A-T 19 9 L P C1 2 * According to EN 10204-2004 : 3.1							
Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment (°C×hr.)			As Welded		
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)					
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.	
	434	552	45	-	-	-	-	-	
AWS Spec.	-	≥520	≥30	-					
Chemical Composition (%)									
Weld Metal	C	Si	Mn	P	S	Cu	Ni	Cr	Mo
	0.03	0.51	1.13	0.014	0.008	0.04	9.35	20.41	0.04
AWS Spec.	≤0.04	≤1.00	0.50 ~2.50	≤0.040	≤0.030	≤0.75	9.00 ~12.00	18.00 ~21.00	≤0.5
* Radiographic Test : ACCEPT * Fillet Weld Test : GOOD									
Others * Face Bend Test : GOOD * FPC No. (CE-Mark): 0035-CPR-C771 * Q'ty: 900 kg									
We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.									
0035-CPR-C771 0035 06 / HWV-402 EN13479				Approved by QA Manager Pham Dinh Anh Vu					

HYUNDAI WELDING VINA CO., LTD.

Road N2, Nhon Trach 5 IZ, Hiep Phuoc Ward, Nhon Trach District, Dong Nai Province, Vietnam.



Comply:
ISO 9001:2015
AWS D1.6:2017

WELDING PROCEDURE

Doc. No.: WPS-DD-FCAW-333

PQR-DD-FCAW-333

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Nov 09, 2024	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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WELDING PROCEDURE SPECIFICATION (WPS)

(AWS D1.6/D1.6M:2017 Structural Welding Code-Stainless Steel)



Company Name
WPS No.
Supporting PQR(s)
CVN Report

DaiDung Metallic Manufacture Construction & Trade Corporation
WPS-DD-FCAW-333.1
PQR-DD-FCAW-333

Rev. No. A

Date: Oct 17, 2024

Page: 1 of 1

BASE METALS	Specification	Type or Grade	AWS Group No.
Base Material	ASTM A240/276/312/403 or equivalent	304/316 Any M-8	Metal Group A,B Table 5.2
Welded To	JIS G3101	SS400	-
Backing Material	-	-	-
Other	-	-	-

BASE METAL THICKNESS	As-Welded (mm)	With PWHT
CJP Groove Welds	-	-
CJP Groove with CVN	-	-
PJP Groove Welds	-	-
Fillet Welds	All	-
DIAMETER	All	-

JOINT DETAILS		JOINT SKETCH	
Joint Type	T, Corner & Lap Joint		
Weld Type	Fillet Weld		
Root Opening	0 - 2 (mm)		
Root Face	-		
Dihedral	Per drawing		
Weld Size	Per drawing		
POSTWELD HEAT TREATMENT		Welded Joint Details as Figures 5.2 AWS D1.6	
Temperature	-		
Time at Temperature	-		
Other	-		

PROCEDURE			
Weld Layer(s)		All	
Weld Pass(es)		All	
Process		FCAW	
Type (Manual, Mechanized, etc.)		Semi-Automatic, Mechanized	
Position		All	
Vertical Progression		-	
Filler Metal (AWS Spec.)		AWS A5.22	
AWS Classification		E309LT1-1/-4; Group B Table 5.3	
Diameter (mm)		1.2	
Manufacturer/Trade Name		Not strictly	
Shielding Gas (Composition)		99.8% CO2	
Flow Rate (LPM)		15 - 25	
Nozzle Size		No. 6	
Preheat Temperature (Min.)		15°C (Ambient)	
Interpass Temperature (Max.)		175°C	
Electrical Characteristics			
Current Type & Polarity		DCEP	
Transfer Mode		-	
Power Source Type (cc, cv, etc.) (Amps)		CV	
Amps (A)		100 - 180	
Volts (V)		24 - 31	
Wire Feed Speed (Amps)			
Travel Speed (mm/min.)		110 - 530	
Maximum Heat Input (kJ/mm)		1.92	
Technique			
Stringer or Weave		Stringer or Weave	
Multi or Single Pass (per side)		Multi or Single Pass	
Oscillation (Mech./Auto.)		-	
Number of Electrodes		1	
Contact Tube to Work Dist. (mm)		15 - 25	
Peening		None	
Interpass Cleaning		Grinding and Wire Brush	
Other	This procedure can be used for build up or repair.		

	Prepared by Welding Coordinator	Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature			
Date	Oct 17, 2024	Oct 17, 2024	





WELDING PROCEDURE SPECIFICATION (WPS)

(AWS D1.6/D1.6M:2017 Structural Welding Code-Stainless Steel)



Company Name: **DaiDung Metallic Manufacture Construction & Trade Corporation**
WPS No.: **WPS-DD-FCAW-333.2** Rev. No.: **A** Date: **Oct 17, 2024** Page: **1 of 1**
Supporting PQR(s): **PQR-DD-FCAW-333**
CVN Report: **-**

BASE METALS	Specification	Type or Grade	AWS Group No.	BASE METAL THICKNESS	As-Welded (mm)	With PWHT
Base Material	ASTM A240/276/312/403 or equivalent	304/316 Any M-8	Metal Group A,B Table 5.2	CJP Groove Welds	-	-
Welded To	JIS G3101	SS400	-	CJP Groove with CVN	-	-
Backing Material	-	-	-	PJP Groove Welds	2-20 mm	-
Other	-	-	-	Fillet Welds	-	-
				DIAMETER	All	-

JOINT DETAILS			JOINT SKETCH	
Joint Type	Butt, T & Corner Joint			
Weld Type	Single V Groove	Single Bevel Groove Weld		
Groove Angle	60° - 70°	45° - 55°		
Root Opening	0 - 2 mm			
Root Face	Per drawing			
Weld Size	Table 5.5			
POSTWELD HEAT TREATMENT			Welded Joint Details as Figures 5.3 AWS D1.6	
Temperature	-			
Time at Temperature	-			
Other	-			

PROCEDURE		PROCEDURE		
Weld Layer(s)		All		
Weld Pass(es)		All		
Process		FCAW		
Type (Manual, Mechanized, etc.)		Semi-Automatic, Mechanized		
Position		All		
Vertical Progression		-		
Filler Metal (AWS Spec.)		AWS A5.22		
AWS Classification		E309LT1-1/-4; Group B Table 5.3		
Diameter	(mm)	1.2		
Manufacturer/Trade Name		Not strictly		
Shielding Gas (Composition)		99.8% CO2		
Flow Rate	(LPM)	15 - 25		
Nozzle Size		No. 6		
Preheat Temperature	(Min.)	15°C (Ambient)		
Interpass Temperature	(Max.)	175°C		
Electrical Characteristics				
Current Type & Polarity		DCEP		
Transfer Mode		-		
Power Source Type (cc, cv, etc.)	(Amps)	CV		
Amps	(A)	100 - 180		
Volts	(V)	24 -31		
Wire Feed Speed		(Amps)		
Travel Speed	(mm/min.)	110 - 530		
Maximum Heat Input	(kJ/mm)	1.92		
Technique				
Stringer or Weave		Stringer or Weave		
Multi or Single Pass (per side)		Multi or Single Pass		
Oscillation (Mech./Auto.)		-		
Number of Electrodes		1		
Contact Tube to Work Dist.	(mm)	15 - 25		
Peening		None		
Interpass Cleaning		Grinding and Wire Brush		
Other		This procedure can be used for build up or repair.		

	Prepared by Welding Coordinator	Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature			
Date	Oct 17, 2024	Oct 17, 2024	





WELDING PROCEDURE SPECIFICATION (WPS)

(AWS D1.6/D1.6M:2017 Structural Welding Code-Stainless Steel)



Company Name: DaiDung Metallic Manufacture Construction & Trade Corporation
WPS No.: WPS-DD-FCAW-333.3
Supporting PQR(s): PQR-DD-FCAW-333
CVN Report: -
Rev. No.: A
Date: Oct 17, 2024
Page: 1 of 2

BASE METALS	Specification	Type or Grade	AWS Group No.
Base Material	ASTM A240/276/312/403 or equivalent	304/316 Any M-8	Metal Group A,B Table 5.2
Welded To	JIS G3101	SS400	-
Backing Material	Ceramic or same base metal	-	-
Other	-	-	-

BASE METAL THICKNESS	As-Welded (mm)	With PWHT
CJP Groove Welds	2 - 20	-
CJP Groove with CVN	-	-
PJP Groove Welds	-	-
Fillet Welds	-	-
DIAMETER	All	-

JOINT DETAILS	
Groove Type	See attached sketch for more details
Groove Angle	See attached sketch for more details
Root Opening	See attached sketch for more details
Root Face	See attached sketch for more details
Backgouging	Yes
Method	Grinding

JOINT SKETCH
See Joint Sketch in page 2 of 2

POSTWELD HEAT TREATMENT	
Temperature	-
Time at Temperature	-
Other	-

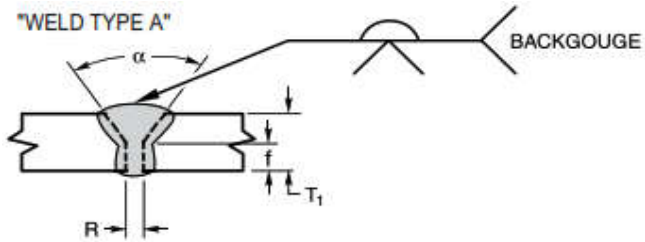
PROCEDURE			
Weld Layer(s)		All	
Weld Pass(es)		All	
Process		FCAW	
Type (Manual, Mechanized, etc.)		Semi-Automatic, Mechanized	
Position		All	
Vertical Progression		-	
Filler Metal (AWS Spec.)		AWS A5.22	
AWS Classification		E309LT1-1/-4; Group B Table 5.3	
Diameter (mm)		1.2	
Manufacturer/Trade Name		Not strictly	
Shielding Gas (Composition)		99.8% CO2	
Flow Rate (LPM)		15 - 25	
Nozzle Size		No. 6	
Preheat Temperature (Min.)		15°C (Ambient)	
Interpass Temperature (Max.)		175°C	
Electrical Characteristics			
Current Type & Polarity		DCEP	
Transfer Mode		-	
Power Source Type (cc, cv, etc.) (Amps)		CV	
Amps (A)		100 - 180	
Volts (V)		24 - 31	
Wire Feed Speed (Amps)			
Travel Speed (mm/min.)		110 - 530	
Maximum Heat Input (kJ/mm)		1.92	
Technique			
Stringer or Weave		Stringer or Weave	
Multi or Single Pass (per side)		Multi or Single Pass	
Oscillation (Mech./Auto.)		-	
Number of Electrodes		1	
Contact Tube to Work Dist. (mm)		15 - 25	
Peening		None	
Interpass Cleaning		Grinding and Wire Brush	
Other		This procedure can be used for build up or repair.	

Prepared by Welding Coordinator		Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature			
Date	Oct 17, 2024	Oct 17, 2024	

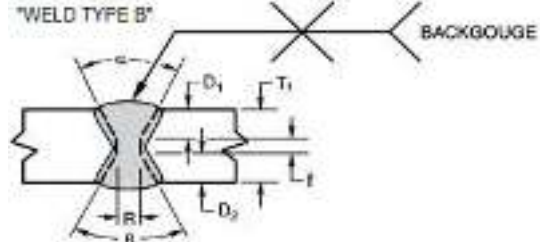


CJP

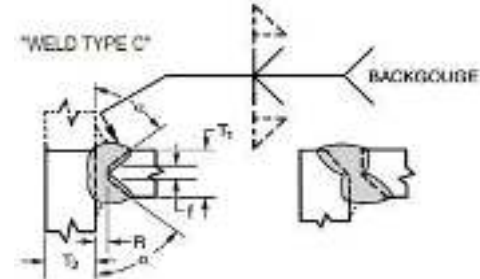
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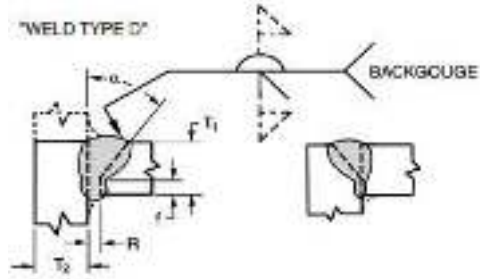
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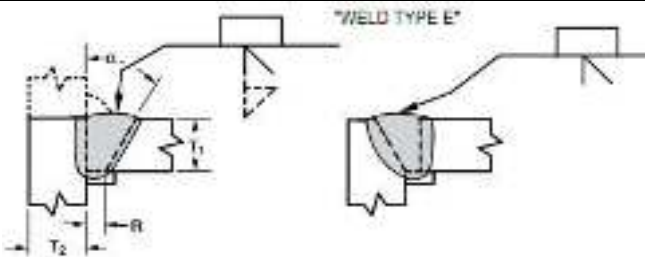
"WELD TYPE C"



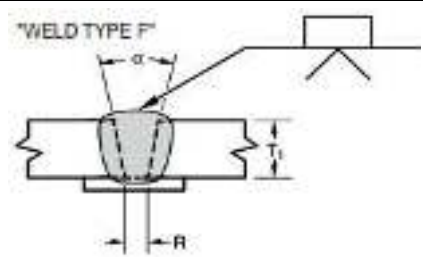
"WELD TYPE D"



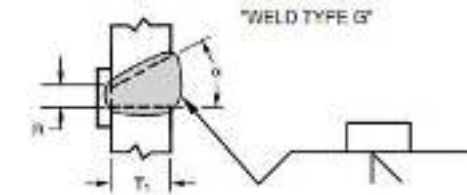
"WELD TYPE E"



"WELD TYPE F"



"WELD TYPE G"

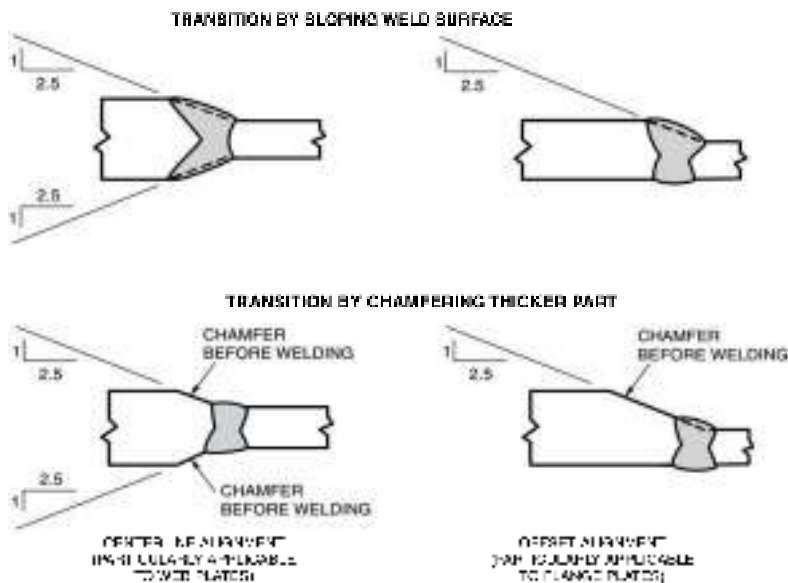


For "WELD TYPE E, F, G":

The permanent backing metal can be used and in conformance with design drawing

Weld Type	Joint Data				Remark
	Root Opening - R (mm)	Root Face - f (mm)	Groove Angel 1 - α	Groove Angle2 - β	
A	0 - 5	0 - 5	$60^{\circ} - 70^{\circ}$	-	Welded Joint Details shall be as described in AWS D1.6, Figure 5.4
B	0 - 5	0 - 5	$60^{\circ} - 70^{\circ}$	$60^{\circ} - 70^{\circ}$	
C	0 - 5	0 - 5	$45^{\circ} - 55^{\circ}$	$45^{\circ} - 55^{\circ}$	
D	0 - 5	0 - 5	$45^{\circ} - 55^{\circ}$	-	
E, F, G	6 - 8	-	$45^{\circ} - 55^{\circ}$	-	

Transition of Splice Joint in Parts of Unequal Section



These Transition shall be as per structure design

Note: Joint data shall be as per the Code AWS D1.6 and in conformance with Project Specification, Drawings.



PROCEDURE QUALIFICATION RECORD (PQR)

(Structural Welding Code-Stainless Steel AWS D1.6/D1.6M:2017)



Company Name: **DaiDung Metallic Manufacture Construction & Trade Corporation**

PQR No.: **PQR-DD-FCAW-333**

Rev. No. **0**

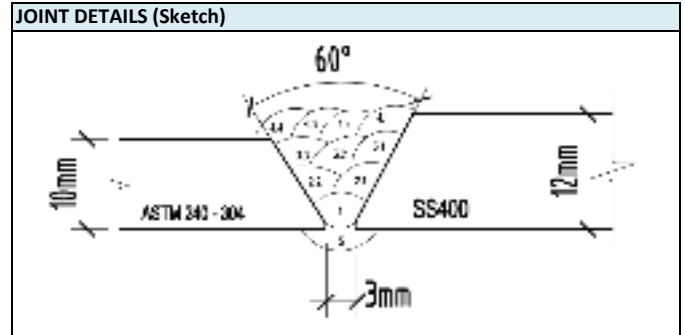
Date: **Oct 17, 2024**

Page: **1 of 2**

BASE METALS	Specification	Type or Grade	AWS Group No.	Thickness	Size (NPS)	Schedule	Diameter
Base Material	ASTM A240	304	Metal Group B	10 mm	-	-	-
Welded To	JIS G3101	SS400	-	12 mm	-	-	-
Backing Material	Ceramic	-	-	-	-	-	-
Other	-	-	-	-	-	-	-

JOINT DETAILS	
Groove Type	Single V Groove Butt Joint
Groove Angle	60°
Root Opening	3-4 mm
Root Face	1 mm
Backgouging	Yes
Method	Grinding

POSTWELD HEAT TREATMENT	
Temperature	N/a
Time at Temperature	-
Other	-



PROCEDURE					
Weld Layer(s)	1	2 - 3	4	5	
Weld Pass(es)	1	2.1 - 3.3	4.1 - 4.4	5	
Process	FCAW	FCAW	FCAW	FCAW	
Type (Manual, Mechanized, etc.)	Semi-Automatic				
Position	2G	2G	2G	2G	
Vertical Progression	-	-	-	-	
Filler Metal (AWS Spec.)	AWS A5.22	AWS A5.22	AWS A5.22	AWS A5.22	
AWS Classification	E309LT1				
Diameter (mm)	Ø 1.2	Ø 1.2	Ø 1.2	Ø 1.2	
Manufacturer/Trade Name	Hyundai SW-309L Cored				
Shielding Gas (Composition)	CO2 (99.8%)	CO2 (99.8%)	CO2 (99.8%)	CO2 (99.8%)	
Flow Rate (LPM)	15-20	15-20	15-20	15-20	
Nozzle Size	No. 6	No. 6	No. 6	No. 6	
Preheat Temperature (Min.)	31 °C (ambient)				
Interpass Temperature (Max.)		160 °C	152	40 °C	
Electrical Characteristics					
Current Type & Polarity	DCEP	DCEP	DCEP	DCEP	
Transfer Mode	-	-	-	-	
Power Source Type (cc, cv, etc.)	CV	CV	CV	CV	
Amps (A)	110 - 120	145 - 163	140 - 160	140 - 160	
Volts (V)	24 - 25	26 - 28	25 - 29	27 - 28	
Wire Feed Speed	-	-	-	-	
Travel Speed (mm/min.)	113	316 - 621	321 - 419	375	
Maximum Heat Input (kJ/mm)	1.6	0.9	0.8	0.7	
Technique					
Stringer or Weave	Weave	Weave	Weave	Weave	
Multi or Single Pass (per side)	Multipass	Multipass	Multipass	Multipass	
Oscillation (Mech./Auto.)	-	-	-	-	
Number of Electrodes	1	1	1	1	
Contact Tube to Work Dist.(mm)	20	20	20	20	
Peening	None	None	None	None	
Interpass Cleaning	Wire Brush	Grinding	Wire Brush	Wire Brush	
Other	-	-	-	-	

Prepared by Welding Coordinator		Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature			
Date	Oct 17, 2024	Oct 17, 2024	



	PROCEDURE QUALIFICATION RECORD (PQR) (Structural Welding Code-Stainless Steel AWS D1.6/D1.6M:2017)	
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PQR No.: **PQR-DD-FCAW-333**

Rev. No.: **0**

Date: **Oct 17, 2024**

Page: **2 of 2**

TEST RESULTS

TESTS

	Type of Tests	Clause/Figure(s) Reference	Acceptance Criteria	Result	Remarks
✓	Visual Inspection	Figure 6.5	6.9.3.1	Acceptable	WPR-DD-FCAW-309
	Radiographic Examination	8.14.1	8.12		
	Ultrasonic Testing	8.14.3	8.13		
	Penetrant Testing	8.14.4	8.10 & Table 8.1		
	Magnetic Particle Testing	8.14.5	8.10 & Table 8.1		
	2 Side Bends	Figure 6.6	6.9.3.2		
	4 Side Bends	Figure 6.6	6.9.3.2		
✓	2 Transverse Root Bends	Figure 6.7 or Figure 6.8	6.9.3.2	Acceptable	INL24/512130-746-3
✓	2 Transverse Face Bends	Figure 6.7 or Figure 6.8	6.9.3.2	Acceptable	INL24/512130-746-3
	2 Longitudinal Face Bends	Figure 6.9	6.9.3.2		
	2 Longitudinal Root Bends	Figure 6.9	6.9.3.2		
✓	2 Tensile Tests	Figure 6.14	6.9.3.3	Acceptable	INL24/512130-746-3
✓	3 Macroetch	Annex I	6.9.3.4	Acceptable	INL24/512130-746-3

TENSILE TEST DETAILS

Specimen Number	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Tensile Load (kN)	Ultimate Unit Stress (MPa)	Type of Failure and Location
TS 1	20.1	9.9	199.0	79.7	401	Ductile/base metal SS400
TS 2	20.1	10.0	200.4	80.4	401	Ductile/base metal SS400

TOUGHNESS TEST DETAILS

Specimen Number	Notch Location	Specimen Size (mm)	Test Temperature	Absorbed Energy (J)	Percent Shear	Lateral Expansion	Average (J)

CERTIFICATION

Welder's Name	ID Number	Stamp Number	Tests Conducted by	
NGUYEN VAN DEM	DDC-WD-0564	-	Laboratory	SGS
-	-	-	Test Number	INL24/512130-746-3
-	-	-	File Number	-

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Clause 6 of AWS D1.6/D1.6M, 2017 *Structural Welding Code-Steel*.

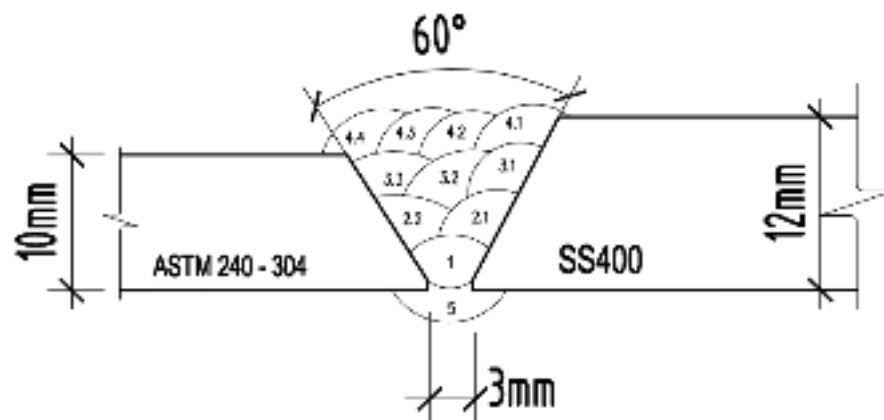
Prepared by Welding Coordinator	Reviewed and Approved by	
Name La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature 		
Date Oct 17, 2024	Oct 17, 2024	



WELDING PARAMETERS RECORD

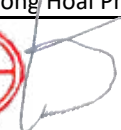
Company name: DaiDung Metallic Manufacture Construction & Trade Corporation
 PQR No.: PQR-DD-FCAW-333
 Test Coupon: DD-FCAW-311
 Test position: 2G
 Base metal:
 - Specification: ASTM A240, TYPE 304 to JIS G3101 SS400
 - Dimension: PL10x200x600 mm to PL12x200x600 mm
 - Heat/Product No.: Y210531025-3 (304) & 321944 (SS400)
 - Certificate No.: 20210717107 (304) & QZA021500806 (SS400)
 Filler metal:
 - Specification: AWS A5.22
 - Classification: E309LT1-1
 - Trade name: Hyundai SW-309L Cored
 - Batch/Lot No.: SW23929F1
 - Certificates No.: HDV-2023-0633
 Gas: CO2 (99.8%)
 Welder Name: NGUYEN VAN DEM
 I.D.: DDC-WD-0564

Joint Sketch:



Root face: 1 mm
 Root gap: 3-4 mm
 Groove angle: 60°

Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
1	FCAW	AWS A5.22 E309LT1-1	1.2	DCEP	110 - 120	24 - 25	5	20	600	113	31 (Ambient)	15-20	1.4 - 1.6	9	4	
2.1	FCAW	AWS A5.22 E309LT1-1	1.2	DCEP	150 - 160	27 - 28	1	21	600	444	43	15-20	0.5 - 0.6	11	3	
2.2	FCAW	AWS A5.22 E309LT1-1	1.2	DCEP	153 - 163	26 - 27	1	26	600	419	68	15-20	0.6 - 0.6	11		
3.1	FCAW	AWS A5.22 E309LT1-1	1.2	DCEP	150 - 160	27 - 28	1	14	600	486	137	15-20	0.5 - 0.6	12		

Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		Volts (V)	min.						sec.	Width	
3.2	FCAW	AWS A5.22 E309LT1-1	1.2	DCEP	145 - 155	27 - 28	0	58	600	621	136	15-20	0.4 - 0.4	12	2	
3.3	FCAW	AWS A5.22/ E309LT1-1	1.2	DCEP	153 - 163	27 - 28	1	54	600	316	160	15-20	0.8 - 0.9	12		
4.1	FCAW	AWS A5.22/ E309LT1-1	1.2	DCEP	150 - 160	27 - 28	1	37	600	371	73	15-20	0.7 - 0.7	12	1	Reinforcement 2mm
4.2	FCAW	AWS A5.22 E309LT1-1	1.2	DCEP	146 - 156	27 - 28	1	33	600	387	111	15-20	0.6 - 0.7	12		
4.3	FCAW	AWS A5.22/ E309LT1-1	1.2	DCEP	140 - 150	28 - 29	1	52	600	321	152	15-20	0.7 - 0.8	12		
4.4	FCAW	AWS A5.22/ E309LT1-1	1.2	DCEP	145 - 155	25 - 26	1	26	600	419	89	15-20	0.5 - 0.6	12		
5	FCAW	AWS A5.22/ E309LT1-1	1.2	DCEP	140 - 160	27 - 28	1	36	600	375	40	15-20	0.6 - 0.7	13		
		Prepared by Welding Coordinator				Reviewed and Approved by										
Name		Nguyen Bao Tinh				Truong Hoai Phong Vu										
Signature		 														
Date		Oct 10, 2024				Oct 10, 2024										

Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
Client's address / Địa chỉ : LÔ SỐ 38, KHU C, ĐƯỜNG D1, KHU CÔNG NGHIỆP AN HẠ, XÃ PHẠM VĂN HAI, HUYỆN BÌNH CHÁNH, THÀNH PHỐ HỒ CHÍ MINH, VIỆT NAM

The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : DD-FCAW-311
Base material: ASTM A240 304 + JIS G3101 SS400
Size: 10x200x600mm to 12x200x600mm
Welding consumable: AWS A5.22 E309LT1-1
Trade name: Hyundai SW-309L Cored (1.2mm)
Type of Joint: Single V groove, Butt joint
Position 2G
Welder: Nguyen Van Dem - DDC-WD-0564

Project / Dự án : LTIA-PBB (PASSENGER BOARDING BRIDGE)
Quantity / Số lượng mẫu : 01 sample / mẫu
Description / Mô tả mẫu : 600 x 200 x 10 (mm) & 600 x 200 x 12 (mm)
Date of receiving / Ngày nhận mẫu : 11-Oct-2024
Date of testing / Ngày thử nghiệm : From / từ 11-Oct-2024 to / đến 16-Oct-2024
Test method / Phương pháp thử : AWS D1.6/D1.6M:2017 Structural Welding Code - Stainless Steel
Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Signed for and on behalf of
SGS Vietnam LTD

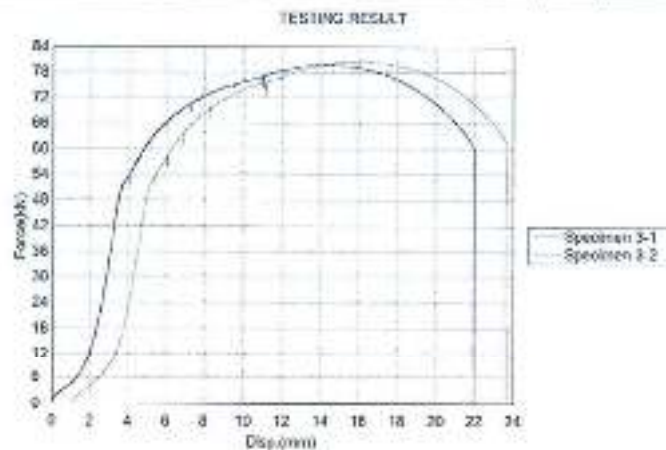
Thay mặt công ty
SGS SGS Vietnam Ltd
Chau Tran Khanh Dung
Laboratory Supervisor
Industrial Services

Please extract your time to give us your feedback as this will help our service better. Thank you!



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Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm	
A. Transverse tension test / Thử kéo theo phương ngang	1	2
• Thickness of specimen / Chiều dày mẫu mm	9.92	9.99
• Width of specimen / Chiều rộng mẫu mm	20.06	20.06
• Cross section area / Tiết diện mm ²	199.0	200.4
• Tensile load / Lực kéo đứt kN	79.7	80.4
• Tensile strength / Giới hạn bền kéo MPa	401	401
• Broken position / Vị trí mẫu đứt	In base metal (SS400)	In base metal (SS400)



The relationship between tensile load and deformation / Biểu đồ tương quan giữa lực thử kéo và biến dạng

Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm			
B. Bend test / Thử uốn	1	2	3	4
• Location bend / Vị trí uốn	Face bend Uốn mặt	Face bend Uốn mặt	Root bend Uốn chân	Root bend Uốn chân
• Thickness of specimen / Chiều dày mm	9.99	9.92	10.01	10.03
• Width of specimen / Chiều rộng mm	39.97	39.96	39.96	39.98
• Bending angle / Góc uốn °/ độ	180	180	180	180
• Plunger diameter / Đường kính gối uốn mm	20	20	20	20
• Result / Kết quả	No Crack Không nứt	No Crack Không nứt	No Crack Không nứt	No Crack Không nứt
• Evaluation / Đánh giá	Pass / Đạt	Pass / Đạt	Pass / Đạt	Pass / Đạt

*** TO BE CONTINUED ***

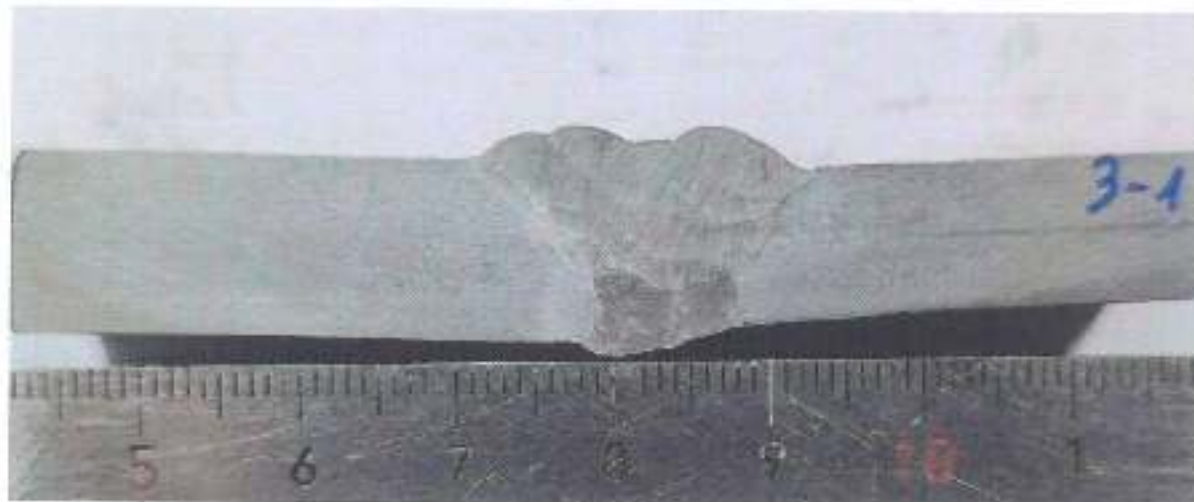
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C. Characteristic / Tên chỉ tiêu : Macro structure / Tổ chức thô đại

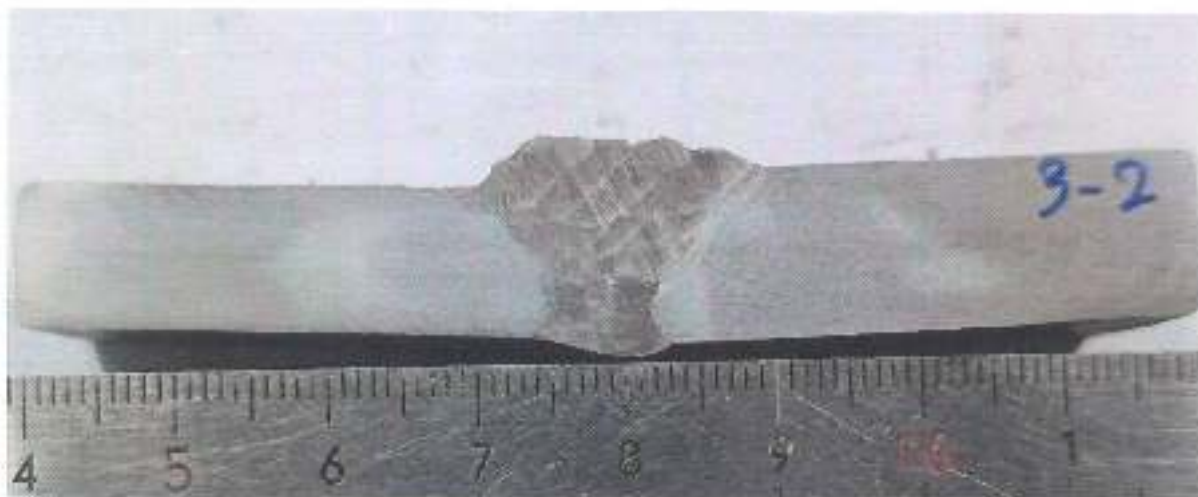
➤ **Specimen 3-1 / Mẫu 3-1 :**

- Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn.
- No defect in weld metal / Trong mối hàn không có khuyết tật.



➤ **Specimen 3-2 / Mẫu 3-2 :**

- Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn.
- No defect in weld metal / Trong mối hàn không có khuyết tật.



*** TO BE CONTINUED ***

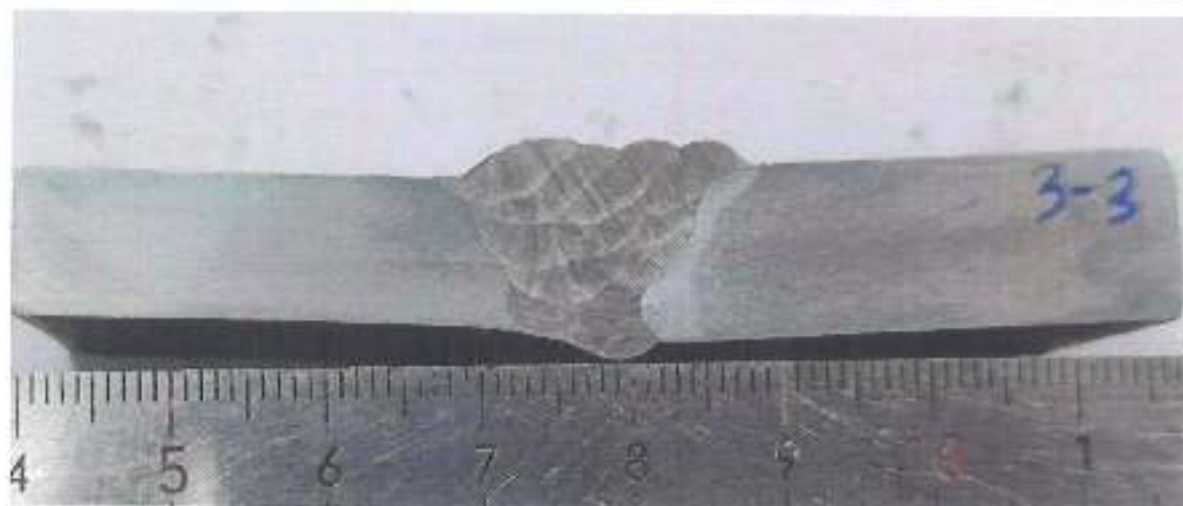
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D. Characteristic / Tên chỉ tiêu : Macro etch ⁽¹⁾ / Tổ chức thô đại

➤ Specimen 3-3 / Mẫu 3-3 :

- Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn.
- No defect in weld metal / Trong mối hàn không có khuyết tật.



Original sample photo(s) / Hình ảnh mẫu nhận:



SGS authenticate the photos on original report only / SGS chỉ xác thực các hình ảnh trong báo cáo chính thức

*** END OF THE REPORT ***

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MILL TEST CERTIFICATE

EN 10204/3.1



CNS-1000000000

ADD: PT. INDU, Desa Fatmaja, Kecamatan Sukadana, Kabupaten
Korupali, Provinsi Sulawesi Tengah, Indonesia

Tel: 0822256751870

Email: liuy@etalingshan.com

Manufacturer: PT. INDONESIA GUNG CHING STEEL AND STAINLESS
STEEL INDUSTRY

Product		VIB-STAR INTERNATIONAL JOINT STOCK COMPANY		Rolling Factory		PHOTO-100-LP-COM		Contract No.		CCTV/VIDEO-100						Certificate No.		20200111007		
Product Name		HOT ROLLED STAINLESS STEEL SHEET		Steel Grade		304		Product Standard		ASTM A240						Description of delivery		For rolling or delivery condition		
No.	Coil No.	QTY	Heat No.	Size (mm)	Weight (kg)	Chemical Composition (%)										Tensile Test				Tolerance
						C	Si	Mn	P	S	Cr	Ni	Al	Fe					R _m (MPa)	
1	20200111007	1	7210000021-0	3.05*1000*1000	21.355	0.042	0.41	1.14	0.003	0.002	18.16	8.25	0.030	0.01	0.01	525.8	208.2	91.5	81.5	
2	20200111007	1	7210000021-1	3.05*1000*1000	21.155	0.045	1.48	1.16	0.005	0.002	18.11	8.04	0.044	0.04	0.04	545.5	204.4	95.8	95.8	
3	20200111007	1	7210000021-2	3.05*1000*1000	21.235	0.040	1.76	1.18	0.002	0.002	18.20	8.01	0.005	0.03	0.03	542.6	205.5	95.8	85.8	
4	20200111007	1	7210000021-3	3.05*1000*1000	21.165	0.004	1.41	0.95	0.002	0.002	18.21	8.02	0.003	0.04	0.04	544.2	205.2	95.7	95.7	
5	20200111007	1	7210000021-4	3.05*1000*1000	21.217	0.007	1.42	0.94	0.004	0.002	18.18	8.02	0.006	0.04	0.04	543.5	205.3	95.7	98.0	
6	20200111007	1	7210000021-5	3.05*1000*1000	21.110	0.051	0.40	0.93	0.007	0.002	18.25	8.04	0.000	0.04	0.04	537.3	205.0	95.4	95.4	
7	20200111007	1	7210000021-6	3.05*1000*1000	21.350	0.045	0.45	1.02	0.004	0.002	18.20	8.02	0.040	0.04	0.04	544.5	202.8	95.9	85.9	
														</						

Remarks

1. Product marking: steel number, heat No, coil No, size.

2. Surface Quality: ok 3. Shape: ok 4. Dimension: ok 5. Radioactivity Tests: ok 6. Non-metallic inclusions: ok 7. Intergranular corrosion: ok

1. We hereby certify that the material herein described has been manufactured, sampled, tested and inspected in accordance with the requirements of above specifications.

2. We hereby certify Product No-weld repair, Free of mercury, Free from radiation contamination.

3. ISO 9001:2015 Quality management system identified by ZJQC.

4. We declare that our testing center has passed ISO/IEC 17025:2017 accreditation review, Accreditation body: CNAS, accreditation capability range: Spectral analysis and tensile testing, certificate number: CNAS L12639.

5. We certify that the material described below fully conforms to IS 6811:2017 chemical composition and Mechanical properties of product as listed in accordance with the scheme of Testing and Inspection outlined in BIS certification marks license No. CNL-0100045662.

6. Issued in agreement with TÜV SÜD Industrie Service GmbH (April 2018), QMS approved acc. to PECCAD 2000, Annex 1, Para. 4.3 by Notified Body 0026 (Certification no. DQR-0036-03-W 13520-01-MUC).

7. Approved acc. to CPR 305/2011 with certification no. 0036-CPR-M-115-2010.

8. Product heat treatment temperature: 1100°C.

9. The certificate should be sealed by quality control department, add date, inspection stamp or authorized official stamp of sales.

Typed by	徐红叶	
Rectifying	利	
Date	2021/7/17	



MILL CERTIFICATE

台塑河靜鋼鐵興業責任有限公司

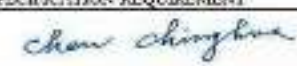
FORMOSA HA TINH STEEL CORPORATION

ADDRESS: Vung Ang Economic Zone, Ky Long Ward, Ky Anh Town,
Ha Tinh Province, Viet Nam

TEL : +84-239-3722-123 Ext.2066 Fax : +84-239-3722-112

PURCHASER	CITICOM COMMERCIAL JOINT STOCK COMPANY	COMMODITY	HOT ROLLED COIL, NON-SKINPASSED		
SPECIFICATION and GRADE	JIS G 3101:2015 SS400	FHS CONTRACT NO.	1H300680064	PURCHASER ORDER NO.	
INSPECTION	FHS MILL INSPECTION	CERTIFICATE NO.	QZA021300806	INVOICE NO.	
END USER NAME	CITICOM COMMERCIAL JOINT STOCK COMPANY	MO ISSUE DATE	2021-05/17	CUSTOMER NO.	MGCTCC

COIL NO.	PRODUCT INFORMATION				HEAT NO.	SAMPLE ID	DIR.	TENSILE TEST				BENDING					CHEMICAL ANALYSIS RESULT %																REMARK																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
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NOTES	COMMODITY on LABEL: HOT ROLLED COIL, NON SKINPASSED SPECIFICATION on LABEL: JIS G 3101:2015 GRADE on LABEL: SS400		
	THIS MILL CERTIFICATE IS ISSUED IN ACCORDANCE WITH EN 10204 TYPE 3.1		
	YPT: YIELD POINT TYPE Tr: TRACE THE ANALYSIS RESULT COMPLIES WITH SPECIFICATION REQUIREMENT		
SURVEYOR TO		WE HEREBY CERTIFY THAT MATERIAL DESCRIBED HEREIN HAS BEEN MANUFACTURED AND TESTED WITH SATISFACTORY RESULTS IN ACCORDANCE WITH THE REQUIREMENT OF THE ABOVE MATERIAL SPECIFICATION.	 ASSISTANT VICE PRESIDENT, TECHNOLOGY DIVISION

CERTIFIED MATERIAL REPORT

CMR No. : HDV – 2023 – 0633 (Rev.2)

Date of Issue : 2023 – 07 – 04



Customer		G-TECH CORPORATION							
Trade Name		SW-309L Cored			Size		1.2 mm x 15 kg		
Lot No.		SW23929F1			Purchaser's Order No.		40-2023/GT/HW/PO		
Specification		AWS A5.22 E309LT1-1 EN ISO 17633-A-T 23 12 L P C1 2 * According to EN 10204-2004 : 3.1							
Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment (°C×hr.)			As Welded		
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)					
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.	
	414	542	45	-	-	-	-	-	
AWS Spec.	-	≥520	≥30	-					
Chemical Composition (%)									
Weld Metal	C	Si	Mn	P	S	Cu	Ni	Cr	Mo
	0.03	0.71	1.33	0.014	0.008	0.04	12.35	22.41	0.04
AWS Spec.	≤0.04	≤1.00	0.50 ~2.50	≤0.040	≤0.030	≤0.75	12.00 ~14.00	22.00 ~25.00	≤0.75
* Radiographic Test : ACCEPT * Fillet Weld Test : GOOD									
Others * Face Bend Test : GOOD * FPC No. (CE-Mark): 0035-CPR-C771 * Q'ty: 900 kg									
We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.									
0035-CPR-C771 0035 06 / HWV-402 EN13479				Approved by QA Manager Pham Dinh Anh Vu					

HYUNDAI WELDING VINA CO., LTD.

Road N2, Nhon Trach 5 IZ, Hiep Phuoc Ward, Nhon Trach District, Dong Nai Province, Vietnam.

Comply:
ISO 9001:2015
AWS D1.6:2017

WELDING PROCEDURE

Doc. No.: WPS-DD-FCAW-334

PQR-DD-FCAW-334

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Nov 09, 2024	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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WELDING PROCEDURE SPECIFICATION (WPS)

(AWS D1.6/D1.6M:2017 Structural Welding Code-Stainless Steel)



Company Name
WPS No.
Supporting PQR(s)
CVN Report

DaiDung Metallic Manufacture Construction & Trade Corporation
WPS-DD-FCAW-334.1
PQR-DD-FCAW-334

Rev. No. A

Date: Oct 17, 2024

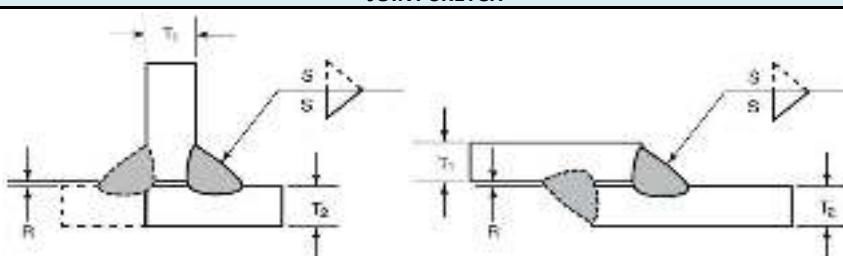
Page: 1 of 1

BASE METALS	Specification	Type or Grade	AWS Group No.
Base Material	ASTM A240/276/312/403 or equivalent	304/316 Any M-8	Metal Group A,B Table 5.2
Welded To	ASTM A572/ A36/ A516	Table 5.3	Any Group I, II
Backing Material	-	-	-
Other	-	-	-

BASE METAL THICKNESS	As-Welded (mm)	With PWHT
CJP Groove Welds	-	-
CJP Groove with CVN	-	-
PJP Groove Welds	-	-
Fillet Welds	All	-
DIAMETER	All	-

JOINT DETAILS	
Joint Type	T, Corner & Lap Joint
Weld Type	Fillet Weld
Root Opening	0 - 2 (mm)
Root Face	-
Dihedral	Per drawing
Weld Size	Per drawing

JOINT SKETCH



POSTWELD HEAT TREATMENT	
Temperature	-
Time at Temperature	-
Other	-

Welded Joint Details as Figures 5.2 AWS D1.6

PROCEDURE			
Weld Layer(s)	All		
Weld Pass(es)	All		
Process	FCAW		
Type (Manual, Mechanized, etc.)	Semi-Automatic, Mechanized		
Position	All		
Vertical Progression	-		
Filler Metal (AWS Spec.)	AWS A5.22		
AWS Classification	E309LT1-1/-4; Group B Table 5.3		
Diameter (mm)	1.2		
Manufacturer/Trade Name	Not strictly		
Shielding Gas (Composition)	99.8% CO2		
Flow Rate (LPM)	15 - 25		
Nozzle Size	No. 6		
Preheat Temperature (Min.)	15°C (Ambient)		
Interpass Temperature (Max.)	175°C		
Electrical Characteristics			
Current Type & Polarity	DCEP		
Transfer Mode	-		
Power Source Type (cc, cv, etc.) (Amps)	CV		
Amps (A)	130 - 180		
Volts (V)	23 - 30		
Wire Feed Speed (Amps)			
Travel Speed (mm/min.)	130 - 460		
Maximum Heat Input (kJ/mm)	1.92		
Technique			
Stringer or Weave	Stringer or Weave		
Multi or Single Pass (per side)	Multi or Single Pass		
Oscillation (Mech./Auto.)	-		
Number of Electrodes	1		
Contact Tube to Work Dist. (mm)	15 - 25		
Peening	None		
Interpass Cleaning	Grinding and Wire Brush		
Other	This procedure can be used for build up or repair.		

	Prepared by Welding Coordinator	Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature			
Date	Oct 17, 2024	Oct 17, 2024	





WELDING PROCEDURE SPECIFICATION (WPS)

(AWS D1.6/D1.6M:2017 Structural Welding Code-Stainless Steel)



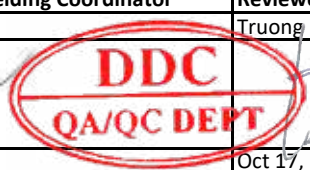
Company Name: DaiDung Metallic Manufacture Construction & Trade Corporation
WPS No.: WPS-DD-FCAW-334.2
Supporting PQR(s): PQR-DD-FCAW-334
CVN Report: -
Rev. No.: A
Date: Oct 17, 2024
Page: 1 of 1

BASE METALS	Specification	Type or Grade	AWS Group No.	BASE METAL THICKNESS	As-Welded (mm)	With PWHT
Base Material	ASTM A240/276/312/403 or equivalent	304/316 Any M-8	Metal Group A,B Table 5.2	CJP Groove Welds	-	-
Welded To	ASTM A572/ A36/ A516	Table 5.3	Any Group I, II	CJP Groove with CVN	-	-
Backing Material	-	-	-	PJP Groove Welds	2-20 mm	-
Other	-	-	-	Fillet Welds	-	-
				DIAMETER	All	-

JOINT DETAILS			JOINT SKETCH	
Joint Type	Butt, T & Corner Joint			
Weld Type	Single V Groove	Single Bevel Groove Weld		
Groove Angle	60° - 70°	45° - 55°		
Root Opening	0 - 2 mm			
Root Face	Per drawing			
Weld Size	Table 5.5			
POSTWELD HEAT TREATMENT			Welded Joint Details as Figures 5.3 AWS D1.6	
Temperature	-			
Time at Temperature	-			
Other	-			

PROCEDURE			
Weld Layer(s)	All		
Weld Pass(es)	All		
Process	FCAW		
Type (Manual, Mechanized, etc.)	Semi-Automatic, Mechanized		
Position	All		
Vertical Progression	-		
Filler Metal (AWS Spec.)	AWS A5.22		
AWS Classification	E309LT1-1/-4; Group B Table 5.3		
Diameter (mm)	1.2		
Manufacturer/Trade Name	Not strictly		
Shielding Gas (Composition)	99.8% CO2		
Flow Rate (LPM)	15 - 25		
Nozzle Size	No. 6		
Preheat Temperature (Min.)	15°C (Ambient)		
Interpass Temperature (Max.)	175°C		
Electrical Characteristics			
Current Type & Polarity	DCEP		
Transfer Mode	-		
Power Source Type (cc, cv, etc.) (Amps)	CV		
Amps (A)	130 - 180		
Volts (V)	23 -30		
Wire Feed Speed	(Amps)		
Travel Speed (mm/min.)	130 - 460		
Maximum Heat Input (kJ/mm)	1.92		
Technique			
Stringer or Weave	Stringer or Weave		
Multi or Single Pass (per side)	Multi or Single Pass		
Oscillation (Mech./Auto.)	-		
Number of Electrodes	1		
Contact Tube to Work Dist. (mm)	15 - 25		
Peening	None		
Interpass Cleaning	Grinding and Wire Brush		
Other	This procedure can be used for build up or repair.		

	Prepared by Welding Coordinator	Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature			
Date	Oct 17, 2024	Oct 17, 2024	





WELDING PROCEDURE SPECIFICATION (WPS)

(AWS D1.6/D1.6M:2017 Structural Welding Code-Stainless Steel)



Company Name: DaiDung Metallic Manufacture Construction & Trade Corporation
WPS No.: WPS-DD-FCAW-334.3
Supporting PQR(s): PQR-DD-FCAW-334
CVN Report: -
Rev. No.: A
Date: Oct 17, 2024
Page: 1 of 2

BASE METALS	Specification	Type or Grade	AWS Group No.
Base Material	ASTM A240/276/312/403 or equivalent	304/316 Any M-8	Metal Group A,B Table 5.2
Welded To	ASTM A572/ A36/ A516	Table 5.3	Any Group I, II
Backing Material	Ceramic or same base metal	-	-
Other	-	-	-

BASE METAL THICKNESS	As-Welded (mm)	With PWHT
CJP Groove Welds	2 - 20	-
CJP Groove with CVN	-	-
PJP Groove Welds	-	-
Fillet Welds	-	-
DIAMETER	All	-

JOINT DETAILS	
Groove Type	See attached sketch for more details
Groove Angle	See attached sketch for more details
Root Opening	See attached sketch for more details
Root Face	See attached sketch for more details
Backgouging	Yes
Method	Grinding

JOINT SKETCH
See Joint Sketch in page 2 of 2

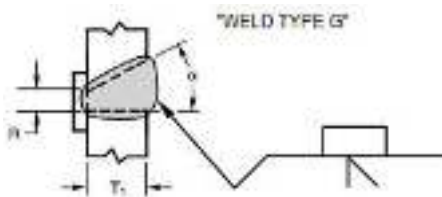
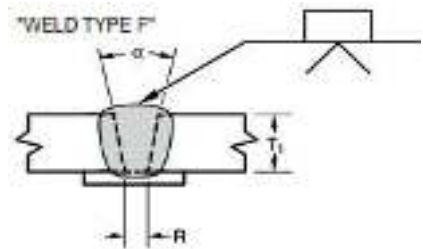
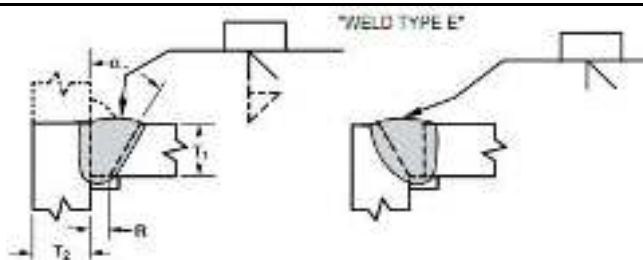
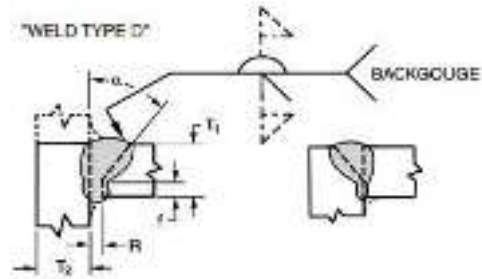
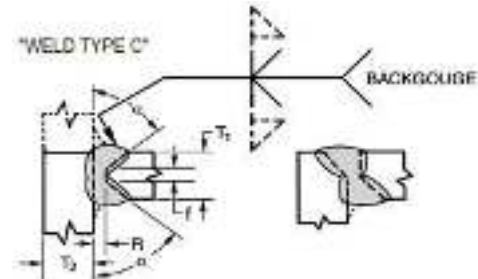
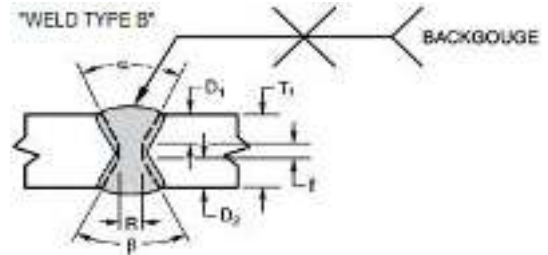
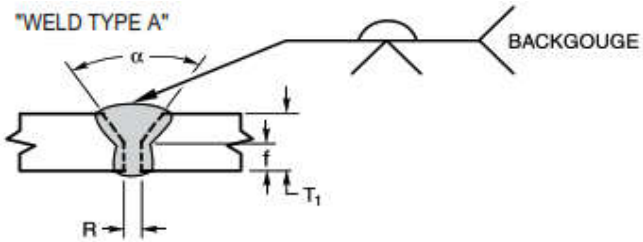
POSTWELD HEAT TREATMENT	
Temperature	-
Time at Temperature	-
Other	-

PROCEDURE			
Weld Layer(s)	All		
Weld Pass(es)	All		
Process	FCAW		
Type (Manual, Mechanized, etc.)	Semi-Automatic, Mechanized		
Position	All		
Vertical Progression	-		
Filler Metal (AWS Spec.)	AWS A5.22		
AWS Classification	E309LT1-1/-4; Group B Table 5.3		
Diameter (mm)	1.2		
Manufacturer/Trade Name	Not strictly		
Shielding Gas (Composition)	99.8% CO2		
Flow Rate (LPM)	15 - 25		
Nozzle Size	No. 6		
Preheat Temperature (Min.)	15°C (Ambient)		
Interpass Temperature (Max.)	175°C		
Electrical Characteristics			
Current Type & Polarity	DCEP		
Transfer Mode	-		
Power Source Type (cc, cv, etc.) (Amps)	CV		
Amps (A)	130 - 180		
Volts (V)	23 - 30		
Wire Feed Speed (Amps)			
Travel Speed (mm/min.)	130 - 460		
Maximum Heat Input (kJ/mm)	1.92		
Technique			
Stringer or Weave	Stringer or Weave		
Multi or Single Pass (per side)	Multi or Single Pass		
Oscillation (Mech./Auto.)	-		
Number of Electrodes	1		
Contact Tube to Work Dist. (mm)	15 - 25		
Peening	None		
Interpass Cleaning	Grinding and Wire Brush		
Other	This procedure can be used for build up or repair.		

Prepared by Welding Coordinator		Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature			
Date	Oct 17, 2024	Oct 17, 2024	



CJP

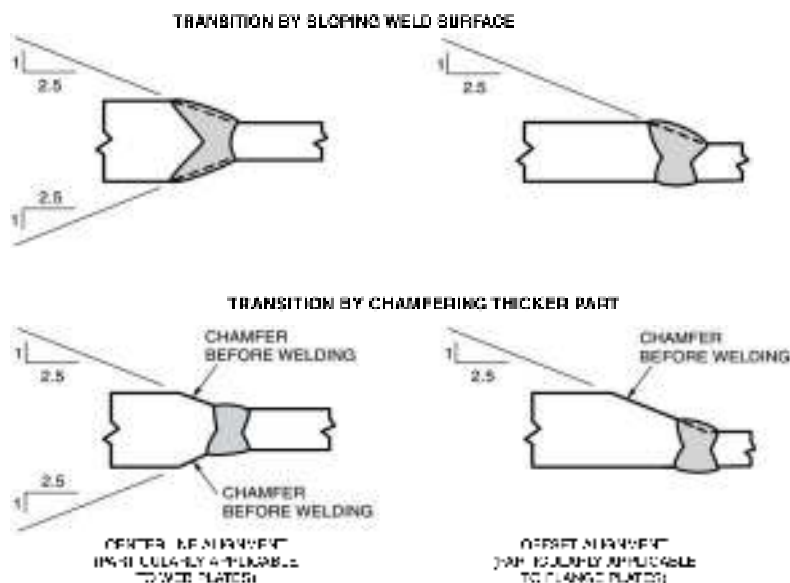


For "WELD TYPE E, F, G":

The permanent backing metal can be used and in conformance with design drawing

Weld Type	Joint Data				Remark
	Root Opening - R (mm)	Root Face - f (mm)	Groove Angel 1 - α	Groove Angle2 - β	
A	0 - 5	0 - 5	$60^{\circ} - 70^{\circ}$	-	Welded Joint Details shall be as described in AWS D1.6, Figure 5.4
B	0 - 5	0 - 5	$60^{\circ} - 70^{\circ}$	$60^{\circ} - 70^{\circ}$	
C	0 - 5	0 - 5	$45^{\circ} - 55^{\circ}$	$45^{\circ} - 55^{\circ}$	
D	0 - 5	0 - 5	$45^{\circ} - 55^{\circ}$	-	
E, F, G	6 - 8	-	$45^{\circ} - 55^{\circ}$	-	

Transition of Splice Joint in Parts of Unequal Section



These Transition shall be as per structure design

Note: Joint data shall be as per the Code AWS D1.6 and in conformance with Project Specification, Drawings.



PROCEDURE QUALIFICATION RECORD (PQR)

(Structural Welding Code-Stainless Steel AWS D1.6/D1.6M:2017)



Company Name: **DaiDung Metallic Manufacture Construction & Trade Corporation**

PQR No.: **PQR-DD-FCAW-334**

Rev. No. **0**

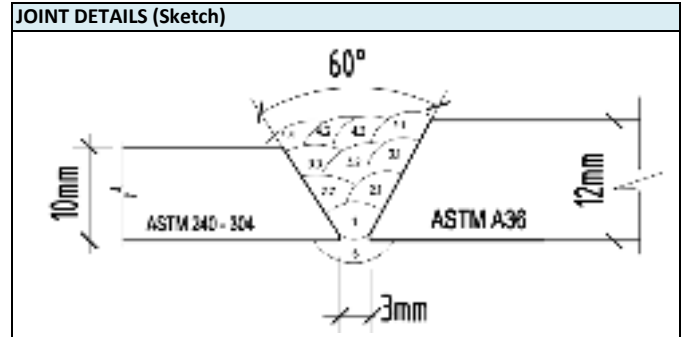
Date: **Oct 17, 2024**

Page: **1 of 2**

BASE METALS	Specification	Type or Grade	AWS Group No.	Thickness	Size (NPS)	Schedule	Diameter
Base Material	ASTM A240	304	Metal Group B	10 mm	-	-	-
Welded To	ASTM A36	-	II	12 mm	-	-	-
Backing Material	Ceramic	-	-	-	-	-	-
Other	-	-	-	-	-	-	-

JOINT DETAILS	
Groove Type	Single V Groove Butt Joint
Groove Angle	60°
Root Opening	3-4 mm
Root Face	1 mm
Backgouging	Yes
Method	Grinding

POSTWELD HEAT TREATMENT	
Temperature	N/a
Time at Temperature	-
Other	-



PROCEDURE					
Weld Layer(s)	1	2 - 3	4	5	
Weld Pass(es)	1	2.1 - 3.3	4.1 - 4.4	5	
Process	FCAW	FCAW	FCAW	FCAW	
Type (Manual, Mechanized, etc.)	Semi-Automatic				
Position	2G	2G	2G	2G	
Vertical Progression	-	-	-	-	
Filler Metal (AWS Spec.)	AWS A5.22	AWS A5.22	AWS A5.22	AWS A5.22	
AWS Classification	E309LT1				
Diameter (mm)	Ø 1.2	Ø 1.2	Ø 1.2	Ø 1.2	
Manufacturer/Trade Name	Hyundai SW-309L Cored				
Shielding Gas (Composition)	CO2 (99.8%)	CO2 (99.8%)	CO2 (99.8%)	CO2 (99.8%)	
Flow Rate (LPM)	15-20	15-20	15-20	15-20	
Nozzle Size	No. 6	No. 6	No. 6	No. 6	
Preheat Temperature (Min.)	33 °C (ambient)				
Interpass Temperature (Max.)		150 °C	164 °C	42 °C	
Electrical Characteristics					
Current Type & Polarity	DCEP	DCEP	DCEP	DCEP	
Transfer Mode	-	-	-	-	
Power Source Type (cc, cv, etc.)	CV	CV	CV	CV	
Amps (A)	140 - 150	140 - 159	140 - 157	140 - 150	
Volts (V)	23 - 24	27 - 29	26 - 29	27 - 28	
Wire Feed Speed	-	-	-	-	
Travel Speed (mm/min.)	133	273 - 545	360 - 429	414	
Maximum Heat Input (kJ/mm)	1.6	1	0.7	0.6	
Technique					
Stringer or Weave	Weave	Weave	Weave	Weave	
Multi or Single Pass (per side)	Multipass	Multipass	Multipass	Multipass	
Oscillation (Mech./Auto.)	-	-	-	-	
Number of Electrodes	1	1	1	1	
Contact Tube to Work Dist.(mm)	20	20	20	20	
Peening	None	None	None	None	
Interpass Cleaning	Wire Brush	Grinding	Wire Brush	Wire Brush	
Other	-	-	-	-	

Prepared by Welding Coordinator		Reviewed and Approved by	
Name	La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature			
Date	Oct 17, 2024	Oct 17, 2024	



	PROCEDURE QUALIFICATION RECORD (PQR) (Structural Welding Code-Stainless Steel AWS D1.6/D1.6M:2017)	
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PQR No.: **PQR-DD-FCAW-334**

Rev. No.: **0**

Date: **Oct 17, 2024**

Page: **2 of 2**

TEST RESULTS

TESTS

	Type of Tests	Clause/Figure(s) Reference	Acceptance Criteria	Result	Remarks
✓	Visual Inspection	Figure 6.5	6.9.3.1	Acceptable	WPR-DD-FCAW-309
	Radiographic Examination	8.14.1	8.12		
	Ultrasonic Testing	8.14.3	8.13		
	Penetrant Testing	8.14.4	8.10 & Table 8.1		
	Magnetic Particle Testing	8.14.5	8.10 & Table 8.1		
	2 Side Bends	Figure 6.6	6.9.3.2		
	4 Side Bends	Figure 6.6	6.9.3.2		
✓	2 Transverse Root Bends	Figure 6.7 or Figure 6.8	6.9.3.2	Acceptable	INL24/512130-746-4
✓	2 Transverse Face Bends	Figure 6.7 or Figure 6.8	6.9.3.2	Acceptable	INL24/512130-746-4
	2 Longitudinal Face Bends	Figure 6.9	6.9.3.2		
	2 Longitudinal Root Bends	Figure 6.9	6.9.3.2		
✓	2 Tensile Tests	Figure 6.14	6.9.3.3	Acceptable	INL24/512130-746-4
✓	3 Macroetch	Annex I	6.9.3.4	Acceptable	INL24/512130-746-4

TENSILE TEST DETAILS

Specimen Number	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Tensile Load (kN)	Ultimate Unit Stress (MPa)	Type of Failure and Location
TS 1	20.0	10.0	199.1	87.7	440	Ductile/base metal A36
TS 2	20.0	9.9	197.6	88.7	449	Ductile/base metal A36

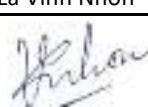
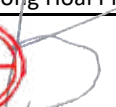
TOUGHNESS TEST DETAILS

Specimen Number	Notch Location	Specimen Size (mm)	Test Temperature	Absorbed Energy (J)	Percent Shear	Lateral Expansion	Average (J)

CERTIFICATION

Welder's Name	ID Number	Stamp Number	Tests Conducted by	
NGUYEN VAN DEM	DDC-WD-0564	-	Laboratory	SGS
-	-	-	Test Number	INL24/512130-746-4
-	-	-	File Number	-

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Clause 6 of AWS D1.6/D1.6M, 2017 *Structural Welding Code-Steel*.

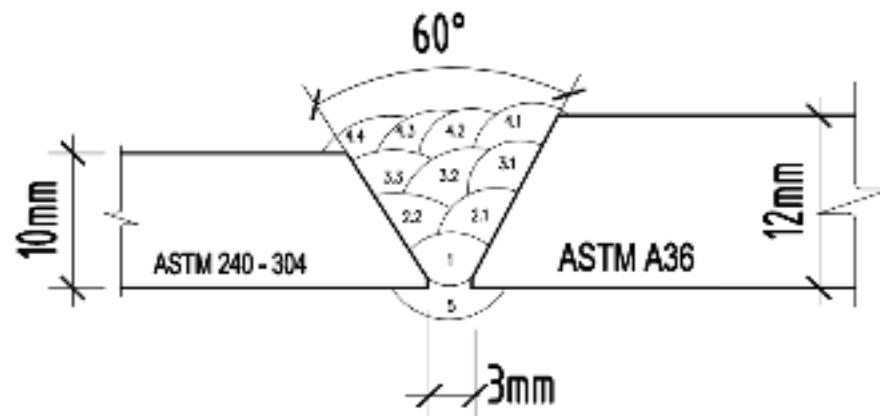
Prepared by Welding Coordinator	Reviewed and Approved by	
Name La Vinh Nhon	Truong Hoai Phong Vu (QA Manager)	
Signature 		
Date Oct 17, 2024	Oct 17, 2024	



WELDING PARAMETERS RECORD

Company name: DaiDung Metallic Manufacture Construction & Trade Corporation
 PQR No.: PQR-DD-FCAW-334
 Test Coupon: DD-FCAW-312
 Test position: 2G
 Base metal:
 - Specification: ASTM A240, TYPE 304 to ASTM A36
 - Dimension: PL10x200x600 mm to PL12x200x600 mm
 - Heat/Product No.: Y210531025-3 (304) & 23110883 (A36)
 - Certificate No.: 20210717107 (304) & S231200164 (A36)
 Filler metal:
 - Specification: AWS A5.22
 - Classification: E309LT1-1
 - Trade name: Hyundai SW-309L Cored
 - Batch/Lot No.: SW23929F1
 - Certificates No.: HDV-2023-0633
 Gas: CO2 (99.8%)
 Welder Name: NGUYEN VAN DEM
 I.D.: DDC-WD-0564

Joint Sketch:



Doc No.: WPR-DD-FCAW-334

Date: Oct 10, 2024

Root face: 1 mm
 Root gap: 3-4 mm
 Groove angle: 60°

Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
1	FCAW	AWS A5.22 E309LT1-1	1.2	DCEP	140 - 150	23 - 24	4	30	600	133	33 (Ambient)	15-20	1.5 - 1.6	9	4	
2.1	FCAW	AWS A5.22 E309LT1-1	1.2	DCEP	151 - 159	27 - 28	2	12	600	273	52	15-20	0.9 - 1	11	3	
2.2	FCAW	AWS A5.22 E309LT1-1	1.2	DCEP	145 - 155	27 - 28	1	42	600	353	96	15-20	0.7 - 0.7	11		
3.1	FCAW	AWS A5.22 E309LT1-1	1.2	DCEP	140 - 150	28 - 29	1	13	600	493	150	15-20	0.5 - 0.5	12		

Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		Volts (V)	min.						sec.	Width	
3.2	FCAW	AWS A5.22 E309LT1-1	1.2	DCEP	146 - 156	28 - 29	1	6	600	545	132	15-20	0.5 - 0.5	12	2	
3.3	FCAW	AWS A5.22/ E309LT1-1	1.2	DCEP	140 - 150	28 - 29	1	9	600	522	143	15-20	0.5 - 0.5	12		
4.1	FCAW	AWS A5.22/ E309LT1-1	1.2	DCEP	145 - 155	26 - 27	1	40	600	360	68	15-20	0.6 - 0.7	12	1	Reinforcement 2mm
4.2	FCAW	AWS A5.22 E309LT1-1	1.2	DCEP	147 - 157	28 - 29	1	35	600	379	138	15-20	0.7 - 0.7	12		
4.3	FCAW	AWS A5.22/ E309LT1-1	1.2	DCEP	145 - 155	28 - 29	1	39	600	364	164	15-20	0.7 - 0.7	12		
4.4	FCAW	AWS A5.22/ E309LT1-1	1.2	DCEP	140 - 150	26 - 27	1	24	600	429	75	15-20	0.5 - 0.6	12		
5	FCAW	AWS A5.22/ E309LT1-1	1.2	DCEP	140 - 150	27 - 28	1	27	600	414	42	15-20	0.5 - 0.6	13		
		Prepared by Welding Coordinator				Reviewed and Approved by										
Name		Nguyen Bao Tinh				Truong Hoai Phong Vu										
Signature		 														
Date		Oct 10, 2024				Oct 10, 2024										

Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
Client's address / Địa chỉ : LÔ SỐ 38, KHU C, ĐƯỜNG D1, KHU CÔNG NGHIỆP AN HẠ, XÃ PHẠM VĂN HAI, HUYỆN BÌNH CHÁNH, THÀNH PHỐ HỒ CHÍ MINH, VIỆT NAM

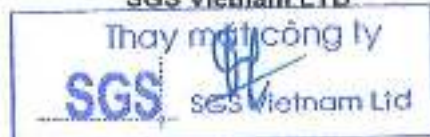
The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : DD-FCAW-312
Base material: ASTM A240 304 + ASTM A36
Size: 10x200x600mm to 12x200x600mm
Welding consumable: AWS A5.22 E309LT1-1
Trade name: Hyundai SW-309L Cored (1.2mm)
Type of Joint: Single V groove, Butt joint
Position 2G
Welder: Nguyen Van Dem - DDC-WD-0564

Project / Dự án : LTIA-PBB (PASSENGER BOARDING BRIDGE)
Quantity / Số lượng mẫu : 01 sample / mẫu
Description / Mô tả mẫu : 600 x 200 x 10 (mm) & 600 x 200 x 12 (mm)
Date of receiving / Ngày nhận mẫu : 11-Oct-2024
Date of testing / Ngày thử nghiệm : From / từ 11-Oct-2024 to / đến 16-Oct-2024
Test method / Phương pháp thử : AWS D1.6/D1.6M:2017 Structural Welding Code - Stainless Steel
Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Signed for and on behalf of
SGS Vietnam LTD.



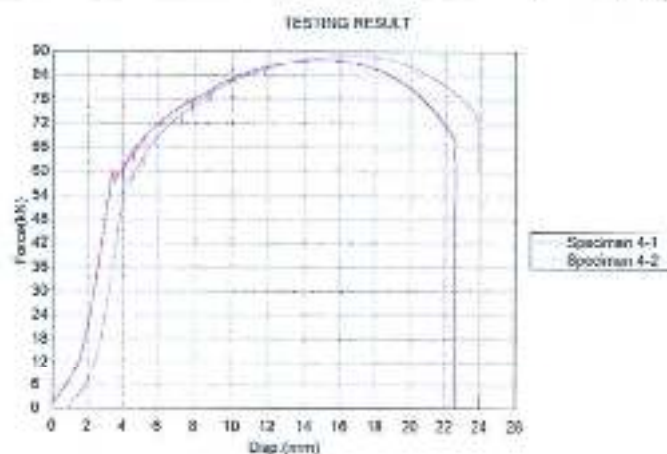
Chau Tran Khanh Dung
Laboratory Supervisor
Industrial Services

Please extract your time to give us your feedback as this will help our service better. Thank you!



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Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm	
A. Transverse tensile test / Thử kéo theo phương ngang	1	2
• Thickness of specimen / Chiều dày mẫu mm	9.96	9.86
• Width of specimen / Chiều rộng mẫu mm	19.99	20.04
• Cross section area / Tiết diện mm ²	199.1	197.6
• Tensile load / Lực kéo đứt kN	87.7	88.7
• Tensile strength / Giới hạn bền kéo MPa	440	449
• Broken position / Vị trí mẫu đứt	In base metal (A36)	In base metal (A36)



The relationship between tensile load and deformation / Biểu đồ tương quan giữa lực thử kéo và biến dạng

Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm			
B. Bend test / Thử uốn	1	2	3	4
• Location bend / Vị trí uốn	Face bend Uốn mặt	Face bend Uốn mặt	Root bend Uốn chân	Root bend Uốn chân
• Thickness of specimen / Chiều dày mm	10.00	10.01	9.99	9.97
• Width of specimen / Chiều rộng mm	39.97	39.99	40.02	40.03
• Bending angle / Góc uốn °/ độ	180	180	180	180
• Plunger diameter / Đường kính gối uốn mm	20	20	20	20
• Result / Kết quả	No Crack Không nứt	No Crack Không nứt	No Crack Không nứt	No Crack Không nứt
• Evaluation / Đánh giá	Pass / Đạt	Pass / Đạt	Pass / Đạt	Pass / Đạt

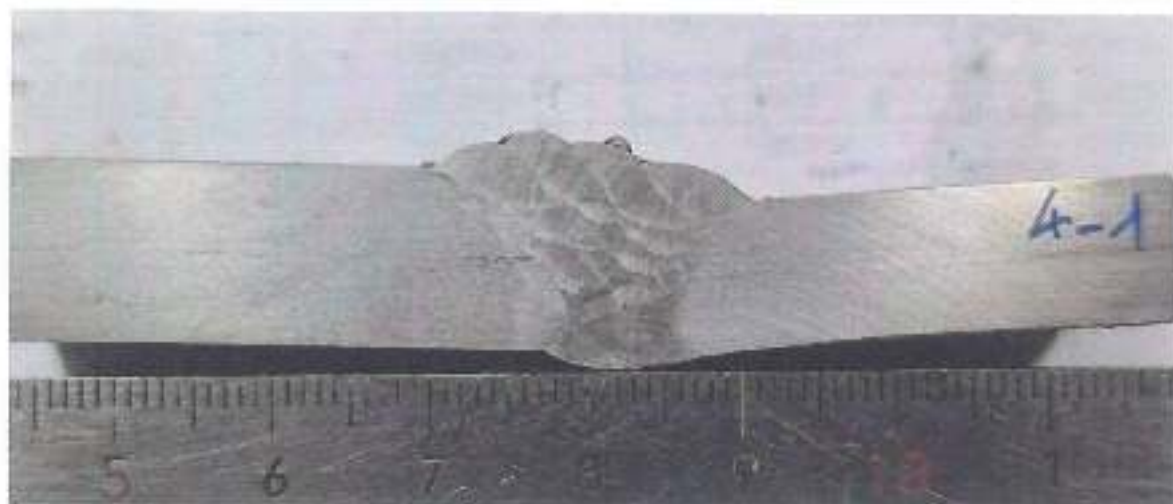
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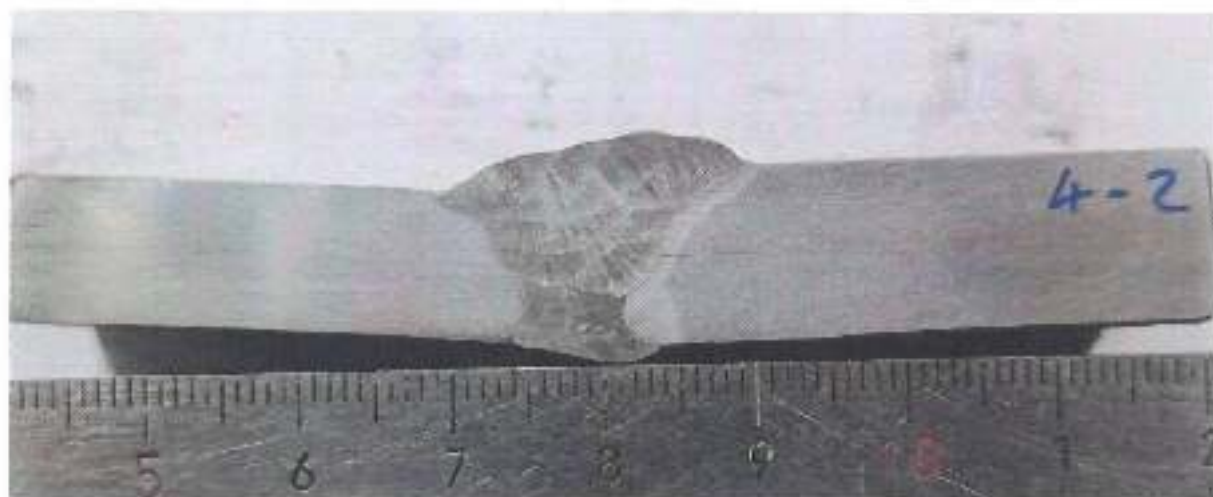
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C. Characteristic / Tên chỉ tiêu : Macro structure / Tổ chức thô đại**➤ Specimen 4-1 / Mẫu 4-1 :**

- Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn.
- No defect in weld metal / Trong mối hàn không có khuyết tật.

**➤ Specimen 4-2 / Mẫu 4-2 :**

- Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn.
- No defect in weld metal / Trong mối hàn không có khuyết tật.



*** TO BE CONTINUED ***

556070

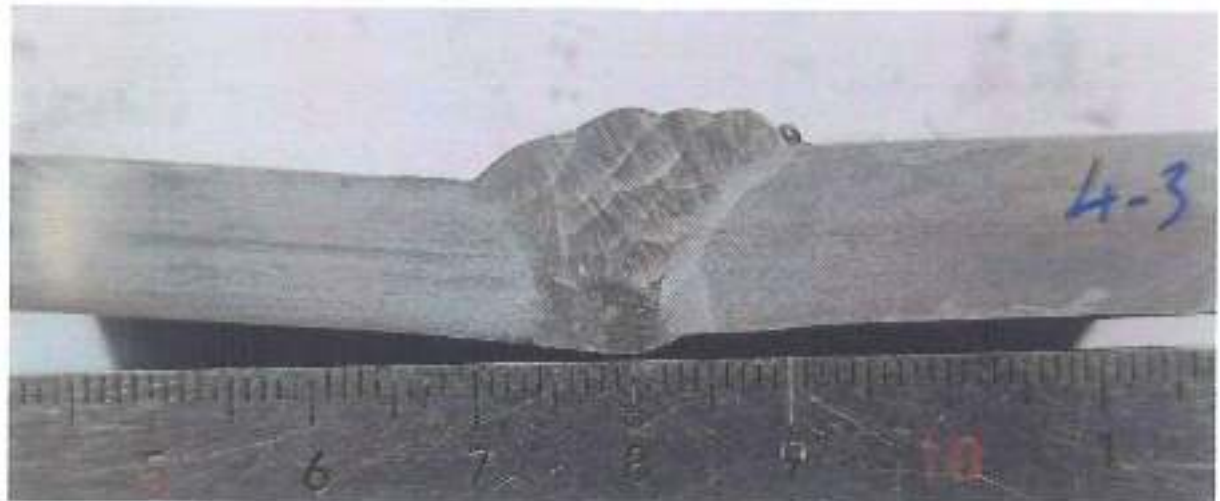
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D. Characteristic / Tên chỉ tiêu : Macro etch □ / Tổ chức thô đại

➤ Specimen 4-3 / Mẫu 4-3 :

- Full penetration and complete fusion weld / Mối hàn thấu và ngấu hoàn toàn.
- No defect in weld metal / Trong mối hàn không có khuyết tật.



Original sample photo(s) / Hình ảnh mẫu nhận:



SGS authenticate the photos on original report only / SGS chỉ xác thực các hình ảnh trong báo cáo chính thức

*** END OF THE REPORT ***

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INSPECTION CERTIFICATE

TEL: 0510-8518801 06188801 FAX: 0510-2618801

序号 No.	批号 LOT No.	牌号 GRADE	规格 SIZE			总重 GWT TGT	重量公差 TOLERANCE	化学成分 CHEMICAL COMPOSITION %										拉伸试验 TENSILE TEST			冲击功 IMPACT KJ/M2	V型冲击功 V-TYPE IMPACT				
			厚度 THICKNESS	宽度 WIDTH	长度 LENGTH			C	SI	Mn	P	S	Cu	Ni	Al	Ba	Caq	Y.S.	T.S.	EL.		温度 TEMPERATURE	1	2	3	AVE
1	23110853	A36	12	2000	12000	20	52.78	10	33	35	25	18						546	445	28	OK	20℃	115	117	97	110
2	23110894	A36	12	2000	12000	26	58.598	17	31	33	22	25						253	442	25	OK	20℃	113	119	84	105
3	23110522	A36	12	2000	12000	12	31.556	18	24	38	22	21						308	442	25	OK	20℃	132	127	125	128
4	23110523	A36	12	2000	12000	11	25.013	17	30	45	30	24						277	424	28	OK	20℃	130	125	121	125
5	23111034	A36	12	2000	12000	8	21.104	16	24	45	23	21						278	441	28	OK	20℃	132	132	88	117
合计 (TOTAL)						77	203.125																			
船级社检验合格证明专用章 CCS TEST CERTIFICATE SEAL	备注 REMARKS	1. 请用户按产品号验收, 如有质量异议请填写好质量异议单, 并凭实物妥善保管好, 切勿动用, PLEASE LET CUSTOMER ACCEPT THE CARGO ACCORDING TO THE PRODUCT LOT NUMBER. IF THERE IS QUALITY CLAIM, PLEASE SEND US OFFICIAL LETTER WITH QUALITY CLAIM. DO NOT USE AND KEEP SAFELY IN KIND. 2. 检验员生效, VALID AFTER OFFICIAL SIGN. 3. 复印无效, THE COPY OF THIS CERTIFICATE IS NOT VALID															制表人 INSPECTOR	陆纯阳	船级社检验合格证明专用章 CCS TEST CERTIFICATE SEAL							
	注释 NOTES	Y.S.=YIELD STRENGTH T.S.=TENSILE STRENGTH EL=ELONGATION RR=REVERSE BEND CB=COLD BEND															质量负责人 QUALITY MANAGER	单书普								
兹证明本表所列产品, 均依材料规范制造及试验, 并且符合规范的要求。 WE HEREBY CERTIFY THAT MATERIAL DESCRIBED HEREIN HAS BEEN MANUFACTURED AND TESTED WITH SATISFACTORY IN ACCORDANCE WITH THE REQUIREMENT OF THE ABOVE MATERIAL SPECIFICATION.																										



MILL TEST CERTIFICATE

EN 10204/3.1



ADD: PT. IMIP, Desa Fatmaja, Kecamatan Sukadadi, Kabupaten Maros, Sulawesi Selatan, Indonesia

Tel: 0822296701870
Email: liuy@etalingshan.com

Manufacturer: PT. INDONESIA GUANG CHENG STEEL AND STAINLESS STEEL INDUSTRY

Product Name		MATERIAL		Rolling Process		MATERIAL		Contract No.		MATERIAL		Certificate No.		20200111007						
Product Name		MATERIAL		Rolling Process		MATERIAL		Contract No.		MATERIAL		Certificate No.		20200111007						
No.	Roll No.	QTY	Heat No.	Size (mm)	Batch (No.)	Chemical Composition (%)										Tensile Test				Reheated
						C	Si	Mn	P	S	Cr	Ni	Al	As	Se	R _m (MPa)	R _{p0.2} (MPa)	F _{0.01} (MPa)	Elongation A (%)	
1	20200111007	1	20200111007	5.00*1000*1000	21.155	0.042	0.41	1.14	0.003	0.002	18.16	0.00	0.000	0.01	0.01	525.8	208.2	7	55.1	81.5
2	20200111007	1	20200111007	5.00*1000*1000	21.155	0.042	0.41	1.14	0.003	0.002	18.16	0.00	0.000	0.01	0.01	525.8	208.2	7	55.1	81.5
3	20200111007	1	20200111007	5.00*1000*1000	21.155	0.042	0.41	1.14	0.003	0.002	18.16	0.00	0.000	0.01	0.01	525.8	208.2	7	55.1	81.5
4	20200111007	1	20200111007	5.00*1000*1000	21.155	0.042	0.41	1.14	0.003	0.002	18.16	0.00	0.000	0.01	0.01	525.8	208.2	7	55.1	81.5
5	20200111007	1	20200111007	5.00*1000*1000	21.155	0.042	0.41	1.14	0.003	0.002	18.16	0.00	0.000	0.01	0.01	525.8	208.2	7	55.1	81.5
6	20200111007	1	20200111007	5.00*1000*1000	21.155	0.042	0.41	1.14	0.003	0.002	18.16	0.00	0.000	0.01	0.01	525.8	208.2	7	55.1	81.5
7	20200111007	1	20200111007	5.00*1000*1000	21.155	0.042	0.41	1.14	0.003	0.002	18.16	0.00	0.000	0.01	0.01	525.8	208.2	7	55.1	81.5
Total		7			21.155															

Remarks

1. Product marking: steel number, heat No, roll No, size.

2. Surface Quality: ok 3. Shape: ok 4. Dimension: ok 5. Radioactivity Tests: ok 6. Non-metallic inclusions: ok 7. Intergranular corrosion: ok

1. We hereby certify that the material herein described has been manufactured, sampled, tested and inspected in accordance with the requirements of above specifications.

2. We hereby certify Product No-weld repair, Free of mercury, Free from radiation contamination.

3. ISO 9001:2015 Quality management system identified by ZJQC.

4. We declare that our testing center has passed ISO/IEC 17025:2017 accreditation review, Accreditation body: CNAS, accreditation capability range: Spectral analysis and tensile testing, certificate number: CNAS L12639.

5. We certify that the material described below fully conforms to IS 6811:2017 chemical composition and Mechanical properties of product as listed in accordance with the scheme of Testing and Inspection outlined in BIS certification marks license No. CNL-0100045662.

6. Issued in agreement with TÜV SÜD Industrie Service GmbH (April 2018), QMS approved acc. to PECCAD 2000, Annex 1, Para. 4.3 by Notified Body 0026 (Certification no. DQR-0036-03-W 13520-01-MUC).

7. Approved acc. to CPR 205/2011 with certification no. 0036-CPR-M-115-2010.

8. Product heat treatment temperature: 1100°C.

9. The certificate should be sealed by quality control department, add date, inspection stamp or authorized official stamp of sales.

Typed by	徐红叶	
Rectifying	利	
Date	2021/7/17	

CERTIFIED MATERIAL REPORT

CMR No. : HDV – 2023 – 0633 (Rev.2)

Date of Issue : 2023 – 07 – 04



Customer		G-TECH CORPORATION							
Trade Name		SW-309L Cored			Size		1.2 mm x 15 kg		
Lot No.		SW23929F1			Purchaser's Order No.		40-2023/GT/HW/PO		
Specification		AWS A5.22 E309LT1-1 EN ISO 17633-A-T 23 12 L P C1 2 * According to EN 10204-2004 : 3.1							
Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment (°C×hr.)			As Welded		
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)					
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.	
	414	542	45	-	-	-	-	-	
AWS Spec.	-	≥520	≥30	-					
Chemical Composition (%)									
Weld Metal	C	Si	Mn	P	S	Cu	Ni	Cr	Mo
	0.03	0.71	1.33	0.014	0.008	0.04	12.35	22.41	0.04
AWS Spec.	≤0.04	≤1.00	0.50 ~2.50	≤0.040	≤0.030	≤0.75	12.00 ~14.00	22.00 ~25.00	≤0.75
* Radiographic Test : ACCEPT * Fillet Weld Test : GOOD									
Others * Face Bend Test : GOOD * FPC No. (CE-Mark): 0035-CPR-C771 * Q'ty: 900 kg									
We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.									
0035-CPR-C771 0035 06 / HWV-402 EN13479				Approved by QA Manager Pham Dinh Anh Vu					

HYUNDAI WELDING VINA CO., LTD.

Road N2, Nhon Trach 5 IZ, Hiep Phuoc Ward, Nhon Trach District, Dong Nai Province, Vietnam.



CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION
● 38, Zone C, D1 St, An Ha Industrial Park, Pham Van Hai, Binh Chanh, HCMC
● sales@daidung.vn ● (84-28) 3868 1689 ● www.daidung.com.vn

Comply:
ISO 9001:2015
AWS D1.3:2018

WELDING PROCEDURE

Doc. No.: WPS-DD-FCAW-324

PQR-DD-FCAW-324

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	Nov 11, 2024	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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WELDING PROCEDURE SPECIFICATION (WPS)

(AWS D1.3/D1.3M: 2018 STRUCTURAL WELDING CODE - SHEET STEEL)



Company Name: DaiDung Metallic Manufacture Construction & Trade Corporation
 WPS No.: WPS-DD-FCAW-324 Rev. No.: 00 Date: Nov 19, 2024 Page: 1 of 3
 Supporting PQR(s): PQR-DD-FCAW-324.1/ .2/ .3/ .4/ .5/ .6 CVN test: N/A

BASE METALS (1.4)	Specification	Type or Grade	AWS D1.3 Group No.	BASE METAL THICKNESS (sheet)	As-Welded (mm)	With PWHT
Sheet steel	ASTM A53... JIS 3101 CS, Low alloy	Gr A, B... SS400 Weldable steel	- Any Group I, II - - Note 1	Square Groove Weld Flare bevel/V Weld	1 - 4 1 - 4	-
Supporting Structural	ASTM A53... JIS 3101 CS, Low alloy	Gr A, B... SS400 Weldable steel	- Any Group I, II - - Note 2	Filet Welds	1 - 4	-
Other	- Coating in weld joint: without - Backing Material: ceramic or same steel sheet			DIAMETER	- Groove: N/A - Fillet: any O.D	

JOINT DETAILS (Table 6.1)	
Type of Joint	Square joint, Flare, T, Lap joint
Butt weld (square)	Both side & one side with backing
Type of Weld	Butt weld, Flare bevel & V groove weld, Fillet weld
Other	See sketch is attached
Backgouging	Yes
Method	Grinding

JOINT SKETCH	
On page 2 & 3 of 3	

POSTWELD HEAT TREATMENT	
Soaking Temperature	-
Soaking Time	-
Other	-

PROCEDURE	Fillet	Square Groove Weld One side both side	Flare bevel/V Groove Weld
Weld Layer(s)	all	all	all
Weld Pass(es) per layer	1 or more	1 or more	1 or more
Process	FCAW	FCAW	FCAW
Type (Manual, Mechanized, etc.)	Semi-Automatic	Semi-Automatic	Semi-Automatic
Position (Table 1.3)	F H	F H	F H
Vertical Progression	-	-	-
Filler Metal (Spec. Table 1.2)	AWS A5.20	AWS A5.20	AWS A5.20
AWS Classification	E71T-1C	E71T-1C	E71T-1C
Diameter (mm)	Ø 1.2	Ø 1.2	Ø 1.2
Manufacturer/Trade Name	Hyundai SC-71LH/ Kuang Tai or Other same classification		
Control of Filler Metal	Drying condition follow Welding Manufacture recommendation		
Shielding Gas (Composition)	Min 99.8% CO ₂	Min 99.8% CO ₂	Min 99.8% CO ₂
Flow Rate (recommend) (LPM)	14 - 21	14 - 21	14 - 21
Nozzle Size (recommend)	No. 6	No. 6	No. 6
Preheat Temperature (t ≤ 20mm) (Min.)	Min 20°C See Note 3	Min 20°C See Note 3	Min 20°C See Note 3
Interpass Temperature (Max.)	250 °C	250 °C	250 °C
Electrical Characteristics			
Current Type & Polarity	DCEP	DCEP	DCEP
Transfer Mode	-	-	-
Power Source Type (cc, cv, etc.)	CV	CV	CV
Amps (A)	86 - 120	86 - 120	86 - 120
Volts (V)	17.5 - 22	17.5 - 23	17.5 - 22
Travel Speed (weld time) (mm/min.)	162 - 220	180 - 210	198 - 260
Wire Feed Speed	(Amps)	(Amps)	(Amps)
Heat Input (kJ/mm)	-	-	-
Technique			
Stringer or Weave	Stringer or Weave	Stringer or Weave	Stringer or Weave
Multi or Single Pass (per side)	Multi or Single Pass	Multi or Single Pass	Multi or Single Pass
Number of Electrodes	1	1	1
Contact Tube to Work Dist. (recommend) (mm)	7 - 16	7 - 16	7 - 16
Peening	None	None	None
Interpass Cleaning	Grinding and Wire Brush	Grinding and Wire Brush	Grinding and Wire Brush

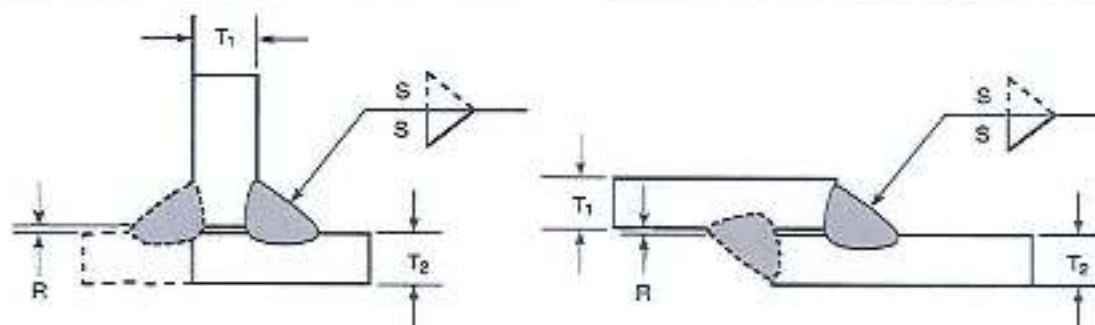
Other: This procedure can be used for repair.
 NOTE Note 1: application for weldable CS & low alloy steel having a minimum specified tensile strength less than 483 Mpa
 Note 2: application for weldable CS & low alloy steel having a minimum specified yield point less than 550 Mpa
 Note 3: minimum preheat follow table 5.8 AWS D1.1/D1.1M. Unlit steel will be follow equivalent steel in table 5.8 AWS D1.1/D1.1M

Category	Thickness < 20mm			
B	Preheat min 20°C			
C	Preheat min 20°C			
D, E	Preheat min 20°C			

Name	DaiDung Metallic Manufacture Construction and Trade Corporation	Examining Body Vice Director of Quatest 3
Signature	Nguyễn Quỳnh Sơ	Nguyễn Anh Triết
Date	Nov 19, 2024	Nov 19, 2024



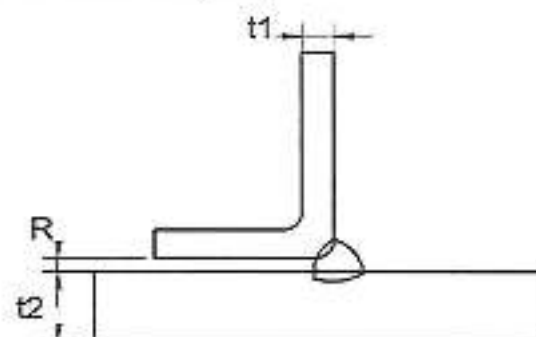
Fillet weld
T-joint (6.2C-D)
Corner joint (6.2C-D)
Lap joint (6.2A-B)



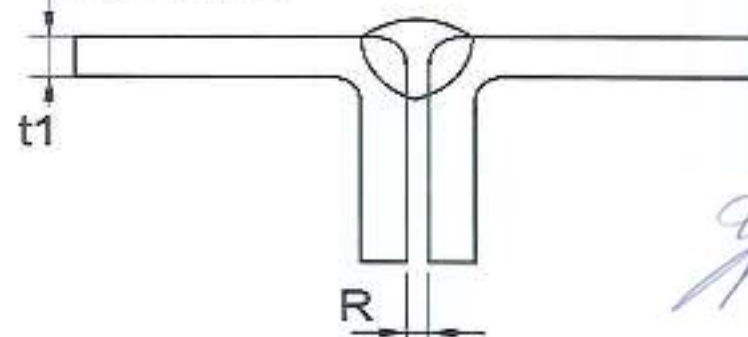
Fillet

Weld Process	Joint Designation Table 6.1	Base Metal thickness sheet	Supporting Structure	Root Gap			Root Face			Groove Angle (degree)			Depth of Groove			Position	Note
				R (mm)			f (mm)			α (°)			D1, D2 (mm)				
				Size	Tolerance		Size	Tolerance		Size	Tolerance		Size	Tolerance			
					Recommend	Fit up Limited		Recommend	Fit up Limited		Recommend	Fit up Limited		recommend	Fit up Limited		
FCAW	6.2C-D	1-4	Qualified	0	+1, -0	Max 0.5 t									F, H	a, b	
FCAW	6.2A-B	1-4	Qualified	0	+1, -0	Max 0.5 t									F, H	a, b	

Flare Bevel Groove weld (6.3 A-B)



Flare V groove weld (6.3C)

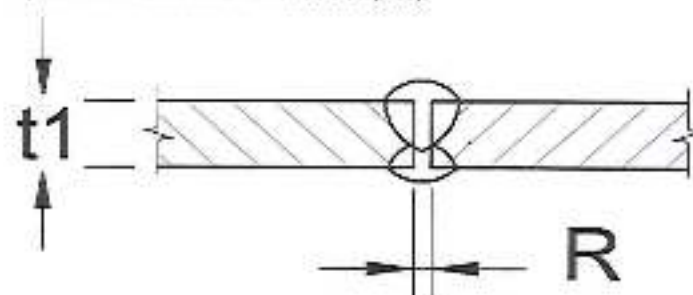


Handwritten signature

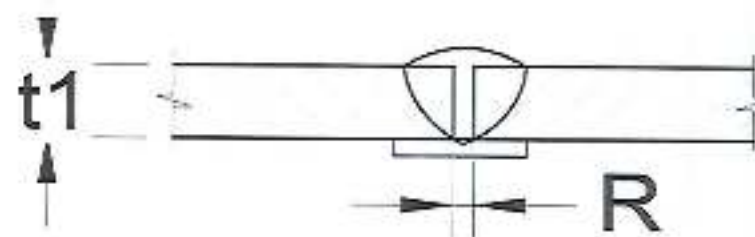
Flare Groove Weld

Weld Process	Joint Designation on Table 5.1	Base Metal thickness as sheet	Supporting Structure	Root Gap			Root Face			Groove Angle (degree)			Depth of Groove			Position	Note
				R (mm)			f (mm)			α (°)			D1, D2 (mm)				
				Size	Tolerance		Size	Tolerance		Size	Tolerance		Size	Tolerance			
					Recommend	Fit up Limited		Recommend	Fit up Limited		Recommend	Fit up Limited		recommend	Fit up Limited		
FCAW	5.3 A-B	1-4	Qualified	0	+1, -0	Max 0.5t									F, H	a, b	
FCAW	6.3 C	1-4	N/A	0	+1, -0	Max 0.5t									F, H	a, b	

SQUARE JOINT BOTH SIDE - BACKGUAGE (S.1B)



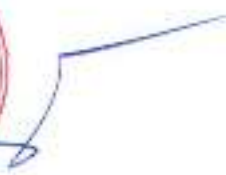
SQUARE JOINT ONE SIDE WITH BACKING

QIP
Square
Joint

Weld Process	Joint Designation Table 6.1	Base Metal thickness sheet	Supporting Structure	Root Gap			Root Face			Groove Angle (degree)			Depth of Groove			Position	Note
				R (mm)			f (mm)			α (°)			D1, D2 (mm)				
				Size	Tolerance		Size	Tolerance		Size	Tolerance		Size	Tolerance			
					Recommend	Fit up Limited		Recommend	Fit up Limited		Recommend	Fit up Limited		recommend	Fit up Limited		
FCAW	6.1B	1-4	N/A	0	+1, -0	Max 1									F	b	
FCAW	Backing	1-4	N/A	2	+1, -0	Max 3									F	b	

Note:

- t: thickness of steel sheet
- Actual run sequence will depend on (welds size), travel speed and welder skill etc.
- a: weld size refer to fabrication drawing or weld map
- b: The orientation of the two members in the joints refer to fabrication drawing

Dai Dung Metallic Manufacture Construction and Trade Corporation		Examining Body Vice Director of Quatest 3	
Name	Nguyen Quynh Sa	Nguyen Anh Trist	
Signature			
Date	19-Nov-2024	Nov 19, 2024	



Số: KT3-0444813
CK4/8

Ngày: 19/11/2024
Trang số: 01/03

BẢO CÁO CHẤP NHẬN QUY TRÌNH HÀN
WELDING PROCEDURE QUALIFICATION RECORD (PQR)
Số PQR/ PQR No.: DDC-FCAW-324.1

Tên công ty/ Company name : DaiDung Metallic Manufacture Construction & Trade Corporation
Địa chỉ/ Address : 38, Zone C, D1 Street, An Ha Industrial Park, Pham Van Hai
Commune, Binh Chanh District, HoChiMinh City
Quá trình hàn/ Welding Process : Hàn hồ quang dây kim loại lõi thuốc trong khí hoạt tính / FCAW
Phương pháp hàn/ Type : Bán tự động/ Semiautomatic
Tiêu chuẩn áp dụng/ Applicable code : AWS D1.3/D1.3M:2018
Tên thợ hàn/ Welder's name (ID) : PHAN THANH PHUONG (082097014779)/ DDC-WD-0393

ĐẶC TÍNH MỐI NỐI/ JOINT DESIGN USED

Loại mối hàn/ Type: Flare-bevel-groove weld
Hàn một phía/ Welded from one side ☒
Đệm lót/ Backing ☐
Vật liệu/ Backing material: -
Khe hở chân/ Root opening: Max. 0,5 mm
Chiều cao chân/ Root face: -

Hàn hai phía/ Welded from both sides ☐
Tạo rãnh mặt sau/ Back gouging ☐
Phương pháp/ Method: -
Góc vát/ Groove Angle: -
Thông tin khác/ Other: Xem sơ đồ/ See sketch

KIM LOẠI CƠ BẢN/ BASE METALS

Đặc tính kỹ thuật/ Material spec.: JIS G3101
Loại/ Grade: SS400

Chiều dày/ Thickness: 2,0 mm
Đường kính/ Diameter (OD): -

KIM LOẠI BÓ SUNG/ FILLER METALS

Đặc tính kỹ thuật/ AWS Spec.: AWS A5.20
Loại/ AWS Classification: E71T-1C

Nhãn hiệu/ Trade name: HYUNDAI SC-71LH
Đường kính/ Diameter: 1,2 mm

BẢO VỆ MỐI HÀN/ SHIELDING

Thuốc bảo vệ/ Flux: ☐
Loại/ Class: -
Đặc tính kỹ thuật/ Specification: -

Khí bảo vệ/ Gas: ☒
Thành phần/ Composition: 99,9% CO₂
Lưu lượng/ Flow rate: (15 - 17) L/min
Lỗ miệng ống khí/ Gas cup size: 15 mm

NUNG NÓNG TRƯỚC KHI HÀN/ PREHEAT

Nhiệt độ nung nóng: 31 °C (Ambient)
Preheat Temp.

Nhiệt độ giữa các lớp/ Interpass temp.:
Min -; Max -

THẾ HÀN/ POSITION

Thế hàn rãnh/ Position of Groove: 2G/H
Thế hàn góc/ Fillet Position: -

Hàn ở vị trí đứng/ Vertical Progression:
Hàn leo/ Up ☐ Hàn tự/ Down ☐

ĐẶC TÍNH DÒNG ĐIỆN HÀN/ ELECTRICAL CHARACTERISTICS

Loại dòng điện/ Current:
AC ☐ DCEP ☒ DCEN ☐ Pulsed ☐
Kiểu chuyển dịch kim loại hàn: Cầu
Transfer mode: Globular

Loại nguồn điện/ Power source: CV
Điện cực Vonfram/ Tungsten electrode (GTAW):
Loại/ Type: -
Đường kính/ Diameter: -

Ngày: 19/11/2024
Trang số: 02/03

Điện cực/ *Electrode:*

Số lượng điện cực/ *Number of Electrodes* 1

Di chuyển đường thẳng/ *Stringer Bead* ☐

Khoảng cách các điện cực/ *Electrodes Spacing*:

Di chuyển đường zig zac/ Weave Bead ☒

Neang / Lateral:

Khoảng cách ống kẹp/vật hàn: (7 – 10) mm
Contact tube to work distance

Doc / Longitudinal:

Góc nghiêng diện cực/ *Angle:*

Hàn một lớp/ *Single Pass (per slide)* ☒

Làm sạch xỉ hàn: Bàn chải

Hàn nhiều lớp/ *Multi-pass (per side)* ☐

Slag removal: Brushing

Kiểu cấp dây/*Wire feed type:* _____

XỬ LÝ NHIỆT SAU KHI HÀN/ POSTWELD HEAT TREATMENT (N/A)

Nhiệt độ/ Temperature: - Thời gian duy trì/ Time: -

CHI TIẾT QUÁ TRÌNH HÀN/ WELDING PROCEDURE

Lớp hàn <i>Layers</i>	Quá trình hàn <i>Process</i>	Kim loại hàn <i>Filler metals</i>		Dòng điện hàn <i>Current</i>			Tốc độ hàn <i>Travel speed,</i> cm/min	Nhiệt cấp vào/ <i>Heat input</i> kJ/mm	Chi tiết mối nối <i>Joint Details</i>
		Loại <i>Class</i>	Đường kính <i>Diam.,</i> mm	Phân cực <i>Polarity</i>	Điện thế <i>Volts,</i> V	Cường độ <i>Amps,</i> A			
1	FCAW	E71T-1C	1,2	DCEP	19 – 20	95 – 110	18 – 20	0,54 – 0,73	

KẾT QUẢ KIỂM TRA MỖI HÀN/ TEST RESULTS

1. Kiểm tra ngoại quan/ *Visual examination*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-04448BCK4/1)
2. Kiểm tra chụp ảnh bức xạ/ *Radiographic test*: Không áp dụng/ *Not applicable*.
3. Thử uốn/ *Bend test* (Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-04448BCK4/2)

Mẫu/ Specimen No	Vị trí hỏng Failure location	Kết quả/ Result	Ghi chú/ Remarks
FCAW-H-FBG-S2.0-SS400 P-T-PHUONG, S1	Vật liệu cơ bản/ Base metal	Đạt/ Pass	
FCAW-H-FBG-S2.0-SS400 P-T-PHUONG, S2	Vật liệu cơ bản/ Base metal	Đạt/ Pass	

Số: KT3-04448B
CK4/8

Ngày: 19/11/2024
Trang số: 03/03

4. Các thử nghiệm khác/ Other tests: Không áp dụng/ Not applicable.

Chứng nhận rằng, mối hàn được chuẩn bị, hàn và kiểm tra phù hợp với yêu cầu quy định tại điều 6, tiêu chuẩn AWS D1.3/D1.3M, (2018) – Structural Welding Code – Sheet Steel / *Certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in conformance with the requirements of Clause 6 of AWS D1.3/D1.3M, (2018) – Structural Welding Code – Sheet Steel.*

	DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION	EXAMINING BODY VICE DIRECTOR OF QUATEST 3
DATE	19/11/2024	19/11/2024
SIGN		
NAME		Nguyễn Anh Triết



Số: KT3-04448B
CK4/9

Ngày: 19/11/2024
Trang số: 01/03

BẢO CÁO CHẤP NHẬN QUY TRÌNH HÀN
WELDING PROCEDURE QUALIFICATION RECORD (PQR)
Số PQR/ PQR No.: DDC-FCAW-324.2

Tên công ty/ Company name : DaiDung Metallic Manufacture Construction & Trade Corporation
Địa chỉ/ Address : 38, Zone C, D1 Street, An Ha Industrial Park, Pham Van Hai
Commune, Binh Chanh District, HoChiMinh City
Quá trình hàn/ Welding Process : Hàn hồ quang dây kim loại lõi thuốc trong khi hoạt tính / FCAW
Phương pháp hàn/ Type : Bán tự động/ Semiautomatic
Tiêu chuẩn áp dụng/ Applicable code : AWS D1.3/D1.3M:2018
Tên thợ hàn/ Welder's name (ID) : PHAN THANH PHUONG (082097014779)/DDC-WD-0393

ĐẶC TÍNH MỐI NỐI/ JOINT DESIGN USED

Loại mối hàn/ Type:.....Flare-bevel-groove weld
Hàn một phía/ Welded from one side ☒
Đệm lót/ Backing ☐
Vật liệu/ Backing material:..... -
Khe hở chân/ Root opening:..... Max. 0,5 mm
Chiều cao chân/ Root face:..... -

Hàn hai phía/ Welded from both sides ☐
Tạo rãnh mặt sau/ Back gouging ☐
Phương pháp/ Method:..... -
Góc vát/ Groove Angle:..... -
Thông tin khác/ Other:..... Xem sơ đồ/ See sketch

KIM LOẠI CƠ BẢN/ BASE METALS

Đặc tính kỹ thuật/ Material spec.:..... JIS G3101
Loại/ Grade:..... SS400

Chiều dày/ Thickness:..... 2,0 mm
Đường kính/ Diameter (OD):..... -

KIM LOẠI BỔ SUNG/ FILLER METALS

Đặc tính kỹ thuật/ AWS Spec.:..... AWS A5.20
Loại/ AWS Classification:..... E71T-1C

Nhãn hiệu/ Trade name:..... HYUNDAI SC-71LH
Đường kính/ Diameter:..... 1,2 mm

BẢO VỆ MỐI HÀN/ SHIELDING

Thuốc bảo vệ/ Flux:..... ☐
Loại/ Class:..... -
Đặc tính kỹ thuật/ Specification:..... -

Khí bảo vệ/ Gas:..... ☒
Thành phần/ Composition:..... 99,9% CO₂
Lưu lượng/ Flow rate:..... (15 - 17) L/min
Lỗ miệng ống khí/ Gas cup size:..... 15 mm

NUNG NÓNG TRƯỚC KHI HÀN/ PREHEAT

Nhiệt độ nung nóng:..... 31 °C (Ambient)
Preheat Temp.

Nhiệt độ giữa các lớp/ Interpass temp.:
Min..... -; Max..... -

THỂ HÀN/ POSITION

Thể hàn rãnh/ Position of Groove:..... 1G/F
Thể hàn góc/ Fillet Position:..... -

Hàn ở vị trí đứng/ Vertical Progression:
Hàn leo/ Up ☐ Hàn tự/ Down ☐

ĐẶC TÍNH DÒNG ĐIỆN HÀN/ ELECTRICAL CHARACTERISTICS

Loại dòng điện/ Current:
AC ☐ DCEP ☒ DCEN ☐ Pulsed ☐
Kiểu chuyển dịch kim loại hàn:..... Cầu
Transfer mode:..... Globular

Loại nguồn điện/ Power source:..... CV
Điện cực Vonfram/ Tungsten electrode (GTAW):
Loại/ Type:..... -
Đường kính/ Diameter:..... -

Ngày: 19/11/2024
Trang số: 02/03

Diện cực/ *Electrode*:

Di chuyển đường thẳng/ *Stringer Bead* ☐
 Di chuyển đường zig zag/ *Weave Bead* ☒
 Khoảng cách ống kẹp/vật hàn: (7 – 10) mm
Contact tube to work distance
 Hàn một lớp/ *Single Pass (per side)* ☒
 Hàn nhiều lớp/ *Multi-pass (per side)* ☐
 Kiểu cấp dây/ *Wire feed type*:

Số lượng điện cực/ *Number of Electrodes* 1Khoảng cách các điện cực/ *Electrodes Spacing*:

Neang / Lateral:

Doc / Longitudinal:

Góc nghiêng diện cực/ *Angle:*

Lâm sach xī hàn: Bàn chái

Slag removal: Brushing

Nhiệt độ/ *Temperature*: - Thời gian duy trì/ *Time*: -

Lớp hàn <i>Layers</i>	Quá trình hàn <i>Process</i>	Kim loại hàn <i>Filler metals</i>		Dòng điện hàn <i>Current</i>			Tốc độ hàn <i>Travel speed,</i> cm/min	Nhiệt cấp vào/ <i>Heat input</i> kJ/mm	Chi tiết mối nối <i>Joint Details</i>
		Loại <i>Class</i>	Đường kính <i>Diam.,</i> mm	Phân cực <i>Polarity</i>	Điện thế <i>Volts,</i> V	Cường độ <i>Amps,</i> A			
1	FCAW	E71T-1C	1,2	DCEP	19 – 20	95 – 110	18 – 20	0,54 – 0,73	

1. Kiểm tra ngoại quan/ *Visual examination*: Đạt yêu cầu / *Acceptable*.
(Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-04448BCK4/1)

2. Kiểm tra chụp ảnh bức xạ/ *Radiographic test*: Không áp dụng/ *Not applicable*.

3. Thử uốn/ *Bend test* (Kết quả thử nghiệm tham khảo số/ *Test report N°*: KT3-04448BCK4/3)

Mẫu/ Specimen No	Vị trí hỏng Failure location	Kết quả/ Result	Ghi chú/ Remarks
FCAW-F-FBG-S2.0-SS400 P-T-PHUONG, S1	Vật liệu cơ bản/ Base metal	Đạt/ Pass	
FCAW-F-FBG-S2.0-SS400 P-T-PHUONG, S2	Vật liệu cơ bản/ Base metal	Đạt/ Pass	

1. Không được viết vào bất kỳ phần nào của bài này nếu không có sự đồng ý của Trung tâm Kĩ Thuật 3/ The Board shall not be responsible except in full, to those the written approval of District 3.

Số: KT3-04448B
CK4/9

Ngày: 19/11/2024
Trang số: 03/03

4. Các thử nghiệm khác/ Other tests: Không áp dụng/ Not applicable.

Chúng nhận rằng, mỗi hàn được chuẩn bị, hàn và kiểm tra phù hợp với yêu cầu quy định tại điều 6, tiêu chuẩn AWS D1.3/D1.3M, (2018) – Structural Welding Code – Sheet Steel / *Certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in conformance with the requirements of Clause 6 of AWS D1.3/D1.3M, (2018) – Structural Welding Code – Sheet Steel.*

	DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION	EXAMINING BODY VICE DIRECTOR OF QUATEST 3
DATE	19/11/2024	19/11/2024
SIGN		
NAME		Nguyễn Anh Triết



Số: KT3-04448B
CK4/I0

Ngày: 19/11/2024
Trang số: 01/03

BÁO CÁO CHẤP NHẬN QUY TRÌNH HÀN
WELDING PROCEDURE QUALIFICATION RECORD (PQR)
Số PQR/ PQR No.: DDC-FCAW-324.3

Tên công ty/ Company name : DaiDung Metallic Manufacture Construction & Trade Corporation
Địa chỉ/ Address : 38, Zone C, D1 Street, An Ha Industrial Park, Pham Van Hai
Commune, Binh Chanh District, HoChiMinh City
Quá trình hàn/ Welding Process : Hàn hồ quang dây kim loại lõi thuốc trong khí hoạt tính / FCAW
Phương pháp hàn/ Type : Bán tự động/ Semiautomatic
Tiêu chuẩn áp dụng/ Applicable code : AWS D1.3/D1.3M:2018
Tên thợ hàn/ Welder's name (ID) : PHAN THANH PHUONG (082097014779)/ DDC-WD-0393

ĐẶC TÍNH MỐI NỐI/ JOINT DESIGN USED

Loại mối hàn/ Type: T-Joint- Fillet weld
Hàn một phía/ Welded from one side ☒
Dệm lót/ Backing ☐
Vật liệu/ Backing material: -
Khe hở chân/ Root opening: Max. 0,5 mm
Chiều cao chân/ Root face: -

Hàn hai phía/ Welded from both sides ☐
Tạo rãnh mặt sau/ Back gouging ☐
Phương pháp/ Method: -
Góc vát/ Groove Angle: -
Thông tin khác/ Other: Xem sơ đồ/ See sketch

KIM LOẠI CƠ BẢN/ BASE METALS

Đặc tính kỹ thuật/ Material spec.: JIS G3101
Loại/ Grade: SS400

Chiều dày/ Thickness: 2,0 mm
Đường kính/ Diameter (OD): -

KIM LOẠI BỔ SUNG/ FILLER METALS

Đặc tính kỹ thuật/ AWS Spec.: AWS A5.20
Loại/ AWS Classification: E71T-1C

Nhãn hiệu/ Trade name: HYUNDAI SC-71LH
Đường kính/ Diameter: 1,2 mm

BẢO VỆ MỐI HÀN/ SHIELDING

Thuốc bảo vệ/ Flux: ☐
Loại/ Class: -
Đặc tính kỹ thuật/ Specification: -

Khí bảo vệ/ Gas: ☒
Thành phần/ Composition: 99,9% CO₂
Lưu lượng/ Flow rate: (15 – 17) L/min
Lỗ miệng ống khí/ Gas cup size: 15 mm

NUNG NÓNG TRƯỚC KHI HÀN/ PREHEAT

Nhiệt độ nung nóng: 32 °C (Ambient)
Preheat Temp.

Nhiệt độ giữa các lớp/ Interpass temp.:
Min: -; Max: -

THẾ HÀN/ POSITION

Thế hàn rãnh/ Position of Groove: -
Thế hàn góc/ Fillet Position: 2F/H

Hàn ở vị trí đứng/ Vertical Progression:
Hàn leo/ Up ☐ Hàn tự/ Down ☐

ĐẶC TÍNH DÒNG ĐIỆN HÀN/ ELECTRICAL CHARACTERISTICS

Loại dòng điện/ Current:
AC ☐ DCEP ☒ DCEN ☐ Pulsed ☐
Kiểu chuyển dịch kim loại hàn: Cầu
Transfer mode: Globular

Loại nguồn điện/ Power source: CV
Điện cực Vonfram/ Tungsten electrode (GTAW):
Loại/ Type: -
Đường kính/ Diameter: -

Số: KT3-04448B
CK4/10

Ngày: 19/11/2024
Trang số: 02/03

KỸ THUẬT HÀN/ TECHNIQUE

Điện cực/ Electrode:

Di chuyển đường thẳng/ Stringer Bead ☐

Di chuyển đường zic zac/ Weave Bead ☒

Khoảng cách ống kẹp/vật hàn: (7 – 10) mm
Contact tube to work distance

Hàn một lớp/ Single Pass (per side) ☒

Hàn nhiều lớp/ Multi-pass (per side) ☐

Kiểu cấp dây/ Wire feed type: –

Số lượng điện cực/ Number of Electrodes 1

Khoảng cách các điện cực/ Electrodes Spacing:

Ngang / Lateral: –

Dọc / Longitudinal: –

Góc nghiêng điện cực/ Angle: –

Làm sạch xỉ hàn: Bàn chải

Slag removal: Brushing

XỬ LÝ NHIỆT SAU KHI HÀN/ POSTWELD HEAT TREATMENT (N/A)

Nhiệt độ/ Temperature: –

Thời gian duy trì/ Time: –

CHI TIẾT QUÁ TRÌNH HÀN/ WELDING PROCEDURE

Lớp hàn Layers	Quá trình hàn Process	Kim loại hàn Filler metals		Dòng điện hàn Current			Tốc độ hàn Travel speed, cm/min	Nhiệt cấp vào/ Heat input kJ/mm	Chi tiết mối nối Joint Details
		Loại Class	Đường kính Diam., mm	Phân cực Polarity	Điện thế Volts, V	Cường độ Amps, A			
1	FCAW	E71T-1C	1,2	DCEP	19 – 20	95 – 110	18 – 20	0,54 – 0,73	

KẾT QUẢ KIỂM TRA MÔI HÀN/ TEST RESULTS

- Kiểm tra ngoại quan/ Visual examination: Đạt yêu cầu / Acceptable.
(Kết quả thử nghiệm tham khảo số/ Test report N°: KT3-04448BCK4/1)
- Kiểm tra chụp ảnh bức xạ/ Radiographic test: Không áp dụng/ Not applicable.
- Thử uốn/ Bend test (Kết quả thử nghiệm tham khảo số/ Test report N°: KT3-04448BCK4/4)

Mẫu/ Specimen N°	Vị trí hỏng Failure location	Kết quả/ Result	Đánh giá/ Assessment
FCAW-2F-S2.0-SS400 P-T-PHUONG, S1	Vật liệu cơ bản/ Base metal	- No weld metal crack - Completed fusion at the root of the joint	Đạt/ Pass
FCAW-2F-S2.0-SS400 P-T-PHUONG, S2	Vật liệu cơ bản/ Base metal	- No weld metal crack - Completed fusion at the root of the joint	Đạt/ Pass

Số: KT3-04448B
CK4/10

Ngày: 19/11/2024
Trang số: 03/03

4. Các thử nghiệm khác/ Other tests: Không áp dụng/ Not applicable.

Chúng nhận rằng, mối hàn được chuẩn bị, hàn và kiểm tra phù hợp với yêu cầu quy định tại điều 6, tiêu chuẩn AWS D1.3/D1.3M, (2018) – Structural Welding Code – Sheet Steel / *Certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in conformance with the requirements of Clause 6 of AWS D1.3/D1.3M, (2018) – Structural Welding Code – Sheet Steel.*

	DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION	EXAMINING BODY VICE DIRECTOR OF QUATEST 3
DATE	19/11/2024	19/11/2024
SIGN		
NAME		Nguyễn Anh Triết

CHỖ ĐÓNG CHỮ

Số: KT3-04448B
CK4/11

Ngày: 19/11/2024
Trang số: 01/03

BẢO CÁO CHẤP NHẬN QUY TRÌNH HÀN
WELDING PROCEDURE QUALIFICATION RECORD (PQR)
Số PQR/ PQR No.: DDC-FCAW-324.4

Tên công ty/ Company name : DaiDung Metallic Manufacture Construction & Trade Corporation
Địa chỉ/ Address : 38, Zone C, D1 Street, An Ha Industrial Park, Pham Van Hai
Commune, Binh Chanh District, HoChiMinh City
Quá trình hàn/ Welding Process : Hàn hồ quang dây kim loại lõi thuốc trong khí hoạt tính / FCAW
Phương pháp hàn/ Type : Bán tự động/ Semiautomatic
Tiêu chuẩn áp dụng/ Applicable code : AWS D1.3/D1.3M:2018
Tên thợ hàn/ Welder's name (ID) : PHAN THANH PHUONG (082097014779)/ DDC-WD-0393

ĐẶC TÍNH MỐI NÚT JOINT DESIGN USED

Loại mối hàn/ Type:.....T-joint- Fillet weld
Hàn một phía/ Welded from one side.....☒
Đệm lót/ Backing.....☐
Vật liệu/ Backing material:.....-
Khe hở chân/ Root opening:.....Max. 0,5 mm
Chiều cao chân/ Root face:.....-

Hàn hai phía/ Welded from both sides.....☐
Tạo rãnh mặt sau/ Back gouging.....☐
Phương pháp/ Method:.....-
Góc vát/ Groove Angle:.....-
Thông tin khác/ Other:.....Xem sơ đồ/ See sketch

KIM LOẠI CƠ BẢN/ BASE METALS

Đặc tính kỹ thuật/ Material spec.:.....JIS G3101
Loại/ Grade:.....SS400

Chiều dày/ Thickness:.....2,0 mm
Đường kính/ Diameter (OD):.....-

KIM LOẠI BỔ SUNG/ FILLER METALS

Đặc tính kỹ thuật/ AWS Spec.:.....AWS A5.20
Loại/ AWS Classification:.....E71T-1C

Nhãn hiệu/ Trade name:.....HYUNDAI SC-71LH
Đường kính/ Diameter:.....1,2 mm

BẢO VỆ MỐI HÀN/ SHIELDING

Thuốc bảo vệ/ Flux:.....☐
Loại/ Class:.....-
Đặc tính kỹ thuật/ Specification:.....-

Khí bảo vệ/ Gas:.....☒
Thành phần/ Composition:.....99,9% CO₂
Lưu lượng/ Flow rate:.....(15 - 17) L/min
Lỗ miệng ống khí/ Gas cup size:.....15 mm

NUNG NÓNG TRƯỚC KHI HÀN/ PREHEAT

Nhiệt độ nung nóng:.....32 °C (Ambient)
Preheat Temp.

Nhiệt độ giữa các lớp/ Interpass temp.:
Min.....-; Max.....-

THẾ HÀN/ POSITION

Thế hàn rãnh/ Position of Groove:.....-
Thế hàn góc/ Fillet Position:.....1F

Hàn ở vị trí đứng/ Vertical Progression:
Hàn leo/ Up ☐ Hàn tự/ Down ☐

ĐẶC TÍNH DÒNG ĐIỆN HÀN/ ELECTRICAL CHARACTERISTICS

Loại dòng điện/ Current:
AC ☐ DCEP ☒ DCEN ☐ Pulsed ☐
Kiểu chuyển dịch kim loại hàn:.....Cầu
Transfer mode:.....Globular

Loại nguồn điện/ Power source:.....CV
Điện cực Vonfram/ Tungsten electrode (GTAW):
Loại/ Type:.....-
Đường kính/ Diameter:.....-

Số: KT3-04448B
CK4/11

Ngày: 19/11/2024
Trang số: 02/03

KỸ THUẬT HÀN/ TECHNIQUE

Điện cực/ Electrode:

Di chuyển đường thẳng/ Stringer Bead ☐

Di chuyển đường zig zag/ Weave Bead ☒

Khoảng cách ống kẹp/vật hàn: (7 – 10) mm
Contact tube to work distance

Hàn một lớp/ Single Pass (per side) ☒

Hàn nhiều lớp/ Multi-pass (per side) ☐

Kiểu cấp dây/ Wire feed type: -

Số lượng điện cực/ Number of Electrodes 1

Khoảng cách các điện cực/ Electrodes Spacing:

Ngang / Lateral: -

Dọc / Longitudinal: -

Góc nghiêng điện cực/ Angle: -

Làm sạch xỉ hàn: Bàn chải

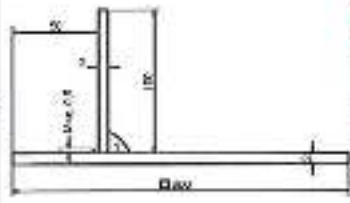
Slag removal: Brushing

XỬ LÝ NHIỆT SAU KHI HÀN/ POSTWELD HEAT TREATMENT (N/A)

Nhiệt độ/ Temperature: -

Thời gian duy trì/ Time: -

CHI TIẾT QUÁ TRÌNH HÀN/ WELDING PROCEDURE

Lớp hàn Layers	Quá trình hàn Process	Kim loại hàn Filler metals		Dòng điện hàn Current			Tốc độ hàn Travel speed, cm/min	Nhiệt cấp vào/ Heat input kJ/mm	Chi tiết mối nối Joint Details
		Loại Class	Đường kính Diam., mm	Phân cực Polarity	Điện thế Volts, V	Cường độ Amps, A			
1	FCAW	E71T-1C	1,2	DCEP	19 – 20	95 – 110	18 – 20	0,54 – 0,73	

KẾT QUẢ KIỂM TRA MỐI HÀN/ TEST RESULTS

- Kiểm tra ngoại quan/ Visual examination: Đạt yêu cầu / Acceptable.
(Kết quả thử nghiệm tham khảo số/ Test report N°: KT3-04448BCK4/1)
- Kiểm tra chụp ảnh bức xạ/ Radiographic test: Không áp dụng/ Not applicable.
- Thử uốn/ Bend test (Kết quả thử nghiệm tham khảo số/ Test report N°: KT3-04448BCK4/5)

Mẫu/ Specimen N°	Vị trí hỏng Failure location	Kết quả/ Result	Đánh giá/ Assessment
FCAW-1F-S2.0-SS400 P-T-PHUONG, S1	Vật liệu cơ bản/ Base metal	- No weld metal crack - Completed fusion at the root of the joint	Đạt/ Pass
FCAW-1F-S2.0-SS400 P-T-PHUONG, S2	Vật liệu cơ bản/ Base metal	- No weld metal crack - Completed fusion at the root of the joint	Đạt/ Pass

Số: KT3-04448B
CK4/11

Ngày: 19/11/2024
Trang số: 03/03

4. Các thử nghiệm khác/ Other tests: Không áp dụng/ Not applicable.

Chứng nhận rằng, mỗi hàn được chuẩn bị, hàn và kiểm tra phù hợp với yêu cầu quy định tại điều 6, tiêu chuẩn AWS D1.3/D1.3M, (2018) – Structural Welding Code – Sheet Steel / Certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in conformance with the requirements of Clause 6 of AWS D1.3/D1.3M, (2018) – Structural Welding Code – Sheet Steel.

	DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION	EXAMINING BODY VICE DIRECTOR OF QUATEST 3
DATE	19/11/2024	19/11/2024
SIGN		
NAME		Nguyễn Anh Triết



Số: KT3-04448B
CK4/12

Ngày: 18/11/2024
Trang số: 01/03

BÁO CÁO CHẤP NHẬN QUY TRÌNH HÀN
WELDING PROCEDURE QUALIFICATION RECORD (PQR)
Số PQR/ PQR No.: DDC-FCAW-324.5

Tên công ty/ Company name : DaiDung Metallic Manufacture Construction & Trade Corporation
Địa chỉ/ Address : 38, Zone C, D1 Street, An Ha Industrial Park, Pham Van Hai Commune, Binh Chanh District, HoChiMinh City
Quá trình hàn/ Welding Process : Hàn hồ quang dây kim loại lõi thuốc trong khí hoạt tính / FCAW
Phương pháp hàn/ Type : Bán tự động/ Semiautomatic
Tiêu chuẩn áp dụng/ Applicable code : AWS D1.3/D1.3M:2018
Tên thợ hàn/ Welder's name (ID) : PHAN THANH PHUONG (082097014779)/ DDC-WD-0393

ĐẶC TÍNH MỐI NỐI/ JOINT DESIGN USED

Loại mối hàn/ Type: Square- groove weld
Hàn một phía/ Welded from one side ☐
Đệm lót/ Backing ☐
Vật liệu/ Backing material: -
Khe hở chân/ Root opening: Max. 0,5 mm
Chiều cao chân/ Root face: -

Hàn hai phía/ Welded from both sides ☒
Tạo rãnh mặt sau/ Back gouging ☒
Phương pháp/ Method: Mài Grinding
Góc vát/ Groove Angle:
Thông tin khác/ Other: Xem sơ đồ/ See sketch

KIM LOẠI CƠ BẢN/ BASE METALS

Đặc tính kỹ thuật/ Material spec.: JIS G3101
Loại/ Grade: S3400

Chiều dày/ Thickness: 2,0 mm
Đường kính/ Diameter (OD): -

KIM LOẠI BỔ SUNG/ FILLER METALS

Đặc tính kỹ thuật/ AWS Spec.: AWS A5.20
Loại/ AWS Classification: E71T-1C

Nhãn hiệu/ Trade name: HYUNDAI SC-71LH
Đường kính/ Diameter: 1,2 mm

BẢO VỆ MỐI HÀN/ SHIELDING

Thuốc bảo vệ/ Flux: ☐
Loại/ Class: -
Đặc tính kỹ thuật/ Specification: -

Khí bảo vệ/ Gas: ☒
Thành phần/ Composition: 99,9% CO₂
Lưu lượng/ Flow rate: (15 - 17) L/min
Lỗ miệng ống khí/ Gas cup size: 15 mm

NUNG NÓNG TRƯỚC KHI HÀN/ PREHEAT

Nhiệt độ nung nóng: 32 °C (Ambient)
Preheat Temp.

Nhiệt độ giữa các lớp/ Interpass temp.:
Min -; Max -

THẾ HÀN/ POSITION

Thế hàn rãnh/ Position of Groove: 1G/F
Thế hàn góc/ Fillet Position: -

Hàn ở vị trí đứng/ Vertical Progression:
Hàn leo/ Up ☐ Hàn tự/ Down ☐

ĐẶC TÍNH DÒNG ĐIỆN HÀN/ ELECTRICAL CHARACTERISTICS

Loại dòng điện/ Current:
AC ☐ DCEP ☒ DCEN ☐ Pulsed ☐
Kiểm chuyển dịch kim loại hàn: Cầu
Transfer mode: Globular

Loại nguồn điện/ Power source: CV
Điện cực Vonfram/ Tungsten electrode (GTAW):
Loại/ Type:
Đường kính/ Diameter: -

Số: KT3-04448B
CK4/12

Ngày: 18/11/2024
Trang số: 02/03

KỸ THUẬT HÀN/ TECHNIQUE

Điện cực/ Electrode:

Di chuyển đường thẳng/ Stringer Bead ☐

Di chuyển đường zic zac/ Weave Bead ☒

Khoảng cách ống kẹp/vật hàn: (7 – 10) mm
Contact tube to work distance

Hàn một lớp/ Single Pass (per side) ☒

Hàn nhiều lớp/ Multi-pass (per side) ☐

Kiểu cấp dây/Wire feed type: -

Số lượng điện cực/ Number of Electrodes 1

Khoảng cách các điện cực/ Electrodes Spacing:

Ngang / Lateral: -

Dọc / Longitudinal: -

Góc nghiêng điện cực/ Angle: -

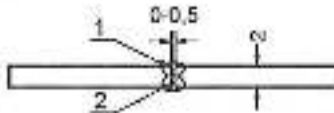
Làm sạch xỉ hàn: Bàn chải

Slag removal: Brushing

XỬ LÝ NHIỆT SAU KHI HÀN/ POSTWELD HEAT TREATMENT (N/A)

Nhiệt độ/ Temperature: - Thời gian duy trì/ Time: -

CHI TIẾT QUÁ TRÌNH HÀN/ WELDING PROCEDURE

Lớp hàn Layers	Quá trình hàn Process	Kim loại hàn Filler metals		Dòng điện hàn Current			Tốc độ hàn Travel speed, cm/min	Nhiệt cấp vào/ Heat input kJ/mm	Chi tiết mối nối Joint Details
		Loại Class	Đường kính Diam., mm	Phân cực Polarity	Điện thế Volts, V	Cường độ Amps, A			
1	FCAW	E71T-1C	1,2	DCEP	19 – 21	95 – 110	22 – 24	0,45 – 0,63	
2	FCAW	E71T-1C	1,2	DCEP	19 – 21	95 – 110	22 – 24	0,45 – 0,63	

KẾT QUẢ KIỂM TRA MỐI HÀN/ TEST RESULTS

- Kiểm tra ngoại quan/ Visual examination: Đạt yêu cầu / Acceptable.
(Kết quả thử nghiệm tham khảo số/ Test report N^o: KT3-04448BCK4/1)
- Kiểm tra chụp ảnh bức xạ/ Radiographic test: Không áp dụng/ Not applicable.
- Thử uốn/ Bend test (Kết quả thử nghiệm tham khảo số/ Test report N^o: KT3-04448BCK4/6)

Mẫu/ Specimen N ^o	Kiểu uốn/ Type of bend	Kết quả/ Result	Ghi chú/ Remarks
FCAW-1G-S2.0-SS400-BS, P-T-PHUONG, S1	Uốn mặt/ Face bend	Đạt/ Pass	
FCAW-1G-S2.0-SS400-BS, P-T-PHUONG, S2	Uốn chân/ Root bend	Đạt/ Pass	

Số: KT3-04448B
CK4/12

Ngày: 18/11/2024
Trang số: 03/03

4. Các thử nghiệm khác/ Other tests: Không áp dụng/ Not applicable.

Chúng nhận rằng, mỗi hàn được chuẩn bị, hàn và kiểm tra phù hợp với yêu cầu quy định tại điều 6, tiêu chuẩn AWS D1.3/D1.3M, (2018) – Structural Welding Code – Sheet Steel / *Certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in conformance with the requirements of Clause 6 of AWS D1.3/D1.3M, (2018) – Structural Welding Code – Sheet Steel.*

	DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION	EXAMINING BODY VICE DIRECTOR OF QUATEST 3
DATE	19/11/2024	19/11/2024
SIGN		
NAME		Nguyễn Anh Triết

Số: KT3-04448B
CK4/13

Ngày: 19/11/2024
Trang số: 01/03

BẢO CÁO CHẤP NHẬN QUY TRÌNH HÀN
WELDING PROCEDURE QUALIFICATION RECORD (PQR)
SỐ PQR/ PQR No.: DDC-FCAW-324.6

Tên công ty/ Company name : DaiDung Metallic Manufacture Construction & Trade Corporation
Địa chỉ/ Address : 38, Zone C, D1 Street, An Ha Industrial Park, Pham Van Hai
Commune, Binh Chanh District, HoChiMinh City
Quá trình hàn/ Welding Process : Hàn hồ quang dây kim loại lõi thuốc trong khi hoạt tính / FCAW
Phương pháp hàn/ Type : Bán tự động/ Semiautomatic
Tiêu chuẩn áp dụng/ Applicable code : AWS D1.3/D1.3M:2018
Tên thợ hàn/ Welder's name (ID) : PHAN THANH PHUONG (082097014779)/ DDC-WD-0393

ĐẶC TÍNH MỐI HÀN/ JOINT DESIGN USED

Loại mối hàn/ Type: Square- groove weld
Hàn một phía/ Welded from one side ☒
Đệm lót/ Backing ☒
Vật liệu/ Backing material: JIS G3101, SS400
Khe hở chân/ Root opening: (2,5 – 3,0)mm
Chiều cao chân/ Root face: –

Hàn hai phía/ Welded from both sides ☐
Tạo rãnh mặt sau/ Back gouging ☐
Phương pháp/ Method: –
Góc vát/ Groove Angle: –
Thông tin khác/ Other: Xem sơ đồ/ See sketch

KIM LOẠI CƠ BẢN/ BASE METALS

Đặc tính kỹ thuật/ Material spec.: JIS G3101
Loại/ Grade: SS400

Chiều dày/ Thickness: 2,0 mm
Đường kính/ Diameter (OD): –

KIM LOẠI BỔ SUNG/ FILLER METALS

Đặc tính kỹ thuật/ AWS Spec.: AWS A5.20
Loại/ AWS Classification: E71T-1C

Nhãn hiệu/ Trade name: HYUNDAI SC-71LH
Đường kính/ Diameter: 1,2 mm

BẢO VỆ MỐI HÀN/ SHIELDING

Thuốc bảo vệ/ Flux: ☐
Loại/ Class: –
Đặc tính kỹ thuật/ Specification: –

Khí bảo vệ/ Gas: ☒
Thành phần/ Composition: 99,9% CO₂
Lưu lượng/ Flow rate: (15 – 17) L/min
Lỗ miệng ống khí/ Gas cup size: 15 mm

NUNG NÓNG TRƯỚC KHI HÀN/ PREHEAT

Nhiệt độ nung nóng: 32 °C (Ambient)
Preheat Temp.

Nhiệt độ giữa các lớp/ Interpass temp.:
Min –; Max –

THỂ HÀN/ POSITION

Thể hàn rãnh/ Position of Groove: 1G/F
Thể hàn góc/ Fillet Position: –

Hàn ở vị trí đứng/ Vertical Progression:
Hàn leo/ Up ☐ Hàn tự/ Down ☐

ĐẶC TÍNH DÒNG ĐIỆN HÀN/ ELECTRICAL CHARACTERISTICS

Loại dòng điện/ Current:
AC ☐ DCEP ☒ DCEN ☐ Pulsed ☐
Kiểu chuyển dịch kim loại hàn: Cầu
Transfer mode: Globular

Loại nguồn điện/ Power source: CV
Điện cực Vonfram/ Tungsten electrode (GTAW):
Loại/ Type: –
Đường kính/ Diameter: –

Số: KT3-04448B
CK4/13

Ngày: 19/11/2024
Trang số: 02/03

KỸ THUẬT HÀN/ TECHNIQUE

Điện cực/ Electrode:

Di chuyển đường thẳng/ Stringer Bead ☐

Di chuyển đường zig zag/ Weave Bead ☒

Khoảng cách ống kẹp/vật hàn: (7 - 10) mm
Contact tube to work distance

Hàn một lớp/ Single Pass (per side) ☒

Hàn nhiều lớp/ Multi-pass (per side) ☐

Kiểu cấp dây/ Wire feed type: -

Số lượng điện cực/ Number of Electrodes 1

Khoảng cách các điện cực/ Electrodes Spacing:

Ngang / Lateral: -

Dọc / Longitudinal: -

Góc nghiêng điện cực/ Angle: -

Làm sạch xỉ hàn: Bàn chải

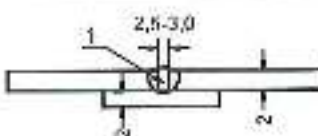
Slag removal: Brushing

XỬ LÝ NHIỆT SAU KHI HÀN/ POSTWELD HEAT TREATMENT (N/A)

Nhiệt độ/ Temperature: -

Thời gian duy trì/ Time: -

CHI TIẾT QUÁ TRÌNH HÀN/ WELDING PROCEDURE

Lớp hàn Layers	Quá trình hàn Process	Kim loại hàn Filler metals		Dòng điện hàn Current			Tốc độ hàn Travel speed, cm/min	Nhiệt cấp vào/ Heat input kJ/mm	Chi tiết mối nối Joint Details
		Loại Class	Đường kính Diam., mm	Phân cực Polarity	Điện thế Volts, V	Cường độ Amps, A			
1	FCAW	E71T-1C	1,2	DCEP	19 - 21	95 - 110	20 - 21	0,52 - 0,69	

KẾT QUẢ KIỂM TRA MỐI HÀN/ TEST RESULTS

- Kiểm tra ngoại quan/ Visual examination: Đạt yêu cầu / Acceptable.
(Kết quả thử nghiệm tham khảo số/ Test report N°: KT3-04448BCK4/1)
- Kiểm tra chụp ảnh bức xạ/ Radiographic test: Không áp dụng/ Not applicable.
- Thử uốn/ Bend test (Kết quả thử nghiệm tham khảo số/ Test report N°: KT3-04448BCK4/7)

Mẫu/ Specimen N°	Kiểu uốn/ Type of bend	Kết quả/ Result	Ghi chú/ Remarks
FCAW-1G-S2.0-SS400-MB P-T-PHUONG, S1	Uốn chân/ Root bend	Đạt/ Pass	
FCAW-1G-S2.0-SS400-MB P-T-PHUONG, S2	Uốn chân/ Root bend	Đạt/ Pass	

Số: KT3-04448B
CK4/13

Ngày: 19/11/2024
Trang số: 03/03

4. Các thử nghiệm khác/ Other tests: Không áp dụng/ Not applicable.

Chúng nhận rằng, mỗi hàn được chuẩn bị, hàn và kiểm tra phù hợp với yêu cầu quy định tại điều 6, tiêu chuẩn AWS D1.3/D1.3M, (2018) – Structural Welding Code – Sheet Steel / *Certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in conformance with the requirements of Clause 6 of AWS D1.3/D1.3M, (2018) – Structural Welding Code – Sheet Steel.*

	DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION	EXAMINING BODY VICE DIRECTOR OF QUATEST 3
DATE	19/11/2024	19/11/2024
SIGN		
NAME		Nguyễn Anh Triết



KT3-04448B
CK4/1

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

19/11/2024
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- Tên mẫu
Name of sample : **MỎI HÀN TRÊN MẪU THÉP TẦM MỎNG.**
WELDS ON STEEL SHEET TEST PIECE.
- Mô tả mẫu
Sample description : Mối hàn trên thép tấm như hình ảnh ở mục 11.
Welds on steel sheet as photographs at item 11.
Kích thước/ *Dimension*: (200 x 200 x 2 x L50 x 100 x 2) mm;
(200 x 200 x 2 x 100 x 100 x2) mm; (200 x 400 x 2) mm.
- Số lượng mẫu
Testing quantity : 06 bộ mẫu/ *Samples.*
- Ngày nhận yêu cầu : 05/11/2024.
Date of receiving
- Ngày thử nghiệm : 05/11/2024.
Date of testing
- Khách hàng
Customer : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.**
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm
Test location : Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.
- Phương pháp thử
Test method : AWS B1.11M/B1.11:2015.
- Điều kiện và kết quả thử nghiệm : Xem chi tiết từ trang 02/03 và 03/03.
Test condition & results **See details on page 02/03 and 03/03.**

TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL
TESTING LAB.


Phạm Văn Út

KT. GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR




Nguyễn Anh Triết

KT3 04448B
CK4/I

PHIẾU KẾT QUẢ THỬ NGHIỆM TEST REPORT

19/11/2024
Page 02/03

10. Điều kiện thử/ Test condition:

☒ Butt weld	☒ Fillet weld	☒ Carbon steel	☐ Stainless steel	Base metal thickness: 2 mm
Base Metal Type/ Grade: JIS G3101, SS400	Filler Metal Class: AWS A5.20, E71T-JC			
Welding Procedure Spec.: SMAW (111)	☒ FCAW (136)	☐ SAW (121)	☐ GTAW	
State of Examination: ☒ After Welding	☐ After Repairing	☐ Prepared edge	☐ After PWHT	
Surface Preparation: ☒ As weld	☒ Brushing	☐ Other:		
Surface Temperature: ☒ (28 ± 3) °C	☐ ... °C			
IR Thermometer: FLUKE 561	Serial: 61560053WS	Next Cal. Due: 08/2025		
Lighting Condition: ☒ Day light > 1000 lux	☐ Artificial > 1000 lux			
Weld Gauge: Multipurpose Gauge	HJC40	Serial: 04830044		

11. Hình ảnh của mẫu/ Photograph of teste pieces:



FCAW-H-FBG-S2.0-SS400 P-T-PHUONG, S1, S2 FCAW-F-FBG-S2.0-SS400 P-T-PHUONG, S1, S2



FCAW-2F-S2.0-SS400 P-T-PHUONG, S1, S2 FCAW-1F-S2.0-SS400 P-T-PHUONG, S1, S2



KT3-04448B
CK4/1

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

19/11/2024
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FCAW-1G-S2.0-SS400-BIS P-T-PHUONG, S1, S2

FCAW-1G-S2.0-SS400-MB P-T-PHUONG, S1, S2

12. Kết quả kiểm tra ngoại quan mỗi hàn/ Visual examination results:

Tên mẫu Name of test piece	Chiều dài kiểm tra Tested length, (mm)	Khuyết tật bề mặt / Surface defect			Đánh giá theo / Assess to AWS D1.3/D1.3M:2018		Ghi chú Remarks
		Loại (*) Type	Chiều dài Length (mm)	Đường kính Diameter (mm)	Đạt Acceptable	Không đạt Unacceptable	
FCAW-H-FBG-S2.0-SS400 P-T-PHUONG, S1, S2	35	-	-	-	✓		
FCAW-F-FBG-S2.0-SS400 P-T-PHUONG, S1, S2	35	-	-	-	✓		
FCAW-2F-S2.0-SS400 P-T-PHUONG, S1, S2	35	-	-	-	✓		
FCAW-1F-S2.0-SS400 P-T-PHUONG, S1, S2	35	-	-	-	✓		
FCAW-1G-S2.0-SS400-BIS P-T-PHUONG, S1, S2	200 x 2 Surfaces	-	-	-	✓		
FCAW-1G-S2.0-SS400-MB P-T-PHUONG, S1, S2	200	-	-	-	✓		

Chú thích / Legends:

IF	: Không ngấu / Incomplete fusion	C	: Nứt / crack	CR	: Lõm chân / Concave Root
OL	: Chạm rập / Overlap	P	: Rỗ / Porosity	U	: Cháy cạnh / Undercut
SI	: Ngậm xỉ / Slag inclusion	R	: Quấn thép / External Reinforcement	CS	: Lõm mặt / Concave Surface
WP	: Biên dạng / Weld bead profile	RO	: Ô xi hóa chân / Root Oxidation	PR	: Quấn thừa / Penetration

1. Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu do khách hàng gửi đến và không phải là giấy chứng nhận sản phẩm.
Test results are valid for the samples submitted (sample(s) only), and thus is not a certificate of product.

2. Tên mẫu, tên khách hàng được ghi theo yêu cầu của nơi gửi mẫu. / Name of sample(s) and customer are written as customer's request.

3. Độ không đảm bảo do mô phỏng được tính từ độ không đảm bảo do chuẩn đo lường với hệ số phủ k = 1, phân bố chuẩn tương ứng với 95 % độ tin cậy.
The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k = 2, at 95 % confidence level.

4. Không được trích xero một phần hoặc kết quả thử nghiệm này nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.
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5. Mọi thắc mắc về kết quả, khách hàng liên hệ theo địa chỉ info@quatest3.com.vn để biết thêm thông tin.

Please contact QUATEST 3 at the email addresses info@quatest3.com.vn for further information about test report.

KT3-04448B
CK4/2

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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- Tên mẫu
Name of sample : **MỎI HÀN TRÊN MẪU THÉP TẤM MỎNG.**
WELDS ON STEEL SHEET TEST PIECE.
- Mô tả mẫu
Sample description : Mỗi hàn trên thép tấm như hình ảnh ở mục 10 và 11.
Welds on steel sheet as photographs at item 10 and 11.
Kích thước/ *Dimension:* (200 x 200 x 2 x 150 x 100 x 2) mm;
- Số lượng mẫu
Testing quantity : 02 mẫu/ *Samples.*
- Ngày nhận yêu cầu
Date of receiving : 05/11/2024.
- Ngày thử nghiệm
Date of testing : 15/11/2024.
- Khách hàng
Customer : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.**
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm
Test Location : Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.
- Phương pháp thử
Test method : AWS D1.3/D1.3M:2018.
- Điều kiện và kết quả thử nghiệm : Xem chi tiết trang 02/02.
Test condition & results *See details on page 02/02.*

TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL
TESTING LAB.

Phạm Văn Út

KT. GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR



Nguyễn Anh Triết

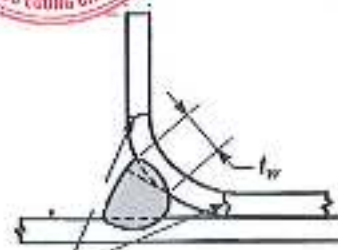
KT3-04448B
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PHIẾU KẾT QUẢ THỬ NGHIỆM TEST REPORT

19/11/2024
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10. Hình ảnh mẫu/ Photograph of test pieces:



FCAW-H-FBG-S2.0-SS400 P-T-PHUONG, S1, S2

11. Kết quả thử uốn/ Bend test results:

Mẫu/ Specimen No	Vị trí hỏng/ Failure location	Kích cỡ mối hàn/ Weld size t_w , mm	Đánh giá theo/ Assess to AWS D1.3/D1.3M:2018
 FCAW-H-FBG-S2.0-SS400 P-T-PHUONG, S1	Vật liệu cơ bản Base metal	3,5	Đạt/ Pass
 FCAW-H-FBG-S2.0-SS400 P-T-PHUONG, S2	Vật liệu cơ bản Base metal	3,5	Đạt/ Pass

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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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- Tên mẫu
Name of sample : **MỎI HÀN TRÊN MẪU THÉP TẮM MỎNG.**
WELDS ON STEEL SHEET TEST PIECE.
- Mô tả mẫu
Sample description : Mỗi hàn trên thép tấm như hình ảnh ở mục 10 và 11.
Welds on steel sheet as photographs at item 10 and 11.
Kích thước/ *Dimension:* (200 x 200 x 2 x L50 x 100 x 2) mm;
- Số lượng mẫu
Testing quantity : 02 mẫu/ *Samples.*
- Ngày nhận yêu cầu
Date of receiving : 05/11/2024.
- Ngày thử nghiệm
Date of testing : 15/11/2024.
- Khách hàng
Customer : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.**
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm
Test Location : Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.
- Phương pháp thử
Test method : AWS D1.3/D1.3M:2018.
- Điều kiện và kết quả thử nghiệm
Test condition & results : Xem chi tiết trang 02/02.
See details on page 02/02.

TRƯỞNG PTN CƠ KHÍ
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Nguyễn Anh Triết

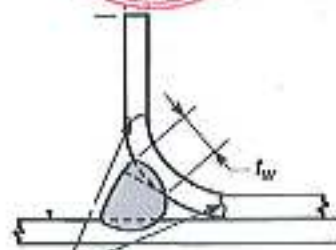
KT3-04448B
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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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10. Hình ảnh mẫu/ Photograph of test pieces:



FCAW-F-FBG-S2.0-SS400 P-T-PHUUONG, S1, S2

11. Kết quả thử uốn/ Bend test results:

Mẫu/ Specimen No	Vị trí hỏng Failure location	Kích cỡ mối hàn Weld size t_w , mm	Đánh giá theo/ Assess to AWS D1.3/D1.3M:2018
 KT3-04448BCK4 FCAW-F-FBG-S2.0-SS400 P-T-PHUUONG S1 VT: AC 05.11.24 FCAW-F-FBG-S2.0-SS400 P-T-PHUUONG, S1	Vật liệu cơ bản Base metal	3,3	Đạt/ Pass
 KT3-04448BCK4 FCAW-F-FBG-S2.0-SS400 P-T-PHUUONG S2 VT: AC 05.11.24 FCAW-F-FBG-S2.0-SS400 P-T-PHUUONG, S2	Vật liệu cơ bản Base metal	3,3	Đạt/ Pass



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PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

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- Tên mẫu
Name of sample : **MÔI HÀN TRÊN MẪU THÉP TẤM MỎNG.**
WELDS ON STEEL SHEET TEST PIECE.
- Mô tả mẫu
Sample description : Mỗi hàn trên thép tấm như hình ảnh ở mục 10 và 11.
Welds on steel sheet as photographs at item 10 and 11.
Kích thước/ *Dimension*: (200 x 200 x 2 x 100 x 100 x 2) mm.
- Số lượng mẫu
Testing quantity : 02 mẫu/ **Samples.**
- Ngày nhận yêu cầu
Date of receiving : 05/11/2024.
- Ngày thử nghiệm
Date of testing : 15/11/2024.
- Khách hàng
Customer : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.**
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm
Test Location : Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.
- Phương pháp thử
Test method : AWS D1.3/D1.3M:2018.
- Điều kiện và kết quả thử nghiệm : Xem chi tiết trang 02/02.
Test condition & results **See details on page 02/02.**

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PHÓ GIÁM ĐỐC/ VICE DIRECTOR



Nguyễn Anh Triết

KT3-04448B
CK4/4

PHIẾU KẾT QUẢ THỬ NGHIỆM TEST REPORT

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10. Hình ảnh mẫu/ Photograph of test pieces:



FCAW-2F-S2.0-SS400 P-T-PHUONG, S1, S2



11. Kết quả thử uốn/ Bend test results:

Mẫu/ Specimen No	Vị trí hỏng Failure location	Kích cỡ mối hàn Weld size t_w , mm	Đánh giá (theo/ Assess to AWS D1.3/D1.3M:2018)
 KT3-04448BCK4 FCAW-2F-S2.0-SS400 P-T-PHUONG S1 110 x 40 x 6.5 mm	Vật liệu cơ bản Base metal	3,0	Đạt/ Pass
 KT3-04448BCK4 FCAW-2F-S2.0-SS400 P-T-PHUONG S2 110 x 40 x 6.5 mm	Vật liệu cơ bản Base metal	3,0	Đạt/ Pass



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PHIẾU KẾT QUẢ THỬ NGHIỆM
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- Tên mẫu
Name of sample : **MỎI HÀN TRÊN MẪU THÉP TẦM MỎNG.**
WELDS ON STEEL SHEET TEST PIECE.
- Mô tả mẫu
Sample description : **Mỏi hàn trên thép tấm như hình ảnh ở mục 10 và 11.**
Welds on steel sheet as photographs at item 10 and 11.
Kích thước/ Dimension: (200 x 200 x 2 x 100 x 100 x 2) mm.
- Số lượng mẫu
Testing quantity : **02 mẫu/ Samples.**
- Ngày nhận yêu cầu
Date of receiving : **05/11/2024.**
- Ngày thử nghiệm
Date of testing : **15/11/2024.**
- Khách hàng
Customer : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.**
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm
Test Location : **Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.**
- Phương pháp thử
Test method : **AWS D1.3/D1.3M:2018.**
- Điều kiện và kết quả thử nghiệm : **Xem chi tiết trang 02/02.**
Test condition & results See details on page 02/02.

TRƯỞNG PTN CƠ KHÍ
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KT. GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR




Nguyễn Anh Triết

KT3-04448B
CK4/5

PHIẾU KẾT QUẢ THỬ NGHIỆM TEST REPORT

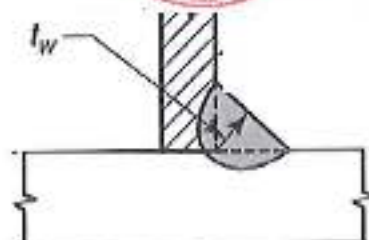
19/11/2024
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10. Hình ảnh mẫu/ Photograph of test pieces:



FCAW-1F-S2.0-SS400 P-T-PHUONG, S1, S2



11. Kết quả thử uốn/ Bend test results:

Mẫu/ Specimen N°	Vị trí hỏng/ Failure location	Kích cỡ mối hàn/ Weld size t_w , mm	Đánh giá theo/ Assess to AWS D1.3/D1.3M:2018
 KT3-04448BCK4 FCAW-1F-S2.0-SS400 P-T-PHUONG S1 VT: AU 05.11.24 FCAW-1F-S2.0-SS400 P-T-PHUONG, S1	Vật liệu cơ bản Base metal	3,3	Đạt/ Pass
 KT3-04448BCKA FCAW-1F-S2.0-SS400 P-T-PHUONG S2 VT: AU 05.11.24 FCAW-1F-S2.0-SS400 P-T-PHUONG, S2	Vật liệu cơ bản Base metal	3,3	Đạt/ Pass

1. Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu thử khách hàng gửi đến và không phải là giấy chứng nhận sản phẩm.
Test results are valid for the sample submitted (sample) only, and this is not a certificate of product.

2. Tên mẫu, tên khách hàng được ghi theo yêu cầu của đơn gửi mẫu. / Name of sample(s) and customer are written as customer's request.

3. Độ chính xác của kết quả thử nghiệm được tính toán dựa trên độ chính xác của chuẩn nghiệm và hệ số phá hủy - 2, phần lớn chuẩn tương ứng với 95 % độ tin cậy.
The reported expanded uncertainty of measurement is stated at the standard uncertainty of measurement multiplied by the coverage factor k = 2, at 95 % confidence level.

4. Không được trích sao một phần hoặc toàn bộ kết quả thử nghiệm này nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.
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5. Mọi thắc mắc về kết quả, kích thước liên hệ theo địa chỉ email: info@quatest3.com.vn để biết thêm thông tin.
Please contact QUATEST 3 at the email addresses info@quatest3.com.vn for further information about test report.

Please contact QUATEST 3 at the email addresses info@quatest3.com.vn for further information about test report.

KT3-04448B
CK4/6

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

19/11/2024
Page 01/02

- Tên mẫu
Name of sample : **MỎI HÀN TRÊN MẪU THÉP TẤM MỎNG.**
WELDS ON STEEL SHEET TEST PIECE.
- Mô tả mẫu
Sample description : **Mỗi hàn trên thép tấm như hình ảnh ở mục 10 và 11.**
Welds on steel sheet as photographs at item 10 and 11.
Kích thước/ Dimension: (200 x 400 x 2) mm.
- Số lượng mẫu
Testing quantity : **02 mẫu/ Samples.**
- Ngày nhận yêu cầu
Date of receiving : **05/11/2024.**
- Ngày thử nghiệm
Date of testing : **15/11/2024.**
- Khách hàng
Customer : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.**
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm
Test Location : **Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.**
- Phương pháp thử
Test method : **AWS D1.3/D1.3M:2018.**
- Điều kiện và kết quả thử nghiệm : **Xem chi tiết trang 02/02.**
Test condition & results See details on page 02/02.

TRƯỞNG PTN CƠ KHÍ
HEAD OF MECHANICAL
TESTING LAB.

Phạm Văn Út

KT. GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR



Nguyễn Anh Triết

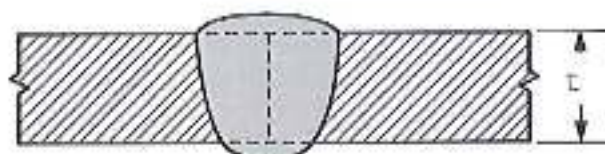
KT3-04448B
CK4/6

PHIẾU KẾT QUẢ THỬ NGHIỆM TEST REPORT

19/11/2024
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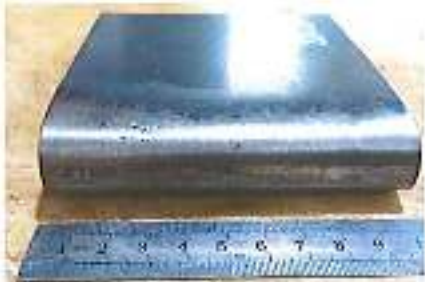


10. Hình ảnh mẫu/ Photograph of test pieces:



FCAW-1G-S2.0-SS400-BS P-T-PHUONG, S1, S2

11. Kết quả thử uốn/ Bend test results:

Mẫu/ Specimen No	Góc uốn Bending angle, độ	Kích cỡ mối hàn Weld size t_w , mm	Đánh giá theo/ Assess to AWS D1.3/D1.3M:2018
 FCAW-1G-S2.0-SS400-BS P-T-PHUONG, S1 (Uốn mặt/ Face bend)	180	2,0	Đạt/ Pass
 FCAW-1G-S2.0-SS400-BS P-T-PHUONG, S2 (Uốn chân/ Root bend)	180	2,0	Đạt/ Pass



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CK4/7

PHIẾU KẾT QUẢ THỬ NGHIỆM
TEST REPORT

19/11/2024
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- Tên mẫu
Name of sample : **MỎI HÀN TRÊN MÀU THÉP TẦM MỎNG.**
WELDS ON STEEL SHEET TEST PIECE.
- Mô tả mẫu
Sample description : Mỗi hàn trên thép tấm như hình ảnh ở mục 10 và 11.
Welds on steel sheet as photographs at item 10 and 11.
Kích thước/ *Dimension*: (200 x 400 x 2) mm.
- Số lượng mẫu
Testing quantity : 02 mẫu/ **Samples.**
- Ngày nhận yêu cầu
Date of receiving : 05/11/2024.
- Ngày thử nghiệm
Date of testing : 15/11/2024.
- Khách hàng
Customer : **CÔNG TY CP CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG.**
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION.
Lô số 38, khu C, đường D1, khu công nghiệp An Hạ, xã Phạm Văn Hai, huyện Bình Chánh, Tp. Hồ Chí Minh.
- Nơi thử nghiệm
Test location : Trung tâm Kỹ Thuật 3, KCN Biên Hòa 1, Tp. Biên Hòa, Đồng Nai.
- Phương pháp thử
Test method : AWS D1.3/D1.3M:2018.
- Điều kiện và kết quả thử nghiệm : Xem chi tiết trang 02/02.
Test condition & results **See details on page 02/02.**

TRƯỞNG PTN CƠ KHÍ
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Phạm Văn Út

KT. GIÁM ĐỐC/ FOR DIRECTOR
PHÓ GIÁM ĐỐC/ VICE DIRECTOR



Nguyễn Anh Triết

1. Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu do khách hàng gửi đến và không phải là giấy chứng nhận sản phẩm.

1. Test results are valid for the sample submitted for testing and this is not a certificate of product.

2. Tên mẫu, mô tả mẫu được ghi trên yêu cầu của khách hàng. / Name of sample(s) and customer are written as customer's request.

3. Độ không đảm bảo do mô hình được tính từ độ không đảm bảo đo chuẩn nhân với hệ số phủ k = 2, phân bố chuẩn tương ứng với 95 % độ tin cậy.

3. The reported expanded uncertainty of measurement is stated at the expanded uncertainty of measurement multiplied by the coverage factor k = 2, at 95 % confidence level.

4. Không được trích sao một phần hoặc kết quả thử nghiệm này nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.

4. This Test Report shall not be reproduced, except in full, without the written permission by Quatest 3.

5. Mọi thắc mắc về kết quả, khách hàng liên hệ theo địa chỉ được ghi trên phiếu này để biết thêm thông tin.

5. Please contact QUATEST 3 at the email addresses info@quatest3.com.vn for further information about test report.

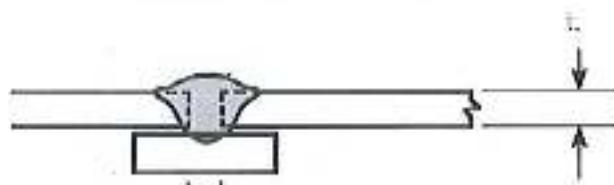
KT3-04448B
CK4/7

PHIẾU KẾT QUẢ THỬ NGHIỆM TEST REPORT



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10. Hình ảnh mẫu/ Photograph of test pieces:



FCAW-1G-S2.0-SS400-MB P-T-PHUONG, S1, S2

11. Kết quả thử uốn/ Bend test results:

Mẫu/ Specimen N°	Góc uốn Bending angle, độ	Kích cỡ mối hàn Weld size t_w , mm	Đánh giá theo/ Assess to AWS D1.3/D1.3M:2018
 FCAW-1G-S2.0-SS400-MB P-T-PHUONG, S1 (Uốn chân/ Root bend)	180	2,0	Đạt/ Pass
 FCAW-1G-S2.0-SS400-MB P-T-PHUONG, S2 (Uốn chân/ Root bend)	180	2,0	Đạt/ Pass

1. Các kết quả thử nghiệm ghi trong phiếu này chỉ có giá trị đối với mẫu do khách hàng gửi đến và không phải là giấy chứng nhận sản phẩm.

Test results are valid for the sample submitted (sample) only; and this is not a certificate of product.

2. Tên mẫu, tên khách hàng được ghi theo yêu cầu của nơi gửi mẫu. (Name of sample) and customer are written as customer's request.

3. Độ không đảm bảo đo mà phòng được tính từ độ không đảm bảo đo chuẩn nhân với hệ số phủ là 2, phần lớn chuẩn tương ứng với 95 % độ tin cậy.

The reported expanded uncertainty of measurement is stated in the measured uncertainty of measurement multiplied by the coverage factor $k = 2$, at 95 % confidence level.

4. Không được trích sao một phần hoặc tất cả kết quả thử nghiệm này nếu không có sự đồng ý bằng văn bản của Trung tâm Kỹ thuật 3.

This Test Report shall not be reproduced, except in full, without the written permission by Quatest 3.

5. Mọi thắc mắc về kết quả, khách hàng liên hệ theo địa chỉ email: quatest3.com.vn để biết thêm thông tin.

Please contact QUATEST 3 at the email address: info@quatest3.com.vn for further information about test report.



MILL CERTIFICATE

台塑河靜鋼鐵興業責任有限公司

FORMOSA HA TINH STEEL CORPORATION

ADDRESS: Vung Ang Economic Zone, Ky Long Ward, Ky Anh Town,

Ho Tinh Province, Viet Nam

TEL: +84-239-3722-121 Ext: 2056 Fax: +84-239-3722-112

Email: fhs.exports@fhs.com.vn

PRODUCT	COMMODITY	HOT ROLLED COIL	
SPECIFICATION and GRADE	ITS CONTRACT NO.	09041201	PURCHASER CREDIT NO.
INSPECTION	CERTIFICATE NO.	Q2304/0025	INVOICE NO.
EXPORTER NAME	MOISU DATE	2009/12	CUSTOMER NO.

COIL NO	PRODUCT INFORMATION				HEAT NO.	DIA	TENSILE TEST				REMARK	CHEMICAL ANALYSIS RESULT %																FORMA																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
	THICK	WIDTH	LENGTH	ACTUAL WEIGHT			Y1	Y2	Y3	EL		C	Si	Mn	P	S	Al	N																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															

COMMODITY as LABEL: HOT ROLLED COIL
 SPECIFICATION as LABEL: JIS G 3003-2004
 GRADE as LABEL: SS400
 THIS MILL CERTIFICATE IS ISSUED IN ACCORDANCE WITH JIS G 3004 TYPE 1

NOTES

Y1: YIELD POINT TYPE

T1: TRACE TEST ANALYSIS RESULT COMPLIES WITH SPECIFICATION REQUIREMENT

Made, initial, passed and original in Vietnam

WE HEREBY CERTIFY THAT MATERIAL DESCRIBED HEREIN HAS BEEN MANUFACTURED AND TESTED WITH SATISFACTORY RESULTS IN ACCORDANCE WITH THE REQUIREMENT OF THE ABOVE MATERIAL SPECIFICATION

SURVEYOR TO

FORMOSA HA TINH STEEL CORPORATION, TECHNOLOGY DIVISION



CERTIFIED MATERIAL REPORT

CMR No. : HDV – 2024 – 0732/2 (Rev.1)

Date of Issue : 2024 – 07 – 26



Customer		G-TECH CORPORATION						
Trade Name		SC-71LH		Size		1.2 mm x 15 kg		
Lot No.		MF24120G71		Purchaser's Order No.		62-2023/GT/HW/PO		
Specification		EN ISO 17632-A-T 42 2 P C1 1 H5 AWS A5.20 E71T-1C * According to EN 10204-2004 : 3.1						
Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment (°C×hr.)			As Welded	
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)				
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.
	527	578	28	-20	79	86	89	85
EN ISO Spec.	≥420	500~640	≥20	≥47				

Chemical Composition (%)

Weld Metal	C	Si	Mn	P	S	Ni	Cr	Mo	V	Cu	Nb
	0.04	0.56	1.42	0.018	0.009	0.02	0.02	0.01	0.01	0.01	0.01
EN ISO Spec.	-	-	≤2.00	-	-	≤0.50	≤0.20	≤0.20	≤0.08	≤0.30	≤0.05

* Hydrogen Content (mℓ/100g) (ISO 3690) : 4.57

* Radiographic Test : ACCEPT

* Fillet Weld Test : GOOD

Others

* AS/NZS ISO 17632-A-T 42 2 P C1 1 H5

* FPC No. (CE-Mark): 0035-CPR-C771

* Q'ty: 14,400 kg

We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.



0035-CPR-C771
0035 10 / HWV-314
EN13479

Approved by

QA Manager Pham Dinh Anh Vu

HYUNDAI WELDING VINA CO., LTD.

Road N2, Nhon Trach 5 IZ, Hiep Phuoc Ward, Nhon Trach District, Dong Nai Province, Vietnam.



CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION
● 38, Zone C, D1 St, An Ha Industrial Park, Pham Van Hai, Binh Chanh, HCMC
● sales@daidung.vn ● (84-28) 3868 1689 ● www.daidung.com.vn

Comply:
ISO 9001:2015
AWS D1.3:2018

WELDING PROCEDURE

Doc. No.: WPS-DD-GMAW-325

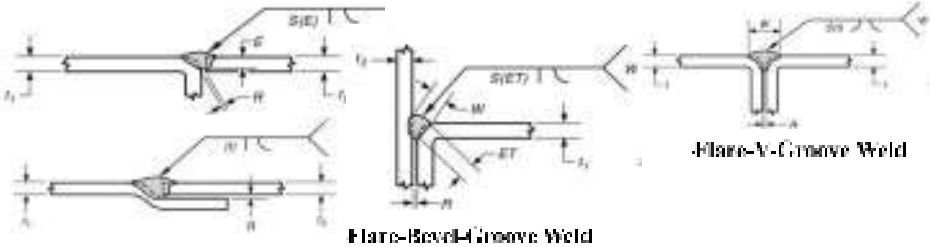
PQR-DD-GMAW-325

Rev	Date	Page	Description	Prepared by	Checked and Approved by
1	Jan 06, 2025	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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Company Name	DaiDung Metallic Manufacture Construction & Trade Corporation	Rev. No.	01	Date:	Nov 18, 2024	Page:	1 of 1
WPS No.	WPS-DD-GMAW-325.1						
Supporting PQR(s)	PQR-DD-GMAW-325.2; 325.3; 325.4.						

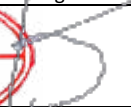
BASE METALS (1.4)	Specification	Type or Grade	AWS Group No.	SHEET METAL THICKNESS	As Welded (mm)	
Sheet Material	ASTM A36, A53, A572,... JIS G3101	Gr A, B 'SS400	Any Group I, II -	Square Groove Weld	1-4	
Supporting Structure	ASTM A36, A53, A572,... JIS G3101	Gr A, B 'SS400	Any Group I, II -	Flare bevel/V Weld	0.8-3.2	
Backing Material	-			DIAMETER	Any	
Coating	Without					

JOINT DETAILS		JOINT SKETCH	
Joint Type	Butt, Corner, Lab Joint.		
Weld Type	Flare-Bevel-Groove Weld		
	Flare-V-Groove Weld		
Root Opening (R)	0 - 1 mm (t/2)		
Weld Size	Per Figure 5.3 A,B,C,D		
	E ≥ 0.5t ₂ ; ET=1-1.5t		
POSTWELD HEAT TREATMENT			
Temperature	-		
Time at Temperature	-		
Other	-		

PROCEDURE			
Weld Layer(s)	All	All	All
Weld Pass(es)	1 or more	1 or more	1 or more
Process	GMAW		
Type (Manual, Mechanized, etc.)	Semi-Automatic		
Position	F	H	V
Vertical Progression	-	-	Upward
Filler Metal (AWS Spec.)	AWS A5.18		
AWS Classification	ER70S-6		
Diameter (mm)	Ø 0.8		
Manufacturer/Trade Name	Kuang Tai KM-70/ Hyundai Welding or Other same classification		
Control of Filler Metal	Drying condition follow Welding Manufacture recommendation		
Shielding Gas (Composition)	CO ₂ (20-25%) + Ar (balance)		
Flow Rate (LPM)	18 - 29	18 - 29	18 - 29
Nozzle Size	No. 6	No. 6	No. 6
Preheat Temperature (Min.)	See Note 3	See Note 3	See Note 3
Interpass Temperature (Max.)	250°C	250°C	250°C
Electrical Characteristics			
Current Type & Polarity	DCEP	DCEP	DCEP
Transfer Mode	Globular	Globular	Globular
Power Source Type (cc, cv, etc.) (Amps)	CV	CV	CV
Amps (A)	77 - 104	77 - 104	72 - 98
Volts (V)	17 - 21	17 - 21	17 - 21
Wire Feed Speed (Amps)	(Amps)	(Amps)	(Amps)
Travel Speed (mm/min.)	172 - 183	172 - 183	135 - 184
Maximum Heat Input (kJ/mm)	1	1	1
Technique			
Stringer or Weave	Stringer or Weave	Stringer or Weave	Stringer or Weave
Multi or Single Pass (per side)	Multi or Single Pass	Multi or Single Pass	Multi or Single Pass
Oscillation (Mech./Auto.)	-	-	-
Number of Electrodes	1	1	1
Contact Tube to Work Dist. (mm)	10 - 25	10 - 25	10 - 25
Peening	None	None	None
Interpass Cleaning	Grinding and Wire Brush	Grinding and Wire Brush	Grinding and Wire Brush
Other	This procedure can be used for repair		

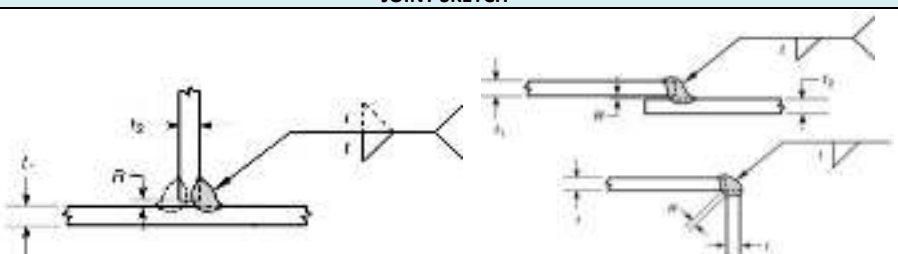
NOTES* Note 1: sheet steel having a minimum specified yield point equal to or less than 80 ksi [550 MPa]
 Note 2: The supporting structural member is listed in D1.1 /D1.1M: 2015 , Table 3.1 , as a Group I or II base metal
 Note 3: minnimum preheat supporting structural member follow table 5.8 AWS D1.1/D1.1M.

Thickness: ≤20	Thickness: 20.1-38mm	Thickness 38.1-65mm	Thickness >65mm
Preheat min 20°C	Preheat min 20°C	Preheat min 65°C/110°C	Preheat min 110/150°C

	Prepared by DDC's QA Welding Coordinator	Reviewed and Approved by	
Name	Nguyen Quynh Sa	Truong Hoai Phong Vu (QA Manager)	
Signature	 		
Date	Nov 18, 2024	Nov 18, 2024	

Company Name	DaiDung Metallic Manufacture Construction & Trade Corporation	Rev. No.	01	Date:	Nov 18, 2024	Page:	1 of 1
WPS No.	WPS-DD-GMAW-325.2						
Supporting PQR(s)	PQR-DD-GMAW-325.1; 325.5						

BASE METALS (1.4)	Specification	Type or Grade	AWS Group No.	SHEET METAL THICKNESS	As Welded (mm)	
Sheet Material	ASTM A36, A53, A572,... JIS G3101, CS, Low alloy steel.	Gr A, B 'SS400	Any Group I, II -	Fillet Welds	0.8 - 3.2	
Supporting Structure	ASTM A36, A53, A572,... JIS G3101, CS, Low alloy steel.	Gr A, B 'SS400	Any Group I, II -			
Backing Material	-			DIAMETER	Any	
Coating	Without					

JOINT DETAILS		JOINT SKETCH	
Joint Type	T, Corner & Lap Joint		
Weld Type	Fillet Weld		
Root Opening (R)	0 - 1 mm		
Weld Size	Per Figure 5.2 A, B, C		
POSTWELD HEAT TREATMENT			
Temperature	-		
Time at Temperature	-		
Other	-		

PROCEDURE			
Weld Layer(s)	All	All	All
Weld Pass(es)	1 or more	1 or more	1 or more
Process	GMAW		
Type (Manual, Mechanized, etc.)	Semi-Automatic		
Position	F	H	V
Vertical Progression	-	-	Upward
Filler Metal (AWS Spec.)	AWS A5.18		
AWS Classification	ER70S-6		
Diameter (mm)	Ø 0.8		
Manufacturer/Trade Name	Kuang Tai KM-70/ Hyundai Welding or Other same classification		
Control of Filler Metal	Drying condition follow Welding Manufacture recommendation		
Shielding Gas (Composition)	CO2 (20-25%) + Ar (balance)		
Flow Rate (LPM)	18 - 29	18 - 29	18 - 29
Nozzle Size	No. 6	No. 6	No. 6
Preheat Temperature (Min.)	See Note 3	See Note 3	See Note 3
Interpass Temperature (Max.)	250°C	250°C	250°C
Electrical Characteristics			
Current Type & Polarity	DCEP	DCEP	DCEP
Transfer Mode	Globular	Globular	Globular
Power Source Type (cc, cv, etc.) (Amps)	CV	CV	CV
Amps (A)	72 - 98	72 - 98	72 - 98
Volts (V)	17 - 21	17 - 21	17 - 21
Wire Feed Speed (Amps)	(Amps)	(Amps)	(Amps)
Travel Speed (mm/min.)	180-230	180-230	205-245
Maximum Heat Input (kJ/mm)	1	1	1
Technique			
Stringer or Weave	Stringer or Weave	Stringer or Weave	Stringer or Weave
Multi or Single Pass (per side)	Multi or Single Pass	Multi or Single Pass	Multi or Single Pass
Oscillation (Mech./Auto.)	-	-	-
Number of Electrodes	1	1	1
Contact Tube to Work Dist. (mm)	10 - 25	10 - 25	10 - 25
Peening	None	None	None
Interpass Cleaning	Grinding and Wire Brush	Grinding and Wire Brush	Grinding and Wire Brush
Other	This procedure can be used for repair		

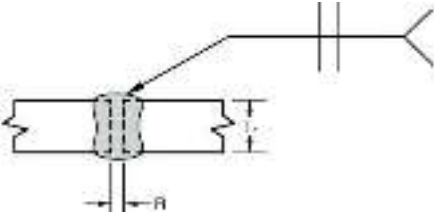
NOTES* Note 1: sheet steel having a minimum specified yield point equal to or less than 80 ksi [550 MPa]
 Note 2: The supporting structural member is listed in D1.1 /D1.1M: 2015 , Table 3.1 , as a Group I or II base metal
 Note 3: minnimum preheat supporting structural member follow table 5.8 AWS D1.1/D1.1M.

Thickness: ≤20	Thickness: 20.1-38mm	Thickness 38.1-65mm	Thickness >65mm
Preheat min 20°C	Preheat min 20°C	Preheat min 65°C/110°C	Preheat min 110/150°C

	Prepared by DDC's QA Welding Coordinator	Reviewed and Approved by	
Name	Nguyen Quynh Sa	Truong Hoai Phong Vu (QA Manager)	
Signature	 		
Date	Nov 18, 2024	Nov 18, 2024	

Company Name	DaiDung Metallic Manufacture Construction & Trade Corporation	Rev. No.	01	Date:	Nov 18, 2024	Page:	1 of 1
WPS No.	WPS-DD-GMAW-325.3						
Supporting PQR(s)	PQR-DD-GMAW-325.3						

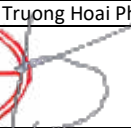
BASE METALS (1.4)	Specification	Type or Grade	AWS Group No.	SHEET METAL THICKNESS	As Welded (mm)	
Sheet Material	ASTM A36, A53, A572,... JIS G3101, CS, Low alloy steel.	Gr A, B 'SS400	Any Group I, II -	Square Groove Weld	1-4	
Supporting Structure	ASTM A36, A53, A572,... JIS G3101, CS, Low alloy steel.	Gr A, B 'SS400	Any Group I, II -			
Backing gouging	Grinding or Air carbon gouging			DIAMETER	Any	
Coating	Without					

JOINT DETAILS		JOINT SKETCH	
Joint Type	Square-groove Butt joint		
Weld Type	Weld both sides		
Root Opening (R)	0 - 1 mm		
Weld Size	equals the thickness t.		
POSTWELD HEAT TREATMENT			
Temperature	-		
Time at Temperature	-		
Other	-		

PROCEDURE			
Weld Layer(s)	All		
Weld Pass(es)	1 or more		
Process	GMAW		
Type (Manual, Mechanized, etc.)	Semi-Automatic		
Position	F		
Vertical Progression	-		
Filler Metal (AWS Spec.)	AWS A5.18		
AWS Classification	ER70S-6		
Diameter (mm)	Ø 0.8		
Manufacturer/Trade Name	Kuang Tai KM-70/ Hyundai Welding or Other same classification		
Control of Filler Metal	Drying condition follow Welding Manufacture recommendation		
Shielding Gas (Composition)	CO2 (20-25%) + Ar (balance)		
Flow Rate (LPM)	18 - 29		
Nozzle Size	No. 6		
Preheat Temperature (Min.)	See Note 3		
Interpass Temperature (Max.)	250°C		
Electrical Characteristics			
Current Type & Polarity	DCEP		
Transfer Mode	Globular		
Power Source Type (cc, cv, etc.) (Amps)	CV		
Amps (A)	72 - 104		
Volts (V)	17 - 21		
Wire Feed Speed (Amps)	(Amps)		
Travel Speed (mm/min.)	240-321		
Maximum Heat Input (kJ/mm)	1		
Technique			
Stringer or Weave	Stringer or Weave		
Multi or Single Pass (per side)	Multi or Single Pass		
Oscillation (Mech./Auto.)	-		
Number of Electrodes	1		
Contact Tube to Work Dist. (mm)	10 - 25		
Peening	None		
Interpass Cleaning	Grinding and Wire Brush		
Other	This procedure can be used for repair		

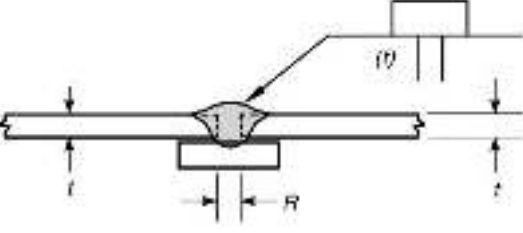
NOTES* Note 1: sheet steel having a minimum specified yield point equal to or less than 80 ksi [550 MPa]
 Note 2: The supporting structural member is listed in D1.1 /D1.1M: 2015 , Table 3.1 , as a Group I or II base metal
 Note 3: minnimum preheat supporting structural member follow table 5.8 AWS D1.1/D1.1M.

Thickness: ≤20	Thickness: 20.1-38mm	Thickness 38.1-65mm	Thickness >65mm
Preheat min 20°C	Preheat min 20°C	Preheat min 65°C/110°C	Preheat min 110/150°C

	Prepared by DDC's QA Welding Coordinator	Reviewed and Approved by	
Name	Nguyen Quynh Sa	Truong Hoai Phong Vu (QA Manager)	
Signature	 		
Date	Nov 18, 2024	Nov 18, 2024	

Company Name	DaiDung Metallic Manufacture Construction & Trade Corporation			Date:	Nov 18, 2024	Page:	1 of 1
WPS No.	WPS-DD-GMAW-325.4	Rev. No.	01				
Supporting PQR(s)	PQR-DD-GMAW-325.3						

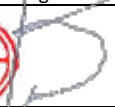
BASE METALS (1.4)	Specification	Type or Grade	AWS Group No.	SHEET METAL THICKNESS	As Welded (mm)	
Sheet Material	ASTM A36, A53, A572,... JIS G3101, CS, Low alloy steel.	Gr A, B 'SS400	Any Group I, II -	Square Groove Weld	1-4	
Supporting Structure	ASTM A36, A53, A572,... JIS G3101, CS, Low alloy steel.	Gr A, B 'SS400	Any Group I, II -			
Backing Material	Same metal sheet material or Ceramic			DIAMETER	Any	
Coating	Without					

JOINT DETAILS		JOINT SKETCH	
Joint Type	Square-groove Butt joint		
Weld Type	Weld with backing		
Root Opening (R)	0 - 1 mm		
Weld Size	equals the thickness t. Per Figure 5.1 A		
POSTWELD HEAT TREATMENT			
Temperature	-		
Time at Temperature	-		
Other	-		

PROCEDURE			
Weld Layer(s)	All		
Weld Pass(es)	1 or more		
Process	GMAW		
Type (Manual, Mechanized, etc.)	Semi-Automatic		
Position	F		
Vertical Progression	-		
Filler Metal (AWS Spec.)	AWS A5.18		
AWS Classification	ER70S-6		
Diameter (mm)	Ø 0.8		
Manufacturer/Trade Name	Kuang Tai KM-70/ Hyundai Welding or Other same classification		
Control of Filler Metal	Drying condition follow Welding Manufacture recommendation		
Shielding Gas (Composition)	CO2 (20-25%) + Ar (balance)		
Flow Rate (LPM)	18 - 29		
Nozzle Size	No. 6		
Preheat Temperature (Min.)	See Note 3		
Interpass Temperature (Max.)	250°C		
Electrical Characteristics			
Current Type & Polarity	DCEP		
Transfer Mode	Globular		
Power Source Type (cc, cv, etc.) (Amps)	CV		
Amps (A)	72 - 104		
Volts (V)	17 - 21		
Wire Feed Speed (Amps)	(Amps)		
Travel Speed (mm/min.)	240-321		
Maximum Heat Input (kJ/mm)	1		
Technique			
Stringer or Weave	Stringer or Weave		
Multi or Single Pass (per side)	Multi or Single Pass		
Oscillation (Mech./Auto.)	-		
Number of Electrodes	1		
Contact Tube to Work Dist. (mm)	10 - 25		
Peening	None		
Interpass Cleaning	Grinding and Wire Brush		
Other	This procedure can be used for repair		

NOTES* Note 1: sheet steel having a minimum specified yield point equal to or less than 80 ksi [550 MPa]
 Note 2: The supporting structural member is listed in D1.1 /D1.1M: 2015 , Table 3.1 , as a Group I or II base metal
 Note 3: minnimum preheat supporting structural member follow table 5.8 AWS D1.1/D1.1M.

Thickness: ≤20	Thickness: 20.1-38mm	Thickness 38.1-65mm	Thickness >65mm
Preheat min 20°C	Preheat min 20°C	Preheat min 65°C/110°C	Preheat min 110/150°C

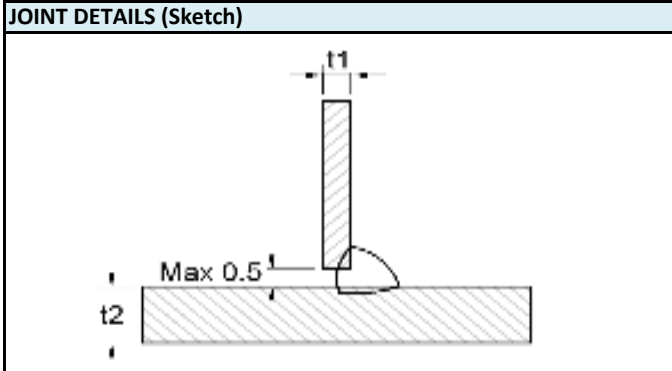
	Prepared by DDC's QA Welding Coordinator	Reviewed and Approved by	
Name	Nguyen Quynh Sa	Truong Hoai Phong Vu (QA Manager)	
Signature	 		
Date	Nov 18, 2024	Nov 18, 2024	

	PROCEDURE QUALIFICATION RECORD (PQR) (AWS D1.3/D1.3M: 2018 STRUCTURAL WELDING CODE - SHEET STEEL)		
	Company Name: DaiDung Metallic Manufacture Construction & Trade Corporation PQR No.: PQR-DD-GMAW-325.1 & 325.5 Rev. No. 00 Date: Nov 18, 2024 Page: 1 of 4		

BASE METALS	Specification	Type or Grade	AWS D1.3 Group No.	Thickness (t) V up (3F)	Thickness (t) H (2F)	Thickness (t)	Diameter
Base Material	JIS 3101	SS400	-	1.6 mm	1.6 mm		-
Welded To	JIS 3101	SS400	-	2.0 mm	1.6 mm		-
Other	Shop primer in Weld Joint: No						

JOINT DETAILS	
Groove Type	Fillet Weld T Joint
Tack Weld include	N/A
Groove Angle	N/A
Root Opening	Max 0.5 mm
Root Face	N/A
Backgouging	N/A
Method	-

POSTWELD HEAT TREATMENT	
Soaking Temperature	-
Soaking Time	-
Other	-



PROCEDURE	DD-GMAW-325.1	DD-GMAW-325.5
Weld Layer(s)	Root/Cap	Root/Cap
Weld Pass(es) per layer	1	1
Process	GMAW	GMAW
Type (Manual, Mechanized, etc.)	Semi-Automatic	Semi-Automatic
Position	H (2F)	V (3F)
Vertical Progression	-	Weld up
Filler Metal (Spec.)	AWS A5.18	AWS A5.18
AWS Classification	ER70S-6	ER70S-6
Diameter (mm)	Ø 0.8	Ø 0.8
Manufacturer/Trade Name	Kuang Tai KM-70	Kuang Tai KM-70
Control of Filler Metal	Follow Welding Manufacture recommendation	Follow Welding Manufacture recommendation
Shielding Gas (Composition)	CO2 (20%) + Ar (80%)	CO2 (20%) + Ar (80%)
Flow Rate (LPM)	20-23	20-23
Nozzle Size	No. 6	No. 6
Preheat Temperature	(ambient)	(ambient)
Interpass Temperature (Max.)	-	-
Electrical Characteristics		
Current Type & Polarity	DCEP	DCEP
Transfer Mode	Globular	Globular
Power Source Type (cc, cv, etc.)	CV	CV
Amps (A)	80-90	80-90
Volts (V)	18.5-19.5	18.5-19.5
Wire Feed Speed	(Amps)	(Amps)
Travel Speed (mm/min.)	200-214	225
Heat Input (kJ/mm)	-	-
Technique		
Stringer or Weave	Weave	Weave
Multi or Single Pass (per side)	Single Pass	Single Pass
Number of Electrodes	1	1
Contact Tube to Work Dist.(mm)	12-18	12-18
Peening	None	None
Interpass Cleaning	-	-
Inspection	Directly	Directly
Other	-	-

	Prepared by DDC's QA Welding	Reviewed and Approved by
Name	Nguyen Quynh Sa	Tr H Phong Vu
Signature		
Date	Nov 18, 2024	Nov 18, 2024

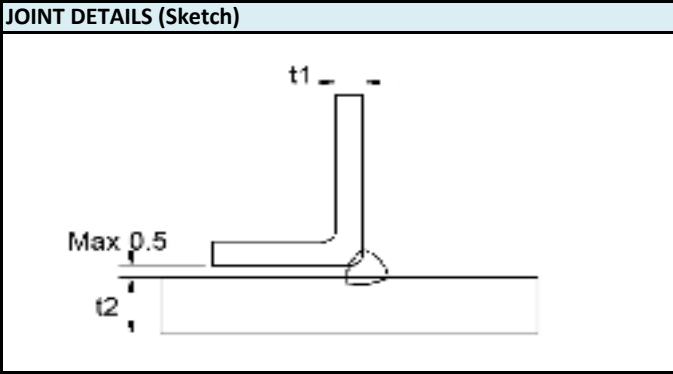
	PROCEDURE QUALIFICATION RECORD (PQR) (AWS D1.3/D1.3M: 2018 STRUCTURAL WELDING CODE - SHEET STEEL)	
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Company Name: DaiDung Metallic Manufacture Construction & Trade Corporation

PQR No.: PQR-DD-GMAW-325.2 & 325.4 Rev. No. 00 Date: Nov 18, 2024 Page: 2 of 4

BASE METALS	Specification	Type or Grade	AWS D1.3 Group No.	Thickness (t) V up	Thickness (t) H	Thickness (t)	Diameter
Base Material	JIS 3101	SS400	-	1.6 mm	1.6 mm		-
Welded To	JIS 3101	SS400	-	2.0 mm	2.0 mm		-
Other	Shop primer in Weld Joint: No						

JOINT DETAILS	
Groove Type	Flare Bevel Groove weld
Tack Weld include	N/A
Groove Angle	N/A
Root Opening	Max 0.5 mm
Root Face	N/A
Backgouging	N/A
Method	-



POSTWELD HEAT TREATMENT	
Soaking Temperature	-
Soaking Time	-
Other	-

PROCEDURE	DD-GMAW-325.2	DD-GMAW-325.4
Weld Layer(s)	Root/Cap	Root/Cap
Weld Pass(es) per layer	1	1
Process	GMAW	GMAW
Type (Manual, Mechanized, etc.)	Semi-Automatic	Semi-Automatic
Position	H	V
Vertical Progression	-	Weld up
Filler Metal (Spec.)	AWS A5.18	AWS A5.18
AWS Classification	ER70S-6	ER70S-6
Diameter (mm)	Ø 0.8	Ø 0.8
Manufacturer/Trade Name	Kuang Tai KM-70	Kuang Tai KM-70
Control of Filler Metal	Follow Welding Manufacture recommendation	Follow Welding Manufacture recommendation
Shielding Gas (Composition)	CO2 (20%) + Ar (80%)	CO2 (20%) + Ar (80%)
Flow Rate (LPM)	20-23	20-23
Nozzle Size	No. 6	No. 6
Preheat Temperature	(ambient)	(ambient)
Interpass Temperature (Max.)	-	-
Electrical Characteristics		
Current Type & Polarity	DCEP	DCEP
Transfer Mode	Globular	Globular
Power Source Type (cc, cv, etc.)	CV	CV
Amps (A)	85-95	80-90
Volts (V)	18.5-19.5	18.5-19.5
Wire Feed Speed	(Amps)	(Amps)
Travel Speed (mm/min.)	171-183	150-168
Heat Input (kJ/mm)	-	-
Technique		
Stringer or Weave	Weave	Weave
Multi or Single Pass (per side)	Single Pass	Single Pass
Number of Electrodes	1	1
Contact Tube to Work Dist.(mm)	12-20	12-20
Peening	None	None
Interpass Cleaning	-	-
Inspection	Directly	Directly
Other	-	-

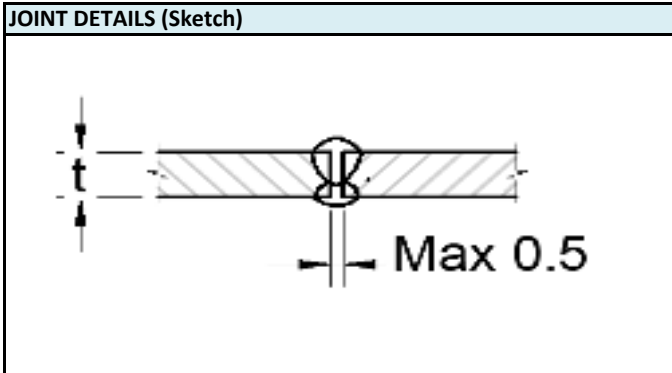
	Prepared by DDC's QA Welding	Reviewed and Approved by
Name	Nguyen Quynh Sa	Tr H Phong Vu
Signature		
Date	Nov 18, 2024	Nov 18, 2024

	PROCEDURE QUALIFICATION RECORD (PQR) (AWS D1.3/D1.3M: 2018 STRUCTURAL WELDING CODE - SHEET STEEL)	
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Company Name: **DaiDung Metallic Manufacture Construction & Trade Corporation**
 PQR No.: **PQR-DD-GMAW-325.3** Rev. No. **00** Date: **Nov 18, 2024** Page: **3 of 4**

BASE METALS	Specification	Type or Grade	AWS D1.3 Group No.	Thickness (t) F (1G)	Thickness (t)	Thickness (t)	Diameter
Base Material	JIS 3101	SS400	-	2.0 mm			-
Welded To	JIS 3101	SS400	-	2.0 mm			-
Other	Shop primer in Weld Joint: No						

JOINT DETAILS	
Groove Type	SQUARE JOINT BOTH SIDE
Tack Weld include	N/A
Groove Angle	N/A
Root Opening	Max 0.5 mm
Root Face	N/A
Backgouging	yes
Method	Grinding



POSTWELD HEAT TREATMENT	
Soaking Temperature	-
Soaking Time	-
Other	-

PROCEDURE	DD-GMAW-325.3 Testing specimen No. 01	DD-GMAW-325.3 Testing specimen No. 02
Weld Layer(s)	Root/Cap	Root/Cap
Weld Pass(es) per layer	1	1
Process	GMAW	GMAW
Type (Manual, Mechanized, etc.)	Semi-Automatic	Semi-Automatic
Position	F (1G)	F (1G)
Vertical Progression	-	-
Filler Metal (Spec.)	AWS A5.18	AWS A5.18
AWS Classification	ER70S-6	ER70S-6
Diameter (mm)	Ø 0.8	Ø 0.8
Manufacturer/Trade Name	Kuang Tai KM-70	Kuang Tai KM-70
Control of Filler Metal	Follow Welding Manufacture recommendation	Follow Welding Manufacture recommendation
Shielding Gas (Composition)	CO2 (20%) + Ar (80%)	CO2 (20%) + Ar (80%)
Flow Rate (LPM)	20-23	20-23
Nozzle Size	No. 6	No. 6
Preheat Temperature	(ambient)	(ambient)
Interpass Temperature (Max.)	50	52
Electrical Characteristics		
Current Type & Polarity	DCEP	DCEP
Transfer Mode	Globular	Globular
Power Source Type (cc, cv, etc.)	CV	CV
Amps (A)	80-90	85-95
Volts (V)	18.5-19.5	18.5-19.5
Wire Feed Speed	(Amps)	(Amps)
Travel Speed (mm/min.)	266-292	272-279
Heat Input (kJ/mm)	-	-

Technique		
Stringer or Weave	Weave	Weave
Multi or Single Pass (per side)	Single Pass	Single Pass
Number of Electrodes	1	1
Contact Tube to Work Dist.(mm)	12-20	12-20
Peening	None	None
Interpass Cleaning	-	-
Inspection	Directly	Directly
Other	-	-

	Prepared by DDC's QA Welding	Reviewed and Approved by
Name	Nguyen Quynh Sa	Tr H Phong Vu
Signature		
Date	Nov 18, 2024	Nov 18, 2024



PROCEDURE QUALIFICATION RECORD (PQR)
(AWS D1.3/D1.3M: 2018 STRUCTURAL WELDING CODE - SHEET STEEL)



PQR No.: **PQR-DD-GMAW-325.1/325.2/325.3/325.4/325.5**

Rev. No.: **00**

Date: **Nov 18, 2024**

Page: **4 of 4**

TEST RESULTS

TESTS (Square Groove weld in Butt Joint)

	Type of Tests	Clause/Figure(s) Reference	Acceptance Criteria	Result	Remarks
✓	Visual Inspection	Clause 6.7.1	Clause 6.7.1.2(2)	Accepted	
✓	Bend	Clause 6.7.1	Clause 6.7.1.2(4)	Accepted	

Welder's Name	ID Number	DDC's Welder ID N.o	Tests Conducted by	
PHAN THANH PHUONG	-	DDC-WD-0393	Laboratory	SGS
-	-	-	Destructive Test No.	INL24/512130-816-5
-	-	-	Testing specimen No.	DD-GMAW-325.3

TESTS (HSS Butt Weld)

	Type of Tests	Clause/Figure(s) Reference	Acceptance Criteria	Result	Remarks
	Visual Inspection	Clause 6.7.2	Clause 6.7.2.2(2a)	-	
	Bend	Clause 6.7.2	Clause 6.7.2.2(2b)	-	

Welder's Name	ID Number	DDC's Welder ID N.o	Tests Conducted by	
-	-	-	Laboratory	-
-	-	-	Destructive Test No.	-
-	-	-	Testing specimen No.	-

TESTS (Fillet)

	Type of Tests	Clause/Figure(s) Reference	Acceptance Criteria	Result	Remarks
✓	Visual Inspection	Clause 6.7.3	Clause 6.7.3.2(3a, b)	Accepted	
✓	Bend	Clause 6.7.3	Clause 6.7.3.2(3d)	Accepted	

Welder's Name	ID Number	DDC's Welder ID N.o	Tests Conducted by	
PHAN THANH PHUONG	-	DDC-WD-0393	Laboratory	SGS
-	-	-	Destructive Test No.	INL24/512130-816-3 & 7
-	-	-	Testing specimen No.	DD-GMAW-325.1 & 325.5

TESTS (Flare Belve Groove Weld)

	Type of Tests	Clause/Figure(s) Reference	Acceptance Criteria	Result	Remarks
✓	Visual Inspection	Clause 6.7.4	Clause 6.7.4.2(3a)	Accepted	
✓	Bend	Clause 6.7.4	Clause 6.7.4.2(3b, c)	Accepted	

Welder's Name	ID Number	DDC's Welder ID N.o	Tests Conducted by	
PHAN THANH PHUONG	-	DDC-WD-0393	Laboratory	SGS
-	-	-	Destructive Test No.	INL24/512130-816-4 & 6
-	-	-	Testing specimen No.	DD-GMAW-325.2 & 325.4

This undersigned certified that the statements in this recode are correct and that the test assemblies were prepared, welded, and tested in conformance with request of Clause 6 of AWS D1.3/D1.3M (2018) Structural Welding Code - Sheet Steel

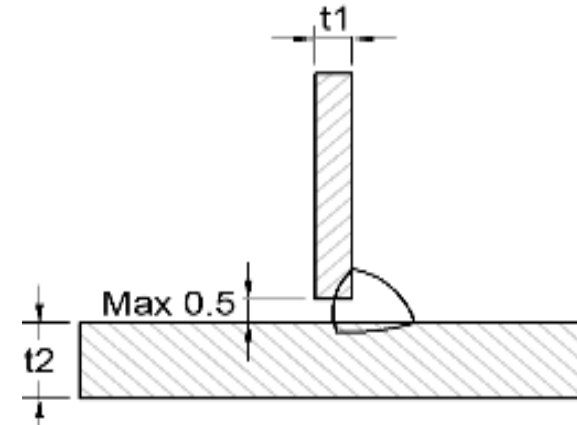
	Prepared by DDC's QA Welding	Reviewed and Approved by
Name	Nguyen Quynh Sa	Tr H Phong Vu
Signature		
Date	Nov 18, 2024	Nov 18, 2024

WELDING PARAMETERS RECORD (Doc No.: WPR-GMAW-325.1)

Company name: DAIDUNG CORPORATION
 PQR No.: PQR-DD-GMAW-325.1
 Test Coupon: DD-GMAW-325.1
 Test position: H (2F)
 Base metal:
 - Specification: JIS 3101 SS400 JIS 3101 SS400
 - Dimension: 150 x 200 x 1.6 mm 100 x 100 x 1.6 mm
 - Heat/Product No.: 25230211 25230211
 - Certificate No.: TF2222101016/QTN TF2222101016/QTN
 Filler metal:
 - Specification: AWS A5.18
 - Classification: ER70S-6
 - Trade name: Kuang Tai KM-70 (Ø0.8 mm)
 - Batch/Lot No.: V24A16012
 - Certificates No.: 24102401
 - Control of Filler Metal: Follow Kuang Tai's recommendation:
 Gas: 80% Ar + 20% CO2
 Welder Name: PHAN THANH PHUONG
 I.D.: DDC-WD-0393

Joint Sketch: below
 Shop primer: No

Date: Nov 8, 2024
 Tack weld included: No



Root face: N/A
 Root gap: 0-0.5 mm
 Groove angle: N/A

Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Leg Length (mm)	Depth	
1	GMAW	AWS A5.18/ ER70S-6	0.8	DCEP	80-90	18.5-19.5		14	50	214	32 (Ambient)	20-23	0.41-0.49	4-5		Coupon 1





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✉ sales@daidung.vn ☎ (84-28) 3868 1689 🌐 www.daidung.com.vn

WELDING PARAMETERS RECORD (Doc No.: WPR-GMAW-325.1)

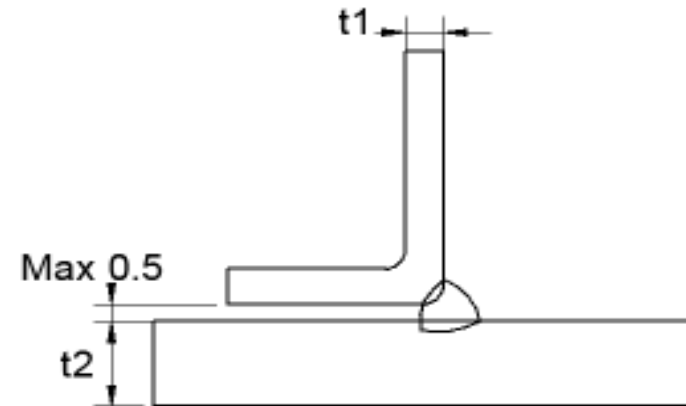
Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Leg Length (mm)	Depth	
1	GMAW	AWS A5.18/ ER70S-6	0.8	DCEP	80-90	18.5-19.5		15	50	200	32 (Ambient)	20-23	0.44-0.53	4-5		Coupon 2
		Conducted by QA Welding							Reviewed and Approved by							
Name		NGUYEN QUYNH SA							Tr H Phong Vu							
Signature																
Date		Nov 8, 2024							November 8, 2024							

WELDING PARAMETERS RECORD (Doc No.: WPR-GMAW-325.2)

Company name	DAIDUNG CORPORATION	
PQR No.	PQR-DD-GMAW-325.2	
Test Coupon	DD-GMAW-325.2	
Test position	H	
Base metal:		
- Specification	JIS 3101 SS400	JIS 3101 SS400
- Dimension	150 x 200 x 2.0 mm	100 x 100 x 1.6 mm
- Heat/Product No.	2542518000	25230211
- Certificaion No.	QZA024700689	TF2222101016/QTN
Filler metal:		
- Specification	AWS A5.18	
- Classification	ER70S-6	
- Trade name	Kuang Tai KM-70 (Ø0.8 mm)	
- Batch/Lot No.	V24A16012	
- Certificates No.	24102401	
- Control of Filler Metal	Follow Kuang Tai's recommendation:	
Gas:	80% Ar + 20% CO2	
Welder Name:	PHAN THANH PHUONG	
I.D:	DDC-WD-0393	

Joint Sketch: below
 Shop primer: No

Date: Nov 08, 2024
 Tack weld included: No



Root face: N/A
 Root gap: 0-0.5 mm
 Groove angle: N/A

Layer/ pass	Process	Filler metal		Electric Current			Times			Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Leg Length (mm)	Depth	
1	GMAW	AWS A5.18/ ER70S-6	0.8	DCEP	85-95	18.5-19.5		14	40	171	32 (Ambient)	20-23	0.55-0.65	4.0		Coupon 1



WELDING PARAMETERS RECORD (Doc No.: WPR-GMAW-325.2)

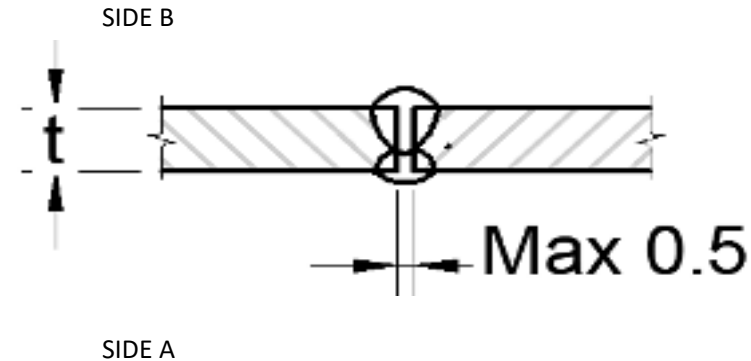
Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark	
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Leg Length (mm)	Depth		
1	GMAW	AWS A5.18/ ER70S-6	0.8	DCEP	85-95	18.5-19.5		18	55	183	32 (Ambient)	20-23	0.51-0.61	4.0		Coupon 2	
		Conducted by QA Welding								Reviewed and Approved by							
Name		NGUYEN QUYNH SA								Tr H Phong Vu							
Signature																	
Date		Jun 13, 2024								June 13, 2024							

WELDING PARAMETERS RECORD (Doc No.: WPR-GMAW-325.3)

Company name: DAIDUNG CORPORATION
 PQR No.: PQR-DD-GMAW-325.3
 Test Coupon: DD-GMAW-325.3
 Test position: F (1G)
 Base metal:
 - Specification: JIS 3101 SS400
 - Dimension: 400 x 200 x 2.0 mm
 - Heat/Product No.: 2542518000
 - Certificate No.: QZA024700689
 Filler metal:
 - Specification: AWS A5.18
 - Classification: ER70S-6
 - Trade name: Kuang Tai KM-70 (Ø0.8 mm)
 - Batch/Lot No.: V24A16012
 - Certificates No.: 24102401
 - Control of Filler Metal: Follow Kuang Tai's recommendation:
 Gas: 80% Ar + 20% CO2
 Welder Name: PHAN THANH PHUONG
 I.D.: DDC-WD-0393

Joint Sketch: below
 Shop primer: No

Date: Nov 08, 2024
 Tack weld included: No



Root face: N/A
 Root gap: 0-0.5 mm
 Groove angle: N/A

Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width (mm)	Depth	
SIDE A																
1	GMAW	AWS A5.18/ ER70S-6	0.8	DCEP	80-90	18.5-19.5		45	200	267	32 (Ambient)	20-23	0.33-0.39	6-7.5		Coupon 1
SIDE B																
2	GMAW	AWS A5.18/ ER70S-6	0.8	DCEP	80-90	18.5-19.5		41	200	293	50	20-23	0.3-0.36	6-7.5		Coupon 1



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WELDING PARAMETERS RECORD (Doc No.: WPR-GMAW-325.3)

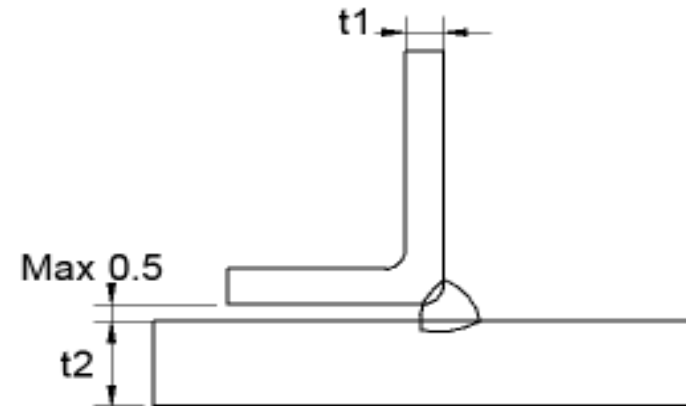
Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width (mm)	Depth	
SIDE A																
1	GMAW	AWS A5.18/ ER70S-6	0.8	DCEP	85-95	18.5-19.5		43	200	279	32 (Ambient)	20-23	0.34-0.4	6-7.5		Coupon 2
SIDE B																
2	GMAW	AWS A5.18/ ER70S-6	0.8	DCEP	85-95	18.5-19.5		44	200	273	52	20-23	0.35-0.41	6-7.5		Coupon 2
		Conducted by QA Welding								Reviewed and Approved by						
Name		NGUYEN QUYNH SA								Tr H Phong Vu						
Signature																
Date		Jun 13, 2024								June 13, 2024						

WELDING PARAMETERS RECORD (Doc No.: WPR-GMAW-325.4)

Company name	DAIDUNG CORPORATION	
PQR No.	PQR-DD-GMAW-325.4	
Test Coupon	DD-GMAW-325.4	
Test position	V (weld up)	
Base metal:		
- Specification	JIS 3101 SS400	JIS 3101 SS400
- Dimension	150 x 200 x 2.0 mm	100 x 100 x 1.6 mm
- Heat/Product No.	2542518000	25230211
- Certificaion No.	QZA024700689	TF2222101016/QTN
Filler metal:		
- Specification	AWS A5.18	
- Classification	ER70S-6	
- Trade name	Kuang Tai KM-70 (Ø0.8 mm)	
- Batch/Lot No.	V24A16012	
- Certificates No.	24102401	
- Control of Filler Metal	Follow Kuang Tai's recommendation:	
Gas:	80% Ar + 20% CO2	
Welder Name:	PHAN THANH PHUONG	
I.D:	DDC-WD-0393	

Joint Sketch: below
 Shop primer: No

Date: Nov 08, 2024
 Tack weld included: No



Root face: N/A
 Root gap: 0-0.5 mm
 Groove angle: N/A

Layer/ pass	Process	Filler metal		Electric Current			Times			Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Leg Length (mm)	Depth	
1	GMAW	AWS A5.18/ ER70S-6	0.8	DCEP	80-90	18.5-19.5		18	45	150	33 (Ambient)	20-23	0.59-0.7	4.0		Coupon 1





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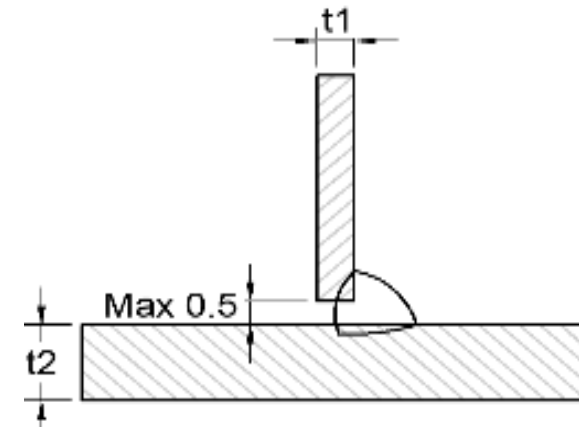
WELDING PARAMETERS RECORD (Doc No.: WPR-GMAW-325.4)

Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark	
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Leg Length (mm)	Depth		
1	GMAW	AWS A5.18/ ER70S-6	0.8	DCEP	80-90	18.5-19.5		16	45	169	33 (Ambient)	20-23	0.53-0.62	4.0		Coupon 2	
		Conducted by QA Welding								Reviewed and Approved by							
Name		NGUYEN QUYNH SA								Tr H Phong Vu							
Signature																	
Date		Jun 13, 2024								June 13, 2024							

WELDING PARAMETERS RECORD (Doc No.: WPR-GMAW-325.5)

Company name: DAIDUNG CORPORATION
 PQR No.: PQR-DD-GMAW-325.5
 Test Coupon: DD-GMAW-325.5
 Test position: V (weld up)
 Base metal:
 - Specification: JIS 3101 SS400 JIS 3101 SS400
 - Dimension: 150 x 200 x 2.0 mm 100 x 100 x 1.6 mm
 - Heat/Product No.: 2542518000 25230211
 - Certificate No.: QZA024700689 TF2222101016/QTN
 Filler metal:
 - Specification: AWS A5.18
 - Classification: ER70S-6
 - Trade name: Kuang Tai KM-70 (Ø0.8 mm)
 - Batch/Lot No.: V24A16012
 - Certificates No.: 24102401
 - Control of Filler Metal: Follow Kuang Tai's recommendation:
 Gas: 80% Ar + 20% CO2
 Welder Name: PHAN THANH PHUONG
 I.D.: DDC-WD-0393

Joint Sketch: below
 Shop primer: No
 Date: Nov 8, 2024
 Tack weld included: No



Root face: N/A
 Root gap: 0-0.5 mm
 Groove angle: N/A

Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Leg Length (mm)	Depth	
1	GMAW	AWS A5.18/ ER70S-6	0.8	DCEP	80-90	18.5-19.5		12	45	225	33 (Ambient)	20-23	0.39-0.47	4-5		Coupon 1





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WELDING PARAMETERS RECORD (Doc No.: WPR-GMAW-325.5)

Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Leg Length (mm)	Depth	
1	GMAW	AWS A5.18/ ER70S-6	0.8	DCEP	80-90	18.5-19.5		16	60	225	33 (Ambient)	20-23	0.39-0.47	4-5		Coupon 2
		Conducted by QA Welding							Reviewed and Approved by							
Name		NGUYEN QUYNH SA							Tr H Phong Vu							
Signature																
Date		Nov 8, 2024							November 8, 2024							

Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
Client's address / Địa chỉ : LÔ SỐ 38, KHU C, ĐƯỜNG D1, KHU CÔNG NGHIỆP AN HẠ, XÃ PHẠM VĂN HAI, HUYỆN BÌNH CHÁNH, THÀNH PHỐ HỒ CHÍ MINH, VIỆT NAM

The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : Mẫu hàn DD-GMAW-325.1
- Position H (T Joint Fillet weld)
- Base metal JIS G3101 SS400
- Size: T 100 x 100 x1.6 + 150 x 200 x1.6mm (2pcs)
- Filler metal: : Kuang Tai KM-70 (0.8mm)
- Welder ID: DDC-WD-0393
Project / Dự án : CẦU ỐNG LÔNG DI ĐỘNG SÂN BAY LONG THÀNH
Item / Hàng mục : Procedure Qualification Record (PQR) Test
Quantity / Số lượng mẫu : 01 sample / mẫu (02 pcs)
Date of receiving / Ngày nhận mẫu : 12-Nov-2024
Date of testing / Ngày thử nghiệm : From / từ 12-Nov-2024 to / đến 15-Nov-2024
Test method / Phương pháp thử : AWS D1.3/D1.3M:2018 Structural Welding Code – Sheet Steel
Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Signed for and on behalf of
SGS Vietnam LTD

Thay mặt công ty

SGS Vietnam Ltd

Chau Tran Khanh Dung
Laboratory Supervisor
Industrial Services

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Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm
Bend test / Thử uốn	Specimen 3.1 / Mẫu 3.1
1. Step 1: Visual inspection of acceptable welds after bending <i>Bước 1: Kiểm tra trực quan mối hàn sau khi uốn</i>	- No weld metal cracks shall be visually detected after bending / Không có vết nứt kim loại hàn phát hiện bằng mắt thường sau khi uốn Accept / Chấp nhận
2. Step 2: Completely separated by either bending the sheets or by hammering a wedge until either the weld or sheet metal fails <i>Bước 2: Tách hoàn toàn bằng cách uốn cong các tấm hoặc đóng búa vào cho đến khi mối hàn hoặc tấm kim loại bị hỏng</i>	- The sheet metal is failed / Tấm kim loại bị hỏng - The fracture surface shall show complete fusion at the root of the joint / Bề mặt gãy cho thấy sự ngấu hoàn toàn tới chân Accept / Chấp nhận
Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm
Bend test / Thử uốn	Specimen 3.2 / Mẫu 3.2
1. Step 1: Visual inspection of acceptable welds after bending <i>Bước 1: Kiểm tra trực quan mối hàn sau khi uốn</i>	- No weld metal cracks shall be visually detected after bending / Không có vết nứt kim loại hàn phát hiện bằng mắt thường sau khi uốn Accept / Chấp nhận
2. Step 2: Completely separated by either bending the sheets or by hammering a wedge until either the weld or sheet metal fails <i>Bước 2: Tách hoàn toàn bằng cách uốn cong các tấm hoặc đóng búa vào cho đến khi mối hàn hoặc tấm kim loại bị hỏng</i>	- The sheet metal is failed / Tấm kim loại bị hỏng - The fracture surface shall show complete fusion at the root of the joint / Bề mặt gãy cho thấy sự ngấu hoàn toàn tới chân Accept / Chấp nhận

Original sample photo(s) / Hình ảnh mẫu nhận:



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Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
Client's address / Địa chỉ : LÔ SỐ 38, KHU C, ĐƯỜNG D1, KHU CÔNG NGHIỆP AN HẠ, XÃ PHẠM VĂN HAI, HUYỆN BÌNH CHÁNH, THÀNH PHỐ HỒ CHÍ MINH, VIỆT NAM

The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : Mẫu hàn DD-GMAW-325.2
- Position H (Flare bevel-groove weld)
- Base metal JIS G3101 SS400
- Size: 100 x 100 x1.6 + 150 x 200 x2mm (2pcs)
- Filler metal: : Kuang Tai KM-70 (0.8mm)
- Welder ID: DDC-WD-0393

Project / Dự án : CẦU ÔNG LÔNG DI ĐỘNG SÂN BAY LONG THÀNH

Item / Hàng mục : Procedure Qualification Record (PQR) Test

Quantity / Số lượng mẫu : 01 sample / mẫu (02 pcs)

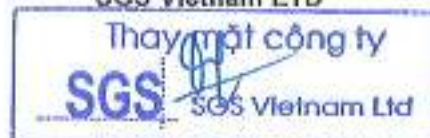
Date of receiving / Ngày nhận mẫu : 12-Nov-2024

Date of testing / Ngày thử nghiệm : From / từ 12-Nov-2024 to / đến 15-Nov-2024

Test method / Phương pháp thử : AWS D1.3/D1.3M:2018 Structural Welding Code – Sheet Steel

Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Signed for and on behalf of
SGS Vietnam LTD



Chau Tran Khanh Dung
Laboratory Supervisor
Industrial Services

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Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm
Bend test / Thử uốn	Specimen 6.1 / Mẫu 6.1
Completely separated by either bending the sheets or by hammering a wedge until either the weld or sheet metal fails <i>Tách hoàn toàn bằng cách uốn cong các tấm hoặc đóng búa vào cho đến khi mối hàn hoặc tấm kim loại bị hỏng</i>	- Failure occurs in the base metal, and no failure occurs in the weld metal / <i>Hỏng xảy ra ở vật liệu cơ bản, và không xảy ra hỏng ở vật liệu mối hàn</i> - Weld size / <i>Kích thước mối hàn</i> : $t_w = 5.0 \text{ mm}$
	Accept / Chấp nhận
Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm
Bend test / Thử uốn	Specimen 6.2 / Mẫu 6.2
Completely separated by either bending the sheets or by hammering a wedge until either the weld or sheet metal fails <i>Tách hoàn toàn bằng cách uốn cong các tấm hoặc đóng búa vào cho đến khi mối hàn hoặc tấm kim loại bị hỏng</i>	- Failure occurs in the base metal, and no failure occurs in the weld metal / <i>Hỏng xảy ra ở vật liệu cơ bản, và không xảy ra hỏng ở vật liệu mối hàn</i> - Weld size / <i>Kích thước mối hàn</i> : $t_w = 5.5 \text{ mm}$
	Accept / Chấp nhận

Original sample photo(s) / Hình ảnh mẫu nhận:



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Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
Client's address / Địa chỉ : LÔ SỐ 38, KHU C, ĐƯỜNG D1, KHU CÔNG NGHIỆP AN HẠ, XÃ PHẠM VĂN HAI, HUYỆN BÌNH CHÁNH, THÀNH PHỐ HỒ CHÍ MINH, VIỆT NAM

The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : Mẫu hàn DD-GMAW-325.3
- Position F (CJP – Both side, Square joint)
- Base metal JIS G3101 SS400
- Size: 400 x 200 x2mm (2pcs)
- Filler metal: : Kuang Tai KM-70 (0.8mm)
- Welder ID: DDC-WD-0393

Project / Dự án : CẦU ÔNG LÔNG DI ĐỘNG SÂN BAY LONG THÀNH

Item / Hàng mục : Procedure Qualification Record (PQR) Test

Quantity / Số lượng mẫu : 01 sample / mẫu (02 pcs)

Date of receiving / Ngày nhận mẫu : 12-Nov-2024

Date of testing / Ngày thử nghiệm : From / từ 12-Nov-2024 to / đến 15-Nov-2024

Test method / Phương pháp thử : AWS D1.3/D1.3M:2018 Structural Welding Code – Sheet Steel

Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Signed for and on behalf of
SGS Vietnam LTD

Thay mặt công ty

SGS Vietnam Ltd

Chau Tran Khanh Dung
Laboratory Supervisor
Industrial Services

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Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm	
Bend test / Thử uốn	Specimen 1 Mẫu 1	Specimen 2 Mẫu 2
• Thickness of specimen / Chiều dày mm	2.00	2.00
• Width of specimen / Chiều rộng mm	185	185
• Bending angle / Góc uốn °/ độ	180	180
• Test result / Kết quả	No Crack Không nứt	No Crack Không nứt
• Evaluation / Đánh giá	Accept / Chấp nhận	Accept / Chấp nhận

Original sample photo(s) / Hình ảnh mẫu nhận:



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Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
Client's address / Địa chỉ : LÔ SỐ 38, KHU C, ĐƯỜNG D1, KHU CÔNG NGHIỆP AN HẠ, XÃ PHẠM VĂN HAI, HUYỆN BÌNH CHÁNH, THÀNH PHỐ HỒ CHÍ MINH, VIỆT NAM

The following sample(s) was/were submitted and identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : Mẫu hàn DD-GMAW-325.4
- Position V (up weld) (Flare bevel-groove weld)
- Base metal JIS G3101 SS400
- Size: 100 x 100 x1.6 + 150 x 200 x2mm (2pcs)
- Filler metal: : Kuang Tai KM-70 (0.8mm)
- Welder ID: DDC-WD-0393
Project / Dự án : CẦU ỐNG LÒNG DI ĐỘNG SÂN BAY LONG THÀNH
Item / Hàng mục : Procedure Qualification Record (PQR) Test
Quantity / Số lượng mẫu : 01 sample / mẫu (02 pcs)
Date of receiving / Ngày nhận mẫu : 12-Nov-2024
Date of testing / Ngày thử nghiệm : From / từ 12-Nov-2024 to / đến 15-Nov-2024
Test method / Phương pháp thử : AWS D1.3/D1.3M:2018 Structural Welding Code – Sheet Steel
Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Signed for and on behalf of
SGS Vietnam LTD

Thay mặt công ty
 SGS Vietnam Ltd

Chau Tran Khanh Dung
Laboratory Supervisor
Industrial Services

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Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm
Bend test / Thử uốn	Specimen 4.1 / Mẫu 4.1
Completely separated by either bending the sheets or by hammering a wedge until either the weld or sheet metal fails <i>Tách hoàn toàn bằng cách uốn cong các tấm hoặc đóng búa vào cho đến khi mối hàn hoặc tấm kim loại bị hỏng</i>	- Failure occurs in the base metal, and no failure occurs in the weld metal / <i>Hỏng xảy ra ở vật liệu cơ bản, và không xảy ra hỏng ở vật liệu mối hàn</i> - Weld size / <i>Kích thước mối hàn</i> : $t_w = 5.0 \text{ mm}$
	Accept / Chấp nhận
Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm
Bend test / Thử uốn	Specimen 4.2 / Mẫu 4.2
Completely separated by either bending the sheets or by hammering a wedge until either the weld or sheet metal fails <i>Tách hoàn toàn bằng cách uốn cong các tấm hoặc đóng búa vào cho đến khi mối hàn hoặc tấm kim loại bị hỏng</i>	- Failure occurs in the base metal, and no failure occurs in the weld metal / <i>Hỏng xảy ra ở vật liệu cơ bản, và không xảy ra hỏng ở vật liệu mối hàn</i> - Weld size / <i>Kích thước mối hàn</i> : $t_w = 5.0 \text{ mm}$
	Accept / Chấp nhận

Original sample photo(s) / Hình ảnh mẫu nhận:



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Client's name / Tên khách hàng : CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
Client's address / Địa chỉ : LÔ SỐ 38, KHU C, ĐƯỜNG D1, KHU CÔNG NGHIỆP AN HẠ, XÃ PHẠM VĂN HAI, HUYỆN BÌNH CHÁNH, THÀNH PHỐ HỒ CHÍ MINH, VIỆT NAM

The following sample(s) was/were submitted and Identified by the client as:

Các thông tin nhận dạng mẫu do khách hàng cung cấp:

Name of sample / Tên mẫu : Mẫu hàn DD-GMAW-325.5
- Position V (up weld) (T Joint Fillet weld)
- Base metal JIS G3101 SS400
- Size: T 100 x 100 x1.6 + 150 x 200 x2.0mm (2pcs)
- Filler metal: : Kuang Tai KM-70 (0.8mm)
- Welder ID: DDC-WD-0393

Project / Dự án : CẦU ÔNG LÔNG DI ĐỘNG SÂN BAY LONG THÀNH
Item / Hạng mục : Procedure Qualification Record (PQR) Test
Quantity / Số lượng mẫu : 01 sample / mẫu (02 pcs)
Date of receiving / Ngày nhận mẫu : 12-Nov-2024
Date of testing / Ngày thử nghiệm : From / từ 12-Nov-2024 to / đến 15-Nov-2024
Test method / Phương pháp thử : AWS D1.3/D1.3M:2018 Structural Welding Code – Sheet Steel
Test result / Kết quả thử nghiệm : See next page(s) / Xem trang sau

Signed for and on behalf of
SGS Vietnam LTD



Chau Tran Khanh Dung
Laboratory Supervisor
Industrial Services

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Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm
Bend test / Thử uốn	Specimen 7.1 / Mẫu 7.1
1. Step 1: Visual inspection of acceptable welds after bending <i>Bước 1: Kiểm tra trực quan mối hàn sau khi uốn</i>	- No weld metal cracks shall be visually detected after bending / Không có vết nứt kim loại hàn phát hiện bằng mắt thường sau khi uốn Accept / Chấp nhận
2. Step 2: Completely separated by either bending the sheets or by hammering a wedge until either the weld or sheet metal falls <i>Bước 2: Tách hoàn toàn bằng cách uốn cong các tấm hoặc đóng búa vào cho đến khi mối hàn hoặc tấm kim loại bị hỏng</i>	- The sheet metal is failed / Tấm kim loại bị hỏng - The fracture surface shall show complete fusion at the root of the joint / Bề mặt gãy cho thấy sự ngấu hoàn toàn tới chân Accept / Chấp nhận
Characteristic / Tên chỉ tiêu	Test result / Kết quả thử nghiệm
Bend test / Thử uốn	Specimen 7.2 / Mẫu 7.2
1. Step 1: Visual inspection of acceptable welds after bending <i>Bước 1: Kiểm tra trực quan mối hàn sau khi uốn</i>	- No weld metal cracks shall be visually detected after bending / Không có vết nứt kim loại hàn phát hiện bằng mắt thường sau khi uốn Accept / Chấp nhận
2. Step 2: Completely separated by either bending the sheets or by hammering a wedge until either the weld or sheet metal falls <i>Bước 2: Tách hoàn toàn bằng cách uốn cong các tấm hoặc đóng búa vào cho đến khi mối hàn hoặc tấm kim loại bị hỏng</i>	- The sheet metal is failed / Tấm kim loại bị hỏng - The fracture surface shall show complete fusion at the root of the joint / Bề mặt gãy cho thấy sự ngấu hoàn toàn tới chân Accept / Chấp nhận

Original sample photo(s) / Hình ảnh mẫu nhận:



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MILL CERTIFICATE

台塑河靜鋼鐵興業責任有限公司

FORMOSA HA TINH STEEL CORPORATION

ADDRESS: Vung Ang Economic Zone, Ky Long Ward, Ky Anh Town,

Ho Tinh Province, Viet Nam

TEL: +84-239-3722-121 Ext: 2056 Fax: +84-239-3722-112

Email: fhs.exports@fhs.com.vn

PRODUCT	COMMODITY	HOT ROLLED COIL	
SPECIFICATION and GRADE	ITS CONTRACT NO.	09041201	PURCHASER CREDIT NO.
INSPECTION	CERTIFICATE NO.	Q2304/0025	INVOICE NO.
EXPORTER NAME	MOBILE DATE	2009/12	CUSTOMER NO.

COIL NO	PRODUCT INFORMATION				HEAT NO.	DIA	TENSILE TEST				REMARK	CHEMICAL ANALYSIS RESULT %																FORMA																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
	THICK	WIDTH	LENGTH	ACTUAL WEIGHT			Y1	Y2	Y3	EL		C	Si	Mn	P	S	Al	N																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															

COMMODITY as LABEL: HOT ROLLED COIL
 SPECIFICATION as LABEL: JIS G 3003-2004
 GRADE as LABEL: SS400
 THIS MILL CERTIFICATE IS ISSUED IN ACCORDANCE WITH JIS G 3004 TYPE 1

NOTES

Y1: YIELD POINT TYPE

T1: TRACE TEST ANALYSIS RESULT COMPLIES WITH SPECIFICATION REQUIREMENT

Made, initial, passed and original in Vietnam

WE HEREBY CERTIFY THAT MATERIAL DESCRIBED HEREIN HAS BEEN MANUFACTURED AND TESTED WITH SATISFACTORY RESULTS IN ACCORDANCE WITH THE REQUIREMENT OF THE ABOVE MATERIAL SPECIFICATION.

SURVEYOR TO

FORMOSA STEEL CORPORATION, TECHNOLOGY DIVISION





INSPECTION CERTIFICATE

EN 10204 3.1

KUANG TAI METAL (VIETNAM) CO., LTD.

NHON TRACH IND PARK 3, NHON TRACH, DONG NAI, VIET NAM
TEL:+84 251-3560097 FAX:+84-251-3560096 E-mail:ktvnqa@kuangtai.com

Customer						Commodity	KM-70		Mfg. Date	2024/10		Lot No.	V24A16012		
Specification & Classification	AWS A5.18 ER70S-6 ASME SECT II PART C- 2023 SFA-5.18 ER70S-6					Dimension	0.8 mm * 15 Kg		Issue Date	24-Oct-24		Certificate No.	24102401		
■ Chemical Composition for Solid Electrode and Rod (%) / □ Weld Metal Chemical Composition (%)															
Element	C	Mn	Si	P	S	Ni	Cr	Mo	V	Cu					
Requirement	0.06~ 0.15	1.40~ 1.85	0.80~ 1.15	MAX 0.025	MAX 0.035	MAX 0.15	MAX 0.15	MAX 0.15	MAX 0.03	MAX 0.50					
Test Result	0.071	1.46	0.86	0.011	0.012	0.013	0.02	0.0021	0.0031	0.015					
Category	Tension Test				Impact Test						Welding Condition				
	Yield * Strength N/mm ²	Yield ** Strength N/mm ²	Tensile Strength N/mm ²	Elongation %	Test Temp. °C	Charpy V-Notch Absorbed Energy J (AVG)					Type of Current	Welding Current A	Arc Voltage V	Shielding Gas or Flux	Postweld Heat Treatment
Requirement	MIN 400	NS	MIN 490	MIN 22	-30	MIN 27					DCEP	NS	NS	C1	NS
Test Result	446	NS	540	27	-30	92	87	101	70	104 (93)	DCEP	NS	NS	C1	NS
Bend Test			Hardness Test	Fillet Weld Test	Radiographic Test	Moisture %	Diffusible Hydrogen mL/100gDM (AVG)	Heat No.	Mn+Ni+Cr+Mo+V						
Face	Root	Side													
NR	NR	NR	NR	NR	ACCEPTABLE	NR	NR	1083082	NS						
We hereby certify that this report is correct and that all test results are in compliance with the specification described herein. The material described above is free of radiation and free of mercury.					Remarks *0.2% offset ** Lower Yield point NS=not specified NR=not required C1=100%CO ₂ M21= 20-25% CO ₂ + Ar (Balance)					Certificate:					
ISO 9001:2015 CERTIFICATE NO. NO.:18/2292					Manager Of Quality Assurance 										



CÔNG TY TNHH VƯƠNG PHÁT

Mã số thuế: 0302790587

Địa chỉ: 190 - 192 Trương Định Hối, Phường 16, Quận 8, Thành phố Hồ Chí Minh, Việt Nam

Điện thoại: (028) 3 8367873

Số tài khoản: 6113709 - Ngân hàng TMCP Á Châu

QUALITY CONFIRMATION LETTER GIẤY CHỨNG NHẬN CHẤT LƯỢNG

DATE (NGÀY)	07/11/2024
CERTIFICATION NO. (GIẤY CHỨNG NHẬN SỐ)	VP-DD-MIX-07112409
CUSTOMER (KHÁCH HÀNG)	Công Ty Cổ Phần Cơ Khí Xây Dựng Thương Mại Đại Đồng
PRODUCT'S NAME (TÊN SẢN PHẨM)	KHÍ ARGON TRỌN
CYLINDER SIZE (DUNG TÍCH CHAI)	40 LÍT
FILLING PRESSURE (ÁP SUẤT NẠP)	150 BAR
GAS CONTENT (DUNG LƯỢNG KHÍ)	9 KG
PRODUCT GRADE (LOẠI SẢN PHẨM)	CÔNG NGHIỆP

VƯƠNG PHÁT committed that the product delivered in conformity with the specification below
VƯƠNG PHÁT cam kết rằng sản phẩm được giao tuân theo đặc tính kỹ thuật dưới đây:

PURITY (ĐỘ TINH KHIẾT)	80% ARGON + 20% CO2
PURCHASE ORDER NO (MÃ SỐ ĐƠN ĐẶT HÀNG)	DDC-PO24/05-2285 ARGON TRỌN
QUANTITY OF CYLINDERS (SỐ LƯỢNG CHAI CHỨA KHÍ)	6

ISSUED BY

NGƯỜI CẤP

CÔNG TY TNHH VƯƠNG PHÁT

Đ/C: 190-192 Trương Định Hối, P.16, Q.8, TP.HCM

MST: 0302790587

SĐT: 094.6666.224 - 078.3556.866

Đào Thị Ngọc Trâm

	WELDING PROCEDURE SPECIFICATION (WPS) (AWS D1.3/D1.3M:2018 Structural Welding Code - Sheet Steel)	
--	---	--

Company Name	DaiDung Metallic Manufacture Construction & Trade Corporation	Rev. No.	0	Date:	Nov 18, 2024	Page: 1 of 1
WPS No.	WPS-DD-GMAW-325.5					
Supporting PQR(s)	PQR-DD-GMAW-325.6					

BASE METALS (1.4)	Specification	Type or Grade	AWS Group No.	SHEET METAL THICKNESS	As Welded (mm)	
Sheet Material	ASTM A36, A53, A572,... JIS G3101, CS, Low alloy steel.	Gr A, B 'SS400	Any Group I, II	Square Groove Weld	1-4	
Supporting Structure	ASTM A36, A53, A572,... JIS G3101, CS, Low alloy steel.	Gr A, B 'SS400	Any Group I, II			
Backing Material	-			DIAMETER	Any	
Coating	Without					

JOINT DETAILS	JOINT SKETCH
Joint Type: Square-groove Butt joint, Corner joint Weld Type: Weld 1 side without backing Root Opening (R): 0 - 1 mm Weld Size: E ≥ 0.5 t Per Figure 4.1; 5.1B; 5.3A	
POSTWELD HEAT TREATMENT Temperature: - Time at Temperature: - Other: -	

PROCEDURE	All	All	
Weld Layer(s)	1 or more	1 or more	
Weld Pass(es)			
Process		GMAW	
Type (Manual, Mechanized, etc.)		Semi-Automatic	
Position	F	H	
Vertical Progression	-		
Filler Metal (AWS Spec.)		AWS A5.18	
AWS Classification		ER70S-6	
Diameter (mm)		Ø 0.8	
Manufacturer/Trade Name		Kuang Tai KM-70/ Hyundai Welding or Other same classification	
Control of Filler Metal		Drying condition follow Welding Manufacture recommendation	
Shielding Gas (Composition)		CO2 (20-25%) + Ar (balance)	
Flow Rate (LPM)	18 - 29	18 - 29	
Nozzle Size	No. 6	No. 6	
Preheat Temperature (Min.)	See Note 3	See Note 3	
Interpass Temperature (Max.)	250°C	250°C	
Electrical Characteristics			
Current Type & Polarity	DCEP	DCEP	
Transfer Mode	Globular	Globular	
Power Source Type (cc, cv, etc.)	CV	CV	
Amps (A)	55 - 85	55 - 85	
Volts (V)	17 - 21	17 - 21	
Wire Feed Speed (Amps)			
Travel Speed (mm/min.)	400-550	400-550	
Maximum Heat Input (kJ/mm)	0.4	0.4	
Technique			
Stringer or Weave	Stringer or Weave	Stringer or Weave	
Multi or Single Pass (per side)	Multi or Single Pass	Multi or Single Pass	
Oscillation (Mech./Auto.)	-	-	
Number of Electrodes	1	1	
Contact Tube to Work Dist (mm)	10 - 25	10 - 25	
Peening	None	None	
Interpass Cleaning	Grinding and Wire Brush	Grinding and Wire Brush	
Other	This procedure can be used for repair		

NOTES* Note 1: sheet steel having a minimum specified yield point equal to or less than 80 ksi [550 MPa]
 Note 2: The supporting structural member is listed in D1.1 /D1.1M: 2015 , Table 3.1 , as a Group I or II base metal
 Note 3: minimum preheat supporting structural member follow table 5.8 AWS D1.1/D1.1M.

Thickness: ≤20	Thickness: 20.1-38mm	Thickness 38.1-65mm	Thickness >65mm
Preheat min 20°C	Preheat min 20°C	Preheat min 65°C/110°C	Preheat min 110/150°C

Prepared by DDC's QA Welding Coordinator	Reviewed and Approved by	
Name: Nguyen Quynh Sa	Trubng Hoài Phong Vu (QA Manager)	
Signature:		
Date: Nov 18, 2024	Nov 18, 2024	





PROCEDURE QUALIFICATION RECORD (PQR) (AWS D1.3/D1.3M: 2018 STRUCTURAL WELDING CODE - SHEET STEEL)



Company Name: **Daidung Metallic Manufacture Construction & Trade Corporation**

PQR No.: **PQR-DD-GMAW-325.6**

Rev. No.: **00**

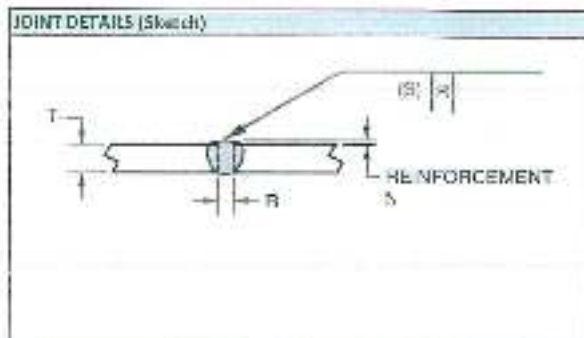
Date: **Nov 18, 2024**

Page: **1 of 2**

BASE METALS	Specification	Type or Grade	AWS D1.3 Group No	Thickness (t)	Thickness (t)	Thickness (t)	Diameter
Base Material	JIS 3101	SS400	-	F (3G)	2.0 mm		-
Welded To	JIS 3101	SS400	-	2.0 mm			-
Other:	Shop primer in Weld Joint: No						

JOINT DETAILS	
Joint Type	SQUARE JOINT WELD ONF SIDE
Toe Weld Include	N/A
Groove Angle	N/A
Root Opening	Max 0.5 mm
Root Face	-
Backgouging	N/A
Method	Grinding

POSTWELD HEAT TREATMENT	
Soaking Temperature	-
Soaking Time	-
Other:	-



PROCEDURE	DD-GMAW-325.6 (1G)	DD-GMAW-325.6 (2G)
Weld Layer(s)	Root/Cap	Root/Cap
Weld Pass(es) per layer	1	1
Process	GMAW	GMAW
Type (Manual, Mechanized, etc.)	Semi-Automatic	Semi Automatic
Position	F (3G)	H (2G)
Vertical Progression	-	-
Filler Metal (Spec.)	AWS A5.18	AWS A5.18
AWS Classification	ER70S-6	ER70S-6
Diameter (mm)	Ø 0.8	Ø 0.8
Manufacturer/Trade Name	Kuang Tai KM-70	Kuang Tai KM-70
Control of Filler Metal	Follow Welding Manufacture recommendation	Follow Welding Manufacture recommendation
Shielding Gas (Composition)	CO ₂ (20%) + Ar (80%)	CO ₂ (20%) + Ar (80%)
Flow Rate (l/min)	20-25	20-25
Nozzle Size	No. 6	No. 6
Preheat Temperature	30°C (ambient)	30°C (ambient)
Interpass Temperature (Max)	-	-
Electrical Characteristics		
Current Type & Polarity	DCEP	DCEP
Transfer Mode	Globular	Globular
Power Source Type (cc, cv, etc.)	CV	CV
Amps (A)	60-70	70-80
Volts (V)	18-19	19-20
Wire Feed Speed	(Amps)	(Amps)
Travel Speed (mm/min)	522	462
Heat Input (kJ/mm)	0.2	0.2
Technique		
Stringer or Weave	Weave	Weave
Multi or Single Pass (per side)	Single Pass	Single Pass
Number of Electrodes	1	1
Contact Tube to Work (Distance)	12-20	12-20
Peening	None	None
Interpass Cleaning	-	-
Inspection	Directly	Directly
Other:	-	-

Prepared by DDC's QA Welding		Reviewed and Approved by	
Name	Nguyen Quynh Sa	Name	Truong Hoa Phung on
Signature		Signature	
Date	Nov 18, 2024	Date	Nov 18, 2024





PROCEDURE QUALIFICATION RECORD (PQR)
(AWS D1.3/D1.3M: 2018 STRUCTURAL WELDING CODE - SHEET STEEL)



PQR No.: PQR-DD-GMAW-325.6

Date: Nov 18, 2024

Rev. No.: 00

Page: 2 of 2

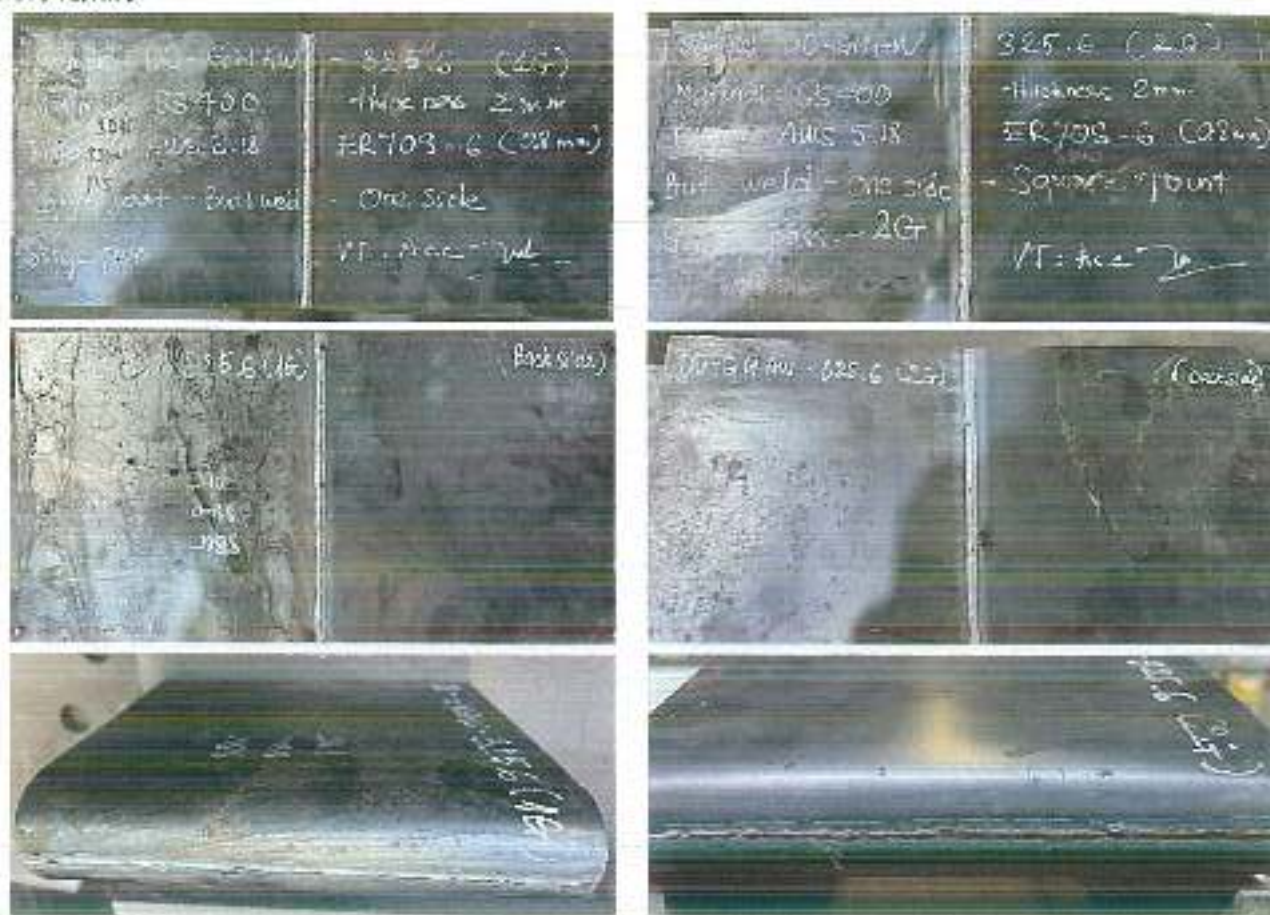
TEST RESULTS

TESTS (Square Groove Joint, Butt Weld- One side without badding)

Type of Tests	Clause/Figure(s) Reference	Acceptance Criteria	Result	Remarks
✓ Visual Inspection	Clause 6.7.1	Clause 6.7.1.2(2)	Accepted	
✓ Bend	Clause 6.7.1	Clause 6.7.1.2(4)	Accepted	

Welder's Name	ID Number	DDC's Welder ID No.	Tests Conducted by:	
NGUYEN THANH TUNG	-	DDC-WD-0862	Manufacture	DDC
-	-	-	Destructive Test No.	Accepted
-	-	-	Testing specimen No.	DD-GMAW-325.6

PHOTO TESTING



- (a) No cracks are visually detected after bending.
(b) Square groove welds have penetration as requirement of the sheet base metal.

This undersigned certified that the statements in this record are correct and that the test assemblies were prepared, welded, and tested in conformance with request of Clause 6 of AWS D1.3/D1.3M (2018) Structural Welding Code - Sheet Steel

	Prepared by DDC's QA Welding	Reviewed and Approved by
Name	Nguyen Quynh Sa	Truong Huai Phong Vu
Signature	 	 
Date	Nov 18, 2024	Nov 18, 2024

**CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG**

DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION

• 38, Zone C, D1 St, An Ha Industrial Park, Pham Van Hai, Binh Chanh, HCMC

• sales@daidung.vn • (84-28) 3868 1689 • www.daidung.com.vn

WELDING PARAMETERS RECORD

(Doc No.: WPR-GMAW-325.3)

Company name: DAIDUNG CORPORATION
PQR No.: PQR-DD-GMAW-325.6
Test Coupon: DD-GMAW-325.6
Test position: F (1G); H (2G)

Base metal:

- Specification: JIS 3101 SS400
- Dimension: 400 x 200 x 2.0 mm
- Heat/Product No.: 2542518000
- Certificate No.: QZA024700689

Filler metal:

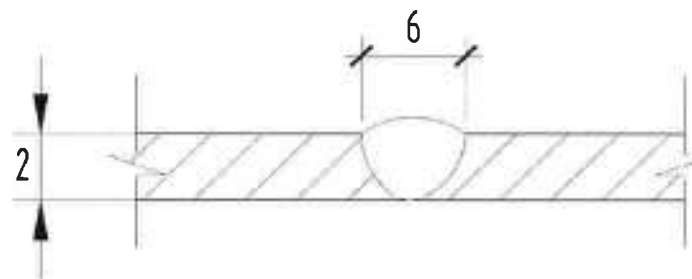
- Specification: AWS A5.18
- Classification: ER70S-6
- Trade name: Kuang Tai KM-70 (Ø0.8 mm)
- Batch/Lot No.: V24A16012
- Certificate No.: 24102401
- Control of Filler Metal: Follow Kuang Tai's recommendation:

Joint Sketch: below

Shop primer: No

Date: 18-Nov-2024

Tack weld included: No



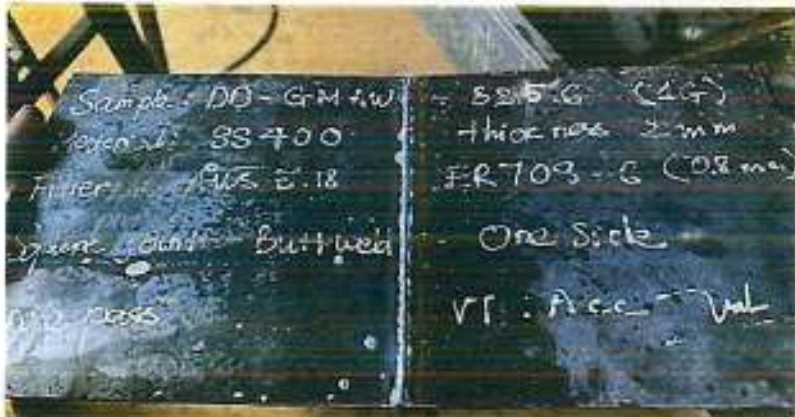
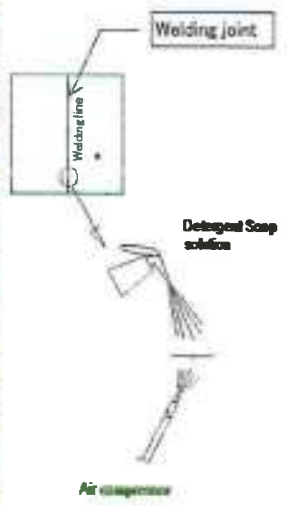
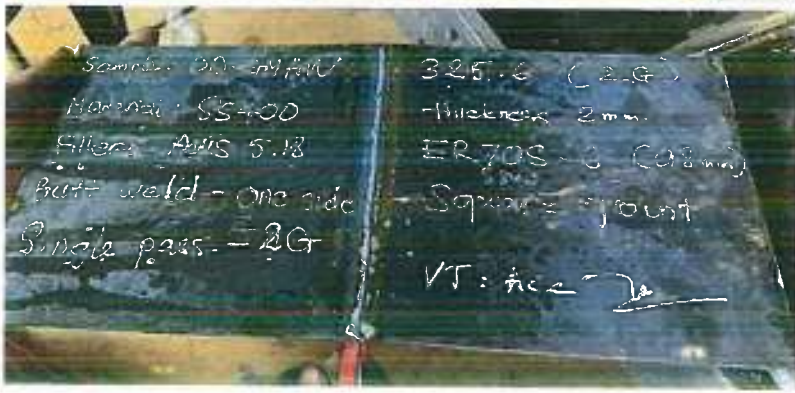
Gas: 80% Ar + 20% CO₂
Welder Name: NGUYEN THANH TUNG
I.D.: DDC-WD-0863

Root face: -
Root gap: 0-0.5 mm
Groove angle: N/A

Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width (mm)	Depth	
1	GMAW	AWS A5.18/ ER70S-6	0.8	DCEP	60-70	18-19		23	200	522	30 (Ambient)	20-23	0.12-0.15	5-6		1G
1	GMAW	AWS A5.18/ ER70S-6	0.8	DCEP	70-80	19-20		26	200	462	30 (Ambient)	20-23	0.17-0.21	5-6		2G
Conducted by QA Welding																
Name		NGUYEN QUYNH SA									Reviewed and Approved by					
Signature																
Date		18-Nov-2024									18-Nov-2024					



Established: 22/May/2015
Revised:

		ShinMaywa Industries, Ltd. Quality Assurance Dept. Parking Systems Div.	
AIR LEAK TEST INSPECTION RECORD		Project Name :	LONG THANH INTERNATIONAL AIRPORT (PASSENGER BOARDING BRIDGE)
DD-GMAW-325.6 (1G)&(2G)		Work Order :	
		PBB Model :	Common
Drawing reference No :	PQR-DD-GMAW-325.6	Gate No :	-
Serial No :		Report No :	LEAK-DD-GMAW-325.6
Testing Method	Apply soap water to welding portions of panel joints and blow air from behind panel.		
Test Percentage	100%		
Air Compressor	Must be above 7 bar		
Acceptance Criteria	No air bubbles observed	Judgment	☒ PASS ☐ REJECT
 			
			
Comment:			
Inspected by QC/Mng	Witness by TPI	Witness by TPI	
			
DATE: 18.Nov.2024	DATE: 18.Nov.2024	DATE: 18.Nov.2024	



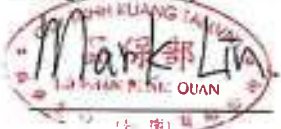
INSPECTION CERTIFICATE

EN 10204 3.1

KUANG TAI METAL (VIETNAM) CO., LTD.

NHON TRACH IND PARK 3, NHON TRACH, DONG NAI, VIET NAM

TEL:+84 251-3560097 FAX:+84-251-3560096 E-mail:ktvnqa@kuangtai.com

Customer						Commodity	KM-70			Mfg. Date	2024/10		Lot No.	V24A16012		
Specification & Classification	AWS A5.18 ER70S-6 ASME SECT II PART C- 2023 SFA-5.18 ER70S-6					Dimension	0.8 mm * 15 Kg			Issue Date	24-Oct-24		Certificate No.	24102401		
■ Chemical Composition for Solid Electrode and Rod (%) / □ Weld Metal Chemical Composition (%)																
Element	C	Mn	Si	P	S	Ni	Cr	Mo	V	Cu						
Requirement	0.06~ 0.15	1.40~ 1.85	0.80~ 1.15	MAX 0.025	MAX 0.035	MAX 0.15	MAX 0.15	MAX 0.15	MAX 0.03	MAX 0.50						
Test Result	0.071	1.46	0.86	0.011	0.012	0.013	0.02	0.0021	0.0031	0.015						
Category	Tension Test				Impact Test						Welding Condition					
	Yield * Strength N/mm ²	Yield ** Strength N/mm ²	Tensile Strength N/mm ²	Elongation %	Test Temp. °C	Charpy V-Notch Absorbed Energy J (AVG.)					Type of Current	Welding Current A	Arc Voltage V	Shielding Gas or Flux	Postweld Heat Treatment	
Requirement	MIN 400	NS	MIN 490	MIN 22	-30	MIN 27					DCEP	NS	NS	C1	NS	
Test Result	446	NS	540	27	-30	92	87	101	70	104	(93)	DCEP	NS	NS	C1 NS	
Bend Test			Hardness Test	Fillet Weld Test	Radiographic Test	Moisture %	Diffusible Hydrogen mL/100gDM (AVG.)		Heat No.		Mn+Ni+Cr+Mo+V					
Face	Root	Side														
NR	NR	NR	NR	NR	ACCEPTABLE	NR	NR		1083082		NS					
We hereby certify that this report is correct and that all test results are in compliance with the specification described herein. The material described above is free of radiation and free of mercury.						Remarks *0.2% offset ** Lower Yield point NS=not specified NR=not required C1=100%CO ₂ M21= 20~25% CO ₂ + Ar (Balance)						Certificate:				
ISO 9001:2015 CERTIFICATE NO. NO.:18/2292 Manager Of Quality Assurance  QUAN																



CÔNG TY TNHH VƯỢNG PHÁT

Mã số thuế: 0302790587

Địa chỉ: 190 - 192 Trương Đình Hội, Phường 16, Quận 8, Thành phố Hồ Chí Minh, Việt Nam

Điện thoại: (028) 3 876 7873

Số tài khoản: 6113709 - Ngân hàng TMCP A Châu

QUALITY CONFIRMATION LETTER GIẤY CHỨNG NHẬN CHẤT LƯỢNG

DATE (NGÀY)	07-11-2024
CERTIFICATION NO. (GIẤY CHỨNG NHẬN SỐ)	VP-DD-MIX-07112409
CUSTOMER (KHÁCH HÀNG)	Công Ty Cổ Phần Cơ Khí Xây Dựng Thương Mại Đại Đồng
PRODUCT'S NAME (TÊN SẢN PHẨM)	KHÍ ARGON TRỌN
CYLINDER SIZE (DUNG TÍCH CHAI)	40 LÍT
FILLING PRESSURE (ÁP SUẤT NẠP)	150 BAR
GAS CONTENT (DUNG LƯỢNG KHÍ)	9 KG
PRODUCT GRADE (LOẠI SẢN PHẨM)	CÔNG NGHIỆP

VUÔNG PHÁT committed that the product delivered in conformity with the specification below
VUÔNG PHÁT cam kết rằng sản phẩm được giao tuân theo đặc tính kỹ thuật dưới đây:

PURITY (ĐỘ TINH KHIẾT)	80% ARGON + 20% CO2
PURCHASE ORDER NO (MÃ SỐ ĐƠN ĐẶT HÀNG)	DDC PO24/05-2285 ARGONTRON
QUANTITY OF CYLINDERS (SỐ LƯỢNG CHAI CHỨA KHÍ)	6

ISSUED BY

NGƯỜI CẤP

CÔNG TY TNHH VƯỢNG PHÁT

Đ/C: 190-192 Trương Đình Hội, P.16, Q.8, TP.HCM

MST: 0302790587

SĐT: 094 666 224 - 078 3556 866

Đào Thị Ngọc Trâm



Comply:
ISO 9001:2015
AWS D1.1: 2025

WELDING PROCEDURE

Doc No.: WPS-DD-FCAW-283

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	May 03, 2024	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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	WELDING PROCEDURE SPECIFICATION (WPS) (AWS D1.1/D1.1M:2020 Structural Welding Code-Steel)	
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Company Name WPS No. Supporting PQR(s) CVN Report	DaiDung Metallic Manufacture Construction & Trade Corporation WPS-DD-FCAW-283 PQR-DD-FCAW-283.1 & 2 No	Rev. No. 01 Date: May 3, 2025	Page: 1 of 11
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BASE METALS	Specification	Type or Grade	AWS Group No.	BASE METAL THICKNESS	As-Welded (mm)	With PWHT
Base Material	JIS G3444 JIS G3101	STK 400 SS400	-	CJP Groove Welds (Square Groove)	3 - 10	-
Welded To	JIS G3444 JIS G3101	STK 400 SS400	-	CJP Groove Welds (Other)	3 - 20	-
Backing Material	Steel backing or Backing Welds refer Section 7.9			PJP Groove Welds	3 - Unlimited	-
Other	- Qualify any fillet weld size, qualify any box size - Dihedral angle ≥ 30° - Except T,K,Y - CJP			Fillet Welds	Any thickness	-
				DIAMETER OD (Outside diameter)	- 26.7 ≤O.D ≤ 114.3 for CJP - Any O.D for fillet, PJP	

JOINT DETAILS	
Groove Type	See attached sketch for more details
Groove Angle	See attached sketch for more details
Root Opening	See attached sketch for more details
Root Face	See attached sketch for more details
Backgouging	Yes
Method	Grinding or Air carbon arc

POSTWELD HEAT TREATMENT	
Temperature	-
Time at Temperature	-
Other	-

JOINT DETAILS (Sketch)
See Joint Details in page 2 to 11 of 11 (PJP: Max Groove Depth 5.5 mm,TKY: unlimited)

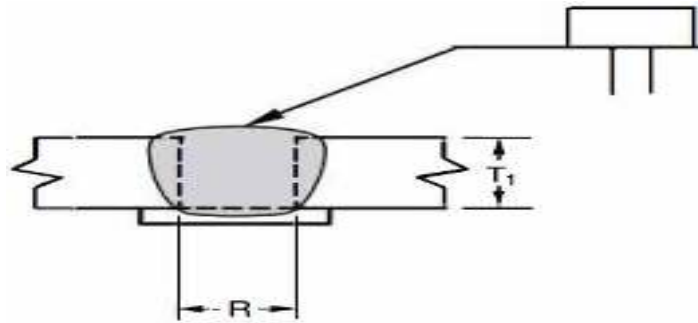
PROCEDURE		
Weld Layer(s)	All	
Weld Pass(es)	All	
Process	FCAW	
Type (Manual, Mechanized, etc.)	Semi-Automatic/Mechanized	
Position	All	
Vertical Progression	Uphill	
Filler Metal (AWS Spec.)	A5.20	
AWS Classification	E71T-1C	
Diameter (mm)	Max. 1.2	
Manufacturer/Trade Name	Not strictly	
Shielding Gas (Composition)	99.8% CO ₂	
Flow Rate (LPM)	16 - 25	
Nozzle Size (mm)	15 or 19	
Preheat Temperature (Min.)	See NOTES* below	
Interpass Temperature (Max.)	250 °C	
Electrical Characteristics		
Current Type & Polarity	DCEP	
Transfer Mode	-	
Power Source Type (cc, cv, etc.)	CV	
Amps (A)	110 - 185	
Volts (V)	18 - 22	
Wire Feed Speed (Amps)		
Travel Speed (mm/min.)	137 - 325	
Maximum Heat Input (kJ/mm)	-	
Technique		
Stringer or Weave	Stringer or Weave	
Multi or Single Pass (per side)	Multi or Single Pass	
Oscillation (Mech./Auto.)	-	
Number of Electrodes	1	
Contact Tube to Work Dist. (mm)	15 - 25	
Peening	None	
Interpass Cleaning	Grinding and Wire Brush	
Other	This procedure can be used for repair	

NOTES* For Thickness 3 mm thru 38 mm incl.: Min. 20°C
 Over 38 mm thru 65 mm incl.: Min. 65°C
 Over 65 mm: Min. 110°C

	Prepared by QA Welding	Reviewed and Approved by	
Name	TRAN MINH PHUNG	LA VINH NHON (D.QA Manager)	
Signature			
Date	May 3, 2025	May 3, 2025	

Square-groove weld (1)

Butt joint (B)



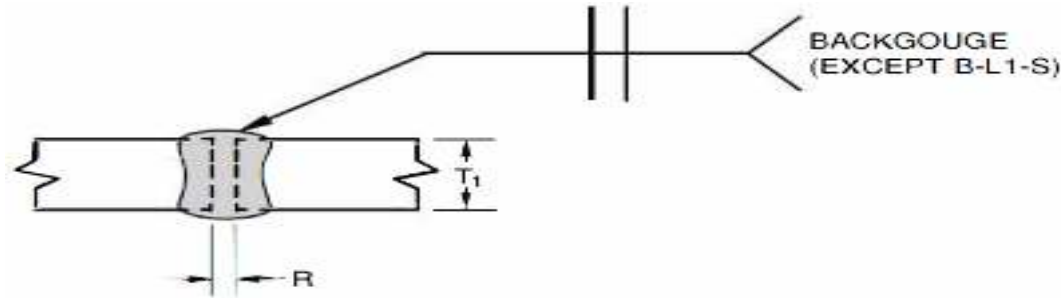
Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-L1a-GF	≤ 10		R=T1	T1-((T1)+2)							F, H, V	j

Note: j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

CJP

Square-groove weld (1)

Butt joint (B)

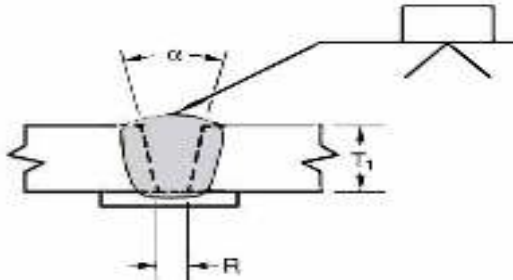


Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-L1b-GF	≤ 10		0-5	≤ 5							F, H, V	d, j

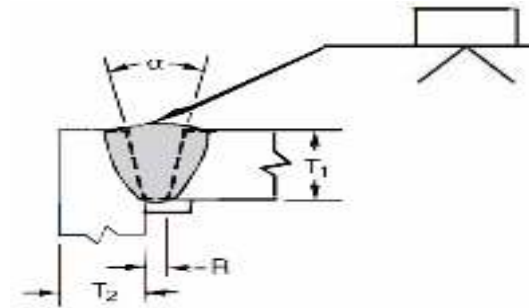
Note: d: Backgouge root to sound metal before welding second side.

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

Single V- groove weld (2)
Butt joint (B)
B-U2a-GF



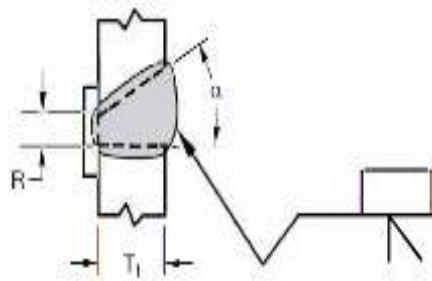
Single V- groove weld (2)
Corner joint (C)
C-U2a-GF



Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-U2a-GF	U		5	≥ 5	0	≤ 2	30	≥ 30			F, V, OH	j
	C-U2a-GF	U	U	5	≥ 5	0	≤ 2	30	≥ 30			F, V, OH	

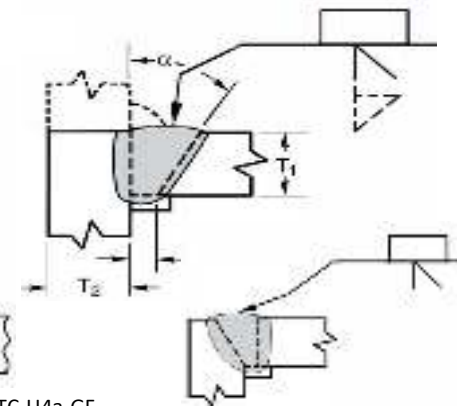
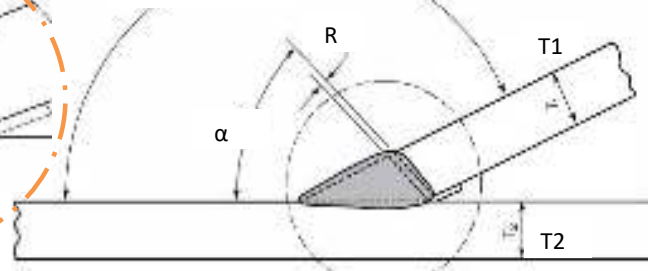
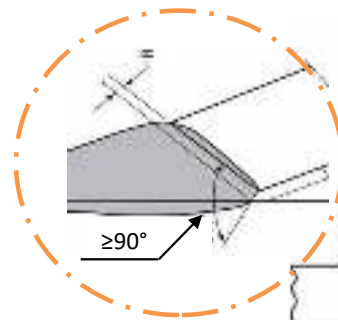
Note: j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

Single bevel- groove weld (4)
Butt joint (B)



Single bevel- groove weld (4)
T-joint (T) Corner joint (C)

$90^\circ \leq \psi \leq 170^\circ$



B-U4a-GF

Note o below

TC-U4a-GF

Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-U4a-GF	U		5	≥ 5	0	≤ 2	30	≥ 30			ALL	c, j
	TC-U4a-GF	U	U	5	≥ 5	0	≤ 2	30	≥ 30			ALL	g, k, o

Note: c: cyclic load application places restrictions on the use of this detail for butt joints in the flat position (See 4.18.2)

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

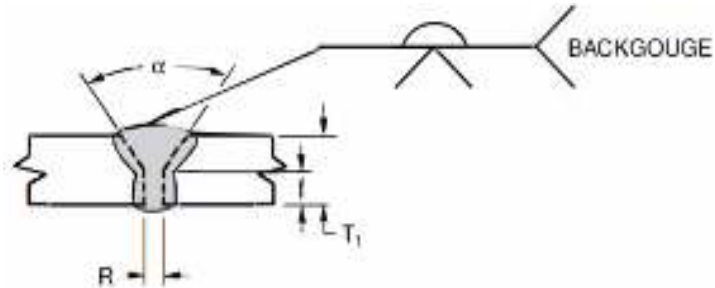
g: If fillet welds are used in statically loaded structures to reinforce groove welds in corner and T-joints, these shall be equal to $T1/4$, but need not exceed 10mm

k: For corner joints, the outside groove preparation may be in either or both members, provided the basic groove configuration is not changed and adequate edge distance is maintained to support the welding operations without excessive edge melting.

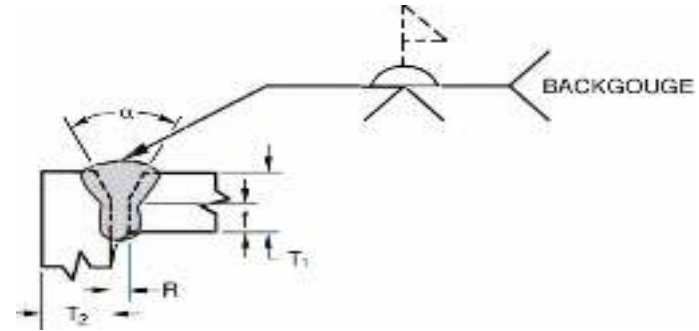
o: For corner and T-joints the member orientation may vary from 90° to less than or equal to 170° provided the groove angle and root opening are maintained, and the angle between the groove faces and the steel backing is at least 90°. See Figure 5.5 AWS D1.1/D1.1M

CJP

Single V- groove weld (2)
Butt joint (B) B-U2-GF



Single V- groove weld (2)
Corner joint (C)
C-U2-GF



Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-U2-GF	U		0-3	≤ 5	0-3	≤ 5	60	≥ 60			ALL	d, j
	C-U2-GF	U	U	0-3	≤ 5	0-3	≤ 5	60	≥ 60			ALL	d, g, j

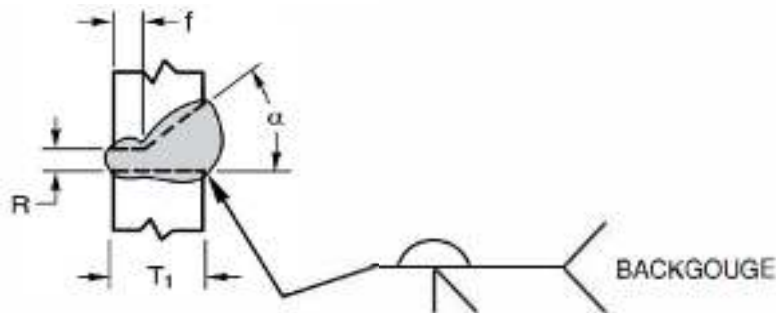
Note: d: Backgouge root to sound metal before welding second side.

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

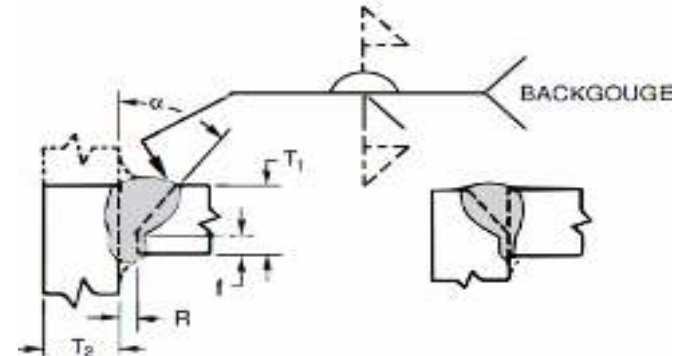
g: If fillet welds are used in statically loaded structures to reinforce groove welds in corner and T-joints, these shall be equal to T1 /4, but need not exceed 10mm

CJP

Single bevel- groove weld (4)
Butt joint (B) B-U4b-GF



Single bevel- groove weld (4)
T-joint (T)
Corner joint (C)
TC-U4b-GF



Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-U4b-GF	U		0-3	≤ 5	0-3	≤ 5	45	≥ 45			ALL	c, d, j
	TC-U4b-GF	U	U	0-3	≤ 5	0-3	≤ 5	45	≥ 45			ALL	d,g,j,k

Note: c: cyclic load application places restrictions on the use of this detail for butt joints in the flat position (See 4.18.2)

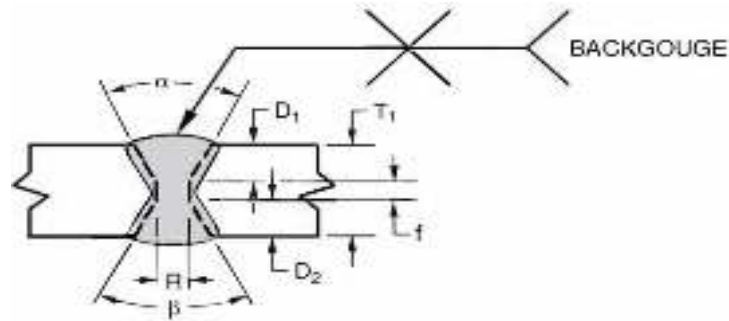
d: Backgouge root to sound metal before welding second side.

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

g: If fillet welds are used in statically loaded structures to reinforce groove welds in corner and T-joints, these shall be equal to T1 /4, but need not exceed 10mm

k: For corner joints, the outside groove preparation may be in either or both members, provided the basic groove configuration is not changed and adequate edge distance is maintained to support the welding operations without excessive edge melting.

Double V- groove weld (3)
Butt joint (B)



For B-U3c-S only		
T ₁		D ₁
Over 50	to 60	35
60	80	45
80	90	55
90	100	60
100	120	70
120	140	80
140	160	95
For T ₁ > 160 or T ₁ < 50 D ₁ = 2/3(T ₁ - 6)		

Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α, β (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-U3-GF	U		0-3	≤ 5	0-3	≤ 5	60	≥ 60	D1 D2	Refer to B-U3c-S	ALL	d, h, j

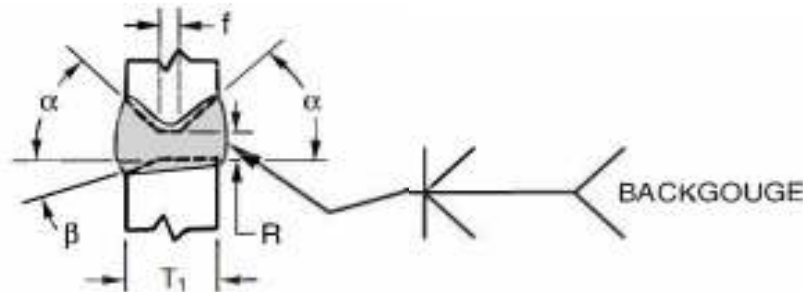
Note: d: Backgouge root to sound metal before welding second side.

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

h: Double-groove welds may have grooves of unequal depth, but the depth of the shallower groove shall be no less than one-fourth of the thickness of the thinner part joined.

Double bevel groove weld (5)

Butt joint (B) B-U5-GF

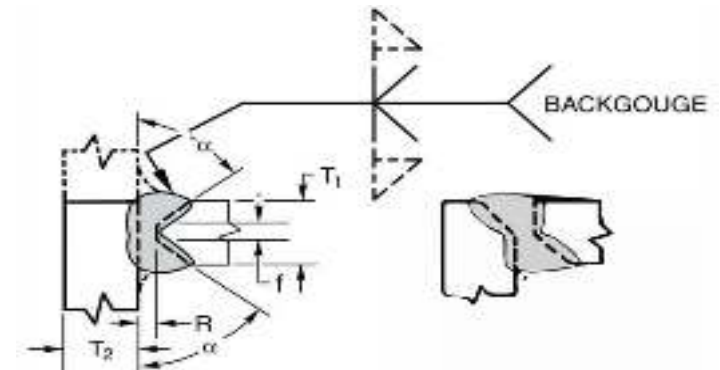


Double bevel groove weld (5)

T-joint (T)

Corner joint (C)

TC-U5-GF



Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α, β (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-U5-GF	U		0-3	≤ 5	0-3	≤ 5	α : 45 β : 0-15	$\alpha + \beta \geq 45$			ALL	c, d, h, j
FCAW	TC-U5-GF	U	U	0-3	≤ 5	0-3	≤ 5	45	≥ 45			ALL	d, g, h, j, k

Note: c: cyclic load application places restrictions on the use of this detail for butt joints in the flat position (See 4.18.2)

d: Backgouge root to sound metal before welding second side.

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

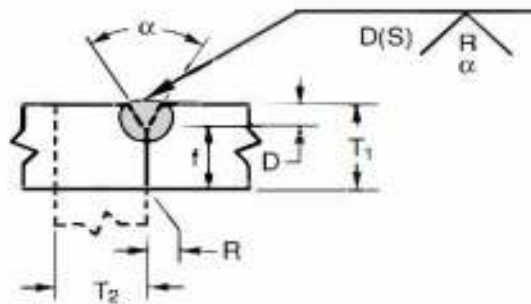
g: If fillet welds are used in statically loaded structures to reinforce groove welds in corner and T-joints, these shall be equal to T1 /4, but need not exceed 10mm

h: Double-groove welds may have grooves of unequal depth, but the depth of the shallower groove shall be no less than one-fourth of the thickness of the thinner part joined.

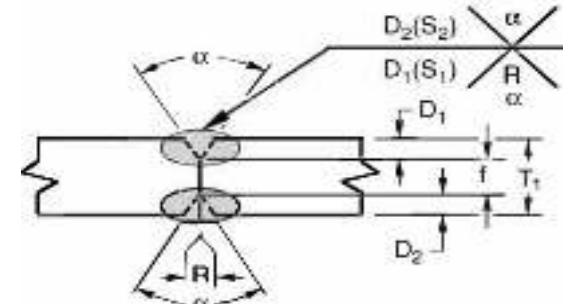
k: For corner joints, the outside groove preparation may be in either or both members, provided the basic groove configuration is not changed and adequate edge distance is maintained to support the welding operations without excessive edge melting.

CJP

Single V-groove weld (2)
Butt joint (B)
Corner joint (C)
BC-P2-GF

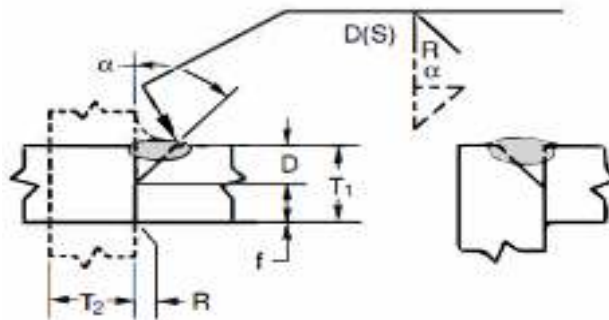


Double V-groove weld (3)
Butt joint (B)
B-P3-GF

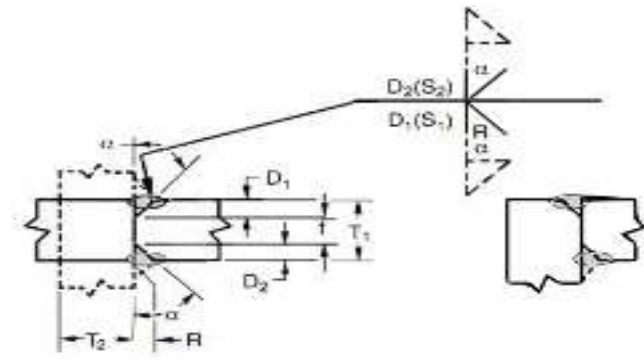


Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove	Weld size S	Position	Note
				R (mm)		f (mm)		α (°)					
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	D, D1, D2 (mm)	recommend		
		T1	T2										
FCAW	BC-P2-GF	≥6	U	0	≤ 2	≥3	≥3	60	≥60	≤ 5.5	D	ALL	b, f, j
FCAW	B-P3-GF	≥12		0	≤ 2	≥3	≥3	60	≥60	≤ 5.5	D1+D2	ALL	f, i, j

Single-bevel-groove weld (5)
Butt joint (B)
T-joint (T)
Corner joint (C)
BTC-P4-GF



Double-bevel-groove weld (5)
Butt joint (B)
T-joint (T)
Corner joint (C)
BTC-P5-GF



Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove	Weld size S	Position	Note
				R (mm)		f (mm)		α (°)					
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	D, D1, D2 (mm)	recommend		
		T1	T2										
FCAW	BTC-P4-GF	≥6	U	0	≤ 2	≥3	≥3	45	≥45	≤ 5.5	D D-3	F, H V, OH	b, f, g, j, k
FCAW	BTC-P5-GF	≥12	U	0	≤ 2	≥3	≥3	45	≥45	≤ 5.5	D1 + D2 (D1 + D2) - 6	F, H V, OH	f, g, i, j, k

Note for PJP b: Joint shall be welded from one side only.

f: Minimum weld size (S) and Depth of groove (D) as specified on drawings.

g: If fillet welds are used in statically loaded structures to reinforce groove welds in corner and T-joints, these shall be equal to $T_1 / 4$, but need not exceed 10mm

i: Double-groove welds may have grooves of unequal depth, provided these conform to the limitations of Note f. Also, the weld size (S) applies individually to each groove.

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

k: For corner joints, the outside groove preparation may be in either or both members, provided the basic groove configuration is not changed and adequate edge distance is maintained to support the welding operations without excessive edge melting.

PJP

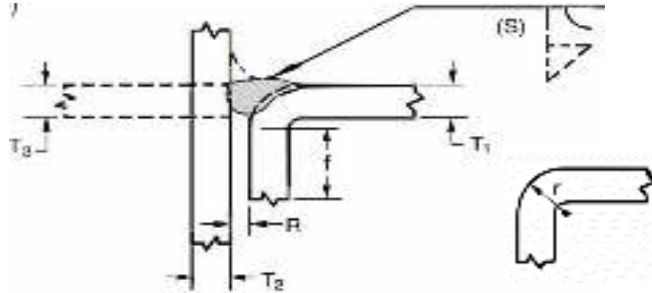
Flare -bevel-groove weld (10)

Butt joint (B)

T-joint (T)

Corner joint (C)

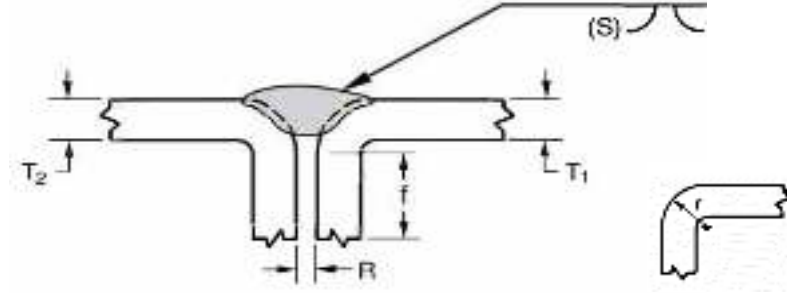
BTC-P10-GF



Flare -V-groove weld (11)

Butt joint (B)

B-P11-GF



PJP

Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Radius		Depth of Groove	Weld size S	Position	Note
		(U: unlimited)		R (mm)		f (mm)		r (mm)					
		T1	T2	Recommend	Limited	Recommend	Limited	Recommend	Limited	D, D1, D2 (mm)	recommend		
FCAW	BTC-P10-GF	≥5	U	0	≤ 2	≥5	≥5	≥ (0.75 T1)	≥ (0.75 T1)		(5r)/ 8	ALL	g, j, l, m
FCAW	B-P11-GF	≥5	≥T1	0	≤ 2	≥5	≥5	≥ (1.5 T1)	≥ (1.5 T1)		0.75 r	ALL	j, l, m, n

Note

g: If fillet welds are used in statically loaded structures to reinforce groove welds in corner and T-joints, these shall be equal to T1 /4, but need not exceed 10mm

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

l: Weld size .(_S_) shall be based on joints welded flush

m: For flare-V-groove welds and flare-bevel-groove welds to rectangular tubular sections, r shall be as two times the wall thickness

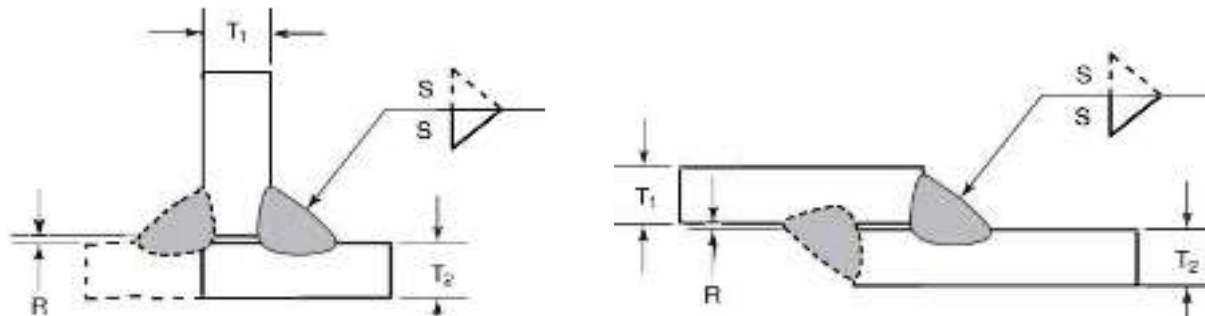
n: For flare-V-groove welds to surfaces with different radii r, the smaller r shall be used

Fillet weld (12)

T-joint (T)

Corner joint (C)

Lap joint (L)

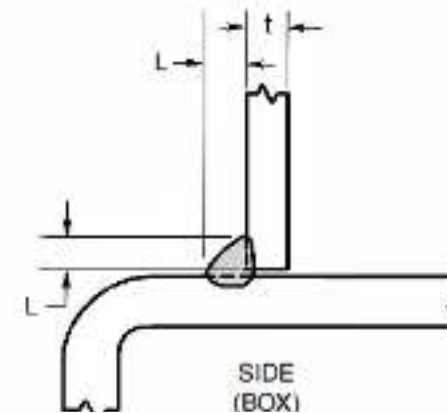
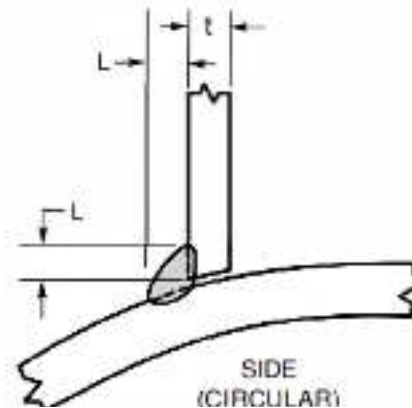
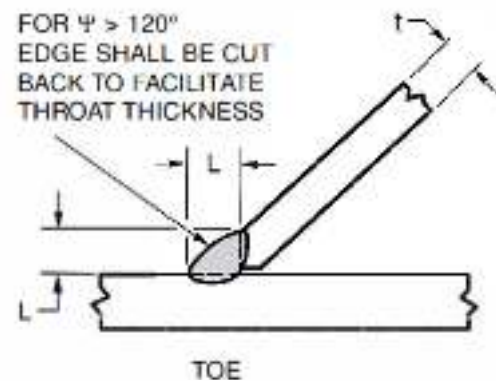
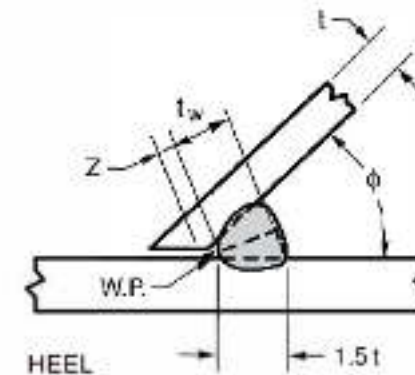
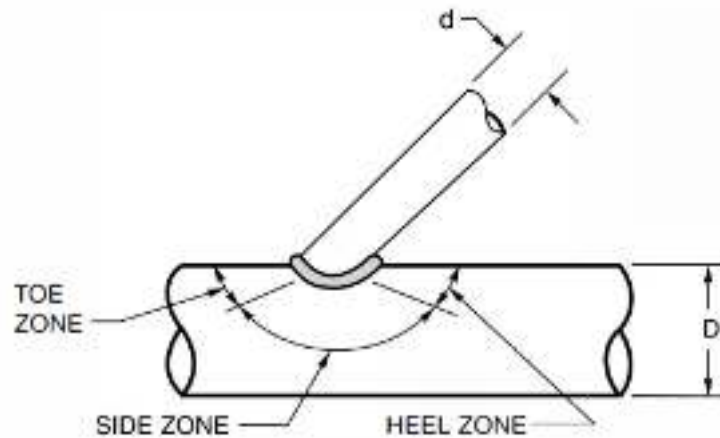
FILLET
PLATE

Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Radius		Depth of Groove	Weld size S	Position	Note
		(U: unlimited)		R (mm)		f (mm)		r (mm)					
		T1	T2	Recommend	Limited	Recommend	Limited	Recommend	Limited	D, D1, D2 (mm)	recommend		
		FCAW	TC-F12-GF	<75	<75	0	≤ 2						
FCAW	TC-F12a-GF	≥75	≥75	0	≤ 2							ALL	1
FCAW	L-F12-GF	<75	<75	0	≤ 2							ALL	1
FCAW	L-F12a-GF	≥75	≥75	0	≤ 2							ALL	1

Note

1: Fillet weld size refer to fabrication drawing or weld map

Figure 10.5 (AWS D1.1/D1.1M) Fillet weld Prequalified Tubular Joint



MIN L FOR

	$E = 0.7t$	$E = t$	$E = 1.07t$
HEEL $< 60^\circ$	$1.5t$	$1.5t$	LARGER OF $1.5t$ OR $1.4t + Z$
SIDE $\leq 100^\circ$	t	$1.4t$	$1.5t$
SIDE $100 - 110^\circ$	$1.1t$	$1.6t$	$1.75t$
SIDE $110 - 120^\circ$	$1.2t$	$1.8t$	$2.0t$
TOE $> 120^\circ$	t BEVEL	$1.4t$ BEVEL	FULL BEVEL 60–90° GROOVE

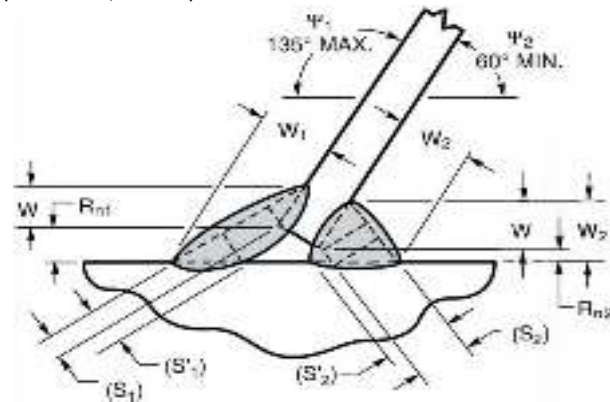
Notes:

1. t = thickness of thinner part.
2. L = minimum size (see 10.2.1 which may require increased weld size for combinations other than 36 ksi [250 MPa] base metal and 70 ksi [485 MPa] electrodes).
3. Root opening 0 in to 3/16 in [5 mm] (see 7.21).
4. Not prequalified for $\phi < 30^\circ$. For $\phi < 60^\circ$, the Z loss dimensions in Table 10.5 apply. See Table 10.12 for welder qualification position requirements.
5. See 10.5.2.2 for limitations on $\beta = d/D$.
6. Ψ = dihedral angle.

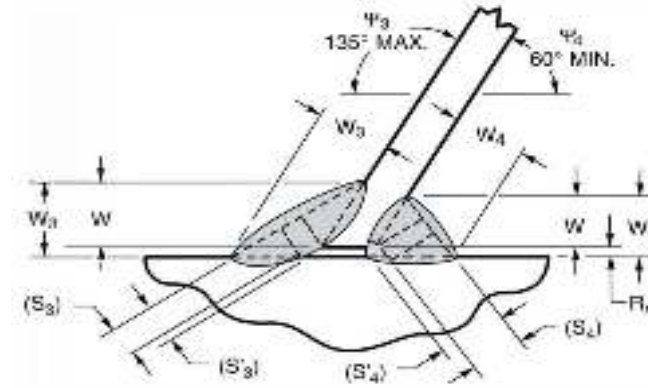
FILLET
PIPE

Figure 5.4 (AWS D1.1/D1.1M) Skewed T - Joint

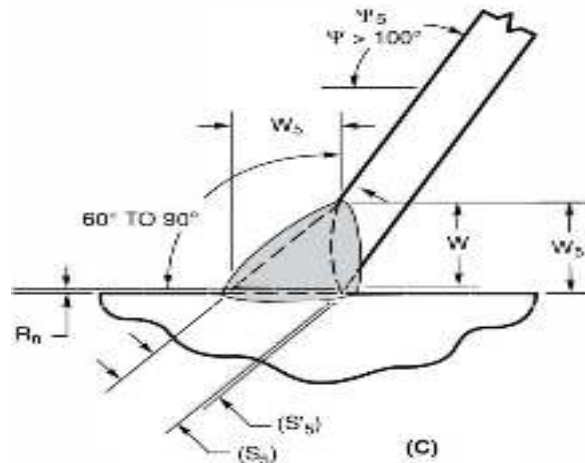
DETAIL A



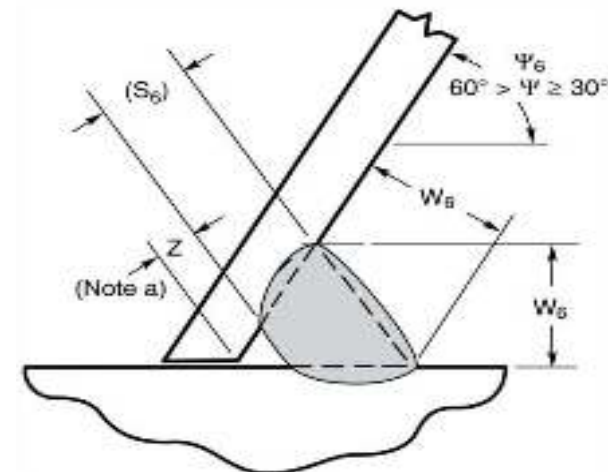
DETAIL B



DETAIL C



DETAIL D

SKEWED
(Plate)

Note a: Detail D, apply Z loss dimension of table 4.2 to determine effective throat

b: Detail D, shall not be under 30°. Welder qualify see table 6.10

1: S_n, S'_n weld size dependent on magnitude of Root open R_n

2: t is thickness of thinner part

3: Minimum weld size of Skewed T-joint for detail A, B, C shall be conform with table 7.7

Table 4.2
Z Loss Dimension (Nontubular) (See 4.4.3.3)

Dihedral Angle Ψ	Position of Welding—V or OH			Position of Welding—H or F		
	Process	Z(in)	Z(mm)	Process	Z(in)	Z(mm)
$60^\circ > \Psi > 45^\circ$	SMAW	1/8	3	SMAW	1/8	3
	FCAW-S	1/8	3	FCAW-S	0	0
	FCAW-G	1/8	3	FCAW-G	0	0
	GMAW	N/A	N/A	GMAW	0	0
$45^\circ > \Psi > 30^\circ$	SMAW	1/8	6	SMAW	1/4	6
	FCAW-S	1/8	6	FCAW-S	1/8	3
	FCAW-G	3/8	10	FCAW-G	1/4	6
	GMAW	N/A	N/A	GMAW	1/4	6

Table 7.7
Minimum Fillet Weld Sizes (See 7.13)

Base Metal Thickness (T) ^a		Minimum Size of Fillet Weld ^b	
in	mm	in	mm
T ≤ 1/4	T ≤ 6	1/8	3
1/4 < T ≤ 1/2	6 < T ≤ 12	3/16	5
1/2 < T ≤ 3/4	12 < T ≤ 20	1/4	6
3/4 < T	20 < T	5/16	8

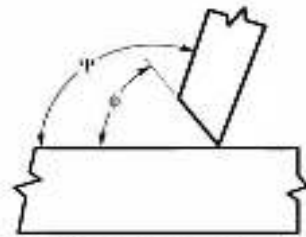
^a For nonlow hydrogen processes without preheat calculated in conformance with 6.8.4, T equals thickness of the thicker part joined; single-pass welds shall be used.

For nonlow hydrogen processes using procedures established to prevent cracking in conformance with 6.8.4 and for low-hydrogen processes, T equals thickness of the thinner part joined; single-pass requirement shall not apply.

^b Except that the weld size need not exceed the thickness of the thinner part joined.

^c Minimum size for cyclically loaded structures shall be 3/16 in (5 mm).

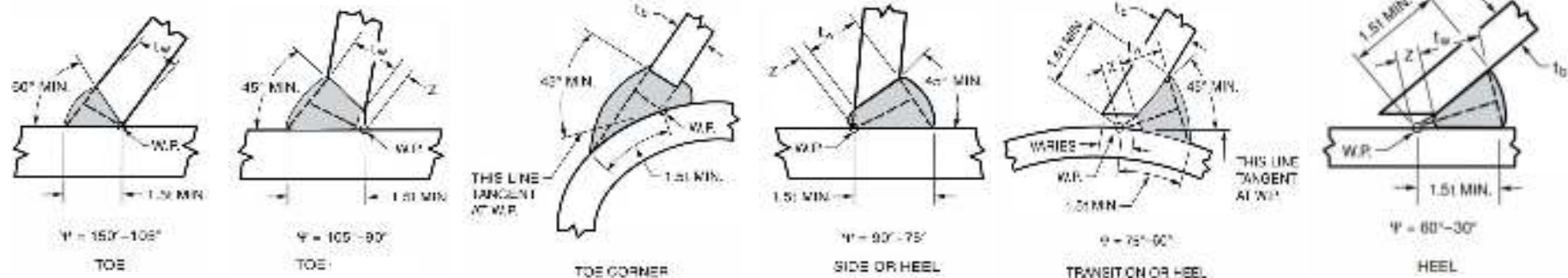
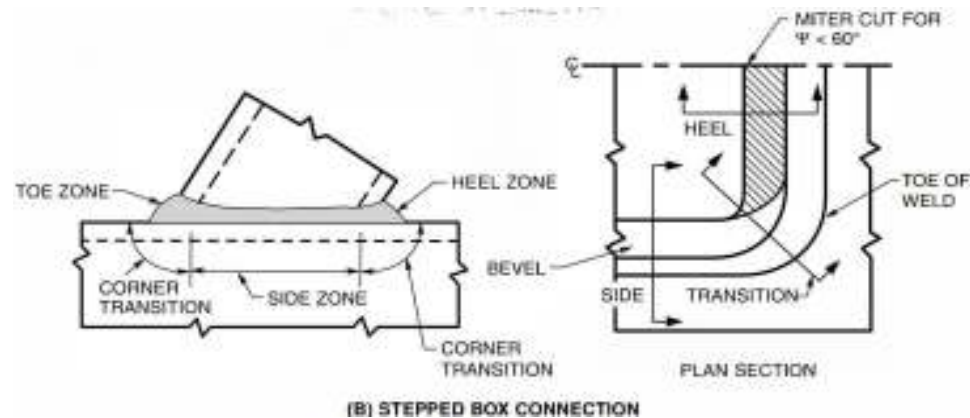
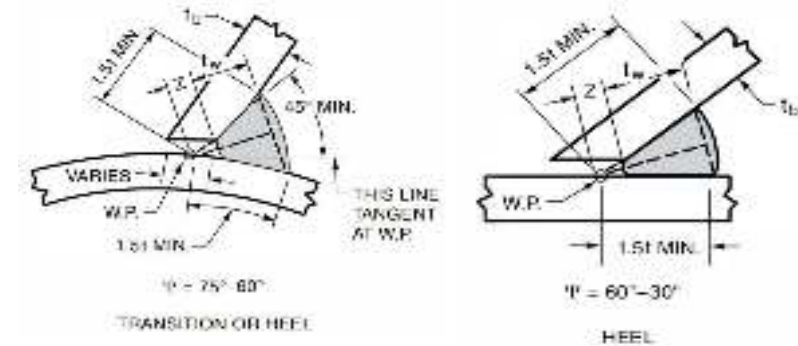
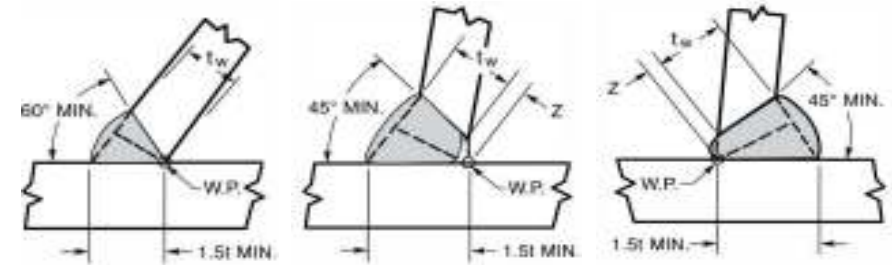
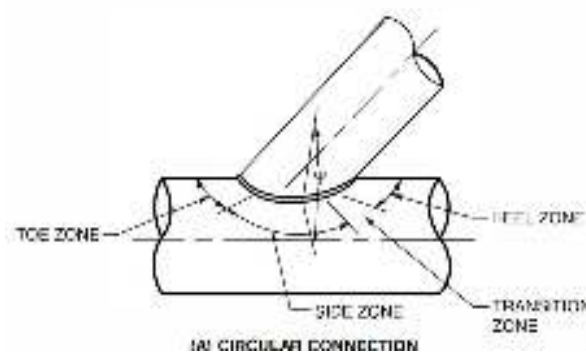
Figure 10.6 (AWS D1.1/D1.1M) PJP prequalify joint detail for T K Y tubular connection



SKETCH FOR ANGULAR DEFINITION

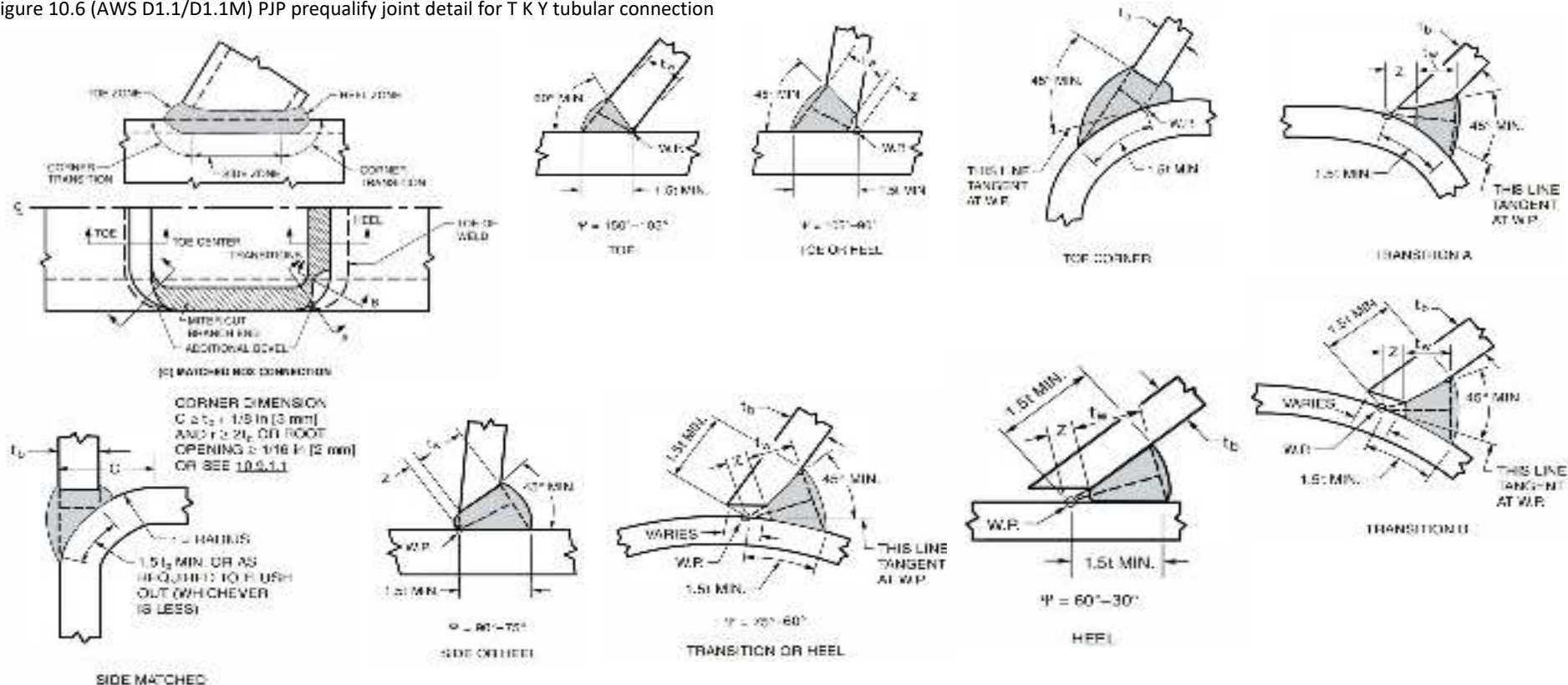
$$150^\circ > \Psi > 30^\circ$$

$$80^\circ > \phi \geq 30^\circ$$



PJP - TKY
connection
(Pipe, Box)

Figure 10.6 (AWS D1.1/D1.1M) PJP prequalify joint detail for T K Y tubular connection



PJP - TKY
connection
(Pipe, Box)

	Prepared by QA Welding	Reviewed and Approved by	Client
Name	TRAN MINH PHUNG	LA VINH NHON (D.QA Manager)	
Signature			
Date	May 3, 2025	May 3, 2025	



PROCEDURE QUALIFICATION RECORD (PQR)

(AWS D1.1/D1.1M:2020 Structural Welding Code-Steel)



Company Name: **DaiDung Metallic Manufacture Construction & Trade Corporation**

PQR No.: **PQR-DD-FCAW-283.1 & 2**

Rev. No. **A**

Date: **May 3, 2024**

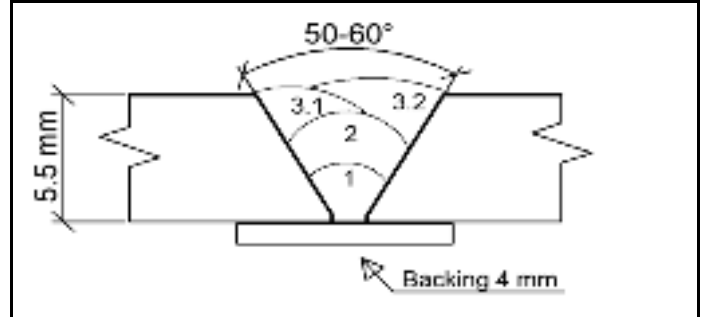
Page: **1 of 2**

BASE METALS	Specification	Type or Grade	AWS Group No.	Thickness	Size (NPS)	Schedule	Diameter
Base Material	JIS G3444	STK 400	-	5.5 mm	DN 80 mm	-	88.9 mm
Welded To	JIS G3444	STK 400	-	5.5 mm	DN 80 mm	-	88.9 mm
Backing Material	JIS G3101	SS 400	-	4 mm			
Other	-						

JOINT DETAILS

Groove Type	Double V Groove Butt Joint
Groove Angle	50-60°
Root Opening	5-6 mm
Root Face	Max 1 mm
Backgouging	N/A
Method	-

JOINT DETAILS (Sketch)



POSTWELD HEAT TREATMENT

Temperature	-
Time at Temperature	-
Other	-

PROCEDURE	PQR-DD-FCAW-283.1			PQR-DD-FCAW-283.2		
Weld Layer(s)	1	2	3	1	2	3
Weld Pass(es)	1	1	2	1	1	2
Process	FCAW	FCAW	FCAW	FCAW	FCAW	FCAW
Type (Manual, Mechanized, etc.)	Semi-Automatic					
Position	6G	6G	6G	6G	6G	6G
Vertical Progression	Uphill	Uphill	Uphill	Uphill	Uphill	Uphill
Filler Metal (AWS Spec.)	AWS A5.20	AWS A5.20	AWS A5.20	AWS A5.20	AWS A5.20	AWS A5.20
AWS Classification	E71T-1C	E71T-1C	E71T-1C	E71T-1C	E71T-1C	E71T-1C
Diameter (mm)	Ø 1.2	Ø 1.2	Ø 1.2	Ø 1.2	Ø 1.2	Ø 1.2
Manufacturer/Trade Name	Hyundai SC 71LH					
Shielding Gas (Composition)	CO2 (99.8%)	CO2 (99.8%)	CO2 (99.8%)	CO2 (99.8%)	CO2 (99.8%)	CO2 (99.8%)
Flow Rate (LPM)	20-22	20-22	20-22	20-22	20-22	20-22
Nozzle Size	No. 6	No. 6	No. 6	No. 6	No. 6	No. 6
Preheat Temperature (Min.)	34 °C (ambient)			34 °C (ambient)		
Interpass Temperature (Max.)	45 °C	90 °C	110 °C	47 °C	91 °C	109 °C
Electrical Characteristics						
Current Type & Polarity	DCEP	DCEP	DCEP	DCEP	DCEP	DCEP
Transfer Mode	-	-	-	-	-	-
Power Source Type (cc, cv, etc.)	CV	CV	CV	CV	CV	CV
Amps (A)	120-160	120-160	120-170	120-160	120-160	120-170
Volts (V)	19-21	19-21	19-21	19-21	19-21	19-21
Wire Feed Speed	(Amps)	(Amps)	(Amps)	(Amps)	(Amps)	(Amps)
Travel Speed (mm/min.)	215-270	171-174	182-209	220-262	167-174	178-204
Maximum Heat Input (kJ/mm)	0.9	1.2	1.1	0.9	1.2	1.1
Technique						
Stringer or Weave	Weave	Weave	Weave	Weave	Weave	Weave
Multi or Single Pass (per side)	Single	Single	Multipass	Single	Single	Multipass
Oscillation (Mech./Auto.)	-	-	-	-	-	-
Number of Electrodes	1	1	1	1	1	1
Contact Tube to Work Dist.(mm)	20	20	20	20	20	20
Peening	None	None	None	None	None	None
Interpass Cleaning	Grinding	Wire Brush	Grinding	Grinding	Grinding	Wire Brush
Other	-	-	-	-	-	-

	Prepared by QA Welding	Reviewed and Approved by
Name	Nguyen Quynh Sa	Truong Hoai Phong Vu (QA Manager)
Signature		
Date	May 3, 2024	May 3, 2024



	PROCEDURE QUALIFICATION RECORD (PQR) (AWS D1.1/D1.1M:2020 Structural Welding Code-Steel)	
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PQR No.: PQR-DD-FCAW-283.1 & 2	Rev. No.: A	Date: May 3, 2024
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Page: **2 of 2**

TEST RESULTS

TESTS

	Type of Tests	Clause/Figure(s) Reference	Acceptance Criteria	Result	Remarks
✓	Visual Inspection	6.10.1	6.10.1	Acceptable	
✓	Radiographic Examination	6.10.2.1	6.10.2.2	Acceptable	
	Ultrasonic Testing	6.10.2.1	6.10.2.2	-	
✓	2 Transverse Root Bends	6.10.3.1/Fig. 6.8	6.10.3.3	Acceptable	
✓	2 Transverse Face Bends	6.10.3.1/Fig. 6.8	6.10.3.3	Acceptable	
	2 Longitudinal Root Bends	6.10.3.2/Fig. 6.8	6.10.3.3	-	
	2 Longitudinal Face Bends	6.10.3.2/Fig. 6.8	6.10.3.3	-	
	2 Side Bends	6.10.3.1/Fig. 6.9	6.10.3.3	-	
	4 Side Bends	6.10.3.1/Fig. 6.9	6.10.3.3	-	
✓	2 Tensile Tests	6.10.3.4/Fig. 6.10	6.10.3.5	Acceptable	
	All-Weld-Metal Tensions	6.10.3.6/Figs. 6.14 and 6.18	6.15.1.3(2)	-	
	3 Macroetch	6.10.4	6.10.4.1	-	
	4 Macroetch	6.10.4	6.10.4.1	-	
	CVN Tests	6 Part D/Fig 6.28	6.27.7	-	

TENSILE TEST DETAILS

Specimen Number	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Tensile Load (kN)	Ultimate Unit Stress (MPa)	Type of Failure and Location
Sample 1	20.4	5.9	120.4	62.1	510	Ductile/Base metal
Sample 2	20.4	5.9	120.4	63.2	520	Ductile/Base metal

TOUGHNESS TEST DETAILS

Specimen Number	Notch Location	Specimen Size (mm)	Test Temperature	Absorbed Energy (J)	Percent Shear	Lateral Expansion	Average (J)

CERTIFICATION

Welder's Name	ID Number	Stamp Number	Tests Conducted by	
NGUYỄN VĂN CHƯƠNG	-	DDC-WD-0672	Laboratory	IQTC
-	-	-	Test Number	2304/2/VL-IQTC
-	-	-	File Number	-

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Clause 10 of AWS D1.1/D1.1M, 2020 *Structural Welding Code-Steel*.

	Prepared by QA Welding	Reviewed and Approved by
Name	Nguyen Quynh Sa	Truong Hoai Phong Vu (QA Manager)
Signature		
Date	May 3, 2024	May 3, 2024

WELDING PARAMETERS RECORD (Doc No.: WPR-FCAW-283.1)

Company name: DAIDUNG CORPORATION

PQR No.: PQR-DD-FCAW-283.1

Test Coupon: DD-FCAW-283.1

Test position: 6G

Base metal:

- Specification: JIS G3444 STK400
- Dimension: D88.9 x 5.5 x 160 (mm) x 2 pcs
- Heat/Product No.: 885060
- Certificate No.: 180-2271

Filler metal:

- Specification: AWS A5.20
- Classification: E71T-1C
- Trade name: Hyundai SC71 LH (Ø1.2 mm)
- Batch/Lot No.: 23120L71
- Certificates No.: HDV-2024-0263

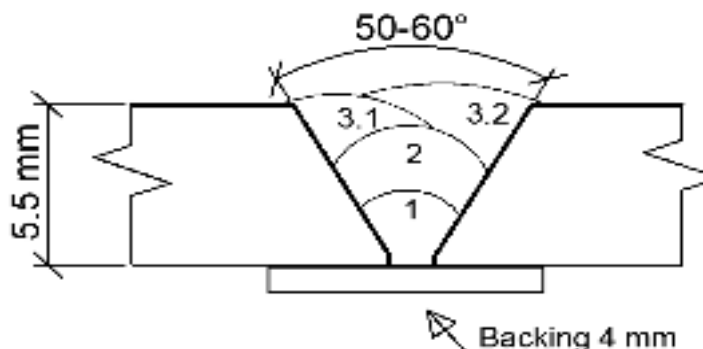
Gas: CO2 (99.8%)

Welder Name: NGUYỄN VĂN CHƯƠng

I.D: DDC-WD-0672

Joint Sketch:

Date: Apr 15, 2024



Root face: 0-1 mm
 Root gap: 5-6 mm
 Groove angle: 50-60°

Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
Side A																
1-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	120-140	19-21	0	31	140	270	34 (Ambient)	20-22	0.5-0.7	8	2.5	
1-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	140-160	19-21	0	39	140	215	45	20-22	0.7-0.9	8	2.5	
2-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	120-140	19-21	0	49	140	171	60	20-22	0.8-1	10	2	
2-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	140-160	19-21	0	48	140	174	90	20-22	0.9-1.2	10	2	



CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG
DAIDUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION
📍 38, Zone C, D1 St, An Ha Industrial Park, Pham Van Hai, Binh Chanh, HCMC
📧 sales@daidung.vn ☎ (84-28) 3868 1689 🌐 www.daidung.com.vn

WELDING PARAMETERS RECORD (Doc No.: WPR-FCAW-283.1)

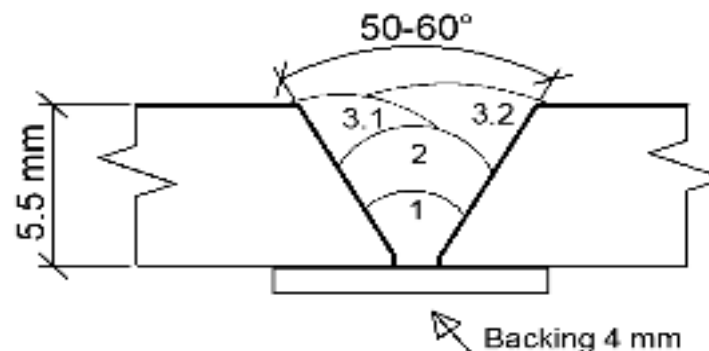
Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
3.1-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	120-140	19-21	0	45	140	186	80	20-22	0.7-0.9	10	1	
3.1-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	140-160	19-21	0	46	140	182	101	20-22	0.9-1.1	10	1	
3.2-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	130-150	19-21	0	40	140	209	97	20-22	0.7-0.9	10	1	
3.2-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	150-170	19-21	0	41	140	204	110	20-22	0.8-1.1	10	1	
Conducted by QA Welding																
Name		NGUYEN QUYNH SA														
Signature																
Date		Apr 15, 2024														

WELDING PARAMETERS RECORD (Doc No.: WPR-FCAW-283.2)

Company name: DAIDUNG CORPORATION
 PQR No.: PQR-DD-FCAW-283.2
 Test Coupon: DD-FCAW-283.2
 Test position: 6G
 Base metal:
 - Specification: JIS G3444 STK400
 - Dimension: D88.9 x 5.5 x 160 (mm) x 2 pcs
 - Heat/Product No.: 885060
 - Certificate No.: 180-2271
 Filler metal:
 - Specification: AWS A5.20
 - Classification: E71T-1C
 - Trade name: Hyundai SC71 LH (Ø1.2 mm)
 - Batch/Lot No.: 23120L71
 - Certificate No.: HDV-2024-0263
 Gas: CO₂ (99.8%)
 Welder Name: NGUYỄN VĂN CHUƠNG
 I.D.: DDC-WD-0672

Joint Sketch:

Date: Apr 15, 2024



Root face: 0-1 mm
 Root gap: 5-6 mm
 Groove angle: 50-60°

Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
Side A																
1-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	120-140	19-21	0	32	140	262	34 (Ambient)	20-22	0.5-0.7	8	2.5	
1-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	140-160	19-21	0	38	140	220	47	20-22	0.7-0.9	8	2.5	
2-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	120-140	19-21	0	48	140	174	61	20-22	0.8-1	10	2	
2-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	140-160	19-21	0	50	140	167	91	20-22	1-1.2	10	2	

WELDING PARAMETERS RECORD (Doc No.: WPR-FCAW-283.2)

Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
3.1-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	120-140	19-21	0	47	140	178	81	20-22	0.8-1	10	1	
3.1-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	140-160	19-21	0	45	140	186	100	20-22	0.9-1.1	10	1	
3.2-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	130-150	19-21	0	41	140	204	93	20-22	0.7-0.9	10	1	
3.2-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	150-170	19-21	0	42	140	199	109	20-22	0.9-1.1	10	1	
Conducted by QA Welding																
Name		NGUYEN QUYNH SA														
Signature																
Date		Apr 15, 2024														



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY

- ✦ Trụ sở chính/ Head office: 27/29 Đường 7A, KDC Bình Tân Đông, Phường Bình Tân Đông B, Quận Bình Tân, TP HCM
✦ Điện thoại/ Tel: 0272 3777 842 Fax: 02723 777640 Email: iqtc@iqtc.vn MST: 0313138782
✦ Phòng thí nghiệm/ Laboratory: Ấp Bình Tiến 2, Xã Đức Hòa Hạ, Huyện Đức Hòa, Tỉnh Long An



- Số TN/Ref: 2304/2/VL-IQTC

- Phiếu yêu cầu số/Request No.: 03-220424/YCTN



KẾT QUẢ THỬ NGHIỆM
TEST REPORT

- Công trình/Project: **NAGASAKI EAST PROJECT**
- Hàng mục/Item: **PROCEDURE QUALIFICATION RECORD (PQR) TEST**
- Địa điểm/Location: **JAPAN**
- Cơ quan yêu cầu/Client: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORP.**
- Đơn vị cung cấp mẫu/Supplier: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORP.**
- Tên mẫu/Name of sample: **Welding steel pipe OD 88.9x5.5x320 mm**
- Quy trình hàn/WPS No.: **WPS-DD-FCAW-283** - Tư thế hàn/Welding position: **6G**
- Yêu cầu kiểm tra/Testing requirements: **Thử kéo & Thử uốn/Tensile test & Bend test**
- Vật liệu/Material: **JIS G 3444 - STK400** - Mã thợ hàn/Welder code: **DDC-WD-0672**
- Số lượng/Quantity: **01 Mẫu/Sample**
- Ngày nhận mẫu/Sampling date: **22/04/2024**
- Ngày thử nghiệm/Testing date: **22/04/2024**
- Thiết bị thử nghiệm/Testing equipment: **Máy thử kéo - nén WAW-1000/ Universal testing machine WAW-1000**
- Phương pháp thử/Test Method: **AWS D1.1:2020**
- Người chứng kiến/Witnesses: **Nguyen Quynh Sa**

Tên chỉ tiêu/Specification	Đơn vị/ Unit	Kết quả/ Results	
		Mẫu 1/ Sample 1	Mẫu 2/ Sample 2
1.Thử kéo/Tensile test			
- Chiều dày mẫu/Thickness of specimen	mm	5.9	5.9
- Chiều rộng mẫu/Width of specimen	mm	20.4	20.4
- Tiết diện mẫu/Cross-sectional area	mm ²	121.6	121.6
- Lực kéo đứt/Tensile load	kN	62.1	63.2
- Giới hạn bền kéo/Tensile strength	MPa	510	520
- Vị trí mẫu đứt/Fracture position		Mỗi hàn/Weld	Mỗi hàn/Weld

Long An, ngày/date 23 tháng/month 04 năm/year 2024

CÔNG TY CP KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JSC (IQTC)

TRƯỞNG PHÒNG THỬ NGHIỆM
LAB MANAGER

Đ/D NHÓM THỬ NGHIỆM
LAB REP.

MAI HUNG KHANH



KSIENG DANG DON TUYEN



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY

✦ Trụ sở chính/ Head office: 27/29 Đường 7A, KDC Bình Trị Đông, Phường Bình Trị Đông B, Quận Bình Tân, TP.HCM
 ✦ Điện thoại/ Tel: 0272 3777 842 Fax: 02723 777640 Email: iqt@iqtc.vn MST: 0313138782
 ✦ Phòng thí nghiệm/ Laboratory: Ấp Bình Tiến 2, Xã Đức Hòa Hạ, Huyện Đức Hòa, Tỉnh Long An



Số TN/Ref: 2304/2/ML-IQTC

Phiếu yêu cầu số/Request No.: 03-220424/YCTN

Tên chỉ tiêu/Specification	Đơn vị/ Unit	Kết quả/Test Results			
		Mẫu 1/ Sample 1	Mẫu 2/ Sample 2	Mẫu 3/ Sample 3	Mẫu 4/ Sample 4
2. Thử uốn/Bend test					
- Vị trí uốn/Location bend		Uốn mặt/ Face bend	Uốn mặt/ Face bend	Uốn chân/ Root bend	Uốn chân/ Root bend
- Chiều dày/Thickness of specimen	mm	5.9	5.9	5.9	5.9
- Chiều rộng/Width of specimen	mm	25.0	25.0	25.0	25.0
- Góc uốn/Bending angle	Độ/°	180	180	180	180
- Đường kính gỏi uốn/Plunger diameter	mm	38	38	38	38
- Kết quả/Test results		Không nứt/ No crack	Không nứt/ No crack	Nứt bề mặt (3+2+2) mm/ Crack (3+2+2) mm of length on the surface	Không nứt/ No crack

Ghi chú/Note:

- * Phiếu kết quả này chỉ có giá trị cho mẫu thử do khách hàng mang tới /This test report is only value for sample submitted by client.
 * Tên mẫu và tên khách hàng được ghi theo yêu cầu của khách hàng /Name of sample and client are written as client's request.



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY

- ✦ Trụ sở chính/ Head office: 27/28 Đường 7A, KDC Bình Trị Đông, Phường Bình Trị Đông B, Quận Bình Tân, TP.HCM
- ✦ Điện thoại/ Tel: 0272 3177 842 Fax: 0272 3177 850 Email: iqtc@iqtc.vn MST: 0313138162
- ✦ Phòng thí nghiệm/ Laboratory: Ấp Bình Nhàn 2, Xã Bình Hòa Hạ, Huyện Bình Hòa, Tỉnh Long An



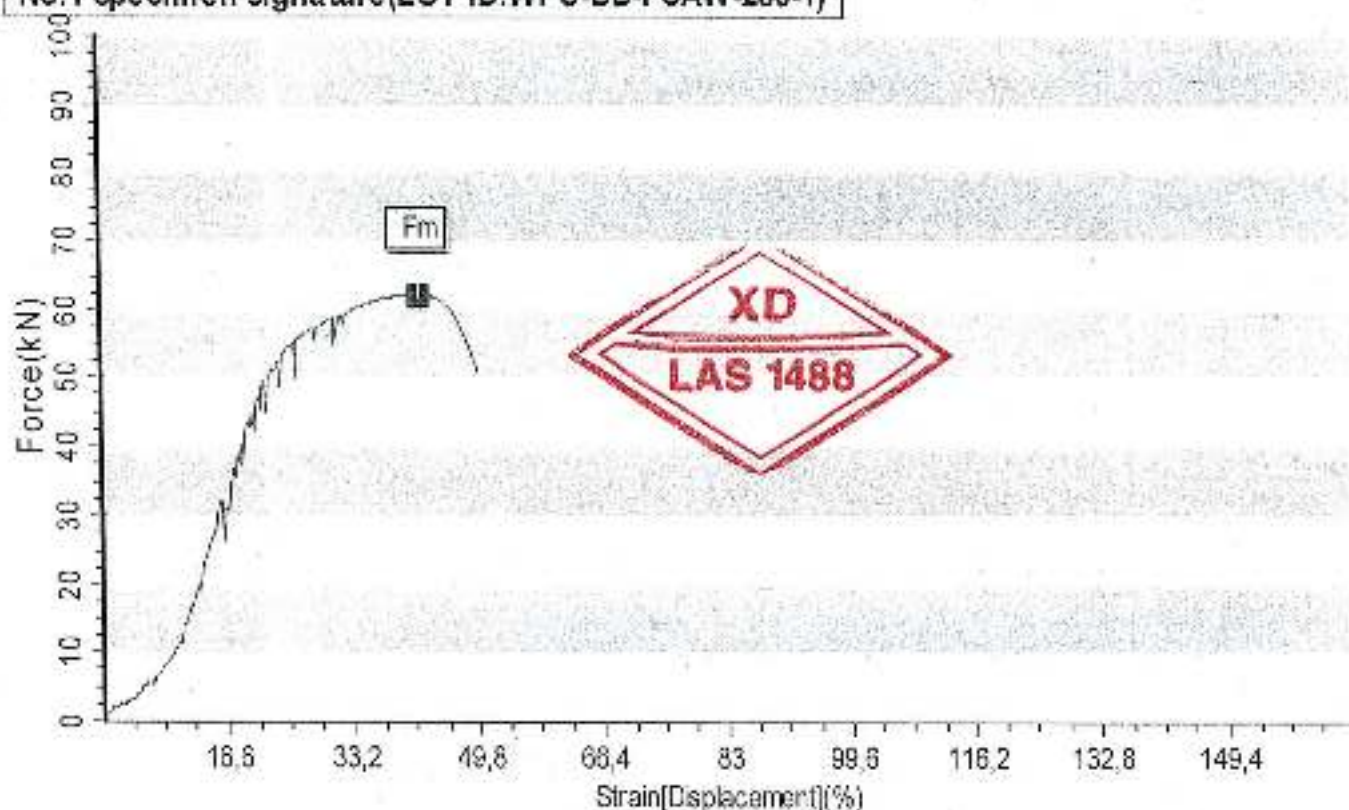
LOT ID: WPS-DD-FCAW-283

Company	DAI DUNG CORPORATION,	Test Date	22/04/2024
Standard	AWS D1.1	Operator	MAI HUNG KHANH
Temperature	25C-27C	Relative Humidity	60%
Machine Model	WAW-1000	Capacity	1000KN

Test Data

No.	b ₀ mm	a ₀ , T _a mm	D ₀ mm	F _m kN	R _m MPa					
01	20,40	5,90	88,90	62,1	610					

No.1 specimen signature(LOT ID:WPS-DD-FCAW-283-1)



CUSTOMER

SUPERVISOR

CONTRACTOR

SUPPLIER

LABORATORY

MAI HUNG KHANH



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY

- ✦ Trụ sở chính/ Head office: 77/79 Đường 74, KDC Bình Trị Đông, Phường Bình Trị Đông B, Quận Bình Tân, TP.HCM
 ✦ Điện thoại/ Tel: 0272 9777 657 Fax: 02728 777660 Email: info@iqtc.vn MST: 0313139702
 ✦ Phòng thí nghiệm/ Laboratory: Ấp Bình Tiến 2, Xã Đức Hòa Hạ, Huyện Đức Hòa, Tỉnh Long An



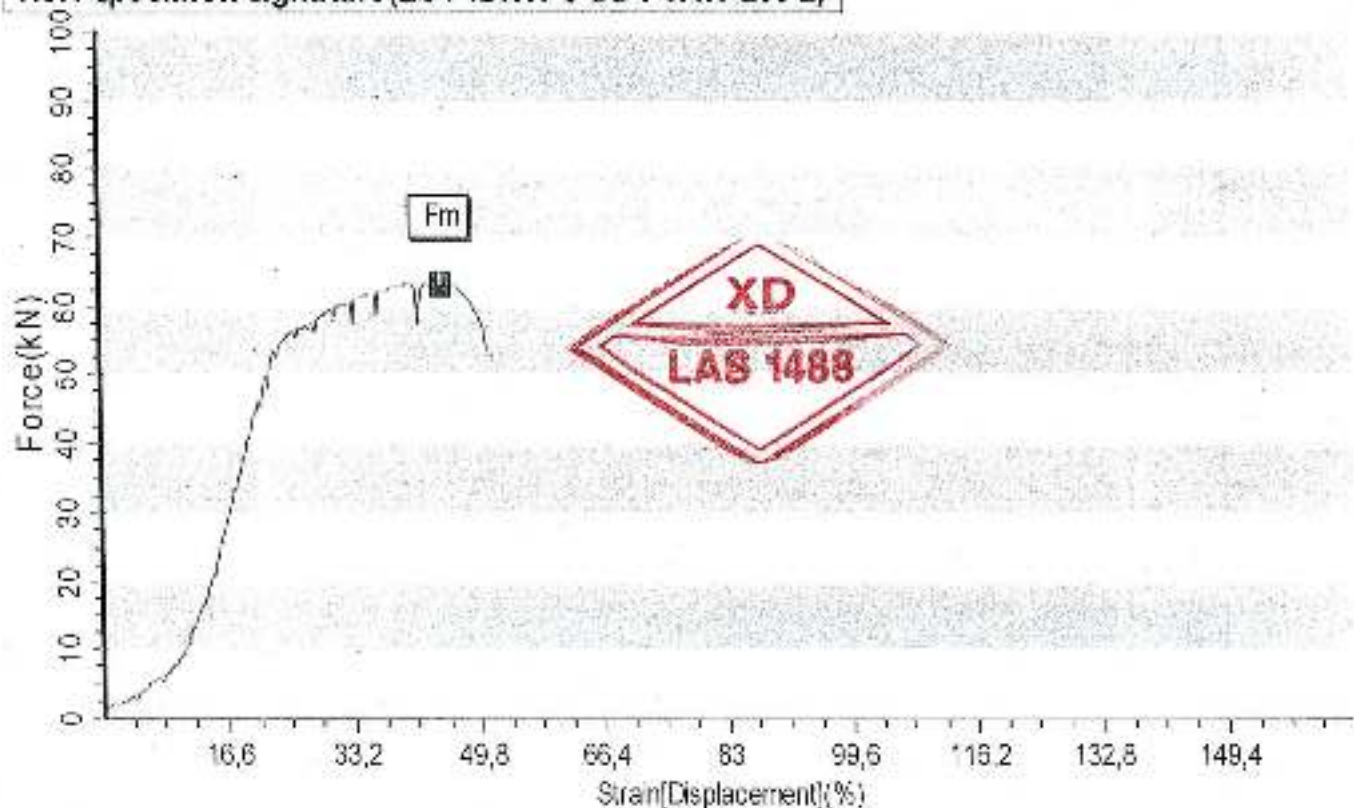
LOT ID: WPS-DD-FCAW-283

Company	DAI DUNG CORPORATION,	Test Date	22/04/2024
Standard	AWS D1.1	Operator	MAI HUNG KHANH
Temperature	25C-27C	Relative Humidity	60%
Machine Model	WAW-1000	Capacity	1000KN

Test Data

No.	bo mm	ao, Ta mm	Do mm	Fm kN	Rm MPa					
01	20,40	5,90	88,90	83,2	620					

No.1 specimen signature (LOT ID: WPS-DD-FCAW-283-2)



CUSTOMER

SUPERVISOR

CONTRACTOR

SUPPLIER

LABORATORY

MAI HUNG KHANH

RADIOGRAPHIC EXAMINATION REPORT

GENERAL INFORMATION

Project Name:	NAGASAKI EAST PROJECT	Request No.:	NDTWPQ-02024.04-17
Client Name:	DAI DUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION	Procedure No.:	ICNDNT-AWS-RT-01
Item:	Procedure Qualification Record (PQR) Test	Revision No.:	01
Application Code:	AWS-D1.1:2020	Drawing No.:	N/A
Acceptance Criteria:	AWS-D1.1:2020	Material:	J-E 03444 STEEL

Examination Technique:	DWSI	IQI Location:	Film Side
Geometric Unsharpness (Jg):	Max. 0.61	Density Range:	2.0 – 4.0
IQI Designation:	ASTM 1A	Film Processing:	Manual
IQI Essential Wire Density:	5	Surface Condition:	As Weld

EQUIPMENT

GAMMA - RAY						X - RAY				
Type	Source Size (mm)	Activity (Ci)	SCD (D) (mm)	OFD (d) (mm)	Qty of Film in Cassette	Exan Voltage (kV)	Focal Spot Size (mm)	SCD (D) (mm)	OFD (d) (mm)	Qty of Film in Cassette
IR - 192	2 x 5	10	88.9	5.5+4	1	-	-	-	-	-

MATERIAL

Manufacture	Type	Film size (cm)	Intensifying Screen type	Thickness of Screen (mm)	
				Front Side	Rear Side
Fuji	D400	10x20	Lead	0.125	0.125

EVALUATION

[illegible]

Abbreviation

Scale: 100 (1000000) km. Scale: 100 (1000000) km.

ACC: Accepted	BT: Burn Through	LP: Lack of Penetration	C: Crack	SI: Slag inclusion	FM: Film Form
REJ: Rejected	RE: Area Exposed	LF: Lack of Fusion	U: Undercut	TI: Tungsten Inclusion	
RS: Reheat	IC: Internal Concavity	CP: Cluster Porosities	P: Porosity	ESI: Erosion Slag Inclusion	

IQTC	QC INSPECTOR	CLIENT	CLIENT
------	--------------	--------	--------

Evaluated: HUYNH HOANG PHUO
RT/LEVEL: 2

Name: _____

Name: _____

same

Signature _____

Signature: _____

Signature _____

Signature _____



Date: 18-Apr-24

Esre

Date _____

1'st st



CÔNG TY CỔ PHẦN 190

Địa chỉ (Address): Lô N11-12 đường số 5, KCN Long Hậu mở rộng,
Xã Long Hậu, Huyện Cần Giuộc, Tỉnh Long An, Việt Nam
Điện thoại (Tel): 028.3873 4716 / 028.3873 4717 * Fax: 028.3873 4714
Email: info@ongthep190.com * Website: ongthep190.com



ISO 9001:2015 Cert. No.: 0483171

GIẤY CHỨNG NHẬN CHẤT LƯỢNG SẢN PHẨM

(Mill Test Certificate of Product Quality)

Ngày phát hành: 07 - 09 - 2023 (Issued date September 07th, 2023)

Certificate No: 180-2271

Tên khách hàng (Customer): CÔNG TY CỔ PHẦN CƠ KHÍ XÂY DỰNG THƯƠNG MẠI ĐẠI DŨNG

Số PO: DDC.PO23/08-0571

16 + 30P

Số TT No	Quy cách ống Specification	Loại ống End type	Mức Thép Grade of MTR	Số lượng (Quantity)		Ngày Sản Xuất (Production date)	COIL NO	Heat No	Cơ tính (Mechanical Properties)			
				Số Cây Pieces	Trọng lượng Weight (kg)				Bền kéo T.S (N/mm ²)	Bền chảy Y.S (N/mm ²)	Dộ giãn dài Elongation (%)	Áp lực làm việc (N/mm ²)
1	101.6 x 5.0 x 12000 mm	BPE	STK400	88	12,578	19/08/2023	2180886000	105012	432	251	37.0	NA
2	88.9 x 5.5 x 6000 mm	BPE	STK400	12	814	19/08/2023	2180886000	885060	432	251	37.0	NA
3	76.0 x 5.0 x 6000 mm	BPE	STK400	8	420	19/08/2023	2180886000	765060	432	251	37.0	NA
Số TT No	Quy cách ống Specification	THÀNH PHẦN HÓA HỌC chemical compositions (%)								Kiểm tra nền phẳng Flatness Test	Kiểm Tra độ kín khít Leak tightness	
		C	Si	Mn	P	S	Al	N				
1	101.6 x 5.0 x 12000 mm	0.17	0.16	0.56	0.020	0.030	0.340	0.044		Good	Good	
2	88.9 x 5.5 x 6000 mm	0.17	0.16	0.56	0.020	0.030	0.340	0.044		Good	Good	
3	76.0 x 5.0 x 6000 mm	0.17	0.16	0.56	0.020	0.030	0.340	0.044		Good	Good	

Ghi chú (Note):

- Giấy chứng nhận chất lượng chỉ cấp một lần.
(This certificate is provided once only).
- BPE: Black Plain Ends pipe (ống thép trơn không ren).

Kết luận (Conclusion):

Chúng tôi cam kết các loại ống trên phù hợp với yêu cầu của tiêu chuẩn JIS G3444-2015

PHỤ TRÁCH CHẤT LƯỢNG
Quality manager



GIÁM ĐỐC
Hà Mạnh Hùng

CERTIFIED MATERIAL REPORT

CMR No. : **HDV – 2024 – 0263**

Date of Issue : 2024 – 02 – 29



Customer		G-TECH CORPORATION								
Trade Name		SC-71LH			Size		1.2 mm x 15 kg			
Lot No.		MF23120L71			Purchaser's Order No.		111-2023/GT/HW/PO			
Specification		AWS A5.20 E71T-1C EN ISO 17632-A-T 42 2 P C1 1 H5 * According to EN 10204-2004 : 3.1								
Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment (°C×hr.)			As Welded			
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)						
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.		
	526	581	27	-20	75	86	84	82		
AWS Spec.	≥390	490~670	≥22	≥27						
Chemical Composition (%)										
Weld Metal	C	Si	Mn	P	S	Ni	Cr	Mo	V	Cu
	0.04	0.58	1.38	0.014	0.010	0.02	0.02	0.01	0.01	0.01
AWS Spec.	≤0.12	≤0.90	≤1.75	≤0.030	≤0.030	≤0.50	≤0.20	≤0.30	≤0.08	≤0.35
* Hydrogen Content (mℓ/100g) (AWS A4.3) : 4.36 * Radiographic Test : ACCEPT * Fillet Weld Test : GOOD										
Others * AS/NZS ISO 17632-A-T 42 2 P C1 1 H5 * FPC No. (CE-Mark): 0035-CPR-C771 * Q'ty: 7,200 kg										
We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.										
 0035-CPR-C771 0035 10 / HWV-314 EN13479 Approved by  QA Manager Pham Dinh Anh Vu										

HYUNDAI WELDING VINA CO., LTD.

Road N2, Nhon Trach 5 IZ, Hiep Phuoc Ward, Nhon Trach District, Dong Nai Province, Vietnam.



Comply:
ISO 9001:2015
AWS D1.1: 2025

WELDING PROCEDURE

Doc No.: WPS-DD-FCAW-284

Rev	Date	Page	Description	Prepared by	Checked and Approved by
0	May 03, 2024	All	Client Review	L.V.Nhon	T.H.P.Vu
					

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	WELDING PROCEDURE SPECIFICATION (WPS) (AWS D1.1/D1.1M:2020 Structural Welding Code-Steel)		
	Company Name: DaiDung Metallic Manufacture Construction & Trade Corporation WPS No.: WPS-DD-FCAW-284 Rev. No.: 1 Date: Jun 3, 2025		

Supporting PQR(s): PQR-DD-FCAW-284.1 & 2 CVN Report: N/A	Page: 1 of 11
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BASE METALS	Specification	Type or Grade	AWS Group No.	BASE METAL THICKNESS	As-Welded (mm)	With PWHT
Base Material	ASTM A53 ASTM A500	Gr B Gr A, B, C	Any Group I	CJP Groove Welds (Square Groove)	3 - 10	-
Welded To	ASTM A53 ASTM A500	Gr B Gr A, B, C	Any Group I	CJP Groove Welds (Other)	3 - 20	-
Backing Material	Steel backing or Backing Welds or Non-Steel Backing (Ceramic, etc.), refer Section 7.9			PJP Groove Welds	3 - Unlimited	-
Other	- Qualify any fillet weld size, qualify any box size - Dihedral angle ≥ 30° - Except T,K,Y - CJP			Fillet Welds	Any thickness	-
				DIAMETER OD (Outside diameter)	- 26.7 ≤O.D ≤ 114.3 for CJP - Any O.D for fillet, PJP	

JOINT DETAILS	
Groove Type	See attached sketch for more details
Groove Angle	See attached sketch for more details
Root Opening	See attached sketch for more details
Root Face	See attached sketch for more details
Backgouging	Yes
Method	Grinding or Air carbon arc

POSTWELD HEAT TREATMENT	
Temperature	-
Time at Temperature	-
Other	-

JOINT DETAILS (Sketch)
See Joint Details in page 2 to 11 of 11 (PJP: Max Groove Depth 5.5 mm,TKY: unlimited)

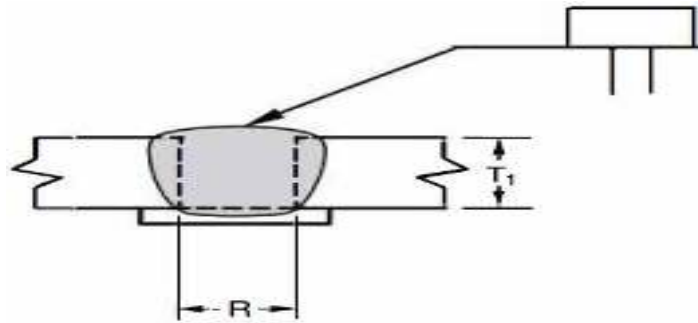
PROCEDURE		
Weld Layer(s)	All	
Weld Pass(es)	All	
Process	FCAW	
Type (Manual, Mechanized, etc.)	Semi-Automatic/Mechanized	
Position	All	
Vertical Progression	Uphill	
Filler Metal (AWS Spec.)	A5.20	
AWS Classification	E71T-1C	
Diameter (mm)	Max. 1.2	
Manufacturer/Trade Name	Not strictly	
Shielding Gas (Composition)	99.8% CO ₂	
Flow Rate (LPM)	16 - 28	
Nozzle Size (mm)	15 or 19	
Preheat Temperature (Min.)	See NOTES* below	
Interpass Temperature (Max.)	250 °C	
Electrical Characteristics		
Current Type & Polarity	DCEP	
Transfer Mode	-	
Power Source Type (cc, cv, etc.)	CV	
Amps (A)	110 - 185	
Volts (V)	19 - 24	
Wire Feed Speed	(Amps)	
Travel Speed (mm/min.)	137 - 325	
Maximum Heat Input (kJ/mm)	-	
Technique		
Stringer or Weave	Stringer or Weave	
Multi or Single Pass (per side)	Multi or Single Pass	
Oscillation (Mech./Auto.)	-	
Number of Electrodes	1	
Contact Tube to Work Dist. (mm)	15 - 25	
Peening	None	
Interpass Cleaning	Grinding and Wire Brush	
Other	This procedure can be used for repair	

NOTES* For Thickness 3 mm thru 38 mm incl.: Min. 20°C
 Over 38 mm thru 65 mm incl.: Min. 65°C
 Over 65 mm: Min. 110°C

	Prepared by QA Welding	Reviewed and Approved by	Client
Name	TRAN MINH PHUNG	LA VINH NHON (D.QA Manager)	
Signature			
Date	Jun 3, 2025	Jun 3, 2025	

Square-groove weld (1)

Butt joint (B)



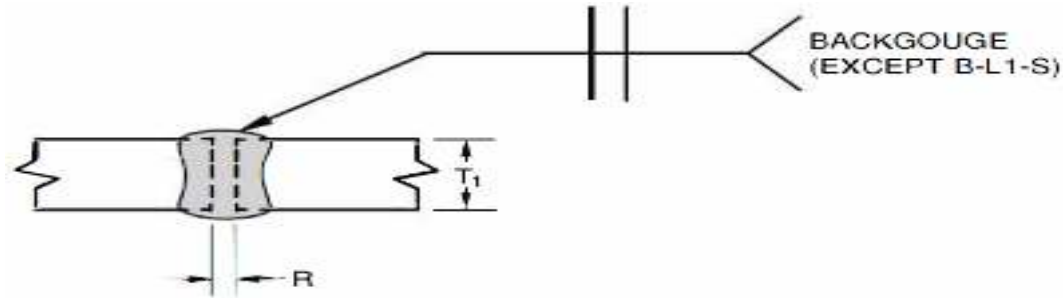
Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-L1a-GF	≤ 10		R=T1	T1-((T1)+2)							F, H, V	j

Note: j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

CJP

Square-groove weld (1)

Butt joint (B)

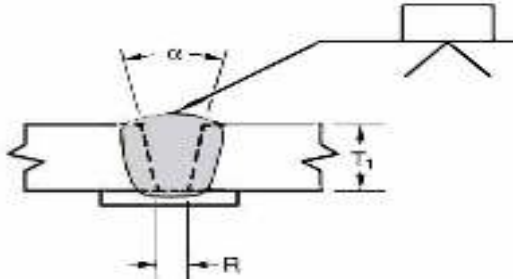


Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-L1b-GF	≤ 10		0-5	≤ 5							F, H, V	d, j

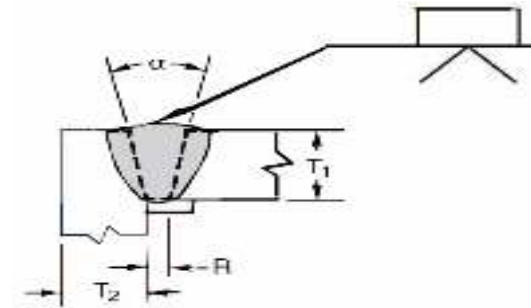
Note: d: Backgouge root to sound metal before welding second side.

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

Single V- groove weld (2)
Butt joint (B)
B-U2a-GF



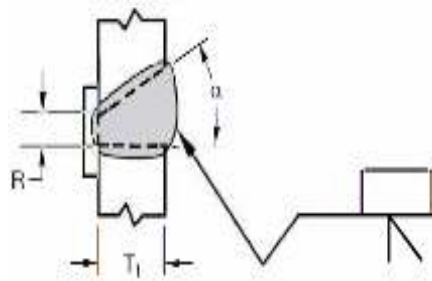
Single V- groove weld (2)
Corner joint (C)
C-U2a-GF



Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-U2a-GF	U		5	≥ 5	0	≤ 2	30	≥ 30			F, V, OH	j
	C-U2a-GF	U	U	5	≥ 5	0	≤ 2	30	≥ 30			F, V, OH	

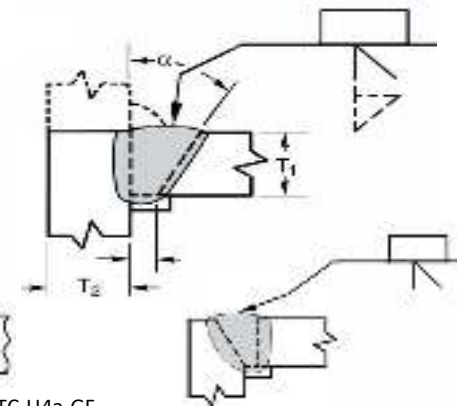
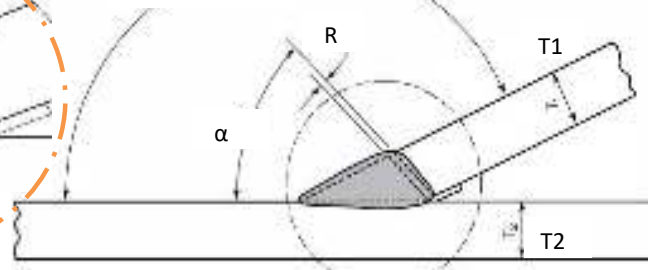
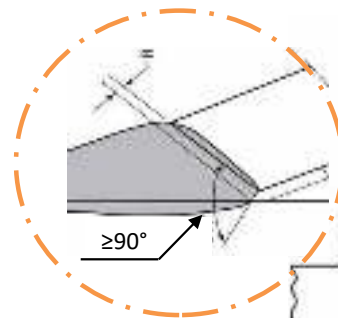
Note: j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

Single bevel- groove weld (4)
Butt joint (B)



Single bevel- groove weld (4)
T-joint (T) Corner joint (C)

$90^\circ \leq \psi \leq 170^\circ$



B-U4a-GF

Note o below

TC-U4a-GF

Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-U4a-GF	U		5	≥ 5	0	≤ 2	30	≥ 30			ALL	c, j
	TC-U4a-GF	U	U	5	≥ 5	0	≤ 2	30	≥ 30			ALL	g, k, o

Note: c: cyclic load application places restrictions on the use of this detail for butt joints in the flat position (See 4.18.2)

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

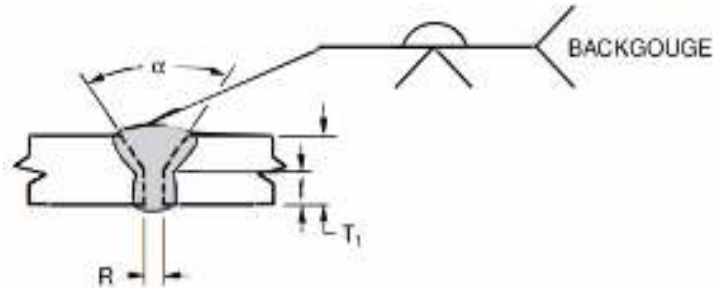
g: If fillet welds are used in statically loaded structures to reinforce groove welds in corner and T-joints, these shall be equal to $T1/4$, but need not exceed 10mm

k: For corner joints, the outside groove preparation may be in either or both members, provided the basic groove configuration is not changed and adequate edge distance is maintained to support the welding operations without excessive edge melting.

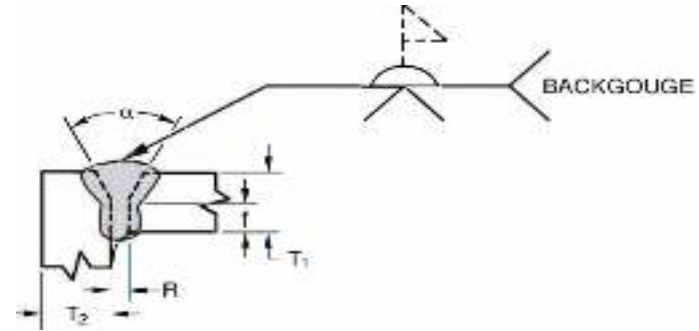
o: For corner and T-joints the member orientation may vary from 90° to less than or equal to 170° provided the groove angle and root opening are maintained, and the angle between the groove faces and the steel backing is at least 90°. See Figure 5.5 AWS D1.1/D1.1M

CJP

Single V- groove weld (2)
Butt joint (B) B-U2-GF



Single V- groove weld (2)
Corner joint (C)
C-U2-GF



Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-U2-GF	U		0-3	≤ 5	0-3	≤ 5	60	≥ 60			ALL	d, j
	C-U2-GF	U	U	0-3	≤ 5	0-3	≤ 5	60	≥ 60			ALL	d, g, j

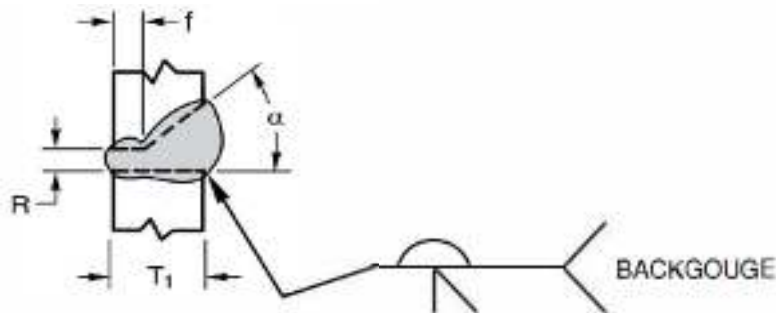
Note: d: Backgouge root to sound metal before welding second side.

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

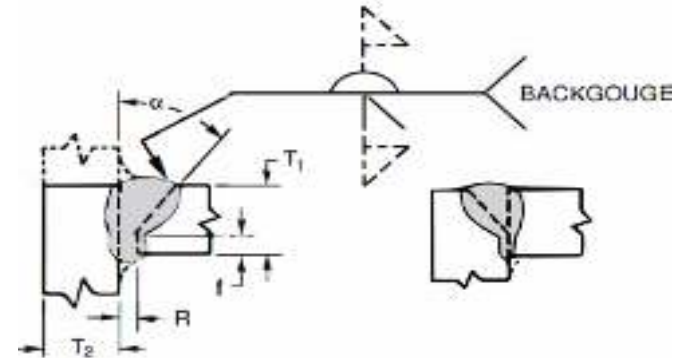
g: If fillet welds are used in statically loaded structures to reinforce groove welds in corner and T-joints, these shall be equal to T1 /4, but need not exceed 10mm

CJP

Single bevel- groove weld (4)
Butt joint (B) B-U4b-GF



Single bevel- groove weld (4)
T-joint (T)
Corner joint (C)
TC-U4b-GF



Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-U4b-GF	U		0-3	≤ 5	0-3	≤ 5	45	≥ 45			ALL	c, d, j
	TC-U4b-GF	U	U	0-3	≤ 5	0-3	≤ 5	45	≥ 45			ALL	d,g,j,k

Note: c: cyclic load application places restrictions on the use of this detail for butt joints in the flat position (See 4.18.2)

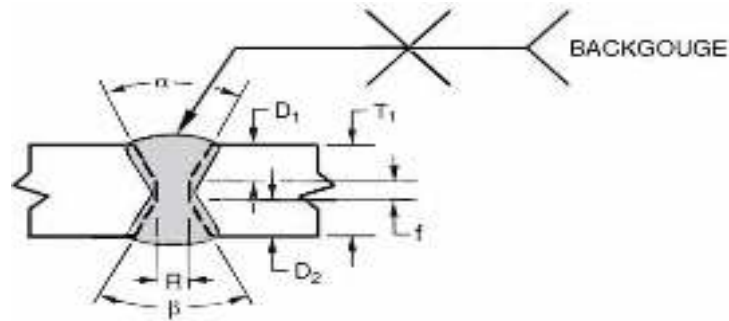
d: Backgouge root to sound metal before welding second side.

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

g: If fillet welds are used in statically loaded structures to reinforce groove welds in corner and T-joints, these shall be equal to T1 /4, but need not exceed 10mm

k: For corner joints, the outside groove preparation may be in either or both members, provided the basic groove configuration is not changed and adequate edge distance is maintained to support the welding operations without excessive edge melting.

Double V- groove weld (3)
Butt joint (B)



For B-U3c-S only		
T ₁		D ₁
Over 50	to 60	35
60	80	45
80	90	55
90	100	60
100	120	70
120	140	80
140	160	95
For T ₁ > 160 or T ₁ < 50 D ₁ = 2/3(T ₁ - 6)		

Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α, β (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-U3-GF	U		0-3	≤ 5	0-3	≤ 5	60	≥ 60	D1 D2	Refer to B-U3c-S	ALL	d, h, j

Note: d: Backgouge root to sound metal before welding second side.

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

h: Double-groove welds may have grooves of unequal depth, but the depth of the shallower groove shall be no less than one-fourth of the thickness of the thinner part joined.

Double bevel groove weld (5)

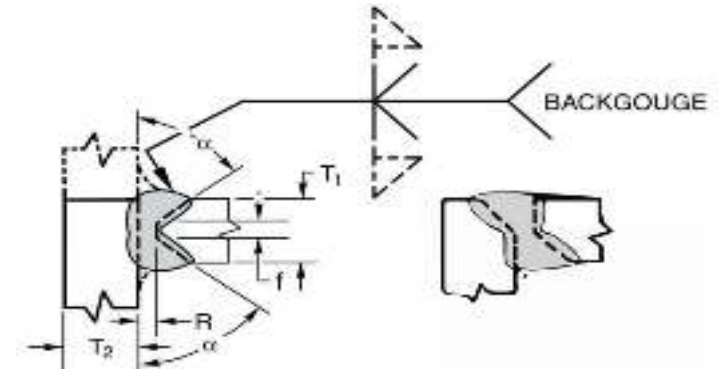
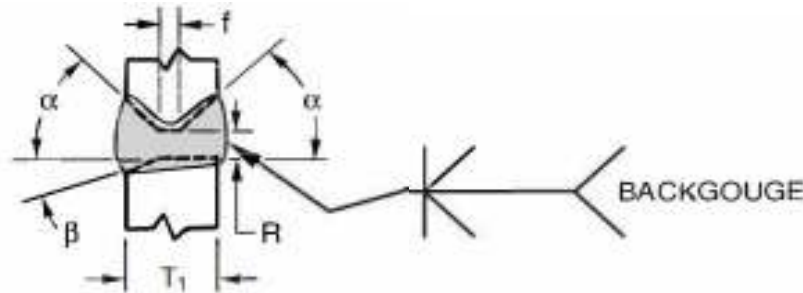
Butt joint (B) B-U5-GF

Double bevel groove weld (5)

T-joint (T)

Corner joint (C)

TC-U5-GF



Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove		Position	Note
				R (mm)		f (mm)		α, β (°)		D1, D2 (mm)			
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	Recommend	Limited		
		T1	T2										
FCAW	B-U5-GF	U		0-3	≤ 5	0-3	≤ 5	α : 45 β :0-15	$\alpha + \beta \geq 45$			ALL	c, d, h, j
FCAW	TC-U5-GF	U	U	0-3	≤ 5	0-3	≤ 5	45	≥ 45			ALL	d, g, h, j, k

Note: c: cyclic load application places restrictions on the use of this detail for butt joints in the flat position (See 4.18.2)

d: Backgouge root to sound metal before welding second side.

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

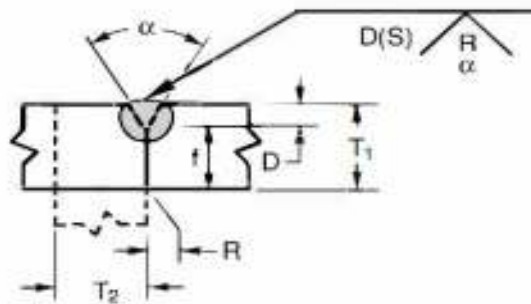
g: If fillet welds are used in statically loaded structures to reinforce groove welds in corner and T-joints, these shall be equal to T1 /4, but need not exceed 10mm

h: Double-groove welds may have grooves of unequal depth, but the depth of the shallower groove shall be no less than one-fourth of the thickness of the thinner part joined.

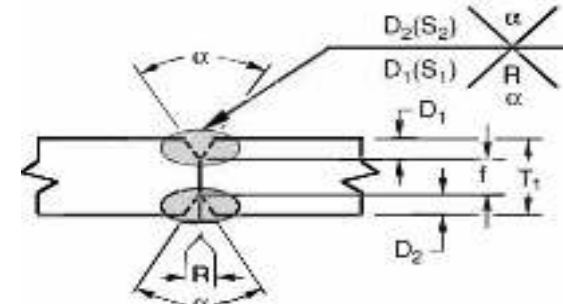
k: For corner joints, the outside groove preparation may be in either or both members, provided the basic groove configuration is not changed and adequate edge distance is maintained to support the welding operations without excessive edge melting.

CJP

Single V- groove weld (2)
Butt joint (B)
Corner joint (C)
BC-P2-GF

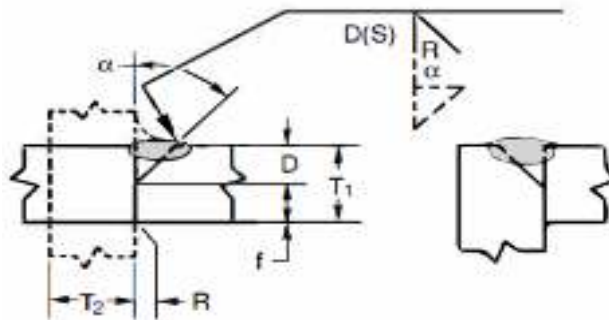


Double V- groove weld (3)
Butt joint (B)
B-P3-GF

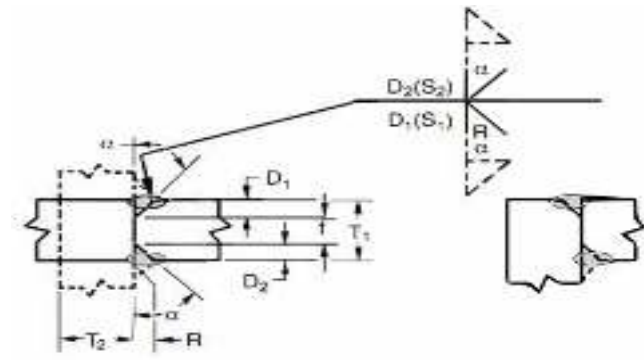


Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove	Weld size S	Position	Note
				R (mm)		f (mm)		α (°)					
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	D, D1, D2 (mm)	recommend		
		T1	T2										
FCAW	BC-P2-GF	≥6	U	0	≤ 2	≥3	≥3	60	≥60	≤ 5.5	D	ALL	b, f, j
FCAW	B-P3-GF	≥12		0	≤ 2	≥3	≥3	60	≥60	≤ 5.5	D1+D2	ALL	f, i, j

Single- bevel -groove weld (5)
Butt joint (B)
T-joint (T)
Corner joint (C)
BTC-P4-GF



Double- bevel -groove weld (5)
Butt joint (B)
T-joint (T)
Corner joint (C)
BTC-P5-GF



Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Groove Angle (degree)		Depth of Groove	Weld size S	Position	Note
				R (mm)		f (mm)		α (°)					
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	D, D1, D2 (mm)	recommend		
		T1	T2										
FCAW	BTC-P4-GF	≥6	U	0	≤ 2	≥3	≥3	45	≥45	≤ 5.5	D D-3	F, H V, OH	b, f, g, j, k
FCAW	BTC-P5-GF	≥12	U	0	≤ 2	≥3	≥3	45	≥45	≤ 5.5	D1 + D2 (D1 + D2) - 6	F, H V, OH	f, g, i, j, k

Note for PJP b: Joint shall be welded from one side only.

f: Minimum weld size (S) and Depth of groove (D) as specified on drawings.

g: If fillet welds are used in statically loaded structures to reinforce groove welds in corner and T-joints, these shall be equal to $T_1 / 4$, but need not exceed 10mm

i: Double-groove welds may have grooves of unequal depth, provided these conform to the limitations of Note f. Also, the weld size (S) applies individually to each groove.

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

k: For corner joints, the outside groove preparation may be in either or both members, provided the basic groove configuration is not changed and adequate edge distance is maintained to support the welding operations without excessive edge melting.

PJP

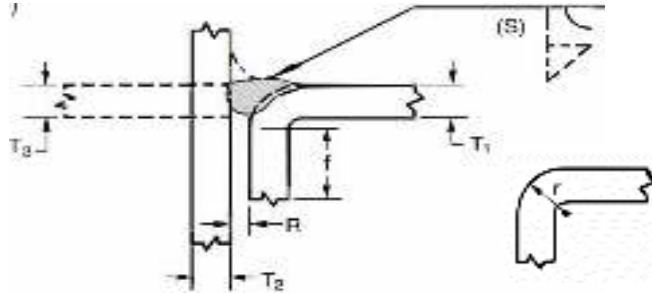
Flare -bevel-groove weld (10)

Butt joint (B)

T-joint (T)

Corner joint (C)

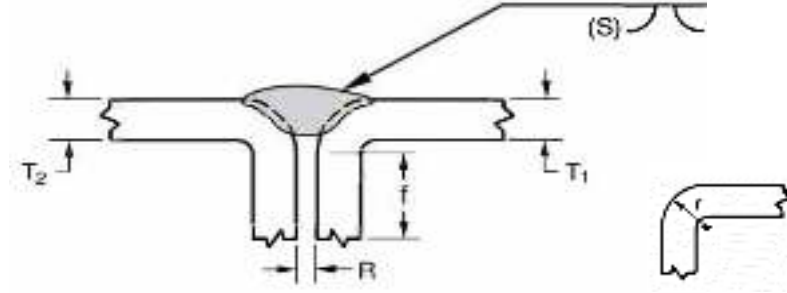
BTC-P10-GF



Flare -V-groove weld (11)

Butt joint (B)

B-P11-GF



PJP

Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Radius		Depth of Groove	Weld size S	Position	Note
		(U: unlimited)		R (mm)		f (mm)		r (mm)					
		T1	T2	Recommend	Limited	Recommend	Limited	Recommend	Limited	D, D1, D2 (mm)	recommend		
FCAW	BTC-P10-GF	≥5	U	0	≤ 2	≥5	≥5	≥ (0.75 T1)	≥ (0.75 T1)		(5r)/ 8	ALL	g, j, l, m
FCAW	B-P11-GF	≥5	≥T1	0	≤ 2	≥5	≥5	≥ (1.5 T1)	≥ (1.5 T1)		0.75 r	ALL	j, l, m, n

Note

g: If fillet welds are used in statically loaded structures to reinforce groove welds in corner and T-joints, these shall be equal to T1 /4, but need not exceed 10mm

j: The orientation of the two members in the joints may vary from 135° to 180° for butt joints, or 45° to 135° for corner joints, or 45° to 90° for T joint

l: Weld size .(_S_) shall be based on joints welded flush

m: For flare-V-groove welds and flare-bevel-groove welds to rectangular tubular sections, r shall be as two times the wall thickness

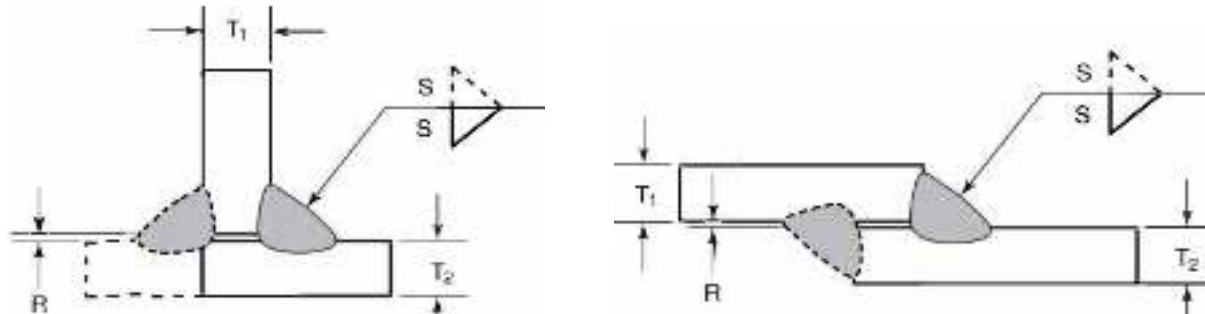
n: For flare-V-groove welds to surfaces with different radii r, the smaller r shall be used

Fillet weld (12)

T-joint (T)

Corner joint (C)

Lap joint (L)

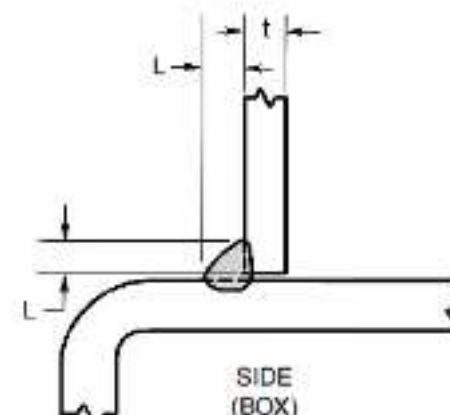
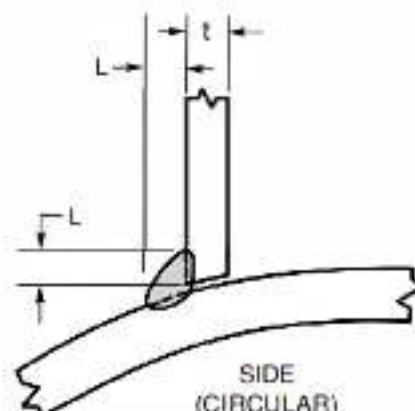
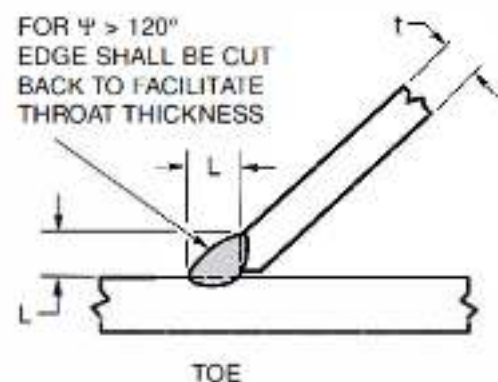
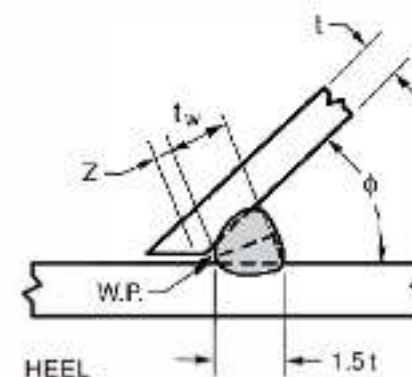
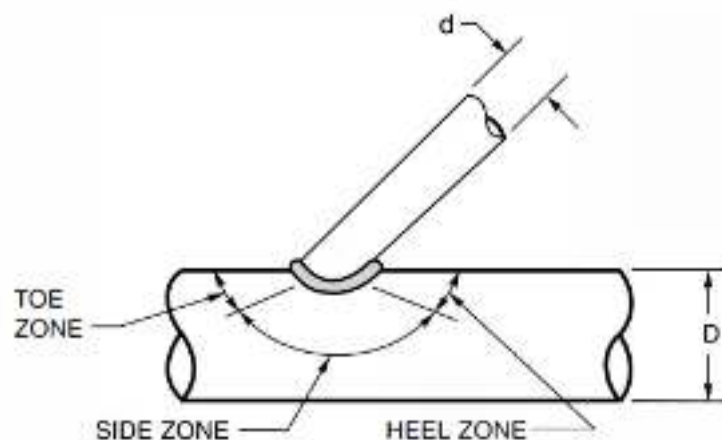
FILLET
PLATE

Weld Process	Joint Designation (AWS D1.1)	Base Metal thickness (mm)		Root Gap		Root Face		Radius		Depth of Groove	Weld size S	Position	Note
				R (mm)		f (mm)		r (mm)					
		(U: unlimited)		Recommend	Limited	Recommend	Limited	Recommend	Limited	D, D1, D2 (mm)	recommend		
		T1	T2										
FCAW	TC-F12-GF	<75	<75	0	≤ 2							ALL	1
FCAW	TC-F12a-GF	≥75	≥75	0	≤ 2							ALL	1
FCAW	L-F12-GF	<75	<75	0	≤ 2							ALL	1
FCAW	L-F12a-GF	≥75	≥75	0	≤ 2							ALL	1

Note

1: Fillet weld size refer to fabrication drawing or weld map

Figure 10.5 (AWS D1.1/D1.1M) Fillet weld Prequalified Tubular Joint



MIN L FOR

	$E = 0.7t$	$E = t$	$E = 1.07t$
HEEL $< 60^\circ$	$1.5t$	$1.5t$	LARGER OF $1.5t$ OR $1.4t + Z$
SIDE $\leq 100^\circ$	t	$1.4t$	$1.5t$
SIDE $100 - 110^\circ$	$1.1t$	$1.6t$	$1.75t$
SIDE $110 - 120^\circ$	$1.2t$	$1.8t$	$2.0t$
TOE $> 120^\circ$	t BEVEL	$1.4t$ BEVEL	FULL BEVEL 60–90° GROOVE

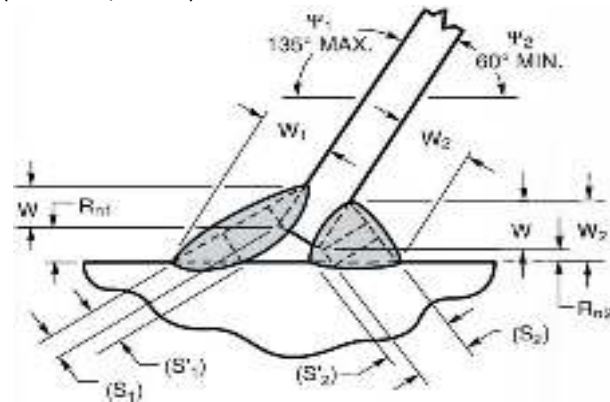
Notes:

1. t = thickness of thinner part.
2. L = minimum size (see 10.2.1 which may require increased weld size for combinations other than 36 ksi [250 MPa] base metal and 70 ksi [485 MPa] electrodes).
3. Root opening 0 in to 3/16 in [5 mm] (see 7.21).
4. Not prequalified for $\phi < 30^\circ$. For $\phi < 60^\circ$, the Z loss dimensions in Table 10.5 apply. See Table 10.12 for welder qualification position requirements.
5. See 10.5.2.2 for limitations on $\beta = d/D$.
6. Ψ = dihedral angle.

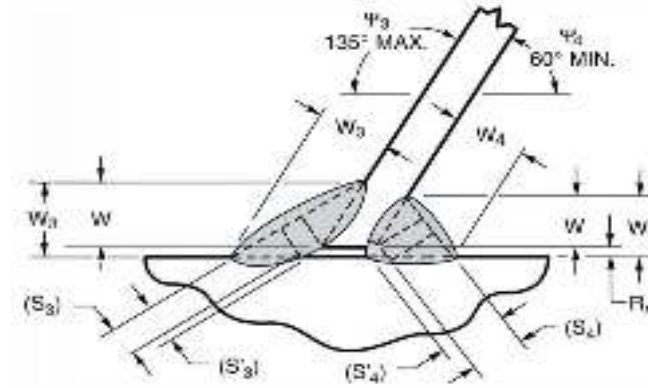
FILLET
PIPE

Figure 5.4 (AWS D1.1/D1.1M) Skewed T - Joint

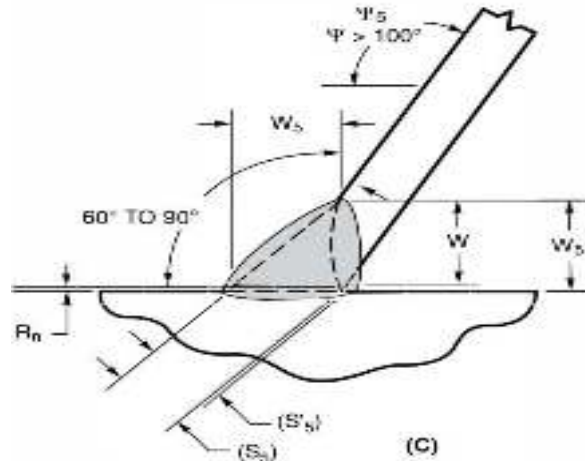
DETAIL A



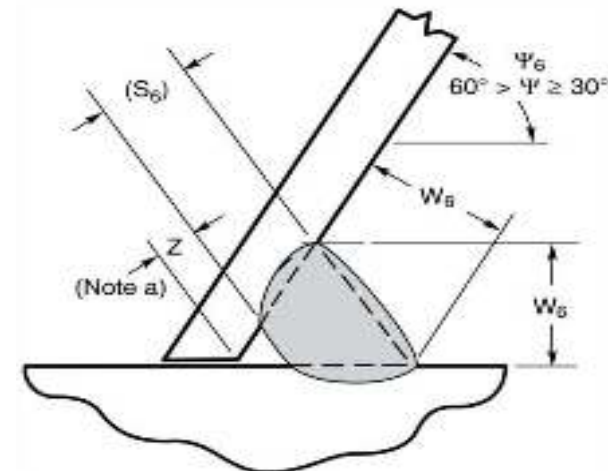
DETAIL B



DETAIL C



DETAIL D

SKEWED
(Plate)

Note a: Detail D, apply Z loss dimension of table 4.2 to determine effective throat

b: Detail D, shall not be under 30°. Welder qualify see table 6.10

1: S_n, S'_n weld size dependent on magnitude of Root open R_n

2: t is thickness of thinner part

3: Minimum weld size of Skewed T-joint for detail A, B, C shall be conform with table 7.7

Table 4.2
Z Loss Dimension (Nontubular) (See 4.4.3.3)

Dihedral Angle Ψ	Position of Welding—V or OH			Position of Welding—H or F		
	Process	Z(in)	Z(mm)	Process	Z(in)	Z(mm)
$60^\circ > \Psi > 45^\circ$	SMAW	1/8	3	SMAW	1/8	3
	FCAW-S	1/8	3	FCAW-S	0	0
	FCAW-G	1/8	3	FCAW-G	0	0
	GMAW	N/A	N/A	GMAW	0	0
$45^\circ > \Psi > 30^\circ$	SMAW	1/4	6	SMAW	1/4	6
	FCAW-S	1/4	6	FCAW-S	1/8	3
	FCAW-G	3/8	10	FCAW-G	1/4	6
	GMAW	N/A	N/A	GMAW	1/4	6

Table 7.7
Minimum Fillet Weld Sizes (See 7.13)

Base Metal Thickness (T) ^a		Minimum Size of Fillet Weld ^b	
in	mm	in	mm
T ≤ 1/4	T ≤ 6	1/8	3
1/4 < T ≤ 1/2	6 < T ≤ 12	3/16	5
1/2 < T ≤ 3/4	12 < T ≤ 20	1/4	6
3/4 < T	20 < T	5/16	8

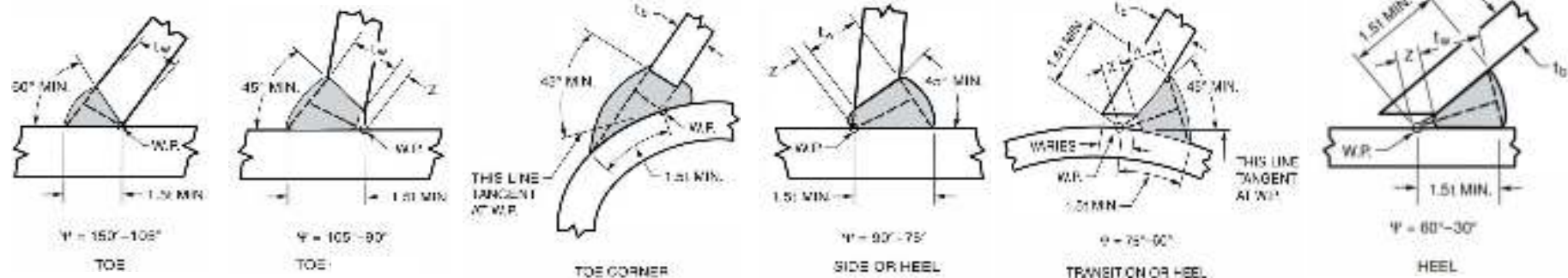
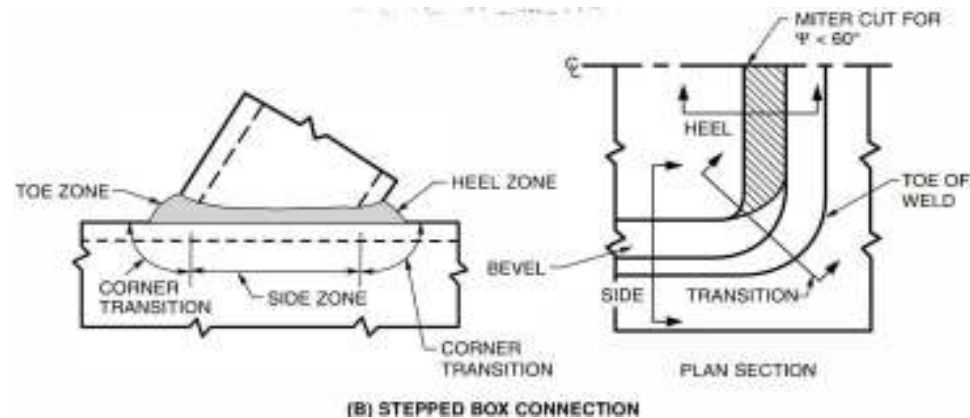
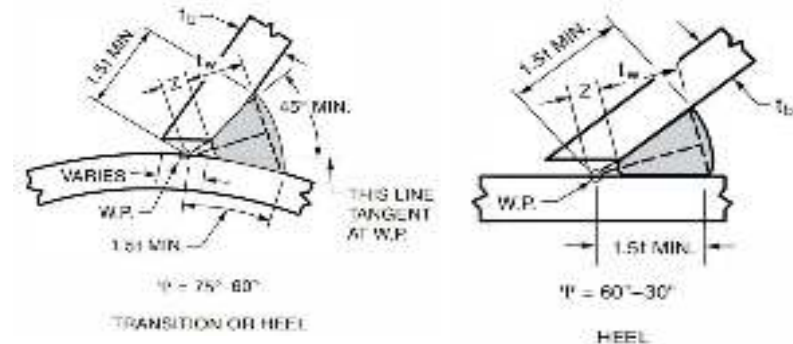
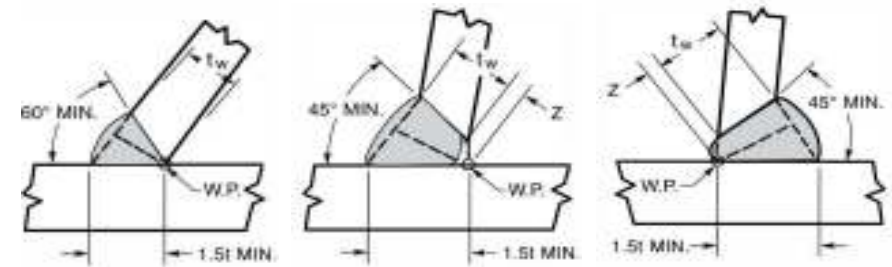
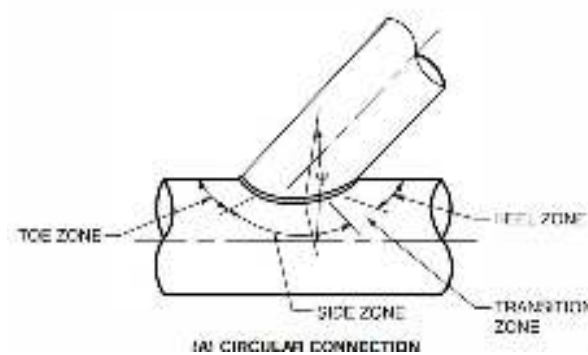
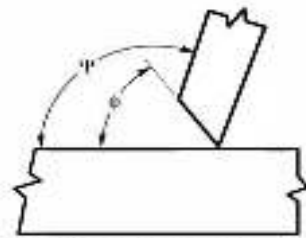
^a For nonlow hydrogen processes without preheat calculated in conformance with 6.8.4, T equals thickness of the thicker part joined; single-pass welds shall be used.

For nonlow hydrogen processes using procedures established to prevent cracking in conformance with 6.8.4 and for low-hydrogen processes, T equals thickness of the thinner part joined; single-pass requirement shall not apply.

^b Except that the weld size need not exceed the thickness of the thinner part joined.

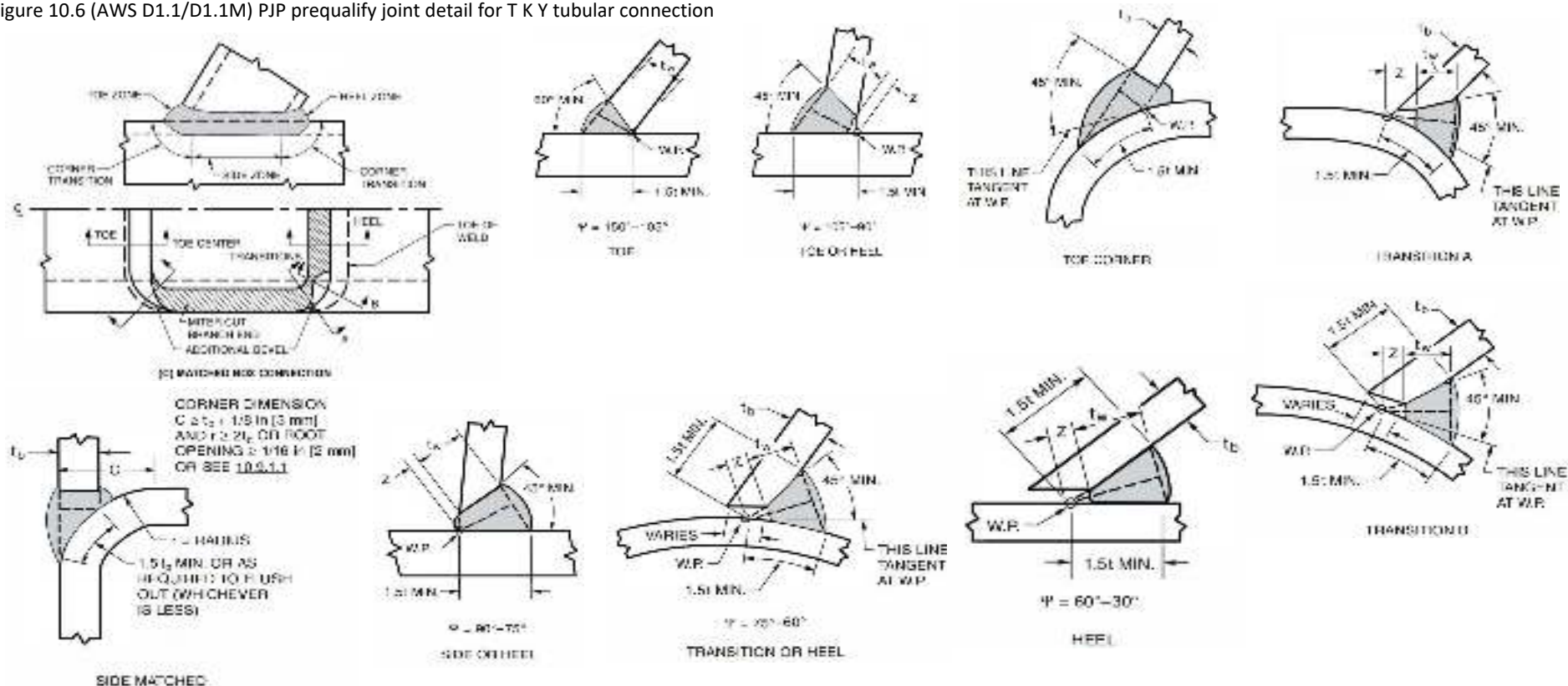
^c Minimum size for cyclically loaded structures shall be 3/16 in (5 mm).

Figure 10.6 (AWS D1.1/D1.1M) PJP prequalify joint detail for T K Y tubular connection



PJP - TKY
connection
(Pipe, Box)

Figure 10.6 (AWS D1.1/D1.1M) PJP prequalify joint detail for T K Y tubular connection



PJP - TKY
connection
(Pipe, Box)

	Prepared by QA Welding	Reviewed and Approved by	Client
Name	TRAN MINH PHUNG	LA VINH NHON (D.QA Manager)	
Signature			
Date	Jun 3, 2025	Jun 3, 2025	



PROCEDURE QUALIFICATION RECORD (PQR)

(AWS D1.1/D1.1M:2020 Structural Welding Code-Steel)



Company Name: **DaiDung Metallic Manufacture Construction & Trade Corporation**

PQR No.: **PQR-DD-FCAW-284.1 & 2**

Rev. No. **A**

Date: **May 3, 2024**

Page: **1 of 2**

BASE METALS	Specification	Type or Grade	AWS Group No.	Thickness	Size (NPS)	Schedule	Diameter
Base Material	ASTM A53	Gr B	I	5.5 mm	DN 80 mm	-	88.9 mm
Welded To	ASTM A53	Gr B	I	5.5 mm	DN 80 mm	-	88.9 mm
Backing Material	JIS G3101	SS 400	-	4 mm			
Other	-						

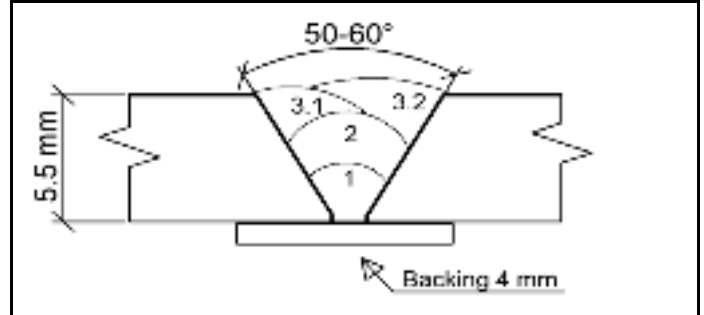
JOINT DETAILS

Groove Type	Double V Groove Butt Joint
Groove Angle	50-60°
Root Opening	5-6 mm
Root Face	Max 1 mm
Backgouging	N/A
Method	-

POSTWELD HEAT TREATMENT

Temperature	-
Time at Temperature	-
Other	-

JOINT DETAILS (Sketch)



PROCEDURE	PQR-DD-FCAW-284.1			PQR-DD-FCAW-284.2		
Weld Layer(s)	1	2	3	1	2	3
Weld Pass(es)	1	1	2	1	1	2
Process	FCAW	FCAW	FCAW	FCAW	FCAW	FCAW
Type (Manual, Mechanized, etc.)	Semi-Automatic					
Position	6G	6G	6G	6G	6G	6G
Vertical Progression	Uphill	Uphill	Uphill	Uphill	Uphill	Uphill
Filler Metal (AWS Spec.)	AWS A5.20	AWS A5.20	AWS A5.20	AWS A5.20	AWS A5.20	AWS A5.20
AWS Classification	E71T-1C	E71T-1C	E71T-1C	E71T-1C	E71T-1C	E71T-1C
Diameter (mm)	Ø 1.2	Ø 1.2	Ø 1.2	Ø 1.2	Ø 1.2	Ø 1.2
Manufacturer/Trade Name	Hyundai SC 71LH					
Shielding Gas (Composition)	CO2 (99.8%)	CO2 (99.8%)	CO2 (99.8%)	CO2 (99.8%)	CO2 (99.8%)	CO2 (99.8%)
Flow Rate (LPM)	20-22	20-22	20-22	20-22	20-22	20-22
Nozzle Size	No. 6	No. 6	No. 6	No. 6	No. 6	No. 6
Preheat Temperature (Min.)	34 °C (ambient)			34 °C (ambient)		
Interpass Temperature (Max.)	44 °C	92 °C	112 °C	48 °C	87 °C	110 °C
Electrical Characteristics						
Current Type & Polarity	DCEP	DCEP	DCEP	DCEP	DCEP	DCEP
Transfer Mode	-	-	-	-	-	-
Power Source Type (cc, cv, etc.)	CV	CV	CV	CV	CV	CV
Amps (A)	120-160	120-160	120-170	120-160	120-160	120-165
Volts (V)	20-22	20-22	20-22	20-22	20-22	20-22
Wire Feed Speed	(Amps)	(Amps)	(Amps)	(Amps)	(Amps)	(Amps)
Travel Speed (mm/min.)	220-262	171-174	186-204	209-254	164-171	178-204
Maximum Heat Input (kJ/mm)	1.0	1.2	1.2	1.0	1.3	1.2
Technique						
Stringer or Weave	Weave	Weave	Weave	Weave	Weave	Weave
Multi or Single Pass (per side)	Single	Single	Multipass	Single	Single	Multipass
Oscillation (Mech./Auto.)	-	-	-	-	-	-
Number of Electrodes	1	1	1	1	1	1
Contact Tube to Work Dist.(mm)	20	20	20	20	20	20
Peening	None	None	None	None	None	None
Interpass Cleaning	Grinding	Wire Brush	Grinding	Grinding	Grinding	Wire Brush
Other	-	-	-	-	-	-

	Prepared by QA Welding	Reviewed and Approved by
Name	Nguyen Quynh Sa	Truong Hoai Phong Vu (QA Manager)
Signature		
Date	May 3, 2024	May 3, 2024



TEST RESULTS

TESTS

	Type of Tests	Clause/Figure(s) Reference	Acceptance Criteria	Result	Remarks
✓	Visual Inspection	6.10.1	6.10.1	Acceptable	
✓	Radiographic Examination	6.10.2.1	6.10.2.2	Acceptable	
	Ultrasonic Testing	6.10.2.1	6.10.2.2	-	
✓	2 Transverse Root Bends	6.10.3.1/Fig. 6.8	6.10.3.3	Acceptable	
✓	2 Transverse Face Bends	6.10.3.1/Fig. 6.8	6.10.3.3	Acceptable	
	2 Longitudinal Root Bends	6.10.3.2/Fig. 6.8	6.10.3.3	-	
	2 Longitudinal Face Bends	6.10.3.2/Fig. 6.8	6.10.3.3	-	
	2 Side Bends	6.10.3.1/Fig. 6.9	6.10.3.3	-	
	4 Side Bends	6.10.3.1/Fig. 6.9	6.10.3.3	-	
✓	2 Tensile Tests	6.10.3.4/Fig. 6.10	6.10.3.5	Acceptable	
	All-Weld-Metal Tensions	6.10.3.6/Figs. 6.14 and 6.18	6.15.1.3(2)	-	
	3 Macroetch	6.10.4 & 6.12.2	6.10.4.1	-	
	4 Macroetch	6.10.4	6.10.4.1	-	
	CVN Tests	6 Part D/Fig 6.28	6.27.7	-	

TENSILE TEST DETAILS

Specimen Number	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Tensile Load (kN)	Ultimate Unit Stress (MPa)	Type of Failure and Location
Sample 1	20.5	5.8	120.1	62.4	520	Ductile/Base metal
Sample 2	20.5	5.8	120.1	63.8	531	Ductile/Base metal

TOUGHNESS TEST DETAILS

Specimen Number	Notch Location	Specimen Size (mm)	Test Temperature	Absorbed Energy (J)	Percent Shear	Lateral Expansion	Average (J)

CERTIFICATION

Welder's Name	ID Number	Stamp Number	Tests Conducted by	
NGUYEN VAN THAO	-	DDC-WD-0673	Laboratory	IQTC
-	-	-	Test Number	2304/2-1/VL-IQTC
-	-	-	File Number	-

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Clause 10 of AWS D1.1/D1.1M, 2020 *Structural Welding Code-Steel*.

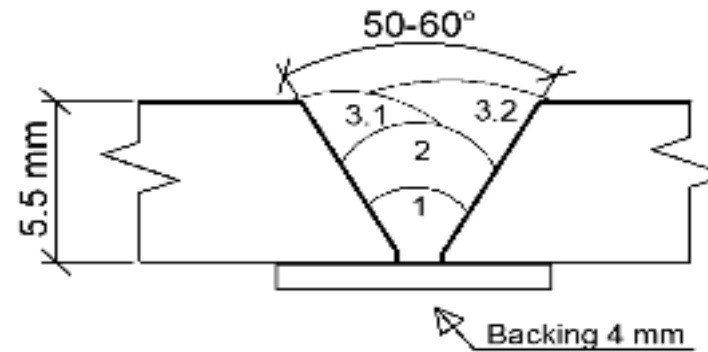
	Prepared by QA Welding	Reviewed and Approved by
Name	Nguyen Quynh Sa	Truong Hoai Phong Vu (QA Manager)
Signature		
Date	May 3, 2024	May 3, 2024

WELDING PARAMETERS RECORD (Doc No.: WPR-FCAW-284.1)

Company name: DAIDUNG CORPORATION
 PQR No.: PQR-DD-FCAW-284.1
 Test Coupon: DD-FCAW-284.1
 Test position: 6G
 Base metal:
 - Specification: ASTM A53 GrB
 - Dimension: D88.9 x 5.5 x 160 (mm) x 2 pcs
 - Heat/Product No.: 22MD3081
 - Certificate No.: QA280923-82(SSV2)
 Filler metal:
 - Specification: AWS A5.20
 - Classification: E71T-1C
 - Trade name: Hyundai SC71 LH (Ø1.2 mm)
 - Batch/Lot No.: 23120L71
 - Certificate No.: HDV-2024-0263
 Gas: CO₂ (99.8%)
 Welder Name: NGUYEN VAN THAO
 I.D.: DDC-WD-0673

Joint Sketch:

Date: Apr 15, 2024



Root face: 0-1 mm
 Root gap: 5-6 mm
 Groove angle: 50-60°

Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
Side A																
1-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	120-140	20-22	0	32	140	262	34 (Ambient)	20-22	0.5-0.7	8	2.5	
1-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	140-160	20-22	0	38	140	220	44	20-22	0.8-1	8	2.5	
2-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	120-140	20-22	0	48	140	174	61	20-22	0.8-1.1	10	2	
2-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	140-160	20-22	0	49	140	171	92	20-22	1-1.2	10	2	



(Doc No.: WPR-FCAW-284.1)

Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
3.1-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	120-140	20-22	0	44	140	190	81	20-22	0.8-1	10	1	
3.1-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	140-160	20-22	0	45	140	186	103	20-22	0.9-1.1	10	1	
3.2-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	135-155	20-22	0	41	140	204	100	20-22	0.8-1	10	1	
3.2-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	150-170	20-22	0	43	140	195	112	20-22	0.9-1.2	10	1	
		Conducted by QA Welding														
Name		NGUYEN QUYNH SA														
Signature																
Date		Apr 15, 2024														



(Doc No.: WPR-FCAW-284.2)

Layer/ pass	Process	Filler metal		Electric Current			Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		Volts (V)	min.						sec.	Width	
Side A																
1-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	120-140	20-22	0	33	140	254	34 (Ambient)	20-22	0.6-0.7	8	2.5	
1-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	140-160	20-22	0	40	140	209	48	20-22	0.8-1	8	2.5	
2-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	120-140	20-22	0	49	140	171	60	20-22	0.8-1.1	10	2	
2-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	140-160	20-22	0	51	140	164	87	20-22	1-1.3	10	2	



(Doc No.: WPR-FCAW-284.2)

Layer/ pass	Process	Filler metal		Electric Current		Volts (V)	Times		Length (mm)	Travel Speed (mm/min.)	Preheat/ Interpass Temp.(°C)	Shielding Gas Flow Rate (L/min.)	Heat input (kJ/mm)	Weld bead		Remark
		Specification/ Classification	Dia. (mm)	Polarity	Current (A)		min.	sec.						Width	Depth	
3.1-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	120-140	20-22	0	46	140	182	81	20-22	0.8-1	10	1	
3.1-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	140-160	20-22	0	47	140	178	102	20-22	0.9-1.2	10	1	
3.2-1	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	130-150	20-22	0	42	140	199	92	20-22	0.8-1	10	1	
3.2-2	FCAW	AWS A5.20/ E71T-1C	1.2	DCEP	145-165	20-22	0	41	140	204	110	20-22	0.9-1.1	10	1	
		Conducted by QA Welding														
Name		NGUYEN QUYNH SA														
Signature																
Date		Apr 15, 2024														

RADIOGRAPHIC EXAMINATION REPORT

GENERAL INFORMATION

Project Name:	NAGASAKI EAST PROJECT	Request No.:	NDT-WFO Q2024-04-17
Client Name:	DAI KUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION	Procedure No.	QTC-NDT-AWSRT-01
Item:	Procedure Qualification Record (PQR) Test	Revision No.:	01
Application Code:	AWS D1.1:2020	Drawing No.	N/A
Acceptance Criteria:	AWS D1.1:2020	Material	ASTM A53 GR B

Examination Technique:	DWSI	IQ Location:	File Side
Geometric Unsharpness (Ug):	Max 0.51	Density Range:	2.0 – 4.0
IQ Designation:	ASTM - 1A	Film Processing:	Manual
IQ Essential Wire identify:	5 Dia (mm)	3.2	Surface Condition: As Weld

EQUIPMENT

GAMMA - RAY						X - RAY				
Type	Source Size (mm)	Activity (Ci)	SOD (D) (mm)	OFD (d) (mm)	Qty of Film in Cassette	Expos. Voltage (kV)	Focal Spot Size (mm)	SOD (D) (mm)	OFD (d) (mm)	Qty of Film in Cassette
IR - 192	2 x 3	10	66.0	5.5-4	1	-	-	-	-	-

MATERIAL

Manufacture	Type	Film size (cm)	Intensifying Screen Type	Thickness of Screen (mm)	
				Front Side	Rear Side
Fuji	DXB	10x20	Lead	0.125	0.125

EVALUATION

[illegible]

Abbreviation:

မြန်မာ့စာပေအသင်းမှ အမွေရရှိခဲ့ပါသည်။

ACC: Accepted	BT: Burn Through	LP: Lack of Penetration	G: Crack	SI: Slag Inclusion	FM: Film Mark
REJ: Rejected	RE: Root Exposed	LF: Lack of Fusion	U: Undercut	TI: Taperless Inclusion	
RS: Reheat	IC: Internal Concavity	CP: Cluster Porosities	P: Porosity	ESI: Enlarged Slag Inclusion	

QTC

QC INSPECTOR

CLIENT

CLIENT

Evaluated: LUYN-HOANG PHU
Rev: 01/2012

Name:

4:00pm

Name	
------	--

Signature: _____

Set it.

Signature _____

Care

Date:

Order



Date: 13-Apr-24

RADIOGRAPHIC EXAMINATION REPORT

GENERAL INFORMATION

Project Name:	NAGASAKI EAST PROJECT	Request No.:	NDT-WFQ-Q2024-04-17
Client Name:	DAI DUNG METALLIC MANUFACTURE CONSTRUCTION AND TRADE CORPORATION	Procedure No.:	IGTC-NDT-AWS-RT-01
Item:	Procedure Qualification Record (PQR) Test	Revision No.:	01
Application Code:	AWS D1.1:2020	Drawing No.:	N/A
Acceptance Criteria:	AWS D1.1:2020	Material:	ASTM A53 GrB

Examination Technique:	DWIS	Q Location:	Film Size
Geometric Unsharpness (Ug):	Max. 0.54	Density Range:	2.0 ~ 4.0
IQ Designation:	ASTM - 1A	Film Processing:	Manual
IQ Essential Wire Identity:	5 Diameter	0.2	Surface Condition:
			As Weld

EQUIPMENT

GAMMA - RAY						X - RAY				
Type	Source Size (mm)	Activity (Ci)	SOO (D) (mm)	OFD (d) (mm)	Qty of Film in Cassette	Exami. Voltage (kV)	Focal Spot Size (mm)	SOO (D) (mm)	OFD (d) (mm)	Qty of Film in Cassette
IR - 192	2 x 3	10	66.0	5.5-4	1	-	-	-	-	-

MATERIAL

Manufacture	Type	Film size (cm)	Intensifying Screen Type	Thickness of Screen (mm)	
				Front Side	Rear Side
Fuji	IXRI	10x20	Lead	0.125	0.125

EVALUATION

[illegible]

Abbreviation:

(note: all diameters are in millimetres)

ACE Accepted

REF: 2593-0

ES: Reduced

BT: Bone Throat

RE: Open Letter

IC: Internal Circuits:

LP: Lack of Penetration

LE: Last of 5.931

CP: Cluster analysis

C. Grack

UNCLASSIFIED

P. Harsanyi

SI: Star inclusion

Eligibility for inclusion

ES: Expanded Senior Services

FM: Film (Analog)

en

1. **Introduction**

IQTC	QC INSPECTOR	CLIENT	CLIENT
Evaluated: HUYNH HOANG PHUC RT LEVEL 2 Signature:   Date: 19-Apr-24	Name: _____ Signature: _____ Date: _____	Name: _____ Signature: _____ Date: _____	Name: _____ Signature: _____ Date: _____





CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY

- ◆ Trụ sở chính/Head office: 27/29 Đường T4, KDC Bình Trị Đông, Phường Bình Trị Đông B, Quận Bình Tân, TP.HCM
◆ Điện thoại/Tel: 0279 3777 842 Fax: 02729 777840 Email: iqtc@iqtc.vn MST: 0313130702
◆ Phòng thí nghiệm/Laboratory: Ấp Sầm Tiền 2, Xã Đức Hòa Hạ, Huyện Đức Hòa, Tỉnh Long An



- Số TN/Ref: 2304/2-1/ML IQTC

- Phiếu yêu cầu số/Request No.: 03-220424/YCTN



KẾT QUẢ THỬ NGHIỆM

TEST REPORT

- Công trình/Project: **NAGASAKI EAST PROJECT**
- Hạng mục/Item: **PROCEDURE QUALIFICATION RECORD (PQR) TEST**
- Địa điểm/Location: **JAPAN**
- Cơ quan yêu cầu/Cient: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORP.**
- Đơn vị cung cấp mẫu/Supplier: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORP.**
- Tên mẫu/Name of sample: **Welding steel pipe OD 88.9x5.5x320 mm**
- Quy trình hàn/WPS No.: **WPS-DD-FCAW-284** - Tư thế hàn/Welding position: **6G**
- Yêu cầu kiểm tra/Testing requirements: **Thử kéo & Thử uốn/Tensile test & Bend test**
- Vật liệu/Material: **ASTM A53 - Gr.B** - Mã thợ hàn/Welder code: **DDC-WD-0673**
- Số lượng/Quantity: **01 Mẫu/Sample**
- Ngày nhận mẫu/Sampling date: **22/04/2024**
- Ngày thử nghiệm/Testing date: **22/04/2024**
- Thiết bị thử nghiệm/Testing equipment: **Máy thử kéo - nén WAW-1000/ Universal testing machine WAW-1000**
- Phương pháp thử/Test Method: **AWS D1.1:2020**
- Người chứng kiến/Witnesses: **Nguyen Quynh Sa**

Tên chỉ tiêu/Specification	Đơn vị/Unit	Kết quả/ Results	
		Mẫu 1/ Sample 1	Mẫu 2/ Sample 2
1.Thử kéo/Tensile test			
- Chiều dày mẫu/Thickness of specimen	mm	5.8	5.8
- Chiều rộng mẫu/Width of specimen	mm	20.5	20.5
- Tiết diện mẫu/Cross-sectional area	mm ²	120.1	120.1
- Lực kéo đứt/Tensile load	kN	62.4	63.8
- Giới hạn bền kéo/Tensile strength	MPa	520	531
- Vị trí mẫu đứt/Fracture position		Kim loại cơ bản/ Base metal	Kim loại cơ bản/ Base metal

Long An, ngày/date 23 tháng/month 04 năm/year 2024

CÔNG TY CP KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JSC (IQTC)

TRƯỞNG PHÒNG THỬ NGHIỆM
LAB MANAGER

Đ/D NHÓM THỬ NGHIỆM
LAB REP.

MAI HUNG KHANH

KS/ENG.DANG DON TUYEN



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY

✦ Trụ sở chính/ Head office: 27/29 Đường TA, KDC Bình Trị Đông, Phường Bình Trị Đông B, Quận Bình Tân, TP.HCM
 ✦ Điện thoại/ Tel: 0272 3777 642 Fax: 02723 777640 Email: iqtc@iqtc.vn MSV: 0313136782
 ✦ Phòng thí nghiệm/ Laboratory: Ấp Bình Tiến 2, Xã Đức Hòa Hạ, Huyện Đức Hòa, Tỉnh Long An



Số TN/Ref: 2304/2-1/ML-IQTC

Phiếu yêu cầu số/Request No.: 03-220424/YCTN

Tên chỉ tiêu/Specification		Đơn vị/ Unit	Kết quả/Test Results			
2. Thử uốn/Bend test			Mẫu 1/ Sample 1	Mẫu 2/ Sample 2	Mẫu 3/ Sample 3	Mẫu 4/ Sample 4
- Vị trí uốn/Location bend			Uốn mặt/ Face bend	Uốn mặt/ Face bend	Uốn chân/ Root bend	Uốn chân/ Root bend
- Chiều dày/Thickness of specimen		mm	5.8	5.8	5.8	5.8
- Chiều rộng/Width of specimen		mm	25.0	25.0	25.0	25.0
- Góc uốn/Bending angle		Độ/ Degree	180	180	180	180
- Đường kính gôi uốn/Plunger diameter		mm	38	38	38	38
- Kết quả/Test results			Không nứt/ No crack	Không nứt/ No crack	Không nứt/ No crack	Không nứt/ No crack

Ghi chú/Note:

* Phiếu kết quả này chỉ có giá trị cho mẫu thử do khách hàng mang tới. /This test report is only value for sample submitted by client

* Tên mẫu và tên khách hàng được ghi theo yêu cầu của khách hàng. /Name of sample and client are written as client's request.



CÔNG TY CỔ PHẦN KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JOINT STOCK COMPANY

- ♦ Trụ sở chính/Head office: 27/29 Đường 74, KDC Bình Trí Đông, Phường Bình Trí Đông B, Quận Bình Tân, TP.HCM
♦ Điện thoại/Tel: 0272 3777 642 Fax: 02725 777640 Email: iqtc@iqtc.vn MSN: 0313130792
♦ Phòng thí nghiệm/Laboratory: Ấp Bình Tiến 2, Xã Đức Hòa Hạ, Huyện Đức Hòa, Tỉnh Long An



- Số TN/Ref: 2604/1/VL-IQTC

- Phiếu yêu cầu số/Request No.: 02-250424/YCTN



KẾT QUẢ THỬ NGHIỆM
TEST REPORT

- Công trình/Project: **NAGASAKI EAST PROJECT**
- Hạng mục/Item: **WELDER QUALIFICATION TEST**
- Địa điểm/Location: **JAPAN**
- Cơ quan yêu cầu/Client: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORP.**
- Đơn vị cung cấp mẫu/Supplier: **DAIDUNG METALLIC MANUFACTURE CONSTRUCTION & TRADE CORP.**
- Tên mẫu/Name of sample: **Welding steel pipe OD 88.9x5.5x320 mm**
- Quy trình hàn/WPS No.: **WPS-DD-FCAW-284** - Tư thế hàn/Welding position: **5G**
- Yêu cầu kiểm tra/Testing requirements: **Thử uốn/Bend test**
- Vật liệu/Material: **ASTM A53 - Gr.B** - Mã thợ hàn/Welder code: **DDC-WD-0338**
- Số lượng/Quantity: **01 Mẫu/Sample**
- Ngày nhận mẫu/Sampling date: **25/04/2024**
- Ngày thử nghiệm/Testing date: **26/04/2024**
- Thiết bị thử nghiệm/Testing equipment: **Máy thử kéo - nén WAW-1000/ Universal testing machine WAW-1000**
- Phương pháp thử/Test Method: **AWS D1.1:2020**
- Người chứng kiến/Witnesses: **Nguyen Quynh Sa**

Tên chỉ tiêu/Specification	Đơn vị/Unit	Kết quả/Test Results			
		Mẫu 1/ Sample 1	Mẫu 2/ Sample 2	Mẫu 3/ Sample 3	Mẫu 4/ Sample 4
1. Thử uốn/Bend test					
- Vị trí uốn/Location bend		Uốn mặt/ Face bend	Uốn mặt/ Face bend	Uốn chân/ Root bend	Uốn chân/ Root bend
- Chiều dày/Thickness of specimen	mm	5.8	5.8	5.8	5.8
- Chiều rộng/Width of specimen	mm	25.0	25.0	25.0	25.0
- Góc uốn/Bending angle	Độ/°	180	180	180	180
- Đường kính gỏi uốn/Plunger diameter	mm	38	38	38	38
- Kết quả/Test results		Không nứt/ No crack	Không nứt/ No crack	Không nứt/ No crack	Không nứt/ No crack

Ghi chú/Note:

- * Phiếu kết quả này chỉ có giá trị cho mẫu thử do khách hàng mang tới /This test report is only valid for sample submitted by client.
* Tên mẫu và tên khách hàng được ghi theo yêu cầu của khách hàng /Name of sample and client are written as client's request.

Long An, ngày/date 26 tháng/month 04 năm/year 2024

CÔNG TY CP KIỂM ĐỊNH CHẤT LƯỢNG QUỐC TẾ
INTERNATIONAL QUALITY TESTING JSC (IQTC)

Đ/D NHÓM THỬ NGHIỆM

LAB REP.

MÃI HUNG KHANH

TRƯỞNG PHÒNG THỬ NGHIỆM

LAB MANAGER



KS/ENG.DANG DON TUYEN

CERTIFIED MATERIAL REPORT

CMR No. : **HDV – 2024 – 0263**

Date of Issue : 2024 – 02 – 29



Customer		G-TECH CORPORATION								
Trade Name		SC-71LH			Size		1.2 mm x 15 kg			
Lot No.		MF23120L71			Purchaser's Order No.		111-2023/GT/HW/PO			
Specification		AWS A5.20 E71T-1C EN ISO 17632-A-T 42 2 P C1 1 H5 * According to EN 10204-2004 : 3.1								
Welding Current	DCEP	Shielding Gas	CO ₂	Post Weld Heat Treatment (°C×hr.)			As Welded			
Mechanical Properties	Tensile Test			Impact Test (CVN - Joule)						
	YS (MPa)	TS (MPa)	EL (%)	Temp. (°C)	X ₁	X ₂	X ₃	Avg.		
	526	581	27	-20	75	86	84	82		
AWS Spec.	≥390	490~670	≥22	≥27						
Chemical Composition (%)										
Weld Metal	C	Si	Mn	P	S	Ni	Cr	Mo	V	Cu
	0.04	0.58	1.38	0.014	0.010	0.02	0.02	0.01	0.01	0.01
AWS Spec.	≤0.12	≤0.90	≤1.75	≤0.030	≤0.030	≤0.50	≤0.20	≤0.30	≤0.08	≤0.35
* Hydrogen Content (mℓ/100g) (AWS A4.3) : 4.36 * Radiographic Test : ACCEPT * Fillet Weld Test : GOOD										
Others * AS/NZS ISO 17632-A-T 42 2 P C1 1 H5 * FPC No. (CE-Mark): 0035-CPR-C771 * Q'ty: 7,200 kg										
We hereby certify that the material covered by this report has been inspected in accordance with and been found to meet the applicable requirements of specification and customer's requirements.										
 0035-CPR-C771 0035 10 / HWV-314 EN13479 Approved by  QA Manager Pham Dinh Anh Vu										

HYUNDAI WELDING VINA CO., LTD.

Road N2, Nhon Trach 5 IZ, Hiep Phuoc Ward, Nhon Trach District, Dong Nai Province, Vietnam.